

习题 1.1

2. 从某图书馆里任取一本书, 事件 A 表示“取到数学类图书”, 事件 B 表示“取到中文版图书”, 事件 C 表示“取到精装图书”;

(1) 试述 ABC 的含义; (2) 何种情况下, $C \subset B$?; (3) 何种情况下, $\bar{A} = B$?

(1) $ABC = \{\text{取到的书是中文版数学类图书, 但非精装版}\}$

(2) 若精装图书全是中文版, 则 $C \subset B$;

(3) 若中文版全是非数学类图书, 则 $\bar{A} = B$;

且非数学类图书全是中文版.

5. 试述下列事件的对立事件:

(1) $A =$ “射击三次皆命中目标”;

(2) $B =$ “甲产品畅销乙产品滞销”;

(3) $C =$ “加工四个零件至少有一个是合格品”.

(1) $A_i = \{\text{第 } i \text{ 次射击命中目标}\}, i=1, 2, 3$; $A = A_1 A_2 A_3$, 则

$\bar{A} = \overline{A_1 A_2 A_3} = \bar{A}_1 \cup \bar{A}_2 \cup \bar{A}_3 = \{\text{至少有一次未命中目标}\}$

(2) 设 $A(C) = \{\text{产品甲(乙)物(畅销)销}\}$, 则有: $B = A \cap C$, 则

$\bar{B} = \overline{A \cap C} = \bar{A} \cup \bar{C} = \{\text{产品甲滞销或产品乙物滞销}\}$

(3) 设 $A_i = \{\text{第 } i \text{ 个零件合格}\}, i=1, 2, 3, 4$; 则有: $C = \bigcup_{i=1}^4 A_i$, 则

$\bar{C} = \overline{\bigcup_{i=1}^4 A_i} = \bigcap_{i=1}^4 \bar{A}_i = \{\text{所有四个零件全不合格}\}$

习题 1.2

2. (1) 袋中有 7 个白球 3 个黑球, 现从中任取 2 个, 试求“所取两球颜色相同”的概率;

(2) 甲袋中有球 5 白 3 黑, 乙袋中有球 4 白 6 黑, 现从两袋中各取一球, 试求“所取两球颜色相同”的概率.

(1) 设 $A(B) = \{\text{取到 2 个白(黑)球}\}$, 则有: A, B 互斥, 且

$$P(\text{两球颜色相同}) = P(A \cup B) = P(A) + P(B) = \frac{|A|}{|S|} + \frac{|B|}{|S|} = \frac{C_7^2}{C_{10}^2} + \frac{C_3^2}{C_{10}^2} = \frac{8}{15}$$

(2) 设 $A(B) = \{\text{从甲(乙)袋中取到白球}\}$, 则有:

$$P(\text{两球颜色相同}) = P(A \bar{B} \cup \bar{A} B) = P(A \bar{B}) + P(\bar{A} B)$$

$$= \frac{5}{8} \times \frac{4}{10} + \frac{3}{8} \times \frac{6}{10} = \frac{38}{80} = \frac{19}{40}$$

$$= A_k$$
$$P(A_k) = \frac{|A_k|}{|S|} = \frac{a}{a+b}$$

27. $|S| = C_{a+b}^a$, $|A_k| = C_{a+b-1}^{a-1}$

$$P(A_k) = \frac{|A_k|}{|S|} = \frac{a}{a+b}$$

其他作法略！

4. (1) n 个人任意地坐成一排, 求“甲、乙两人坐在一起”的概率;
 (2) n 个人随机地围一圆桌而坐, 求“甲、乙相邻”的概率;
 (3) n 个男生、 m 个女生 ($m \leq n+1$) 坐成一排, 求“任意两个女生都不相邻”的概率.

(1) 设 $A = \{\text{甲、乙两人坐在一起的}\}$ ，易见， $|S_2| = A_n^n = n!$ ， $|A| = A_n^2 \cdot A_{n-1}^{n-1}$
 $= 2 \cdot (n-1)!$ ， $P(A) = \frac{|A|}{|S_2|} = \frac{2}{n}$ 。

(2) 设 $A = \{\text{甲, 乙, 丙}\}$ 则 $|\Omega| = \frac{n!}{n} = (n-1)!$

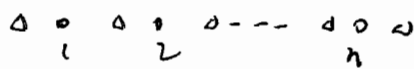
$$|A| = A_n \cdot (n-1)! / n-1 = 2 \cdot (n-2)! \quad , \quad P(A) = \frac{|A|}{|S_1|} = \frac{2}{n-1} .$$



(3). 设 $A = \{\text{所有女生不相邻}\}$, 易见, $|\Omega| = (n+m)!$.

$$|A| = n! \cdot A_{n+2}^m$$

$$P(A) = \frac{1A1}{1521} = \frac{n! \cdot A_{n+1}}{(n+m)!}$$



(1) “两数之和小于 $\frac{6}{5}$ ” 的概率; (2) “两数之积小于 $\frac{1}{4}$ ” 的概率.

记: $A(B) = \{ \text{两数和(积)小于 } \frac{b}{a} (\frac{1}{4}) \}$, 设两数分别为 x, y , 则

$$\Omega = (0, 1) \times (0, 1) = \{(x, y) \mid 0 < x, y < 1\}$$

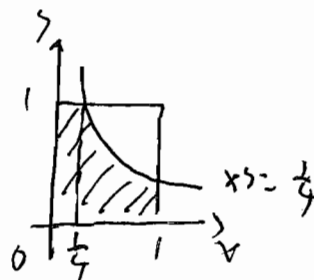
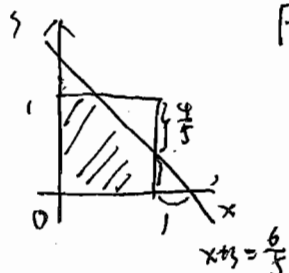
$$A = \{(x, y) \mid x+y < \frac{6}{5}, 0 < x, y < 1\}, B = \{(x, y) \mid xy < \frac{1}{4}, 0 < x, y < 1\}.$$

$$P(A) = \frac{\mu(A)}{\mu(\Omega)} = \frac{1 - \frac{1}{2}(\frac{\varphi}{5})^2}{1 \times 1} = \frac{17}{25}$$

$$p(B) = \frac{\mu(B)}{\mu(\Omega)} = \frac{\frac{1}{4} + \int_{\frac{1}{4}}^1 \frac{1}{4x} dx}{1 \times 1}$$

$$= \frac{1}{4} + \frac{1}{4} \ln 4$$

$$= \frac{1}{4} + \frac{1}{2} \ln 2$$



7. 设 A, B 为两事件, 且 $P(A)=0.4$, $P(B)=0.7$, 问:

(1) 在什么条件下, $P(AB)$ 取得最大值, 最大值是多少?

(2) 在什么条件下, $P(AB)$ 取得最小值, 最小值是多少? 若 $P(B)=0.5$, 结果又如何?

(1) 由 $AB \subset B$, 则有 $P(AB) \leq P(B)$, 也即 $P(AB) \leq \min\{P(A), P(B)\} = 0.4$

若 $A \subset B$, 则 $AB = A$, $P(AB) = P(A) = 0.4$.

(2) 由 $P(AB) = P(A) + P(B) - P(A \cup B) = 1.1 - P(A \cup B) \geq 0.1$.

若 $AB \neq \emptyset$, 且 $A \cup B = \Omega$, 则有 $P(AB) = 0.1$;

若 $P(B)=0.5$, 由 $P(AB) = P(A) + P(B) - P(A \cup B) = 0.9 - P(A \cup B) \geq 0$.

若 $AB = \emptyset$, 则 $P(AB) = 0$.

8. 证明: (1) $P(AB) \geq P(A) + P(B) - 1$;

(2) $P(A_1 A_2 \cdots A_n) \geq P(A_1) + P(A_2) + \cdots + P(A_n) - (n-1)$.

(1) $P(AB) = P(A) + P(B) - P(A \cup B) \geq P(A) + P(B) - 1$;

(2) $n=2$ 时, 易知成立, 假设 $n=k$ 时, $P(A_1 A_2 \cdots A_k) \geq P(A_1) + \cdots + P(A_k) - (k-1)$, 则

$$\begin{aligned} n=k+1 \text{ 时, } P(\bigcap_{i=1}^{k+1} A_i) &= P(\bigcap_{i=1}^k A_i \cap A_{k+1}) \geq P(\bigcap_{i=1}^k A_i) + P(A_{k+1}) - 1 \\ &\geq \sum_{i=1}^k P(A_i) - (k-1) + P(A_{k+1}) - 1 = \sum_{i=1}^{k+1} P(A_i) - k \end{aligned}$$

从而, $\forall n \geq 2$, $P(A_1 A_2 \cdots A_n) \geq P(A_1) + \cdots + P(A_n) - (n-1)$

10. (1) 设事件 A, B, C 同时发生必导致事件 D 发生, 证明:

$$P(A) + P(B) + P(C) \leq 2 + P(D); \quad \underline{ABC \subset D \Rightarrow P(ABC) \leq P(D)}$$

(2) 设 $P(A)=x, P(B)=2x, P(C)=3x$, 且 $P(AB)=P(BC)$, 证明: $x \leq \frac{1}{4}$.

$$\begin{aligned} (1) \quad 2[P(A) + P(B) + P(C)] &= [P(A) + P(B)] + [P(A) + P(C)] + [P(B) + P(C)] \\ &= P(AB) + P(A \cup B) + P(AC) + P(A \cup C) + P(BC) + P(B \cup C) \\ &\leq 3 + P(AB) + P(AC) + P(BC) = 3 + P(AB \cup AC \cup BC) + P(ABC) + P(ABC) \\ &\quad + P(ABC) - P(ABC) \leq 4 + 2P(ABC) \leq 4 + 2P(D) \end{aligned}$$

$$(2) \quad P(BC) = P(B) + P(C) - P(B \cup C) = 5x - P(B \cup C) \geq 5x - 1$$

$$\geq P(AB)$$

$$x = P(A) \geq P(AB)$$

$$\Rightarrow x \leq \frac{1}{4}$$

11. (1) 利用概率方法证明下列恒等式: 设 $a, b (a < b)$ 为任意正整数, 则恒有:

$$1 + \frac{b-a}{b-1} + \frac{(b-a)(b-a-1)}{(b-1)(b-2)} + \dots + \frac{(b-a) \times \dots \times 2 \times 1}{(b-1) \dots (a+1)a} = \frac{b}{a};$$

证: $\frac{a}{b} + \frac{a(b-a)}{b(b-1)} + \frac{a(b-a)(b-a-1)}{b(b-1)(b-2)} + \dots + \frac{(b-a) \times \dots \times 2 \times 1}{b(b-1) \dots (a+1)} = 1$

设袋中有 b 只球, 其中 a 只黑球, $b-a$ 只白球, 现将球一一取出, 取后不放回, 直至取出黑球时为止, 记 $A_i = \{\text{第 } i \text{ 次取出黑球}\}$, $i = 1, 2, \dots, b-a$;

则有: $A_1 \cup \bar{A}_1 A_2 \cup \bar{A}_1 \bar{A}_2 A_3 \cup \dots \cup \bar{A}_1 \bar{A}_2 \dots \bar{A}_{b-a} A_{b-a} = \Omega$ 且

$$\begin{aligned} & P(A_1) + P(\bar{A}_1 A_2) + P(\bar{A}_1 \bar{A}_2 A_3) + \dots + P(\bar{A}_1 \bar{A}_2 \dots \bar{A}_{b-a} A_{b-a}) \\ &= \frac{a}{b} + \frac{a(b-a)}{b(b-1)} + \frac{a(b-a)(b-a-1)}{b(b-1)(b-2)} + \dots + \frac{(b-a) \times \dots \times 2 \times 1}{b(b-1) \dots (a+1)} \end{aligned}$$

$\begin{matrix} 0 & 0 \\ 1 & 2 \\ 0 & 0 & 0 \\ 1 & 2 & 3 \end{matrix}$

$$= 1$$

(2) 试构造概率模型证明恒等式:

$$1 + \frac{N-n}{N} \frac{n+1}{n} + \frac{(N-n)(N-n-1)}{N^2} \frac{n+2}{n} + \dots + \frac{(N-n)(N-n-1) \dots 2 \times 1}{N^{N-n}} \frac{N}{n} = \frac{N}{n}, \quad N > n$$

证: $\frac{n}{N} + \frac{n+1}{N} \frac{N-n}{N} + \frac{(N-n)(N-n-1)}{N \cdot N} \frac{n+2}{N} + \dots + \frac{N-n}{N} \cdot \frac{N-n-1}{N} \dots \frac{1}{N} = 1$

设袋中有 N 只球, 其中 n 只黑球, $N-n$ 只白球; 现将球一一取出, 取后不放回, 且每一球就放回一只黑球, 直至取出黑球为止, 记 $A_i = \{\text{第 } i \text{ 次取出黑球}\}$,

$$P(A_1) = \frac{n}{N}, \quad P(\bar{A}_1 A_2) = P(\bar{A}_1) P(A_2 | \bar{A}_1) = \frac{N-n}{N} \frac{n+1}{N}$$

$$P(\bar{A}_1 \bar{A}_2 A_3) = P(\bar{A}_1) P(\bar{A}_2 | \bar{A}_1) P(A_3 | \bar{A}_1 \bar{A}_2) = \frac{N-n}{N} \frac{N-n-1}{N} \frac{n+2}{N}$$

⋮

$$\begin{aligned} P(\bar{A}_1 \bar{A}_2 \dots \bar{A}_{N-n} A_{N-n+1}) &= P(\bar{A}_1) P(\bar{A}_2 | \bar{A}_1) \dots P(A_{N-n+1} | \bar{A}_1 \bar{A}_2 \dots \bar{A}_{N-n}) \\ &= \frac{N-n}{N} \cdot \frac{N-n-1}{N} \dots \frac{1}{N} \cdot 1 \end{aligned}$$

习题 1.3

$$A_1 \cup \bar{A}_1 A_2 \cup \bar{A}_1 \bar{A}_2 A_3 \cup \dots \cup \bar{A}_1 \bar{A}_2 \dots \bar{A}_{N-n} A_{N-n+1} = \Omega$$

3. 设 $0 < P(A) < 1, 0 < P(B) < 1$, 已知 $P(B|A) > P(B|\bar{A})$, 证明: $P(A|B) > P(A|\bar{B})$.

由 $P(B|A) > P(B|\bar{A})$, 得: $\frac{P(AB)}{P(A)} > \frac{P(\bar{A}B)}{P(\bar{A})} = \frac{P(B) - P(AB)}{1 - P(A)}$ 证得:

$$P(AB)[1 - P(A)] > [P(B) - P(AB)]P(A), \text{ 即: } P(AB) > P(A)P(B), \text{ 且得:}$$

$$P(AB)[1 - P(B)] > [P(A) - P(AB)]P(B), \text{ 也证得:}$$

$$\frac{P(AB)}{P(B)} > \frac{P(A) - P(AB)}{1 - P(B)} = \frac{P(A\bar{B})}{P(\bar{B})}, \text{ 证得:}$$

$$P(A|B) > P(A|\bar{B})$$

4. (1) 设一批产品中一、二、三等品各占60%、35%、5%，从中任取一件，结果不是三等品，求“取到的是一等品”的概率；

记 $A_i = \{\text{取到 } i \text{ 等品}\}$, $i=1, 2, 3$. 由此得, $P(A_1) = \frac{3}{5}$, $P(A_2) = \frac{7}{20}$, $P(A_3) = \frac{1}{20}$.

解法1:

$$P(A_1 | \bar{A}_3) = \frac{P(A_1 \bar{A}_3)}{P(\bar{A}_3)} = \frac{P(A_1)}{1 - P(A_3)} = \dots$$

解法2: $P(A_1 | \bar{A}_3) = \frac{60}{60+35}$ (假设共有100件)

(2) 设10件产品中有4件是不合格品，从中任取两件，已知其中一件是不合格品，求“另一件也是不合格品”的概率。

解法1: 记 $A = \{\text{两件中有一件是不合格品}\}$, $B = \{\text{两件都是不合格品}\}$

$$P(B | A) = \frac{C_4^2}{C_{10}^2 - C_6^2} = \frac{1}{5}.$$

解法2: 记每件产品为一不放回抽取(不放回). 令 $A_i = \{\text{第 } i \text{ 次抽取到不合格品}\}$, $i=1, 2$;

$$P(A_2 | A_1 \cup A_2) = \frac{P(A_1 A_2 \cap (A_1 \cup A_2))}{P(A_1 \cup A_2)} = \frac{P(A_1 A_2)}{P(A_1) + P(A_2) - P(A_1 A_2)} = \frac{1}{5}, \text{ 这里,}$$

$$P(A_1) = \frac{4}{10}, \quad P(A_2) = \frac{C_4^1 C_6^1}{A_{10}^2} = \frac{4}{10}.$$

0 0
1 2

$$P(A_1 A_2) = \frac{A_4^2}{A_{10}^2}$$

5. 在数集 $\{1, 2, \dots, 100\}$ 中随机地取一数，已知取到的数不能被2整除，求“其能被3或5整除”的概率。

记 $A_i = \{\text{取到的数能被 } i \text{ 整除}\}$. 易见, $A_i A_j = A_{[i, j]}$. $[i, j]$ 表示 i, j 的最小公倍数.

且 $P(A_i) = \frac{[\frac{100}{i}]}{100}$, $[x]$ 表示不大于 x 的最大整数. 从而,

$$P(A_3 \cup A_5 | \bar{A}_2) = \frac{P((A_3 \cup A_5) \cap \bar{A}_2)}{P(\bar{A}_2)} = \frac{P(A_3 \bar{A}_2 \cup A_5 \bar{A}_2)}{P(\bar{A}_2)} = \frac{P(A_3 \bar{A}_2) + P(A_5 \bar{A}_2) - P(A_3 A_5 \bar{A}_2)}{P(\bar{A}_2)}$$

$$= \frac{P(A_3) - P(A_3 A_2) + P(A_5) - P(A_5 A_2) - P(A_3 A_5) + P(A_3 A_5 A_2)}{1 - P(A_2)} = \frac{P(A_3) - P(A_6) + P(A_5) - P(A_{10}) - P(A_{15})}{1 - P(A_2)}$$

$$= \frac{\frac{33}{100} - \frac{16}{100} + \frac{20}{100} - \frac{10}{100} - \frac{6}{100} + \frac{3}{100}}{1 - \frac{50}{100}} = \frac{24}{50} = \frac{12}{25}.$$

(5)

6. 一批产品共100件，其中有次品10件，合格品90件；现从中任取一件，取后不放回，接连取三次，试求“第三次才取到合格品”的概率。

设 $A_i = \{\text{第 } i \text{ 次取到合格品}\}$, $i = 1, 2, 3$;

① 依题意: $P(\bar{A}_1 \bar{A}_2 A_3) = P(\bar{A}_1)P(\bar{A}_2|\bar{A}_1)P(A_3|\bar{A}_1 \bar{A}_2) = \frac{10}{100} \times \frac{9}{99} \times \frac{90}{98}$

② 依题意: $P(\bar{A}_1 \bar{A}_2 A_3) = \frac{A_{10}^2 \cdot C_1^1}{A_{100}^3}$

0 2 2
1 2 3

7. 居民甲给居民乙打电话，但忘了其电话号码最后一位数字；因而随机拨号，如果拨完整个电话号码视作完成一次拨号，且假设乙的电话不占线，试求：

- (1) “直到第 k 次才拨通乙的电话”的概率；
(2) “不超过 k 次而拨通乙的电话”的概率。

设 $A_i = \{\text{第 } i \text{ 次拨通}\}$, $i = 1, 2, \dots, 10$. (1) $P(\bar{A}_1 \dots \bar{A}_{k-1} A_k) = P(\bar{A}_1)P(\bar{A}_2|\bar{A}_1) \dots P(A_k|\bar{A}_1 \dots \bar{A}_{k-1})$

$= \frac{9}{10} \times \frac{8}{9} \times \dots \times \frac{1}{11-k} = \frac{1}{10}$. (2) $P(A_1 \cup \bar{A}_1 A_2 \cup \dots \cup \bar{A}_1 \dots \bar{A}_{k-1} A_k)$

$= P(A_1) + P(\bar{A}_1 A_2) + \dots + P(\bar{A}_1 \dots \bar{A}_{k-1} A_k)$

$= \frac{1}{10} + \frac{1}{10} + \dots + \frac{1}{10} = \frac{k}{10}$.

8. 以 A_t 表示“一分子在 $(0, t]$ 内不与其他分子碰撞”，假设“分子在 $(0, t]$ 内不发生碰撞的条件下，在 $(t, t + \Delta t]$ 内发生碰撞”的概率为 $\lambda \Delta t + o(\Delta t)$ ，试求 $P(A_t)$ 。

由此得: $P(A_{t+\Delta t} | A_t) = \lambda \Delta t + o(\Delta t)$ ($\Delta t > 0$ 足够小) 从而: $\frac{P(A_t A_{t+\Delta t})}{P(A_t)} = \frac{P(A_{t+\Delta t} - A_{t+\Delta t} | A_t)}{P(A_t)}$

$\frac{A_t \supset A_{t+\Delta t}}{\Delta t > 0} \quad \frac{P(A_t) - P(A_{t+\Delta t})}{P(A_t)} = \lambda \Delta t + o(\Delta t)$; 令 $f(t) = P(A_t)$, $t > 0$, 从而:

$\frac{f(t) - f(t+\Delta t)}{f(t)} = \lambda \Delta t + o(\Delta t)$. 也即: $\frac{f(t+\Delta t) - f(t)}{\Delta t} = -f(t) [\lambda + \frac{o(\Delta t)}{\Delta t}]$, 令 $\Delta t \rightarrow 0$.

从而: $f'(t) = -\lambda f(t)$; 同理, $f'(t) = -\lambda f(t)$. 从而: $f'(t) = -\lambda f(t)$. $\int \frac{df(t)}{f(t)} = \int -\lambda dt$

$\ln f(t) - \ln C = -\lambda t$, 从而: $f(t) = C \cdot e^{-\lambda t}$. $\forall \{t_n\}$: $t_1 > t_2 > \dots > t_n \rightarrow 0$, 从而:

$\{[0, t_n], n \geq 1\} \searrow \emptyset$, 从而: $\{A_{t_n}, n \geq 1\} \rightarrow \Omega$, 从而: $1 = P(\Omega) = P(\bigcap_{n=1}^{\infty} A_{t_n})$

$= \lim_{n \rightarrow \infty} P(A_{t_n}) = \lim_{n \rightarrow \infty} f(t_n) = \lim_{n \rightarrow \infty} C \cdot e^{-\lambda t_n} = C \cdot 1$; 也即: $P(A_t) = e^{-\lambda t}$.

9. 袋中有4白1红5只球, 现有5人依次从袋中各取一球, 取后不放回, 试求“第 i ($i=1, 2, \dots, 5$) 人取到红球”的概率.

$$\text{解: } P(A_1) = \frac{1}{5}, \quad P(A_2) = P(A_2 \cap (A_1 \cup \bar{A}_1)) = P(A_2 A_1 \cup A_2 \bar{A}_1) \\ = P(A_2 A_1) + P(A_2 \bar{A}_1) = P(A_1)P(A_2|A_1) + P(\bar{A}_1)P(A_2|\bar{A}_1) = \frac{1}{5} \times 0 + \frac{4}{5} \times \frac{1}{4} = \frac{1}{5}$$

$$\text{或 } P(A_i) \xrightarrow{\text{对称性}} \frac{C_4^1}{A_5^1} = \frac{1}{5}, \quad P(A_3) = \frac{A_4^1}{A_5^1} = \frac{1}{5}$$

$$\dots, P(A_5) = \frac{1}{5}.$$

10. 两台车床加工同样的零件, “第一台出现不合格品”的概率是0.03, “第二台出现不合格品”的概率是0.06, 加工出来的零件放在一起, 并且已知第一台加工的零件比第二台加工的零件多一倍,

(1) 试求“任取一个零件是合格品”的概率;

(2) 如果取出的零件是不合格品, 求“它是由第二台车床加工”的概率.

设 $A = \{\text{任取一个零件是合格品}\}$, $B_i = \{\text{任取一个零件是合格品由第 } i \text{ 台车床加工}\}$, $i=1, 2$;

由题设, $P(B_1) = \frac{2}{3}$, $P(B_2) = \frac{1}{3}$, $P(A|B_1) = 0.03$, $P(A|B_2) = 0.06$.

$$(1) P(A) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2) = \frac{2}{3} \times 0.03 + \frac{1}{3} \times 0.06 = 0.02 + 0.02 = 0.04$$

$$P(\bar{A}) = 1 - P(A) = 0.96$$

$$(2) P(\bar{A}|B_2) = \frac{P(B_2)P(\bar{A}|B_2)}{P(B_1)P(\bar{A}|B_1) + P(B_2)P(\bar{A}|B_2)} = \frac{0.02}{0.02 + 0.02}$$

$$= 0.5$$

11. (1) 甲袋中有2只白球1只黑球, 乙袋中有1只白球2只黑球, 今从甲袋中任取一球放入乙袋, 再从乙袋中任取一球, 求“此球是白球”的概率;

设 $A = \{\text{从乙袋中取出白球}\}$, $B = \{\text{从甲袋中取出白球}\}$

或 $A(B) = \{\text{从乙(甲)袋中取出白球}\}$.

$$P(A) = P(A \cap (B \cup \bar{B})) = P(AB \cup A\bar{B}) = P(AB) + P(A\bar{B})$$

$$= P(B)P(A|B) + P(\bar{B})P(A|\bar{B}) = \frac{2}{3} \times \frac{2}{4} + \frac{1}{3} \times \frac{1}{4} = \frac{5}{12}$$

(2) 甲袋中有5只黑球和1只白球，乙袋中有6只黑球，每次从甲乙两袋中随机各取一球交换放入另一袋中，依次取三次，求“白球仍在甲袋中”的概率；

令 $A_i = \{\text{第 } i \text{ 次交换后，白球仍在甲袋中}\}$, $i=1, 2, 3$; 可见, $P(A_1) = \frac{5}{6}$.

$$P(A_2) = P(A_2 | A_1) + P(A_2 | \bar{A}_1) = P(A_1)P(A_2 | A_1) + P(\bar{A}_1)P(A_2 | \bar{A}_1) \\ = \frac{5}{6} \times \frac{5}{6} + \frac{1}{6} \times \frac{1}{6} = \frac{13}{18}$$

$$P(A_3) = P(A_3 | A_2) + P(A_3 | \bar{A}_2) = P(A_2)P(A_3 | A_2) + P(\bar{A}_2)P(A_3 | \bar{A}_2) \\ = \frac{13}{18} \times \frac{5}{6} + \frac{5}{18} \times \frac{1}{6} = \frac{5 \times 14}{18 \times 6} = \frac{35}{54}$$

(3) 在 n 只袋中各有4只白球、6只黑球，而另一袋中有5只白球、5只黑球；今从 $n+1$ 只袋中任选一袋，从中随机取两球，都是白球，在这种情况下，有5只黑球、3只白球留在所选袋中的概率为 $\frac{1}{7}$ ，试求 n 。

记: $A = \{\text{取出2只白球}\}$, $B = \{\text{取出白球来自 } n+1 \text{ 只袋中的 } n \text{ 只袋}\}$, 由此得

$$P(B) = \frac{n}{n+1}, \quad P(\bar{B}) = \frac{1}{n+1}, \quad \text{且}$$

$$P(A) = P(B)P(A|B) + P(\bar{B})P(A|\bar{B}) = \frac{n}{n+1} \cdot \frac{C_4^2}{C_{10}^2} + \frac{1}{n+1} \cdot \frac{C_5^2}{C_{10}^2} = \frac{1}{7}$$

$$\text{得: } 7[6n+10] = 45(n+1), \quad 42n+70 = 45n+45, \quad 3n = 25$$

$$P(\bar{B}|A) = \frac{P(\bar{B})P(A|\bar{B})}{P(A)} = \frac{P(\bar{B}) \cdot P(A|\bar{B})}{P(B)} = \frac{C_5^2}{n \cdot C_4^2 + C_5^2} = \frac{10}{6n+10} = \frac{1}{7}$$

$$n = 10$$

12. (1) 某商店正在销售10台彩电，其中7台是一级品，3台是二级品；某人到商店时，彩电已售出2台，试求“此人能买到一级品”的概率； $\left(\frac{7}{10}\right)$

记 $A = \{\text{此人买到一级品}\}$, $B_i = \{\text{已售出 } i \text{ 台一级品}\}$, $i=0, 1, 2$.

$$P(A) = P(B_0)P(A|B_0) + P(B_1)P(A|B_1) + P(B_2)P(A|B_2)$$

$$= \frac{C_3^2}{C_{10}^2} \cdot \frac{7}{8} + \frac{C_7^1 C_3^1}{C_{10}^2} \cdot \frac{6}{8} + \frac{C_7^2}{C_{10}^2} \cdot \frac{5}{8}$$

$$= \frac{21 + 7 \times 18 + 7 \times 15}{9 \times 5 \times 8} = \frac{7 \times (3 + 18 + 15)}{9 \times 5 \times 8} = \frac{7 \times 36}{9 \times 5 \times 8} = \left(\frac{7}{10}\right)$$

(2) 送检的两批灯泡在运输中各打碎一只；若每批10只，且第一批中有一只次品，第二批中有两只次品；现从剩下的灯泡中任取一只，求“取到次品”的概率。

设 $A = \{\text{取到次品}\}$, $B_i = \{\text{取到第 } i \text{ 批灯泡}\}$, $i = 1, 2$; $P(B_i) = \frac{1}{2}$, $i = 1, 2$;
 $P(A) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2) = \frac{1}{2} [P(A|B_1) + P(A|B_2)] = \frac{1}{2} \times [\frac{1}{10} + \frac{2}{10}] = \frac{3}{20}$.

另法: $P(A|B_1) = P(C_1|B_1)P(A|C_1B_1) + P(C_2|B_1)P(A|C_2B_1)$
 $= \frac{1}{10} \times 0 + \frac{9}{10} \times \frac{1}{9} = \frac{1}{10}$, 同理, $P(A|B_2) = \frac{2}{10}$.

$C_i = \{\text{打碎 } i \text{ 只次品}\}$, $i = 0, 1$;

——条件概率的公式

13. 有两箱零件，第一箱装50件，其中20件是一等品；第二箱装30件，其中18件是一等品；现从两箱中随意挑出一箱，然后从该箱中先后任取两个零件，求：

- (1) “第一次取出一等品”的概率；令 $A_i = \{\text{第 } i \text{ 次取到次品}\}$, $i = 1, 2$;
 (2) “第二次取出一等品”的概率； $B_i = \{\text{零件取自第 } i \text{ 箱}\}$, $i = 1, 2$; $P(B_i) = \frac{1}{2}$, $i = 1, 2$;
 (3) 在第一次取出一等品的条件下，“第二次取出的仍然是一等品”的概率；
 (4) 在第二次取出一等品的条件下，“第一次取出的仍然是一等品”的概率。

(1) $P(A_1) = P(B_1)P(A_1|B_1) + P(B_2)P(A_1|B_2) = \frac{1}{2} \times [\frac{2}{5} + \frac{2}{3}] = 0.5$;

(2) $P(A_2) = P(B_1)P(A_2|B_1) + P(B_2)P(A_2|B_2) = \frac{1}{2} \times [\frac{2}{5} + \frac{2}{3}] = 0.5$;

(3) $P(A_2|A_1) = \frac{P(A_1A_2)}{P(A_1)} = \frac{P(B_1)P(A_1A_2|B_1) + P(B_2)P(A_1A_2|B_2)}{P(B_1)P(A_1|B_1) + P(B_2)P(A_1|B_2)} = \frac{\frac{C_{20}^2}{C_{50}^2} + \frac{C_{18}^2}{C_{30}^2}}{\frac{2}{5} + \frac{2}{3}} = \frac{C_{20}^2}{C_{50}^2} + \frac{C_{18}^2}{C_{30}^2}$

(4) $P(A_1|A_2) = \frac{P(A_1A_2)}{P(A_2)} = \frac{P(B_1)P(A_1A_2|B_1) + P(B_2)P(A_1A_2|B_2)}{P(B_1)P(A_2|B_1) + P(B_2)P(A_2|B_2)} = \frac{C_{20}^2}{C_{50}^2} + \frac{C_{18}^2}{C_{30}^2}$

(14) 设 $A = \{\text{检查出1件次品}\}$, $B_i = \{\text{检查出的产品中有 } i \text{ 件次品}\}$, $i = 0, 1, 2, 3, 4$. 且
 $P(B_0) = 0.37 = P(B_1)$, $P(B_2) = 0.18$, $P(B_3) = 0.06$, $P(B_4) = 0.02$
 可见, $A \subset \bigcup_{i=1}^4 B_i$, 且 $P(A) = P(A \cap \bigcup_{i=1}^4 B_i) = P(\bigcup_{i=1}^4 A \cap B_i) = \sum_{i=1}^4 P(A \cap B_i)$

14. 设某批产品共50件，其中有0,1,2,3,4件次品的概率分别为 0.37, 0.37, 0.18, 0.06, 0.02；现从该批产品中任取10件，检查出1件次品，试求“该批产品中次品不超过2件”的概率。

$= \sum_{i=1}^4 P(B_i)P(A|B_i) = 0.37 \times \frac{C_{49}^9}{C_{50}^{10}} + 0.18 \times \frac{C_{48}^9 C_1^1}{C_{50}^{10}} + 0.06 \times \frac{C_{47}^9 C_2^1}{C_{50}^{10}} + 0.02 \times \frac{C_{46}^9 C_3^1}{C_{50}^{10}}$

$P(B_1 \cup B_2 | A) = P(B_1|A) + P(B_2|A)$

$= \frac{P(A \cap B_1) + P(A \cap B_2)}{P(A)} = \frac{P(B_1)P(A|B_1) + P(B_2)P(A|B_2)}{P(A)}$

≈ 0.8

15. 设有来自三个地区的分别有10名, 15名和25名考生的报名表, 其中女生报名表分别有3份, 7份和5份; 现随机地抽取一个地区的报名表, 从中先后抽出两份,

(1) 求“先抽到的是一份女生表”的概率;

(2) 已知后抽到的是一份男生表, 求“先抽到的是一份女生表”的概率.

记 $A_i = \{\text{第 } i \text{ 次抽到女生表}\}, i=1, 2; B_j = \{\text{报名表取自第 } j \text{ 个地区}\}, j=1, 2, 3; P(B_j) = \frac{1}{3}, j=1, 2, 3;$

$$(1) P(A_1) = P(B_1)P(A_1|B_1) + P(B_2)P(A_1|B_2) + P(B_3)P(A_1|B_3)$$

$$= \frac{1}{3} \times \left[\frac{3}{10} + \frac{7}{15} + \frac{5}{25} \right]$$

$$(2) P(A_1|\bar{A}_2) = \frac{P(A_1\bar{A}_2)}{P(\bar{A}_2)} = \frac{P(B_1)P(A_1\bar{A}_2|B_1) + P(B_2)P(A_1\bar{A}_2|B_2) + P(B_3)P(A_1\bar{A}_2|B_3)}{P(B_1)P(\bar{A}_2|B_1) + P(B_2)P(\bar{A}_2|B_2) + P(B_3)P(\bar{A}_2|B_3)}$$

$$= \frac{\frac{C_3^1 C_7^1}{A_{10}^2} + \frac{C_7^1 C_8^1}{A_{15}^2} + \frac{C_5^1 C_5^1}{A_{25}^2}}{\frac{7}{10} + \frac{8}{15} + \frac{20}{25}}$$

16. 假设有两箱同类零件, 第一箱内装50件, 其中10件是一等品, 第二箱内装30件, 其中18件是一等品; 现从两箱中随机挑选一箱, 再从该箱中先后取出两个零件, 记 $A_i = \{\text{第 } i \text{ 次取出一等品}\}, B_i = \{\text{零件取自第 } i \text{ 箱}\}, P(B_i) = \frac{1}{2}, i=1, 2;$

(1) 求“先取出的零件是一等品”的概率;

(2) 已知先取出的零件是一等品, 求“后取出的零件仍是一等品”的概率

$$(1) P(A_1) = P(B_1)P(A_1|B_1) + P(B_2)P(A_1|B_2) = \frac{1}{2} \times \left[\frac{1}{5} + \frac{3}{5} \right] = \frac{2}{5}$$

$$(2) P(A_2|A_1) = \frac{P(A_1 A_2)}{P(A_1)} = \frac{P(B_1)P(A_1 A_2|B_1) + P(B_2)P(A_1 A_2|B_2)}{P(B_1)P(A_1|B_1) + P(B_2)P(A_1|B_2)} = \frac{\frac{A_{10}^2}{A_{50}^2} + \frac{C_{18}^2}{C_{30}^2}}{\frac{1}{5} + \frac{3}{5}}$$

17. 记 $A = \{\text{顾客买下这箱玻璃杯}\}, B_i = \{\text{所取箱中含有 } i \text{ 只次品}\}, i=0, 1, 2$ 且 $P(B_0) = 0.8, P(B_1) = P(B_2) = 0.1$.

17. 玻璃杯成箱出售, 每箱20只, 假设各箱有0, 1, 2只次品的概率分别为

0.8, 0.1, 0.1; 一个顾客欲购一箱玻璃杯, 在购买时售货员随机取一箱, 顾客开箱随机地查看4只, 若无次品, 就买下这箱玻璃杯, 否则退回; 试求:

(1) “顾客买下这箱玻璃杯”的概率;

(2) “在顾客买下的一箱中, 确实没有次品”的概率.

$$(1) P(A) = P(B_0)P(A|B_0) + P(B_1)P(A|B_1) + P(B_2)P(A|B_2)$$

$$= 0.8 \times 1 + 0.1 \times \frac{C_{19}^4}{C_{20}^4} + 0.1 \times \frac{C_{18}^4}{C_{20}^4}$$

$$(2) P(B_0|A) = \frac{P(A B_0)}{P(A)} = \frac{P(B_0)P(A|B_0)}{P(A)} = \frac{0.8 \times 1}{\dots}$$

18. 证明: $P(A|B) = P(A|BC)P(C|B) + P(A|B\bar{C})P(\bar{C}|B)$.

或: $P_B(A) = P_B(A|C)P_B(C) + P_B(A|\bar{C})P_B(\bar{C})$.

两边乘以 $P(B)$, 即:

$$\begin{aligned} P(AB) &= P(A|BC) \cdot P(BC) + P(A|B\bar{C})P(B\bar{C}) \\ &= P(ABC) + P(AB\bar{C}) \\ &= P(ABC \cup AB\bar{C}) \end{aligned}$$

19. (末步分析法)

(1) 连续 n 次掷一枚硬币, 第一次掷出正面的概率为 a , “第二次以后每次出现与前一次相同面”的概率为 b , 求“第 n 次掷出正面”的概率;

记 $A_i = \{\text{第 } i \text{ 次掷出正面}\}$, $i = 1, 2, \dots, n$; 易见, $P(A_1) = a$, $\forall i \geq 2$.

$$\begin{aligned} P(A_i) &= P(A_{i-1})P(A_i|A_{i-1}) + P(\bar{A}_{i-1})P(A_i|\bar{A}_{i-1}) = P(A_{i-1}) \cdot b + [1 - P(A_{i-1})] \cdot (1-b) \\ &= (2b-1)P(A_{i-1}) + 1-b, \quad \text{令 } P(A_i) = a_i, \quad i \geq 1, \quad \text{则有: } a_1 = a. \end{aligned}$$

$$i \geq 2. \quad a_i = (2b-1)a_{i-1} + 1-b, \quad \text{令 } a_i + x = (2b-1)(a_{i-1} + x), \quad x = -\frac{1}{2} \quad \text{则有:}$$

$$\text{从而, } a_i - \frac{1}{2} = (2b-1)^{i-1} \cdot (a_1 - \frac{1}{2}), \quad \text{由上, } P(A_n) = a_n = (2b-1)^{n-1} \cdot (a - \frac{1}{2}) + \frac{1}{2}.$$

$$= \begin{cases} \frac{1}{2}, & a = \frac{1}{2}; \\ (2b-1)^{n-1} \cdot (a - \frac{1}{2}) + \frac{1}{2}, & a \neq \frac{1}{2}; \end{cases}$$

(2) 设有 n 只袋子, 每只袋中有 a 只黑球和 b 只白球, 现从第一只袋中任取一球放入第二只袋中, 然后从第二只袋中任取一球放入第三只袋子, 依此下去, 问: “从第 n 只袋中任取一球是黑球”的概率.

记 $A_i = \{\text{第 } i \text{ 只袋中取出黑球}\}$, $i = 1, 2, \dots, n$; 易见, $P(A_1) = \frac{a}{a+b}$.

$$\begin{aligned} \forall i \geq 2, \quad P(A_i) &= P(A_{i-1})P(A_i|A_{i-1}) + P(\bar{A}_{i-1})P(A_i|\bar{A}_{i-1}) \\ &= P(A_{i-1}) \cdot \frac{a+1}{a+1+b} + [1 - P(A_{i-1})] \cdot \frac{a}{a+b+1}. \end{aligned}$$

$$= \frac{1}{a+b+1} P(A_{i-1}) + \frac{a}{a+b+1}, \quad \text{令 } P(A_i) = a_i, \quad \text{则}$$

$$a_i + x = \frac{1}{a+b+1} (a_{i-1} + x), \quad \text{则有: } x = -\frac{a}{a+b}; \quad \text{从而,}$$

$$a_n + \frac{a}{a+b} = \frac{1}{a+b+1} (a_{n-1} + \frac{a}{a+b}) = \dots = (\frac{1}{a+b+1})^{n-1} \cdot (a_1 + \frac{a}{a+b}), \quad \text{故}$$

$$P(A_n) = a_n = \frac{a}{a+b}.$$

20. (首步分析法)

(1) (赌徒破产问题) 设某赌徒有赌本 $i (i \geq 1)$ 元, 其对手有赌本 $a-i (\geq 0)$ 元, 每赌一次该赌徒以 $p (q=1-p)$ 的概率赢(输)一元, 赌博一直进行到两赌徒中有一人破产为止, 试求“该赌徒破产”的概率; $P(A_0)=1, P(A_a)=0$.

记 $A_i = \{\text{有 } i \text{ 元赌本的赌徒破产}\}, i=0, 1, \dots, a$; 已知, $P_0=1, P_a=0$, 且

$$\forall i \in \{1, 2, \dots, a-1\}, P(A_i) = P(A_i|A_1) + P(\bar{A}_1)P(A_i|\bar{A}_1)$$

$$B = \{\text{有 } i \text{ 元赌本的赌徒赢了第一局}\}, \text{则有: } P(A_i) = P(B)P(A_i|B) + P(\bar{B})P(A_i|\bar{B})$$

$$= p \cdot P(A_{i-1}) + q \cdot P(A_{i+1}), \text{ 记 } P(A_i) = P_i, i=0, 1, \dots, a; \text{ 则有: } P_0=1, P_a=0, \text{ 且}$$

$$P_i = p \cdot P_{i+1} + q \cdot P_{i-1}, \text{ 则有: } P(P_{i+1} - P_i) = q(P_i - P_{i-1}), \text{ 也即:}$$

$$\begin{cases} P_i - P_{i-1} = \left(\frac{q}{p}\right)(P_{i-1} - P_{i-2}) = \dots = \left(\frac{q}{p}\right)^{i-1}(P_1 - 1) & \text{则有: } P_i - 1 = \left[1 + \frac{q}{p} + \dots + \left(\frac{q}{p}\right)^{i-1}\right](P_1 - 1) \\ P_{i-1} - P_{i-2} = \left(\frac{q}{p}\right)^{i-2}(P_1 - 1) \\ \vdots \\ P_1 - P_0 = \left(\frac{q}{p}\right)^0(P_1 - 1) \end{cases}$$

$$= \frac{1 - \left(\frac{q}{p}\right)^i}{1 - \frac{q}{p}}(P_1 - 1), \text{ 且 } i=a.$$

$$-1 = \frac{1 - \left(\frac{q}{p}\right)^a}{1 - \frac{q}{p}} \cdot (P_1 - 1), \text{ 则有: } P_i = \frac{\left(\frac{q}{p}\right)^i - \left(\frac{q}{p}\right)^a}{1 - \left(\frac{q}{p}\right)^a}.$$

(2) (玻利亚模型) 罐中有 a 只黑球和 b 只白球, 每次从中任取一球并连同 c 只同色球一起放回, 如此反复进行, 试求“第 n 次取球时取出黑球”的概率 ($c=0(-1)$ 时即有(无)放回抽样).

记 $A_i = \{\text{第 } i \text{ 次取出黑球}\}, i=1, 2, \dots, n$; 已知, $P(A_1) = \frac{a}{a+b}$.

$$P(A_2) = P(A_1)P(A_2|A_1) + P(\bar{A}_1)P(A_2|\bar{A}_1) = \frac{a}{a+b} \times \frac{a+c}{a+c+b} + \frac{b}{a+b} \times \frac{a}{a+b+c} = \frac{a}{a+b}; \text{ 且当 } n=1 \text{ 时,}$$

$$P(A_k) = \frac{a}{a+b}; \text{ 当 } n=k+1 \text{ 时, } P(A_{k+1}) = P(A_1)P(A_{k+1}|A_1) + P(\bar{A}_1)P(A_{k+1}|\bar{A}_1)$$

$$= \frac{a}{a+b} \times \frac{a+c}{a+c+b} + \frac{b}{a+b} \times \frac{a}{a+b+c} = \frac{a}{a+b}; \text{ 从而, } \forall n \geq 1, P(A_n) = \frac{a}{a+b}.$$

习题 1.4

2. 甲乙两人独立地对同一目标射击一次, 其命中率分别为 0.8 和 0.7, 现已知目标被击中, 求“甲命中”的概率.

记 $A(B) = \{\text{甲(乙)命中目标}\}$, 由题设, A, B 独立, 且 $P(A)=0.8$.

$$P(B)=0.7, \text{ 则有: } P(A|A \cup B) = \frac{P(A \cap (A \cup B))}{P(A \cup B)} = \frac{P(A)}{P(A) + P(B) - P(A)P(B)}$$

$$= \frac{0.8}{0.8 + 0.7 - 0.56} = \frac{0.8}{0.94} = \frac{40}{47}$$

3. 若事件 A, B 独立, 且两事件“仅 A 发生”与“仅 B 发生”的概率都是 $\frac{1}{4}$, 试求 $P(A)$ 与 $P(B)$.

由题知, $P(A\bar{B}) = P(A)P(\bar{B}) = P(A)[1 - P(B)] = \frac{1}{4}$,
 $P(\bar{A}B) = P(\bar{A})P(B) = [1 - P(A)]P(B) = \frac{1}{4}$,

则有: $P(A) = P(B) = \frac{1}{2}$.

4. 三人独立地破译一个密码, 他们单独译出的概率分别为 $\frac{1}{3}, \frac{1}{4}, \frac{1}{5}$, 求“此

密码被译出”的概率. 记三人分别译为甲、乙、丙, 令 $A(B, C) = \{\text{甲(乙、丙)译出密码}\}$

由题知, A, B, C 独立, 且 $P(A) = \frac{1}{3}, P(B) = \frac{1}{4}, P(C) = \frac{1}{5}$, 则有:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC) = P(A) + P(B) + P(C) - P(A)P(B) - P(A)P(C) - P(B)P(C) + P(A)P(B)P(C) = \dots = \frac{3}{5}$$

另法: $P(\overline{A \cup B \cup C}) = P(\bar{A}\bar{B}\bar{C}) = P(\bar{A})P(\bar{B})P(\bar{C}) = [1 - P(A)][1 - P(B)][1 - P(C)] = \frac{2}{5}$

5. 一射手对同一目标独立地射击四次, 若“至少命中一次”的概率为 $\frac{80}{81}$, 试求该射手进行一次射击的命中率 p

记 $A_i = \{\text{第 } i \text{ 次命中目标}\}$, $i = 1, 2, 3, 4$. 由题知, A_1, A_2, A_3, A_4 独立, 且 $P(A_i) = p, i = 1, 2, 3, 4$.

则有: $P(A_1 \cup A_2 \cup A_3 \cup A_4) = P(\bigcup_{i=1}^4 A_i) = 1 - P(\bigcap_{i=1}^4 \bar{A}_i) = 1 - P(\bar{A}_1 \bar{A}_2 \bar{A}_3 \bar{A}_4) = 1 - (1 - p)^4 = \frac{80}{81}$, $p = \frac{2}{3}$.

另法: 由题知, $b(0; 4, p) = 1 - \frac{80}{81}$, 则有: $C_4^0 \cdot p^0 \cdot (1 - p)^{4-0} = \frac{1}{81}$.

则有 $p = \frac{2}{3}$.

6. 三门高射炮独立地向一飞机射击, 已知“飞机中一弹被击落”的概率为 0.4, “飞机中两弹被击落”的概率为 0.8, 中三弹则必然被击落; 假设每门高射炮的命中率为 0.6, 现三门高射炮各对飞机射击一次, 求“飞机被击落”的概率.

记 $A = \{\text{飞机被击落}\}$, $B_i = \{\text{飞机中 } i \text{ 弹}\}$, $i = 0, 1, 2, 3$, 则有: $P(A|B_0) = 0$, $P(A|B_1) = 0.4$, $P(A|B_2) = 0.8$, $P(A|B_3) = 1$; 令 $C_i = \{\text{第 } i \text{ 门高射炮命中}\}$, 且 $P(B_0) = C_3^0 (0.6)^0 (0.4)^3$, $P(B_1) = C_3^1 (0.6) (0.4)^2$, $P(B_2) = C_3^2 (0.6)^2 (0.4)$, $P(B_3) = C_3^3 (0.6)^3$, $A = \bigcup_{i=1}^3 B_i$,

$$P(A) = P(A \cap (\bigcup_{i=1}^3 B_i)) = P(\bigcup_{i=1}^3 A B_i) = \sum_{i=1}^3 P(A B_i) = \sum_{i=1}^3 P(B_i) \cdot P(A|B_i)$$

$= \dots =$

7. 甲乙两人连续独立地掷 n 次硬币, 试求“甲乙两人掷出的正面数相等”的概率.

记: $B_k(C_k) = \{\text{甲(乙)掷出 } k \text{ 次正面向上的}\}$, $k=0, 1, \dots, n$; 易见, B_k, C_k 独立, $k=0, 1, \dots, n$.
 $P(\text{甲乙掷出的正面数相等}) = P(\bigcup_{k=0}^n B_k C_k) = \sum_{k=0}^n P(B_k C_k) = \sum_{k=0}^n P(B_k) \cdot P(C_k)$
 $= \sum_{k=0}^n C_n^k (\frac{1}{2})^n \cdot C_n^k (\frac{1}{2})^n = (\frac{1}{2})^{2n} \cdot \sum_{k=0}^n C_n^k \cdot C_n^k$ (这里, $C_n^k = C_n^{n-k}$, 且由 $(1+x)^n = (1+x)^n \cdot (1+x)^n$, 从而得到 x^k 的系数
 $= C_n^n (\frac{1}{2})^{2n}$ 分别为: $C_n^n = \sum_{k=0}^n C_n^k C_n^{n-k}$)

9. 甲、乙两选手进行乒乓球单打比赛, 已知每局中“甲获胜”的概率为 0.6, “乙获胜”的概率为 0.4; 比赛可采用三局两胜制或五局三胜制, 问: 何种赛制对甲更有利?

记 $A_i = \{\text{第 } i \text{ 局甲胜}\}$, $i=1, 2, 3, 4, 5$, 易见, $A_1, A_2, \dots, A_5$ 独立, 且 $P(A_i) = 0.6$.
 令 $B = \{\text{甲获胜}\}$. (i) 若采用三局两胜制: $P(B) = P(A_1 A_2 \cup A_1 \bar{A}_2 A_3 \cup \bar{A}_1 A_2 A_3)$
 $= P(A_1 A_2) + P(A_1 \bar{A}_2 A_3) + P(\bar{A}_1 A_2 A_3) = 0.6^2 + 0.6 \cdot 0.4 \cdot 0.6 + 0.4 \cdot 0.6 \cdot 0.6 = 0.6^2 + 2 \cdot 0.4 \cdot 0.6 \cdot 0.6$
 (ii) 若采用五局三胜制: $P(B) = P(A_1 A_2 A_3 \cup [\bigcup_{1 \leq i < j \leq 3} A_i A_j \bar{A}_3] A_4) \cup [\bigcup_{1 \leq i < j \leq 4} \bar{A}_i \bar{A}_j A_k \bar{A}_4 A_5])$
 $= P(A_1 A_2 A_3) + \sum_{1 \leq i < j \leq 3} P(A_i A_j \bar{A}_3 A_4) + \sum_{1 \leq i < j \leq 4} P(\bar{A}_i \bar{A}_j A_k \bar{A}_4 A_5)$
 $= C_3^3 (0.6)^3 + C_3^2 \cdot 0.6^2 \cdot 0.4 \cdot 0.6 + C_4^2 \cdot 0.6^2 \cdot 0.4^2 \cdot 0.6$

10. 某电厂由甲乙两台机组并联向一城市供电; 当一台机组发生故障时, 另一台机组能在这段时间满足城市全部用电需求的概率为 0.85, 设每台机组发生故障的概率为 0.1, 且它们是否发生故障相互独立;

(1) 求“保证城市供电”的概率;

(2) 若已知电厂机组发生故障, 求“供电能满足需求”的概率.

记 $A = \{\text{电厂机组能保证城市供电}\}$, $B_i = \{\text{第 } i \text{ 台机组发生故障}\}$, $i=0, 1, 2$;
 易见, $P(B_0) = C_2^0 \cdot 0.1^0 \cdot (0.9)^2 = 0.81$, $P(B_1) = C_2^1 \cdot 0.1 \cdot 0.9 = 0.18$,
 $P(B_2) = C_2^2 \cdot 0.1^2 \cdot (0.9)^0 = 0.01$, 且 $P(A|B_0) = 1$, $P(A|B_1) = 0.85$, $P(A|B_2) = 0$.
 (1) $P(A) = P(B_0)P(A|B_0) + P(B_1)P(A|B_1) + P(B_2)P(A|B_2)$
 $= 0.81 \times 1 + 0.18 \times 0.85 + 0.01 \times 0 = 0.963$

(2) $P(A|B_1 \cup B_2) = \frac{P(A \cap (B_1 \cup B_2))}{P(B_1 \cup B_2)} = \frac{P(A|B_1)P(B_1) + P(A|B_2)P(B_2)}{P(B_1) + P(B_2)}$
 $= \frac{P(B_1)P(A|B_1) + P(B_2)P(A|B_2)}{P(B_1) + P(B_2)} = \frac{0.18 \times 0.85 + 0.01 \times 0}{0.18 + 0.01} = 0.81$

11. 甲乙丙三人同时各自独立地对同一目标进行射击，三人击中目标的概率分别为0.4, 0.5, 0.7；若一人击中目标时目标被击毁的概率为0.2，两人击中目标时目标被击毁的概率为0.6，三人同时击中目标则目标必定被击毁；

(1) 求“目标被击毁”的概率；

(2) 已知目标被击毁，求“其是由一人击中”的概率；

(3) 已知目标被击毁，求“其是由甲击中”的概率。

令 $A = \{\text{目标被击毁}\}$, $B_i = \{\text{有 } i \text{ 人命中目标}\}$, $i = 0, 1, 2, 3$. 且

$$P(A|B_0) = 0, P(A|B_1) = 0.2, P(A|B_2) = 0.6, P(A|B_3) = 1.$$

令第1, 2, 3人分别为甲, 乙, 丙, 且 $C_i = \{\text{第 } i \text{ 人命中目标}\}$, 由此得, C_1, C_2, C_3

$$\text{独立, 且 } P(C_1) = 0.4, P(C_2) = 0.5, P(C_3) = 0.7,$$

$$B_1 = C_1 \bar{C}_2 \bar{C}_3 \cup \bar{C}_1 C_2 \bar{C}_3 \cup \bar{C}_1 \bar{C}_2 C_3, \quad P(B_1) = P(C_1)P(\bar{C}_2)P(\bar{C}_3) + P(\bar{C}_1)P(C_2)P(\bar{C}_3) + P(\bar{C}_1)P(\bar{C}_2)P(C_3) = 0.36$$

$$B_2 = C_1 C_2 \bar{C}_3 \cup C_1 \bar{C}_2 C_3 \cup \bar{C}_1 C_2 C_3, \quad \dots \quad P(B_2) = \dots = 0.41$$

$$B_3 = C_1 C_2 C_3, \quad \dots \quad P(B_3) = P(C_1)P(C_2)P(C_3) = 0.14$$

$$(1) P(A) = P(A \cap [\bigcup_{i=1}^3 B_i]) = \sum_{i=1}^3 P(B_i)P(A|B_i)$$

$$= P(B_1) \cdot 0.2 + P(B_2) \cdot 0.6 + P(B_3) \cdot 1 = 0.458$$

$$(2) P(B_1|A) = \frac{P(AB_1)}{P(A)} = \frac{P(B_1)P(A|B_1)}{P(A)} = \frac{0.2P(B_1)}{0.2P(B_1) + 0.6P(B_2) + P(B_3)}$$

$$= 0.16$$

$$(3) P(\bar{C}_1|A) = \frac{P(A\bar{C}_1)}{P(A)} \quad P(C_1 \bar{C}_2 \bar{C}_3 | A) = \frac{P(C_1 \bar{C}_2 \bar{C}_3 A)}{P(A)} \quad \text{因为}$$

$$P(A\bar{C}_1) = P(\bar{C}_1)P(A|C_1) \quad P(C_1 \bar{C}_2 \bar{C}_3 A)$$

$$= P(C_1 \bar{C}_2 \bar{C}_3)P(A|C_1 \bar{C}_2 \bar{C}_3)$$

$$= P(C_1)P(\bar{C}_2)P(\bar{C}_3) \cdot 0.2$$

$$= 0.4 \times 0.5 \times 0.3 \times 0.2 = 0.012$$

习题 2.1

2. (1) 设 (离散型) 随机变量 (r.v.) X 的分布函数 (d.f.) 为

$$F(x) = \begin{cases} 0, & x < 0; \\ 1/4, & 0 \leq x < 3; \\ 1/3, & 3 \leq x < 6; \\ 1, & x \geq 6; \end{cases}, \text{ 试求 } P(X < 3), P(X \leq 3), P(X > 1), P(X \geq 1);$$

$\forall x \in \mathbb{R}, P\{X < x\} = F(x-).$

$$P(X < 3) = F(3-) = \lim_{x \rightarrow 3^-} F(x) = \frac{1}{4}.$$

$$P(X \leq 3) = F(3) = \frac{1}{3}; \quad P(X > 1) = 1 - P(X \leq 1) = 1 - F(1) = 1 - \frac{1}{4} = \frac{3}{4}.$$

$$P(X \geq 1) = 1 - P(X < 1) = 1 - F(1-) = 1 - \lim_{x \rightarrow 1^-} \frac{1}{4} = 1 - \frac{1}{4} = \frac{3}{4}.$$

(2) 设 (连续型) 随机变量 X 的分布函数为 $F(x) = \begin{cases} 0, & x < 1; \\ \ln x, & 1 \leq x < e; \\ 1, & x \geq e; \end{cases}$, 试求:

$$P(X < 2), P(0 \leq X \leq 3), P(2 < X < 2.5);$$

$$P(X < 2) = F(2-) = F(2) = \ln 2;$$

$$P(0 \leq X \leq 3) = P(X \leq 3) - P(X < 0) = F(3) - F(0-) = F(3) - F(0) = 1 - 0 = 1.$$

$$\begin{aligned} P(2 < X < 2.5) &= P(X < 2.5) - P(X \leq 2) = F(2.5-) - F(2) = F(2.5) - F(2) \\ &= \ln 2.5 - \ln 2 = \ln \frac{5}{4}. \end{aligned}$$

(3) 已知 (混合型) 随机变量 X 的分布函数为 $F(x) = \begin{cases} 0, & x < 0; \\ x/2, & 0 \leq x < 1; \\ 2/3, & 1 \leq x < 2; \\ 11/12, & 2 \leq x < 3; \\ 1, & x \geq 3; \end{cases}$, 试求:

$$P(X < 3), P(1 \leq X < 3), P\left(X > \frac{1}{2}\right), P(X = 3).$$

$$P(X < 3) = F(3-) = \lim_{x \rightarrow 3^-} F(x) = \frac{11}{12}.$$

$$P(1 \leq X < 3) = P(X < 3) - P(X < 1) = F(3-) - F(1-) = \frac{11}{12} - \frac{1}{2} = \frac{5}{12}.$$

$$P\left(X > \frac{1}{2}\right) = 1 - P\left(X \leq \frac{1}{2}\right) = 1 - F\left(\frac{1}{2}\right) = 1 - \frac{1}{4} = \frac{3}{4}.$$

$$\begin{aligned} P(X = 3) &= P\{X \leq 3\} - P\{X < 3\} = F(3) - F(3-) \\ &= 1 - \frac{11}{12} = \frac{1}{12}. \end{aligned}$$

3. 设随机变量 ξ 的分布函数为 $F(x)$, 试用 $F(x)$ 表示下列事件的概率:

$$\{\xi < 1\}, \{\xi - 2 < 3\}, \{2\xi + 1 > 5\}, \{\xi^2 \leq 4\}, \{\xi^3 < 8\}, \{a\xi + b \leq c\}. \quad \begin{matrix} \{X \leq x\} \\ = \{\omega | X(\omega) \leq x\} \end{matrix}$$

$$P(2\xi + 1 > 5) = P(\xi > 2) = 1 - P(\xi \leq 2) = 1 - F(2).$$

$$P(\xi^3 < 8) = P(\xi < 2) = F(2), \quad P(\xi^2 \leq 4) = P(-2 \leq \xi \leq 2) = P(\xi \leq 2) - P(\xi < -2) = F(2) - F(-2)$$

$$a > 0, \quad P(a\xi + b \leq c) = P(a\xi \leq c - b) = P(\xi \leq \frac{c-b}{a}) = F(\frac{c-b}{a});$$

$$a < 0, \quad P(a\xi + b \leq c) = P(a\xi \leq c - b) = P(\xi \geq \frac{c-b}{a}) = 1 - P(\xi < \frac{c-b}{a}) = 1 - F(\frac{c-b}{a});$$

$$a = 0 \quad \{0\xi + b \leq c\} = \{\omega | 0\xi(\omega) + b \leq c\} = \{\omega | 0 \leq c - b\} = \begin{cases} \emptyset & c < b; \\ \Omega & c \geq b; \end{cases}$$

$$\text{即, } a = 0 \quad P(a\xi + b \leq c) = \begin{cases} 0 & c < b; \\ 1 & c \geq b; \end{cases}$$

5. (1) 设 ξ 的分布函数为: $F(x) = \begin{cases} 0, & x < -1; \\ a + b \arcsin x, & -1 \leq x < 1; \\ 1, & x \geq 1; \end{cases}$ 试确定常数 a, b ;

$$\text{由 } F(-1-) = F(-1+), \text{ 即: } 0 = a + b(-\frac{\pi}{2}).$$

$$\text{由 } F(1-) = F(1+), \text{ 即: } 1 = a + b \cdot \frac{\pi}{2}. \quad \text{解得: } a = \frac{1}{2}, b = \frac{1}{\pi}.$$

(2) 设 ξ 的分布函数为 $F(x) = A + B \arctan x, x \in R$, 试确定常数 A, B .

$$\text{由 } F(+\infty) = 1, \text{ 即: } \lim_{x \rightarrow +\infty} F(x) = \lim_{x \rightarrow +\infty} (A + B \arctan x) = A + B \cdot \frac{\pi}{2} = 1.$$

$$\text{由 } F(-\infty) = 0, \text{ 即: } \lim_{x \rightarrow -\infty} F(x) = \lim_{x \rightarrow -\infty} (A + B \arctan x) = A + B \cdot (-\frac{\pi}{2}) = 0.$$

$$\text{解得: } A = \frac{1}{2}, B = \frac{1}{\pi}.$$

6. (1) 在半径为 R 的圆内任取一点, 求此点到圆心距离 X 的分布函数及概率

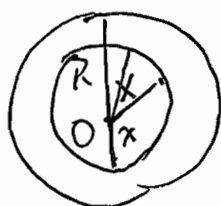
$$P\left(X > \frac{2}{3}R\right);$$

解, $R(X) = [0, R)$. 或 X 的所有可能取值范围为 $[0, R)$,

$$\text{即: } \forall x < 0, \quad F(x) = P(X \leq x) = P(\emptyset) = 0;$$

$$\forall x \geq R, \quad F(x) = P(X \leq x) = P(\Omega) = 1;$$

$$\forall x \in [0, R), \quad F(x) = P(X \leq x) \stackrel{\text{几何法}}{=} \frac{\pi x^2}{\pi R^2} = \frac{x^2}{R^2}.$$

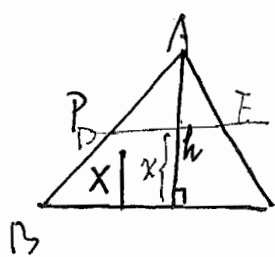


$$F(x) = \begin{cases} 0 & x < 0; \\ \frac{x^2}{R^2} & 0 \leq x < R; \\ 1 & x \geq R; \end{cases}$$

$$P(X > \frac{2}{3}R) = 1 - P(X \leq \frac{2}{3}R)$$

$$= 1 - F(\frac{2}{3}R) = 1 - \frac{4}{9} = \frac{5}{9}.$$

(2) 在 $\triangle ABC$ 内任取一点 P , 记 X 为点 P 到底边 BC 的距离, 试求 X 的分布函数.



设 BC 边上的高为 h , 由此得, $R(X) = (0, h)$. 证:

$\forall x \leq 0$, $F(x) = P(X \leq x) = P(\emptyset) = 0$; $\forall x \geq h$, $F(x) = P(X \leq x) = P(\Omega) = 1$;

$\forall x \in (0, h)$, $F(x) = P(X \leq x) \xrightarrow{\text{几何法}} \frac{S_{\triangle PBC}}{S_{\triangle ABC}} = 1 - \frac{S_{\triangle AEF}}{S_{\triangle ABC}}$

$= 1 - \left(\frac{h-x}{h}\right)^2$. 证:

$$F(x) = \begin{cases} 0 & , x \leq 0; \\ 1 - \left(\frac{h-x}{h}\right)^2 & , 0 < x < h; \\ 1 & , x \geq h; \end{cases}$$

7. (1) 设 $F_1(x), F_2(x)$ 分别是两个随机变量的分布函数, $a, b > 0$ 且 $a+b=1$, 试证明: $F(x) = aF_1(x) + bF_2(x)$ 也是一个分布函数;

证: $F(x)$ 是 $(-\infty, +\infty)$ 上的一个非降连续函数. 且有连续.

$$\begin{aligned} F(+\infty) &= \lim_{x \rightarrow +\infty} F(x) = \lim_{x \rightarrow +\infty} [aF_1(x) + bF_2(x)] = a \cdot F_1(+\infty) + b \cdot F_2(+\infty) \\ &= a \cdot 1 + b \cdot 1 = 1 \end{aligned}$$

$$\begin{aligned} F(-\infty) &= \lim_{x \rightarrow -\infty} F(x) = \lim_{x \rightarrow -\infty} [aF_1(x) + bF_2(x)] = a \cdot F_1(-\infty) + b \cdot F_2(-\infty) \\ &= a \cdot 0 + b \cdot 0 = 0 \end{aligned}$$

(2) 若 $F(x)$ 是一分布函数, 试证: $\forall h > 0, \Phi(x) = \frac{1}{h} \int_x^{x+h} F(t) dt$ 也是一分布函数.

证: $\Phi(x) = \frac{1}{h} \int_x^{x+h} F(t) dt$ 是 x 的连续函数. 且 $\Phi(x) = \frac{1}{h} \int_x^{x+h} F(t) dt \xrightarrow{\text{令 } t=x+u}$

$$\frac{1}{h} \int_0^h F(x+u) du; \quad \forall x_1, x_2 \in \mathbb{R}, x_1 < x_2, \quad \forall F(x+u) \leq F(x_2+u),$$

$$\text{证: } \int_0^h F(x_1+u) du \leq \int_0^h F(x_2+u) du, \text{ 也证: } \Phi(x_1) \leq \Phi(x_2).$$

由 $\lim_{x \rightarrow +\infty} F(x) = 1$, 证: $\forall \varepsilon > 0, \exists \Delta > 0, \forall x > \Delta, F(x) > 1 - \varepsilon$. 从而,

$$\forall \varepsilon > 0, \exists \Delta > 0, \forall x > \Delta, F(x+u) > 1 - \varepsilon. \quad \frac{1}{h} \int_0^h F(x+u) du > \frac{1}{h} \int_0^h (1 - \varepsilon) du$$

$$= 1 - \varepsilon. \text{ 证: } \lim_{x \rightarrow +\infty} \Phi(x) = 1; \text{ 同理可证: } \lim_{x \rightarrow -\infty} \Phi(x) = 0. \dots\dots$$

习题 2.2

2. 现有三只盒子，第一只盒中装有1只白球4只黑球，第二只盒中装有2只白球3只黑球，第三只盒中装有3只白球2只黑球；现任取一只盒子，从中任取3只球，以 X 表示所取到的白球数，试求：

(1) X 的分布列；(2) “取到白球数不少于2”的概率。

(1) 易见, $R(X) = \{0, 1, 2, 3\}$, 记: $A_i = \{\text{取到第 } i \text{ 只盒子}\}, i = 1, 2, 3$. 则有, $P(A_i) = \frac{1}{3}$;

$$P(X=0) = \sum_{i=1}^3 P(A_i) \cdot P(X=0|A_i) = \frac{1}{3} \times [P(X=0|A_1) + P(X=0|A_2) + P(X=0|A_3)] = \frac{1}{3} \times [\frac{C_4^3}{C_5^3} + \frac{C_3^3}{C_5^3} + 0]$$

$$= \frac{1}{6};$$

$$P(X=1) = \sum_{i=1}^3 P(A_i) P(X=1|A_i) = \frac{1}{3} [\frac{C_4^2 C_1^1}{C_5^3} + \frac{C_3^2 C_2^1}{C_5^3} + \frac{C_2^2 C_3^1}{C_5^3}] = \frac{1}{3} \cdot \frac{15}{10} = \frac{1}{2}.$$

$$P(X=2) = \sum_{i=1}^3 P(A_i) P(X=2|A_i) = \frac{1}{3} [0 + \frac{C_4^1 C_2^2}{C_5^3} + \frac{C_3^1 C_3^2}{C_5^3}] = \frac{1}{3} \cdot \frac{9}{10} = \frac{3}{10}.$$

$$P(X=3) = 1 - P(X=0) - P(X=1) - P(X=2) = \frac{1}{30}.$$

或:
$$X \sim \begin{pmatrix} 0 & 1 & 2 & 3 \\ \frac{1}{6} & \frac{1}{2} & \frac{3}{10} & \frac{1}{30} \end{pmatrix}$$

$P(\text{取到白球数不少于2}) = P(X \geq 2) = P(X=2) + P(X=3) = \frac{1}{3}.$

3. 某公司有5个顾问，假定每个顾问提供正确意见的概率为0.6，现公司就某项事宜是否可行征求各顾问的意见，并按多数人的意见作出决策，试求“公司作出正确决策”的概率。

记 $A = \{\text{公司作出正确决策}\}$, X 为提供正确意见的顾问数。

易见, $X \sim B(5, 0.6)$, 且

$$P(A) = P(X \geq 3) = P(X=3) + P(X=4) + P(X=5)$$

$$= C_5^3 \cdot (0.6)^3 (0.4)^2 + C_5^4 \cdot (0.6)^4 (0.4) + C_5^5 \cdot (0.6)^5 = \dots = 0.6826.$$

4. 现有5件产品，其中2件是次品；每次任取一件测试，

(1) 直到两件次品都找出，记 X, Y 分别为找出第一件、第二件次品所用的次数，试求 X, Y 的分布列；

(2) 直到找出两件次品或三件正品为止，试求需要测试次数 Z 的概率分布。

(1) X | 1 2 3 4

P	$\frac{2}{5}$	$\frac{3}{10}$	$\frac{1}{5}$	$\frac{1}{10}$
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易见, $P(X=1) = P(A_1) = \frac{2}{5}$, $P(X=2) = P(\bar{A}_1 A_2) = P(\bar{A}_1) P(A_2|\bar{A}_1) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$; $P(X=3) = P(\bar{A}_1 \bar{A}_2 A_3) = P(\bar{A}_1) P(\bar{A}_2|\bar{A}_1) P(A_3|\bar{A}_1 \bar{A}_2) = \frac{3}{5} \times \frac{2}{4} \times \frac{2}{3} = \frac{1}{5}$.

(2) Y | 2 3 4 5

P	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{3}{10}$	$\frac{2}{5}$
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易见, $P(Y=2) = \frac{1}{10}$, $P(Y=3) = \frac{C_2^1 C_3^1}{C_5^2} = \frac{2 \times 3}{10} = \frac{3}{5}$; $P(Y=4) = \frac{C_2^2 C_3^0}{C_5^3} = \frac{1 \times 1}{10} = \frac{1}{10}$.

(2) Z | 2 3 4

P	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{2}{5}$
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易见, $P(Z=2) = P(A_1 A_2) = \frac{2}{5} \times \frac{1}{4} = \frac{1}{10}$; $P(Z=3) = P(\bar{A}_1 \bar{A}_2 A_3 \cup A_1 \bar{A}_2 A_3 \cup \bar{A}_1 A_2 A_3)$

$$= \frac{2}{5} \times \frac{2}{4} \times \frac{1}{3} + \frac{2}{5} \times \frac{1}{4} \times \frac{1}{3} + \frac{2}{5} \times \frac{2}{4} \times \frac{1}{3} = \frac{3}{10}$$

5. 已知某射手的射击命中率为 $\frac{4}{5}$, 现其对一目标射击, 分别以 X, Y 表示直到第一次、第二次命中为止所进行的射击次数, 试求: “ X 取奇数”、“ $Y=6$ ” 的概率.

解: $X \sim G(\frac{4}{5})$. 即: $P\{X \text{ 取奇数}\} = P(\bigcup_{k=1}^{\infty} \{X=2k-1\}) = \sum_{k=1}^{\infty} P(X=2k-1)$
 $= \sum_{k=1}^{\infty} (\frac{4}{5})^{2k-2} \cdot \frac{4}{5} = \frac{4/5}{1 - \frac{4}{5}} = \frac{4}{5} \times \frac{5}{1} = \frac{4}{5}$.

记 Z 为前五次射击中命中的次数, 则 $Z \sim B(5, \frac{4}{5})$. 令 $A_6 = \{\text{第六次命中}\}$.

$$P(Y=6) = P\{Z=1\} \cap A_6 = P(Z=1) \cdot P(A_6 | Z=1) = P(Z=1) \cdot P(A_6)$$

$$= C_5^1 \cdot (\frac{4}{5})^1 \cdot (1 - \frac{4}{5})^4 \cdot \frac{4}{5} = \frac{16}{5^5}$$

6. 袋中有 5 只球, 编号为 1, 2, 3, 4, 5; 现从中任取 3 只, 以 X 表示 3 只球中的最大号码; (1) 试求 X 的分布列; (2) 写出 X 的分布函数.

解: X 的分布列为:

X	3	4	5
P	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{3}{5}$

即: $P(X=3) = \frac{1}{C_5^3}, P(X=4) = \frac{C_3^2}{C_5^3};$ 从而,

$$F(x) = P(X \leq x) = \begin{cases} 0 & x < 3; \\ \frac{1}{10} & 3 \leq x < 4; \\ \frac{4}{10} & 4 \leq x < 5; \\ 1 & x \geq 5; \end{cases}$$

7. 一汽车沿一街道行驶, 需要经过三个设有红绿信号灯的路口, 每个信号灯亮何灯相互独立, 且红绿两种信号显示的时间相等; 以 X 表示该汽车首次遇到红灯前已通过的路口个数, 试求 X 的概率分布.

解: $X \sim R(X) = \{0, 1, 2, 3\}$. 且

$$P(X=0) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = P(A_1) = \frac{1}{8}$$

$$P(X=1) = P(\bar{A}_1 A_2) = P(\bar{A}_1) P(A_2) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$P(X=2) = P(\bar{A}_1 \bar{A}_2 A_3) = P(\bar{A}_1) P(\bar{A}_2) P(A_3) = \frac{1}{8}$$

$$P(X=3) = P(\bar{A}_1 \bar{A}_2 \bar{A}_3) = P(\bar{A}_1) P(\bar{A}_2) P(\bar{A}_3) = \frac{1}{8}$$

即:

X	0	1	2	3
P	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$

由于每个路口红绿灯的时间相等, 故汽车首次遇到红灯的概率为 $\frac{1}{2}$.
 记 $A_i = \{\text{第 } i \text{ 个路口遇到红灯}\}$.
 $i=1, 2, 3$; 且 $P(A_i) = \frac{1}{2}$.
 A_1, A_2, A_3 独立.

8. (1) 已知随机变量 X 的分布函数为 $F(x) = \begin{cases} 0, & x < 0; \\ 0.5, & 0 \leq x < 1; \\ 0.7, & 1 \leq x < 3; \\ 1, & x \geq 3; \end{cases}$ 试求 X 的分布列;

$$\text{由 } F(0) - F(0-) = P(X=0) = 0.5 - 0 = 0.5,$$

$$F(1) - F(1-) = P(X=1) = 0.7 - 0.5 = 0.2,$$

$$F(3) - F(3-) = P(X=3) = 1 - 0.7 = 0.3, \text{ 则有:}$$

X	0	1	3
P	0.5	0.2	0.3

(2) 已知随机变量 X 的分布列为: $\begin{pmatrix} -1 & 0 & 1 \\ 0.25 & a & b \end{pmatrix}$, 其分布函数为:

$$F(x) = \begin{cases} c, & x < -1; \\ d, & -1 \leq x < 0; \\ 0.75, & 0 \leq x < 1; \\ e, & x \geq 1; \end{cases} \text{ 试求 } a, b, c, d, e.$$

$$\text{解, } 0.25 + a + b = 1, \text{ 即 } a + b = 0.75, \text{ 由 } F(-\infty) = F(-1-) = c, \text{ 即 } c = 0.$$

$$\text{由 } P(X=-1) = F(-1) - F(-1-) = d - 0 = 0.25, \text{ 即 } d = 0.25$$

$$\text{由 } F(+\infty) = 1, \text{ 即 } e = 1.$$

$$\text{由 } P(X=0) = a = F(0) - F(0-) = 0.75 - d = 0.5, \text{ 即 } a = 0.5, b = 0.25.$$

9. (1) 从 1, 2, 3, 4, 5 五个数中任取三个, 按大小顺序排列记为: $x_1 < x_2 < x_3$, 令 $X = x_2$, 试求: X 的分布及 $P(X < 2), P(X > 4)$;

$$\text{解, } \begin{array}{c|ccc} X & 2 & 3 & 4 \\ \hline P & \frac{3}{10} & \frac{4}{10} & \frac{3}{10} \end{array}, \text{ 即 } P(X=2) = \frac{C_3^1}{C_5^3} = \frac{3}{10},$$

$$P(X=3) = \frac{C_2^1 C_3^1}{C_5^3} = \frac{4}{10}, P(X=4) = \frac{C_3^1}{C_5^3};$$

$$\text{故, } P(X < 2) = 0, P(X > 4) = 0.$$

(2) 连续“独立”地掷 n 次骰子, 记 X, Y 分别为 n 个点数的最小、最大值, 试求 X, Y 的分布列.

解. $R(X) = R(Y) = \{1, 2, 3, 4, 5, 6\}$. 且 $\forall k \in \{1, 2, 3, 4, 5, 6\}$

$$P(Y=k) = P(\{Y \leq k\} - \{Y \leq k-1\}) = P(Y \leq k) - P(Y \leq k-1) = \frac{k^n}{6^n} - \frac{(k-1)^n}{6^n} \\ = \left(\frac{k}{6}\right)^n - \left(\frac{k-1}{6}\right)^n;$$

$$P(X=k) = P(\{X \geq k\} - \{X \geq k+1\}) = P(X \geq k) - P(X \geq k+1) = \frac{(7-k)^n}{6^n} - \frac{(6-k)^n}{6^n} \\ = \left(\frac{7-k}{6}\right)^n - \left(\frac{6-k}{6}\right)^n.$$

10. (1) 设 $X \sim P(\lambda)$, 试求 X 的最大可能值, 即: k 取何值时, 概率 $P(X=k)$ 取最大值?

$$\text{由 } \frac{P(X=k)}{P(X=k+1)} = \frac{\frac{\lambda^k}{k!} e^{-\lambda}}{\frac{\lambda^{k+1}}{(k+1)!} e^{-\lambda}} = \frac{k+1}{\lambda} \geq 1, \text{ 且 } \frac{P(X=k)}{P(X=k-1)} = \frac{\frac{\lambda^k}{k!} e^{-\lambda}}{\frac{\lambda^{k-1}}{(k-1)!} e^{-\lambda}} = \frac{\lambda}{k} \geq 1,$$

$$\text{即有 } k \leq \lambda \leq k+1. \text{ 即 } \lambda-1 \leq k \leq \lambda;$$

若 λ 为正整数, 当 $k=\lambda$ 或 $\lambda-1$ 时, $P(X=k)$ 最大!

若 λ 不为正整数, 当 $k=[\lambda]$ 时, $P(X=k)$ 最大!

(2) 某食品店有 4 名售货员, 据统计每名售货员平均在 1 小时内使用电子秤 15 分钟, 问: 该食品店配置几台电子秤较为合适?

记 X 为某个时间使用电子秤的售货员数, 由题设, $X \sim B(4, \frac{1}{4})$.

假如要配置 n 台电子秤, 且 $P(X \leq n) \geq 0.95$, 便可以认为 n 就是较为合适的

$$\text{电子秤台数; 即由 } P(X \leq 2) = C_4^0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^4 + C_4^1 \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^3 + C_4^2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^2$$

$$= \frac{3^4 + 4 \cdot 3^3 + 6 \cdot 3^2}{4^4} = \frac{3^2 \cdot 27}{16 \cdot 16} = \frac{243}{256} \geq 0.95. \text{ 故配置 2 台电子秤合适!}$$

(3) 某公司生产一种配件, 其不合格率为0.02; 试问: 一箱中至少应装多少配件才能以95%的把握保证每箱中有100件合格品?

(见理科教材P40.例2.2.7) 设每箱中至少应装进 $100+n$ 件配件, 记 X 为其中不合格的配件数; 易见, $X \sim P(100+n, 0.02)$, 记: $\lambda = (100+n) \times 0.02 \approx 2$. 由Poisson定理, $X \sim P(2)$, 且由 $P(X \leq n) \approx \sum_{k=0}^n \frac{2^k}{k!} e^{-2} \geq 0.95$, 查Poisson分布表, 取 $n=5$.

11. (1) 自动生产线在调整之后出现次品的概率为0.004, 生产过程中只要一出现次品便立即进行调整, 求在两次调整之间生产的正品数 X 的分布律;

几何分布的两种设法: ① $P(X=k) = (1-p)^{k-1} \cdot p, k=1, 2, \dots$,
则 $X \sim G(p)$;
② $P(X=k) = (1-p)^k \cdot p, k=0, 1, \dots$,
则 $X \sim G(p)$.

假设从上次调整开始, 记: $A_i = \{\text{生产出的第 } i \text{ 件是正品}\}, i=1, 2, \dots$

由此知, $A_1, A_2, \dots$, 独立, 且 $P(A_i) = 0.996$;

$$P(X=k) = P(A_1 \cdots A_k \bar{A}_{k+1}) = P(A_1) \cdots P(A_k) P(\bar{A}_{k+1}) = (0.996)^k \cdot 0.004, k=0, 1, \dots$$

$$\text{即 } X \sim G(0.004)$$

(2) 设 $X \sim \begin{pmatrix} -3 & 0 & 3 & 4 & 7 \\ 0.2 & 0.3 & 0.2 & 0.1 & 0.2 \end{pmatrix}$, 试求:

$$P(|X| < 3), P(-3 < X \leq 3), P(X \geq 2 | X \neq 4), P(X < 4 | X = 0).$$

$$P(|X| < 3) = P(-3 < X < 3) = P(X=0) = 0.3$$

$$P(-3 < X \leq 3) = P(X=0) + P(X=3) = 0.5.$$

$$P(X \geq 2 | X \neq 4) = \frac{P\{X \geq 2 \mid X \neq 4\}}{P(X \neq 4)} = \frac{P(X=3) + P(X=7)}{1 - P(X=4)} = \frac{0.2 + 0.2}{1 - 0.1} = \frac{4}{9}.$$

$$P(X < 4 | X = 0) = 1,$$

12. 已知运载火箭在飞行中进入其仪器舱的宇宙粒子数服从参数为2的 Poisson 分布, 而进入仪器舱的粒子随机落到仪器重要部位的概率为0.1, 求落到仪器重要部位的粒子数的概率分布.

记 X 为进入仪器舱的宇宙粒子数, Y 为落到仪器重要部位的粒子数, 由题意, $X \sim P(2)$, 且当 $X=k$ 时, $Y \sim B(k, 0.1)$. 从而, $\forall k \in \{0, 1, 2, \dots\}$

$$\begin{aligned}
 P(Y=k) &= P\{Y=k \cap [\bigcup_{h=k}^{\infty} \{X=h\}]\} = P\{\bigcup_{h=k}^{\infty} \{X=h, Y=k\}\} = \sum_{h=k}^{\infty} P(X=h, Y=k) \\
 &= \sum_{h=k}^{\infty} P(X=h) \cdot P(Y=k|X=h) = \sum_{h=k}^{\infty} \left(\frac{2^h}{h!} e^{-2}\right) \cdot C_h^k \cdot (0.1)^k \cdot (1-0.1)^{h-k} \\
 &= \frac{(2 \cdot 0.1)^k}{k!} \sum_{h=k}^{\infty} \frac{(2 \cdot 0.9)^{h-k}}{(h-k)!} \cdot e^{-2} \cdot \frac{(2 \cdot 0.1)^k}{k!} \cdot \left[\sum_{l=0}^{\infty} \frac{(2 \cdot 0.9)^l}{l!}\right] e^{-2} = \frac{(2 \cdot 0.1)^k}{k!} e^{-2 \cdot 0.9} \cdot e \\
 &= \frac{(0.2)^k}{k!} e^{-0.2}, \quad k=0, 1, \dots; \quad \therefore Y \sim P(0.2).
 \end{aligned}$$

$$\frac{C_h^k}{k! \cdot (h-k)!} = \frac{n!}{k! \cdot (n-k)!}$$

13. 一盒中装有两枚硬币, 一枚是正品, 一枚是次品 (两面都印有分值), 在盒中随机取一枚, 投掷直至出现分值面, 以 X 表示所需投掷的次数, 试求 X 的分布律.

令 $A = \{\text{取到正品}\}$, 则 $P(A) = \frac{1}{2}$, 且

$$\begin{aligned}
 P(X=1) &= P(A)P(X=1|A) + P(\bar{A})P(X=1|\bar{A}) = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times 1 = \frac{3}{4} \\
 P(X=2) &= P(A)P(X=2|A) + P(\bar{A})P(X=2|\bar{A}) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times 0 = \frac{1}{8} \\
 P(X=3) &= P(A)P(X=3|A) + P(\bar{A})P(X=3|\bar{A}) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times 0 = \frac{1}{16} \\
 &\vdots \\
 \forall k \geq 2, \quad P(X=k) &= P(A)P(X=k|A) + P(\bar{A})P(X=k|\bar{A}) = \frac{1}{2} \times \frac{1}{2^{k-1}} + \frac{1}{2} \times 0 = \frac{1}{2^{k+1}}, \quad \text{从而,}
 \end{aligned}$$

X	1	2	...	k	...
P	$\frac{3}{4}$	$\frac{1}{8}$	...	$\frac{1}{2^{k+1}}$	...

15. 为保证设备正常工作, 需要配备一些维修工; 如果各台设备发生故障是相互独立的, 且每台设备发生故障的概率都是0.01; 若某工厂有同类设备300台, 为了保证设备发生故障而不能及时修理的概率小于0.01, 至少应配备多少维修工?

记 X 为发生故障的设备数, 由题意, $X \sim B(300, 0.01)$. $300 \times 0.01 = 3$, 由 Poisson 定理, $X \sim P(3)$; 故至少配备 n 名维修工, 由

$$P(X > n) < 0.01, \quad \text{查 Poisson 分布表, } n = 8.$$

14. 设实验室器皿中产生甲、乙两类细菌的机会是相等的，以 X 表示产生细菌的

个数，其分布为： $P(X=k) = \frac{\lambda^k}{k!} e^{-\lambda}, k=0,1,2,\dots$ ；试求：

(1) “产生甲类细菌但没有乙类细菌”的概率；

(2) 在已知产生了细菌而没有乙类细菌的条件下，“有两个甲类细菌”的概率。

记 Y 为产生的甲类细菌数。

$$\begin{aligned}
 (1) \quad P(\{\text{产生甲类细菌但没有乙类细菌}\}) &= P(\{X=Y \geq 1\}) \\
 &= P(\bigcup_{k=1}^{\infty} \{X=Y=k\}) = \sum_{k=1}^{\infty} P(X=Y=k) = \sum_{k=1}^{\infty} P(\{X=k\} \cap \{Y=k\}) \\
 &= \sum_{k=1}^{\infty} P(X=k) \cdot P(Y=k|X=k) = \sum_{k=1}^{\infty} \left(\frac{\lambda^k}{k!} e^{-\lambda} \right) \cdot \left(\frac{k}{k} \cdot \left(\frac{1}{2}\right)^k \right) \\
 &= \sum_{k=1}^{\infty} \frac{\left(\frac{\lambda}{2}\right)^k}{k!} e^{-\lambda} = e^{-\lambda} \cdot \left[\sum_{k=0}^{\infty} \frac{\left(\frac{\lambda}{2}\right)^k}{k!} - 1 \right] = e^{-\lambda} \cdot [e^{\frac{\lambda}{2}} - 1] \\
 &= e^{-\frac{\lambda}{2}} - e^{-\lambda}
 \end{aligned}$$

$$\begin{aligned}
 (2) \quad P(Y=2 | X=Y \geq 1) &= \frac{P(\{X=Y \geq 1, Y=2\})}{P(X=Y \geq 1)} = \frac{P(X=Y=2)}{P(X=Y \geq 1)} \\
 &= \frac{\frac{\lambda^2}{2!} e^{-\lambda} \cdot \left(\frac{1}{2}\right)^2}{e^{-\frac{\lambda}{2}} - e^{-\lambda}} = \frac{\lambda^2 \cdot e^{-\lambda}}{8(e^{-\frac{\lambda}{2}} - e^{-\lambda})}
 \end{aligned}$$

习题 2.3

1. 设随机变量 X 的分布函数为: $F(x) = \begin{cases} 0, & x < 0; \\ A \sin x, & 0 \leq x < \pi/2; \\ 1, & x \geq \pi/2; \end{cases}$ 试求:

(1) A ; (2) $P\left(|X| < \frac{\pi}{6}\right)$; (3) X 的概率密度函数.

(1) 由 $F(\frac{\pi}{2}) = F(\frac{\pi}{2}-)$, 可得: $1 = \lim_{x \rightarrow \frac{\pi}{2}-} A \sin x = A$.

(2) $P(|X| < \frac{\pi}{6}) = P(-\frac{\pi}{6} < X < \frac{\pi}{6}) = F(\frac{\pi}{6}) - F(-\frac{\pi}{6}) = F(\frac{\pi}{6}) - 0 = \frac{1}{2} - 0 = \frac{1}{2}$.

(3) 设 $X \sim f(x)$, 可得: $f(x) = \frac{dF(x)}{dx} = \begin{cases} \cos x, & 0 \leq x < \frac{\pi}{2}; \\ 0, & \text{其他}; \end{cases}$

4. 设随机变量 X 的密度函数为 $f(x) = \begin{cases} 1-|x|, & -1 < x < 1; \\ 0, & \text{其他}; \end{cases}$ 试求 $P\left(-\frac{1}{2} \leq X < \frac{1}{2}\right)$

及 X 的分布函数.

$P(-\frac{1}{2} \leq X < \frac{1}{2}) = \int_{-\frac{1}{2}}^{\frac{1}{2}} f(x) dx = 2 \int_0^{\frac{1}{2}} f(x) dx = 2 \int_0^{\frac{1}{2}} (1-x) dx = 1 - \frac{1}{4} = \frac{3}{4}$;

$\forall x \in \mathbb{R}, F(x) = P(X \leq x) = P(-\infty < X \leq x) = \int_{-\infty}^x f(t) dt$

$$= \begin{cases} \int_{-\infty}^x 0 dt = 0, & x \leq -1; \\ \int_{-\infty}^{-1} 0 dt + \int_{-1}^x (1-|t|) dt = x + 1 + \frac{1}{2} |t| \Big|_{-1}^x, & -1 < x < 1; \\ \int_{-\infty}^{-1} 0 dt + \int_{-1}^1 (1-|t|) dt + \int_1^x 0 dt = 1, & x \geq 1; \end{cases}$$

即: $\int |x| dx = \frac{1}{2} x \cdot |x| + C$.

5. 设连续型随机变量 X 的分布函数为 $F(x) = \begin{cases} ae^x, & x < 0; \\ b, & 0 \leq x < 1; \\ 1 - ae^{-(x-1)}, & x \geq 1; \end{cases}$ 试求:

(1) 常数 a, b ; (2) 概率密度函数 $f(x)$; (3) $P\left(X > \frac{1}{2}\right)$.

(1) 由 $F(0-) = F(0)$, 可得: $a = b$;

由 $F(1-) = F(1)$, 可得: $b = 1 - a$, 可得: $a = b = \frac{1}{2}$.

(2) 由 $f(x) = \frac{dF(x)}{dx} = \begin{cases} \frac{1}{2} e^x, & x < 0; \\ \frac{1}{2} e^{-(x-1)}, & x \geq 1; \\ 0, & \text{其他}; \end{cases}$

(3) $P(X > \frac{1}{2}) = 1 - P(X \leq \frac{1}{2}) = 1 - F(\frac{1}{2}) = 1 - b = \frac{1}{2}$.

6. (1) 已知随机变量 X 的概率密度函数为 $f(x) = ce^{-|x|}$, $-\infty < x < +\infty$, 试确定常数 c 并求 X 的分布函数;

由 $\int_{-\infty}^{+\infty} f(x) dx = 1$. 证: $\int_{-\infty}^{+\infty} f(x) dx = 2 \int_0^{+\infty} f(x) dx = 2c \cdot \int_0^{+\infty} e^{-x} dx = 2c = 1$. $c = \frac{1}{2}$.

$\forall x \in \mathbb{R}$, $F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$

$$= \begin{cases} \int_{-\infty}^x ce^t dt = ce^t \Big|_{-\infty}^x = \frac{1}{2}e^x, & x \leq 0; \\ \int_{-\infty}^0 ce^t dt + \int_0^x ce^{-t} dt, & x > 0; \\ = c + c(1 - e^{-x}) = 1 - \frac{1}{2}e^{-x}. \end{cases}$$

(2) 求常数 c , 使得 $f(x) = ce^{1-x-x^2}$ 成为某连续型随机变量 X 的密度函数;

证: 设 $X \sim N(\mu, \sigma^2)$. $X \sim f(x)$, $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, $-\infty < x < +\infty$

证: 由 $\int_{-\infty}^{+\infty} f(x) dx = 1$. 证: $\int_{-\infty}^{+\infty} c \cdot e^{-(x-\frac{1}{2})^2} \cdot e^{\frac{5}{4}} dx \stackrel{\text{令 } x-\frac{1}{2}=t}{=} ce^{\frac{5}{4}} \cdot \int_{-\infty}^{+\infty} e^{-t^2} dt = ce^{\frac{5}{4}} \cdot \sqrt{\pi} = 1$.

证: 由 $f(x) = c \cdot e^{-(x-\frac{1}{2})^2} \cdot e^{\frac{5}{4}} = c \cdot e^{\frac{5}{4}} \cdot e^{-\frac{(x-\frac{1}{2})^2}{2 \cdot (\frac{1}{\sqrt{\pi}})^2}}$, $-\infty < x < +\infty$.

故令 $X \sim f(x)$, 证: $X \sim N(\frac{1}{2}, (\frac{1}{\sqrt{\pi}})^2)$. 证: $c \cdot e^{\frac{5}{4}} = \frac{1}{\sqrt{\pi} \cdot \frac{1}{\sqrt{\pi}}}$, $c = e^{-\frac{5}{4}} \cdot \frac{1}{\sqrt{\pi}}$

(3) 设 $f(x) = \frac{1}{ax^2 + bx + c}$, 为使 $f(x)$ 成为某连续型随机变量 X 在 $(-\infty, +\infty)$ 上

的密度函数, a, b, c 应该满足什么条件?

证: 记: $g(x) = ax^2 + bx + c$, $x \in (-\infty, +\infty)$. 证: $\forall x \in (-\infty, +\infty)$, $g(x) > 0$. 证: $a > 0$,

$\Delta = b^2 - 4ac < 0$. 由 $\int_{-\infty}^{+\infty} f(x) dx = 1$. 证: $\int_{-\infty}^{+\infty} f(x) dx = \int_{-\infty}^{+\infty} \frac{1}{ax^2 + bx + c} dx$

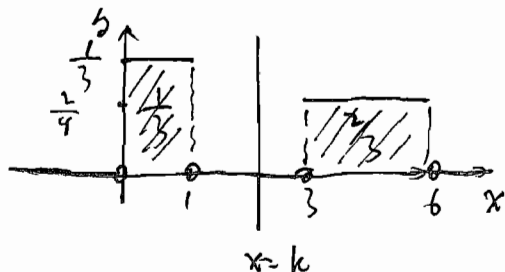
$= \int_{-\infty}^{+\infty} \frac{1}{a(x + \frac{b}{2a})^2 + c - \frac{b^2}{4a}} d(x + \frac{b}{2a})$

$\int \frac{1}{a+u^2} du = \frac{1}{a} \arctan \frac{u}{\sqrt{a}} + C, a > 0$

$\stackrel{\text{令 } x + \frac{b}{2a} = t}{=} \frac{1}{a} \cdot \int_{-\infty}^{+\infty} \frac{1}{t^2 + c - \frac{b^2}{4a}} dt = \frac{1}{a} \cdot \frac{1}{\sqrt{c - \frac{b^2}{4a}}} \cdot \arctan \frac{t}{\sqrt{c - \frac{b^2}{4a}}} \Big|_{-\infty}^{+\infty} = \frac{2\pi}{\sqrt{4ac - b^2}} = 1$.

7. (1) 设随机变量 X 的密度为 $f(x) = \begin{cases} 1/3, 0 \leq x \leq 1; \\ 2/9, 3 \leq x \leq 6; \\ 0, \text{其他}; \end{cases}$ 若 $P(X \geq k) = \frac{2}{3}$, 试确定

k 的取值范围;



$P(X \geq k) = \int_k^{+\infty} f(x) dx = \frac{2}{3}$. 证: 证: $k \in [1, 3]$.

(2) 设随机变量 X, Y 同分布 (又记为: $X \stackrel{d}{=} Y$), 且 X 有概率密度函数为

$$f(x) = \begin{cases} \frac{3}{8}x^2, & 0 < x < 2; \\ 0, & \text{其他}; \end{cases}$$

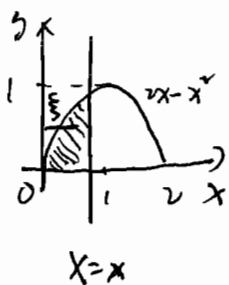
已知事件 $A = \{X > a\}$ 与 $B = \{Y > a\}$ 独立, 且 $P(A \cup B) = \frac{3}{4}$, 试求常数 a .

若令 $Y \sim f_Y(y)$, 则 $f_Y(y) = \begin{cases} \frac{3}{8}y^2, & 0 < y < 2; \\ 0, & \text{其他}; \end{cases}$ 从而, $P(X > a) = \int_a^{+\infty} f_X(x) dx$.

$P(Y > a) = \int_a^{+\infty} f_Y(y) dy$, 因 $P(A) = P(B) = x$, 由 $P(A \cup B) = P(A) + P(B) - P(A)P(B) = 2x - x^2 = \frac{3}{4}$
 $x^2 - 2x + \frac{3}{4} = (x - \frac{1}{2})(x - \frac{3}{2}) = 0$. $x = \frac{1}{2}$. 由 $P(A) = P(X > a) = \int_a^{+\infty} f_X(x) dx = \frac{1}{2}$. 从而:

$$\int_a^{+\infty} f_X(x) dx = \int_a^2 \frac{3}{8}x^2 dx + \int_2^{+\infty} 0 dx = \frac{1}{8}(8 - a^3) = \frac{1}{2}. \quad a = \sqrt[3]{4}.$$

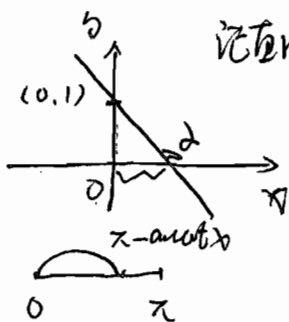
8. (1) 设 A 为曲线 $y = 2x - x^2$ 与 x 轴所围成的区域, 在 A 中任取一点, 求该点到 y 轴的距离 ξ 的分布函数及密度函数;



知 $R(\xi) = (0, 2)$; 从而, $\forall x \leq 0, F_\xi(x) = P\{\xi \leq x\} = P(\emptyset) = 0$;
 $\forall x \geq 2, F_\xi(x) = P\{\xi \leq x\} = P(\Omega) = 1$; $\forall x \in (0, 2), F_\xi(x) = P\{\xi \leq x\}$
 $\xrightarrow{\text{几何法}} \frac{\int_0^x (2t - t^2) dt}{\int_0^2 (2t - t^2) dt} = \frac{x^2 - \frac{1}{3}x^3}{4 - \frac{8}{3}} = \frac{3x^2 - x^3}{4}$. 从而:

$$F_\xi(x) = \begin{cases} 0, & x \leq 0; \\ \frac{3x^2 - x^3}{4}, & 0 < x < 2; \\ 1, & x \geq 2; \end{cases} \quad \text{令 } \xi \sim f_\xi(x), \text{ 从而: } f_\xi(x) = \frac{d}{dx} F_\xi(x) = \begin{cases} \frac{6x - 3x^2}{4}, & 0 < x < 2; \\ 0, & \text{其他}; \end{cases}$$

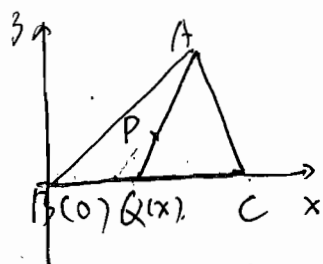
(2) 通过点 $(1, 0)$ 任意作直线与 x 轴相交成 α 角 ($0 < \alpha < \pi$), 求直线在 x 轴上的截距的概率密度函数;



记直线在 x 轴上的截距为 X , 知 $X = -\cot \alpha$, 且 $\alpha \sim U(0, \pi)$.
 从而 $R(X) = (-\infty, +\infty)$. $\forall x \in \mathbb{R}, F_X(x) = P(X \leq x) = P(-\cot \alpha \leq x)$
 $= P(\cot \alpha \geq -x) = P(\alpha \leq \arccot(-x)) = P(\alpha \leq \pi - \arccot x)$
 $= \frac{\pi - \arccot x}{\pi}$. 令 $X \sim f_X(x)$. 从而:

$$f_X(x) = \frac{d}{dx} F_X(x) = \left(\frac{\pi - \arccot x}{\pi} \right)' = \frac{1}{\pi(1+x^2)}, \quad -\infty < x < +\infty.$$

(3) 向任意 $\triangle ABC$ 内随机地抛掷一点 P ，并将 AP 延长交 CB 于 Q ，证明： Q 点服从 CB 上的均匀分布。



设 BC 边长为 a ， $P \in C(a, 0)$ ，记： $BQ = X$ 。

易见， $\forall x \leq 0$ ， $F_X(x) = P(X \leq x) = P(\emptyset) = 0$ ； $\forall x \geq a$ ， $F_X(x) = P(X \leq x) = P(\Omega) = 1$ 。

$\forall x \in [0, a]$ ， $F_X(x) = P(X \leq x) \xrightarrow{\text{几何法}} \frac{S_{\triangle ABQ}}{S_{\triangle ABC}} = \frac{x}{a}$ ， $\therefore$

$$F_X(x) = \begin{cases} 0 & , x < 0; \\ \frac{x}{a} & , 0 \leq x < a; \\ 1 & , x \geq a; \end{cases} \quad \text{令 } X \sim f_X(x), \text{ 则 } f_X(x) = \frac{dF_X(x)}{dx} = \begin{cases} \frac{1}{a} & , 0 < x < a; \\ 0 & , \text{其他} \end{cases} \quad \therefore X \sim U(0, a)$$

9. 假设一设备在时间 t (h) 内发生故障的次数 $N(t)$ 服从参数为 λt 的 Poisson 分布；若以 T 表示相邻两次故障之间的时间间隔，试求：

(1) T 的分布；(2) 故障修复之后，“设备无故障运行 $3h$ ” 的概率；

(3) 如果设备已经无故障运行 $3h$ 的情况下，“再无故障运行 $3h$ ” 的概率。

(1), 易见， $\forall t \leq 0$ ， $F_T(t) = P(T \leq t) = P(\emptyset) = 0$ ；

$\forall t > 0$ ， $F_T(t) = P(T \leq t) = 1 - P(T > t) = 1 - P(N(t) = 0)$ ， $N(t) \sim P(\lambda t)$

$$= 1 - \frac{(\lambda t)^0}{0!} e^{-\lambda t} = 1 - e^{-\lambda t}, \quad \therefore F_T(t) = \begin{cases} 1 - e^{-\lambda t} & , t > 0; \\ 0 & , t \leq 0; \end{cases} \quad \therefore T \sim E(\lambda)$$

$$(2) P(T > 3) = \int_3^{\infty} \lambda e^{-\lambda t} dt = e^{-3\lambda}$$

$$(3) P(T > 3+t | T > 3) \xrightarrow{\text{无记忆性}} P(T > 3) = e^{-3\lambda}.$$

10. (1) 设随机变量 $\xi \sim U[0, 5]$ ，试求“方程 $4x^2 + 4\xi x + \xi + 2 = 0$ 有实根” 的概率；

$$P(\text{方程 } 4x^2 + 4\xi x + \xi + 2 = 0 \text{ 有实根}) = P(\Delta = (4\xi)^2 - 4 \cdot 4(\xi + 2) \geq 0)$$

$$= P(\xi^2 - \xi - 2 = (\xi - 2)(\xi + 1) \geq 0) = P(\xi \geq 2 \text{ 或 } \xi \leq -1) = P(\xi \geq 2) \cup \{\xi \leq -1\}$$

$$= P(\xi \geq 2) + P(\xi \leq -1) = P(\xi \geq 2) \xrightarrow{\text{几何法}} \frac{5-2}{5-0} = \frac{3}{5}.$$

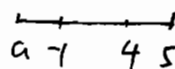


(2) 设 $Y \sim U(a, 5)$, “方程 $4x^2 + 4Yx + 3Y + 4 = 0$ 无实根” 的概率为 0.25, 求常数 a .

$$P(\{4x^2 + 4Yx + 3Y + 4 = 0 \text{ 无实根}\}) = P(\{\Delta = (4Y)^2 - 4 \cdot 4 \cdot (3Y + 4) < 0\})$$

$$= P(\{Y^2 - 3Y - 4 = (Y-4)(Y+1) < 0\}) = P(\{-1 < Y < 4\})$$

若 $a < -1$, $P(\{-1 < Y < 4\}) = \frac{\text{长度}}{\text{总长度}} = \frac{4 - (-1)}{5 - a} = \frac{1}{4}$



$$a = -15;$$

若 $a > -1$ (≤ 4), $P(\{-1 < Y \leq 4\}) = \frac{\text{长度}}{\text{总长度}} = \frac{4-a}{5-a} = \frac{1}{4}$



$$a = \frac{11}{3}$$

11. 设随机变量 ξ 的概率密度函数为 $f(x) = \frac{1}{\sqrt{\pi}} e^{-x^2 + 2x - 1}$, $-\infty < x < +\infty$, 试求

$$P(0 \leq \xi \leq 2).$$

$P(0 \leq \xi \leq 2) = \int_0^2 f(x) dx$... 积分出来!

由 $\xi \sim f(x) = \frac{1}{\sqrt{\pi}} e^{-\frac{(x-1)^2}{2}}$, $-\infty < x < +\infty$, 证 $\xi \sim N(1, (\frac{1}{\sqrt{\pi}})^2)$

$\frac{\xi-1}{\frac{1}{\sqrt{\pi}}} \sim N(0, 1^2)$. 从而, $P(0 \leq \xi \leq 2) = P(\frac{1}{\sqrt{\pi}} \leq \frac{\xi-1}{\frac{1}{\sqrt{\pi}}} \leq \frac{1}{\sqrt{\pi}}) = \Phi(\sqrt{\pi}) - \Phi(-\sqrt{\pi})$
 $= 2\Phi(\sqrt{\pi}) - 1 = \dots$

12. 设随机变量 $X \sim N(3, 2^2)$, 试求:

(1) $P(2 < X \leq 5)$, $P(|X| > 2)$; (2) 确定 c , 使得 $P(X > c) = P(X < c)$;

(3) 设 d 满足 $P(X > d) \geq 0.9$, d 至多为多少?

由 $X \sim N(3, 2^2)$. 证: $\frac{X-3}{2} \sim N(0, 1^2)$; 从而

(1) $P(2 < X \leq 5) = P(\frac{1}{2} < \frac{X-3}{2} \leq \frac{1}{2}) = \Phi(1) - \Phi(-\frac{1}{2}) = \Phi(1) + \Phi(\frac{1}{2}) - 1$;

$P(|X| > 2) = 1 - P(|X| \leq 2) = 1 - P(-\frac{5}{2} \leq \frac{X-3}{2} \leq \frac{1}{2}) = 1 - [\Phi(\frac{1}{2}) - \Phi(-\frac{5}{2})]$
 $= 1 + \Phi(\frac{1}{2}) - \Phi(\frac{5}{2})$

(2) $P(X > c) = P(X \leq c) = P(X < c) = \frac{1}{2}$, 证:

$P(X \leq c) = P(\frac{X-3}{2} \leq \frac{c-3}{2}) = \Phi(\frac{c-3}{2}) = \frac{1}{2}$. $\frac{c-3}{2} = 0$. $c = 3$.

(3) $P(X > d) = 1 - P(X \leq d) = 1 - P(\frac{X-3}{2} \leq \frac{d-3}{2}) = 1 - \Phi(\frac{d-3}{2})$

$= \Phi(-\frac{d-3}{2}) = \Phi(\frac{3-d}{2}) \geq 0.9 = \Phi(1.28)$

$\frac{3-d}{2} \geq 1.28$, $d \leq 0.44$.

13. 由学校到火车站有两条路线, 所需时间随交通堵塞状况有所变化, 若以分钟计, 第一条路线所需时间 $\xi_1 \sim N(50, 10^2)$, 第二条路线所需时间 $\xi_2 \sim N(60, 4^2)$,

如果要求:

(1) 在 70 分钟内赶到火车站;

(2) 在 65 分钟内赶到火车站; 试问: 各应选择哪条路线?

由题意, $\frac{\xi_1 - 50}{10} \sim N(0, 1)$, $\frac{\xi_2 - 60}{4} \sim N(0, 1)$;

(1) $P(\xi_1 \leq 70) = P(\frac{\xi_1 - 50}{10} \leq \frac{70 - 50}{10}) = \Phi(2)$, $P(\xi_2 \leq 70) = P(\frac{\xi_2 - 60}{4} \leq \frac{70 - 60}{4}) = \Phi(2.5)$
 $\Phi(2) < \Phi(2.5)$, 故选择第二条路线.

(2) $P(\xi_1 \leq 65) = P(\frac{\xi_1 - 50}{10} \leq \frac{65 - 50}{10}) = \Phi(1.5)$, $P(\xi_2 \leq 65) = P(\frac{\xi_2 - 60}{4} \leq \frac{65 - 60}{4}) = \Phi(1.25)$
 $\Phi(1.5) > \Phi(1.25)$, 故选择第一条路线.

14. 假设一机器的检修时间 (单位: 小时) 服从 $\lambda = \frac{1}{2}$ 的指数分布, 试求:

(1) “检修时间超过 2 小时” 的概率;

(2) 若已经检修 4 小时, 求 “总共至少 5 小时检修好” 的概率.

记检修时间为 X , 则有 $X \sim E(\frac{1}{2})$.

(1) $P(X > 2) = \int_2^{\infty} \frac{1}{2} e^{-\frac{1}{2}x} dx = e^{-1}$.

(2) $P(X > 5 | X > 4) \stackrel{\text{无记忆性}}{=} P(X > 1) = e^{-\frac{1}{2}}$.

15. (1) 设 $X \sim U(2, 5)$, 试求 “对 X 进行三次独立地观测中, 至少有两次观测值大于 3” 的概率;

为见, $P(X > 3) = \frac{2}{3}$. 记 Y 为观测到 $\{X > 3\}$ 出现的次数, $Y \sim B(3, \frac{2}{3})$.



$P(Y \geq 2) = P(Y=2) + P(Y=3)$

$= C_3^2 (\frac{2}{3})^2 (\frac{1}{3}) + C_3^3 (\frac{2}{3})^3$

(2) 设顾客在某银行的窗口等待服务的时间 X (以分钟记) 服从参数为 $\frac{1}{5}$ 的指数分布,

某顾客在窗口等待服务若超过 10 分钟他就离开; 他一个月要到银行五次, 以 Y 表示一个月内他未等到服务而离开窗口的次数, 试求 $P(Y \geq 1)$.

为见, $P(X > 10) = \int_{10}^{\infty} \frac{1}{5} e^{-\frac{1}{5}x} dx = -e^{-\frac{1}{5}x} \Big|_{10}^{\infty} = e^{-2}$. $Y \sim B(5, e^{-2})$.

$P(Y \geq 1) = 1 - P(Y < 1) = 1 - P(Y=0) = 1 - C_5^0 (e^{-2})^0 (1 - e^{-2})^5$
 $= 1 - (1 - e^{-2})^5$.

16. (1) 对某地考生抽样调查的结果表明：考生的外语成绩（百分制）近似服从 $N(72, \sigma^2)$ ($\sigma > 0$ 未知)；已知96分以上的考生占考生总数的2.3%，试求“考生成绩介于60分与84分之间”的概率；

随机地挑选一名考生，记其外语成绩为 X ，由题设， $X \sim N(72, \sigma^2)$ 。

$$\text{由 } P(X > 96) = 0.023, \text{ 即 } P(X \leq 96) = P\left(\frac{X-72}{\sigma} \leq \frac{24}{\sigma}\right) = 0.977, \text{ 即 } \Phi\left(\frac{24}{\sigma}\right) = 0.977 \\ = \Phi(2), \text{ 即 } \frac{24}{\sigma} = 2, \sigma = 12. \text{ 从而,}$$

$$P(60 < X < 84) = P\left(\frac{72-72}{12} < \frac{X-72}{12} < \frac{12}{12}\right) = \Phi(1) - \Phi(-1) = 2\Phi(1) - 1$$

(2) 用正态分布估计高考录取最低分。某市有9万名高中毕业生参加高考，结果有5.4万名被各类高校录取；已知满分为600分，540分以上者有2025人，360分以下者有13500人；试估计高考录取的最低分。

设高考的最低分为 a ，随机地挑选一名高中毕业生，记其高考成绩为 X ，设 $X \sim N(\mu, \sigma^2)$ 。

$$\text{由题设, } P(X > 540) = \frac{2025}{90000}. \text{ 即 } P(X \leq 540) = P\left(\frac{X-\mu}{\sigma} \leq \frac{540-\mu}{\sigma}\right) = 1 - \frac{2025}{90000} = 0.9775 = \Phi(2.01) \\ \frac{540-\mu}{\sigma} = 2.01, \quad P(X < 360) = P\left(\frac{X-\mu}{\sigma} < \frac{360-\mu}{\sigma}\right) = \Phi\left(\frac{360-\mu}{\sigma}\right) = \frac{13500}{90000}$$

$$\text{即: } \Phi\left(\frac{\mu-360}{\sigma}\right) = \frac{76500}{90000} = 0.85 = \Phi(1.04), \text{ 即 } \frac{\mu-360}{\sigma} = 1.04$$

$$\text{由 } P(X \geq a) = \frac{5.4}{9} = 0.6, \text{ 即 } P(X \leq a) = \Phi\left(\frac{a-\mu}{\sigma}\right) = 0.4 \Rightarrow \Phi\left(\frac{\mu-a}{\sigma}\right) = 0.6$$

17. 用正态分布设计公交大巴车门的高度；

设计要求：男子与车门顶端碰头的机会必须控制在1%以下；

参数提供：通过大范围的抽样调查，已知中国男性的平均身高为173cm，标准差为9cm。

随机地挑选一名男性，记其身高为 X (cm)，由题设， $X \sim N(173, 9^2)$ 。

设公交车的车门高度为 h (cm)，基于设计要求，

$$\text{由 } P(X > h) \leq 0.1, \text{ 即 } P(X \leq h) \geq 0.99, \text{ 即}$$

$$P(X \leq h) = P\left(\frac{X-173}{9} \leq \frac{h-173}{9}\right) = \Phi\left(\frac{h-173}{9}\right) \geq 0.99 = \Phi(2.33)$$

$$\text{即 } \frac{h-173}{9} \geq 2.33, \quad h \geq 173 + 20.97 = 193.97, \text{ 故至少}$$

车门高度为194cm。

18. 设某电子元件在工作中其两端电压 $V \sim N(220, 20^2)$, 当 $V \in [200, 240]$, 失效的概率为 0.05; 当 $V < 200$, 失效的概率为 0.1; 当 $V > 240$, 失效的概率为 0.5; 求:

(1) “此元件失效” 的概率;

(2) “当元件失效时, 电压超过 240” 的概率.

设 $A = \{\text{元件失效}\}$. 由题知, $P(A|200 \leq V \leq 240) = 0.05$, $P(A|V < 200) = 0.1$, $P(A|V > 240) = 0.5$;

$$\begin{aligned} (1) P(A) &= P(200 \leq V \leq 240) \cdot P(A|200 \leq V \leq 240) + P(V < 200) \cdot P(A|V < 200) + \\ &\quad P(V > 240) \cdot P(A|V > 240) \\ &= [\Phi(1) - \Phi(-1)] \cdot 0.05 + \Phi(-1) \cdot 0.1 + [1 - \Phi(1)] \cdot 0.5 \\ &= [2\Phi(1) - 1] \cdot 0.05 + [1 - \Phi(1)] \cdot 0.1 + [1 - \Phi(1)] \cdot 0.5 \end{aligned}$$

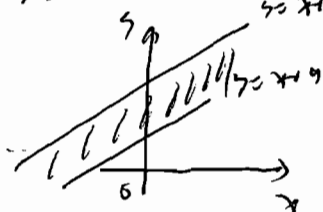
$$\begin{aligned} (2) P(V > 240 | A) &= \frac{P\{V > 240 \cap A\}}{P(A)} = \frac{P(V > 240) \cdot P(A|V > 240)}{P(A)} \\ &= \frac{[1 - \Phi(1)] \cdot 0.5}{[2\Phi(1) - 1] \cdot 0.05 + [1 - \Phi(1)] \cdot 0.1 + [1 - \Phi(1)] \cdot 0.5} \end{aligned}$$

19. 设 $F(x)$ 是连续型随机变量 X 的分布函数, 证明: 对于任意的实数 $a, b, a < b$,

下面等式成立: $\int_{-\infty}^{+\infty} [F(x+b) - F(x+a)] dx = b - a$.

证 $X \sim f(x)$. 则 $F(x+b) - F(x+a) = \int_{x+a}^{x+b} f(y) dy$. 从而,

$$\begin{aligned} \int_{-\infty}^{+\infty} [F(x+b) - F(x+a)] dx &= \int_{-\infty}^{+\infty} \left[\int_{x+a}^{x+b} f(y) dy \right] dx \xrightarrow{\text{交换次序}} \int_{-\infty}^{+\infty} \left[\int_{y-b}^{y-a} f(y) dx \right] dy \\ &= \int_{-\infty}^{+\infty} (b-a) \cdot f(y) dy = (b-a) \cdot \int_{-\infty}^{+\infty} f(y) dy = b-a. \end{aligned}$$



习题 2.4

1. (2) 设随机变量 X 的概率分布为: $P(X=k) = \frac{1}{2^k}, k=1, 2, \dots$; 试求 $Y = \sin\left(\frac{\pi}{2}X\right)$ 的分布律.

易见, $R(Y) = \{-1, 0, 1\}$. 且 $\{Y=0\} = \bigcup_{k=1}^{\infty} \{X=2k\}$

$$P(Y=0) = P\left(\bigcup_{k=1}^{\infty} \{X=2k\}\right) = \sum_{k=1}^{\infty} P(X=2k) = \sum_{k=1}^{\infty} \frac{1}{2^{2k}} = \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{1}{3}.$$

$$P(Y=1) = P\left(\bigcup_{k=0}^{\infty} \{X=4k+1\}\right) = \sum_{k=0}^{\infty} P(X=4k+1) = \sum_{k=0}^{\infty} \frac{1}{2^{4k+1}} = \frac{\frac{1}{2}}{1 - \frac{1}{16}} = \frac{8}{15}.$$

$$P(Y=-1) = 1 - P(Y=0) - P(Y=1) = \frac{2}{15}, \text{ 即:}$$

Y	-1	0	1
P	$\frac{2}{15}$	$\frac{1}{3}$	$\frac{8}{15}$

2. (1) 设随机变量 $X \sim U(-1, 2)$, 记 $Y = \begin{cases} 1, X \geq 0; \\ -1, X < 0; \end{cases}$, 试求 Y 的分布列;

记: $f(x) = \begin{cases} 1, x \geq 0; \\ -1, x < 0. \end{cases}$ 则 $Y = f(X) = \begin{cases} 1, X \geq 0 \\ -1, X < 0 \end{cases}$ 且 $P(Y=1) = P(\{X \geq 0\})$

$$= \frac{2}{3}, \text{ 从而, } \begin{array}{c|cc} Y & -1 & 1 \\ \hline P & \frac{1}{3} & \frac{2}{3} \end{array}$$

(2) 设随机变量 $\xi \sim U[0, 1]$, 试求 $X = [n\xi] + 1$ 与 $Y = \left[\frac{\ln \xi}{\ln q} \right] + 1 (0 < q < 1)$ 的分布.

① 求 $X = [n\xi] + 1$ 的分布: 易见, $R(X) = \{1, 2, \dots, n+1\}$, 且

$$P(X = n+1) = P([n\xi] + 1 = n+1) = P([n\xi] = n) = P(\xi = 1) = 0, \quad \forall k \in \{1, 2, \dots, n\}$$

$$P(X = k) = P([n\xi] + 1 = k) = P([n\xi] = k-1) = P(k-1 \leq n\xi < k) = P\left(\frac{k-1}{n} \leq \xi < \frac{k}{n}\right)$$

$$= \frac{1}{n}, \text{ 从而: } X \sim \left(\frac{1}{n}, \frac{2}{n}, \dots, \frac{n}{n} \right)$$

② $\forall k \geq 1, (k=1, 2, \dots), P(Y=k) = P\left(\left[\frac{\ln \xi}{\ln q}\right] + 1 = k\right) = P\left(\left[\frac{\ln \xi}{\ln q}\right] = k-1\right)$

$$= P\left(k-1 \leq \frac{\ln \xi}{\ln q} < k\right) = P(k \ln q < \ln \xi \leq (k-1) \ln q) = P(q^k \leq \xi < q^{k-1}) = q^{k-1} - q^k$$

$$= q^{k-1} (1-q) \stackrel{\text{记: } p=1-q}{=} (1-p)^{k-1} p, \text{ 从而: } Y \sim G(p).$$

3. (1) 设随机变量 $X \sim U(0, 1)$, 试求 $1-X$ 的分布;

解: $\overline{0 \quad 1} \quad X \sim U(0, 1), \quad 1-X \sim U(0, 1).$

记: $Y = 1-X$, 由 $R(X) = (0, 1), R(Y) = (0, 1), \forall s \leq 0, F_Y(s) = P(Y \leq s) = P(\emptyset) = 0$

$\forall s \geq 1, F_Y(s) = P(Y \leq s) = P(\Omega) = 1, \forall s \in (0, 1), F_Y(s) = P(Y \leq s) = P(1-X \leq s)$

$$= P(X \geq 1-s) \stackrel{\text{几何法}}{=} \frac{1-(1-s)}{1-0} = s. \text{ 从而 } F_Y(s) = \begin{cases} 0, & s \leq 0; \\ s, & 0 < s < 1; \\ 1, & s \geq 1; \end{cases}$$

令 $Y \sim f_Y(s)$, 则 $f_Y(s) = \frac{d}{ds} F_Y(s) = \begin{cases} 1, & 0 < s < 1; \\ 0, & \text{其他} \end{cases}$ 从而 $Y = 1-X \sim U(0, 1).$

(2) 设随机变量 $X \sim E(2)$, 试证: $Y_1 = e^{-2X}$ 与 $Y_2 = 1 - e^{-2X}$ 均服从 $(0,1)$ 上的均匀分布.

易知, $R(X) = (0, +\infty)$, $R(Y_1) = (0, 1)$, 从而, $\forall y \leq 0$, $F_1(y) = P(Y_1 \leq y) = P(\emptyset) = 0$;
 $\forall y \geq 1$, $F_1(y) = P(Y_1 \leq y) = P(\Omega) = 1$; $\forall y \in (0, 1)$, $F_1(y) = P(Y_1 \leq y) = P(e^{-2X} \leq y)$
 $= P(-2X \leq \ln y) = P(X \geq -\frac{1}{2} \ln y) = \int_{-\frac{1}{2} \ln y}^{+\infty} 2e^{-2x} dx = -e^{-2x} \Big|_{-\frac{1}{2} \ln y}^{+\infty} = y$, 证:

$$F_1(y) = \begin{cases} 0 & , y \leq 0; \\ y & , 0 < y < 1; \\ 1 & , y \geq 1; \end{cases} \quad \text{证: } Y_1 \sim U(0, 1); \quad \forall y \in (0, 1), \quad Y_2 = 1 - Y_1 \sim U(0, 1)$$

4. 若随机变量 $\ln X \sim N(\mu, \sigma^2)$, 则称 X 服从对数正态分布;

(1) 试求 X 的概率密度函数 $f_X(x)$; (2) 若 $\ln X \sim N(1, 4^2)$, 求 $P\left(\frac{1}{e} \leq X \leq e^3\right)$.

$$(2) \quad P\left(\frac{1}{e} \leq X \leq e^3\right) = P(-1 \leq \ln X \leq 3) = P\left(-\frac{2}{4} \leq \frac{\ln X - 1}{4} \leq \frac{2}{4}\right) = \Phi\left(\frac{1}{2}\right) - \Phi\left(-\frac{1}{2}\right) \\ = 2\Phi\left(\frac{1}{2}\right) - 1$$

$$(1) \quad \text{记: } Y = \ln X, \quad \text{则 } X = e^Y, \quad \text{且 } Y \sim N(\mu, \sigma^2),$$

$$\text{若 } Y \sim f_Y(y), \quad \text{则 } f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\mu)^2}{2\sigma^2}}, \quad -\infty < y < +\infty,$$

易知, $\forall x > 0$,

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \cdot f_Y[\ln x] \cdot |(\ln x)'| = \frac{1}{\sqrt{2\pi}\sigma x} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}, \quad x > 0.$$

$$\text{证: } f_X(x) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma x} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}} & , x > 0; \\ 0 & , x \leq 0. \end{cases}$$

5. 设随机变量 $X \sim U(0,1)$, 试求以下 Y 的密度函数:

(1) $Y = -2\ln X$; (2) $Y = 3X + 1$; (3) $Y = e^X$; (4) $Y = |\ln X| = -\ln X$

(1) 易见, $R(X) = X(\Omega) = (0, 1)$, $R(Y) = (0, +\infty)$. 从而, $\forall y \leq 0$, $F_Y(y) = P\{Y \leq y\} = P(\emptyset) = 0$.
 $\forall y > 0$, $F_Y(y) = P\{Y \leq y\} = P\{-2\ln X \leq y\} = P\{X \geq e^{-\frac{y}{2}}\} = 1 - e^{-\frac{y}{2}}$. 

即: $F_Y(y) = \begin{cases} 1 - e^{-\frac{y}{2}}, & y > 0; \\ 0, & y \leq 0; \end{cases}$ 令 $Y \sim f_Y(y)$, 则有: $f_Y(y) = \frac{d}{dy} F_Y(y) = \begin{cases} \frac{1}{2} e^{-\frac{y}{2}}, & y > 0; \\ 0, & y \leq 0; \end{cases}$

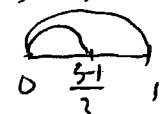
即: $Y \sim E(\frac{1}{2})$.

另法: 设 $X \sim f_X(x)$, $Y \sim f_Y(y)$, $y = g(x) = -2\ln x$, $x \in (0, 1)$, 其反函数为: $x = e^{-\frac{y}{2}}$, $y \in (0, +\infty)$

即: $f_X(x) = \begin{cases} 1, & 0 < x < 1; \\ 0, & \text{其他}; \end{cases}$ 

$f_Y(y) = f_X[h(y)] \cdot |h'(y)|$, $y > 0$.
 $= 1 \cdot \frac{1}{2} e^{-\frac{y}{2}}$, $y > 0$
 $= \begin{cases} \frac{1}{2} e^{-\frac{y}{2}}, & y > 0; \\ 0, & y \leq 0; \end{cases}$

(2) 易见 $R(X) = (0, 1)$, $R(Y) = (1, 4)$. 从而, $\forall y \leq 1$, $F_Y(y) = P\{Y \leq y\} = P(\emptyset) = 0$;

$\forall y \geq 4$, $F_Y(y) = P\{Y \leq y\} = P(\Omega) = 1$; $\forall y \in (1, 4)$, $F_Y(y) = P\{Y \leq y\} = P\{3X + 1 \leq y\}$
 $= P\{X \leq \frac{y-1}{3}\} = \frac{y-1}{3}$, 即: 

$F_Y(y) = \begin{cases} 0, & y \leq 1; \\ \frac{y-1}{3}, & 1 < y < 4; \\ 1, & y \geq 4; \end{cases}$ 令 $Y \sim f_Y(y)$, 则有: $f_Y(y) = \frac{d}{dy} F_Y(y) = \begin{cases} \frac{1}{3}, & 1 < y < 4; \\ 0, & \text{其他} \end{cases}$

即: $Y \sim U(1, 4)$.

另法: 设 $X \sim f_X(x)$, $Y \sim f_Y(y)$, $y = g(x) = 3x + 1$, $x \in (0, 1)$, 其反函数为: $x = h(y)$

$= \frac{y-1}{3}$, $y \in (1, 4)$; 即:

$f_Y(y) = f_X[h(y)] \cdot |h'(y)|$, $y \in (1, 4)$

$= 1 \cdot \frac{1}{3}$, $1 < y < 4$

$= \begin{cases} \frac{1}{3}, & y \in (1, 4); \\ 0, & \text{其他} \end{cases}$

6. (1) 设随机变量 X 的密度函数为 $f(x) = \begin{cases} \frac{1}{3\sqrt[3]{x^2}}, & x \in [1, 8]; \\ 0, & \text{其他}; \end{cases}$, $F(x)$ 为 X 的分布

函数, 试求随机变量 $Y = F(X)$ 的分布函数;

解: $R(X) = [1, 8]$, $\forall x < 1, F(x) = P\{X \leq x\} = P(\emptyset) = 0$; $\forall x \geq 8, F(x) = P\{X \leq x\} = P(\Omega) = 1$;

$$\forall x \in (1, 8), F(x) = P\{X \leq x\} = \int_{-\infty}^x f(t) dt = \left(\int_{-\infty}^1 + \int_1^x \right) f(t) dt = \int_1^x \frac{1}{3\sqrt[3]{t^2}} dt = \sqrt[3]{t} \Big|_1^x = \sqrt[3]{x} - 1;$$

$$\text{即: } F(x) = \begin{cases} 0, & x < 1; \\ \sqrt[3]{x} - 1, & 1 \leq x < 8; \\ 1, & x \geq 8. \end{cases} \quad \text{由 } Y = F(X), \text{ 知 } R(Y) = [\sqrt[3]{x} - 1, \text{ 知 } R(Y) = [0, 1].$$

$\forall y < 0, F_Y(y) = P\{Y \leq y\} = P(\emptyset) = 0$; $\forall y \geq 1, F_Y(y) = P\{Y \leq y\} = P(\Omega) = 1$;

$\forall y \in (0, 1), F_Y(y) = P\{Y \leq y\} = P\{F(X) \leq y\} = P\{\sqrt[3]{X} - 1 \leq y\} = P\{X \leq (1+y)^3\}$

$$= F((1+y)^3) = \sqrt[3]{(1+y)^3} - 1 = y. \quad \text{即: } F_Y(y) = \begin{cases} 0, & y < 0; \\ y, & 0 \leq y < 1; \\ 1, & y \geq 1; \end{cases} \quad \text{即: } Y \sim U(0, 1).$$

(2) 设随机变量 X 的分布函数 $F(x)$ 为严格单调连续函数, 试求 $Y = F(X)$ 的分布函数 $F_Y(y)$;

解: $R(Y) = (0, 1)$, $\forall y \leq 0, F_Y(y) = 0$; $\forall y \geq 1, F_Y(y) = 1$;

$\forall y \in (0, 1), F_Y(y) = P\{Y \leq y\} = P\{F(X) \leq y\} = P\{X \leq F^{-1}(y)\}$

$$= F[F^{-1}(y)] = y. \quad \text{即:}$$

$$F_Y(y) = \begin{cases} 0, & y \leq 0; \\ y, & 0 < y < 1; \\ 1, & y \geq 1; \end{cases} \quad \text{即: } Y \sim U(0, 1).$$

注: 即使 $F(x)$ 非严格单调, 依然会有 $Y = F(X) \sim U(0, 1)$.

(3) 设随机变量 X 的分布函数 $F(x)$ 为严格单调连续函数, 试求 $Z = -2\ln F(X)$ 的概率分布.

证 1: 由 $F(X) \sim U(0, 1)$, 即 $Z = -2\ln F(X) \sim E(\frac{1}{2})$.

证 2: 易见, $R(Z) = (0, +\infty)$. $\forall z \leq 0, F_Z(z) = P\{Z \leq z\} = P\{\emptyset\} = 0$; $\forall z > 0$,

$$\begin{aligned} F_Z(z) &= P\{Z \leq z\} = P\{-2\ln F(X) \leq z\} = P\{\ln F(X) \geq -\frac{z}{2}\} = P\{F(X) \geq e^{-\frac{z}{2}}\} \\ &= P\{X \geq F^{-1}(e^{-\frac{z}{2}})\} = P\{X > F^{-1}(e^{-\frac{z}{2}})\} = 1 - P\{X \leq F^{-1}(e^{-\frac{z}{2}})\} \\ &= 1 - F[F^{-1}(e^{-\frac{z}{2}})] = 1 - e^{-\frac{z}{2}}; \quad \text{即:} \end{aligned}$$

$$F_Z(z) = \begin{cases} 1 - e^{-\frac{z}{2}}, & z > 0; \\ 0, & z \leq 0; \end{cases} \quad \text{即: } Z \sim E(\frac{1}{2}).$$

7. (1) 设随机变量 $X \sim N(0, 1)$, $\Phi(x)$ 表示 X 的分布函数, 证明: 随机变量

$$Y = X + |X| \text{ 的分布函数 } F_Y(y) = \begin{cases} \Phi\left(\frac{y}{2}\right), & y \geq 0; \\ 0, & y < 0; \end{cases}$$

易见, $R(Y) = [0, +\infty)$; 即 $\forall y < 0, F_Y(y) = P\{Y \leq y\} = P\{\emptyset\} = 0$; $\forall y \geq 0$,

$$\begin{aligned} F_Y(y) &= P\{Y \leq y\} = P\{X \geq 0\} P\{Y \leq y | X \geq 0\} + P\{X < 0\} P\{Y \leq y | X < 0\} \\ &= P\{X \geq 0\} P\{2X \leq y | X \geq 0\} + P\{X < 0\} P\{0 \leq y | X < 0\} \\ &= P\{X \geq 0\} \cdot P\{X \leq \frac{y}{2} | X \geq 0\} + P\{X < 0\} = P\{0 \leq X \leq \frac{y}{2}\} + P\{X < 0\} \\ &= P\{X \leq \frac{y}{2}\} = \Phi\left(\frac{y}{2}\right). \quad \text{即: } F_Y(y) = \begin{cases} \Phi\left(\frac{y}{2}\right), & y \geq 0; \\ 0, & y < 0; \end{cases} \end{aligned}$$

Y 的概率密度函数

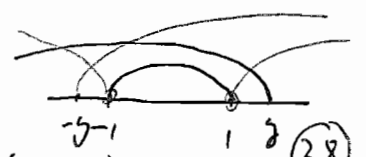
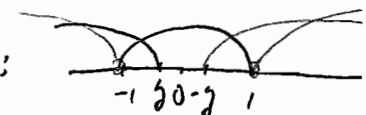
(2) 设随机变量 $\xi \sim N(0, 1^2)$, $\eta = \xi$ 或 $\eta = -\xi$ 视 $|\xi| \leq 1$ 或 $|\xi| > 1$ 而定, 试求 η 的分布.

设 $g(x) = \begin{cases} x, & |x| \leq 1; \\ -x, & |x| > 1; \end{cases}$ 即 $\eta = g(\xi) = \begin{cases} \xi, & |\xi| \leq 1; \\ -\xi, & |\xi| > 1; \end{cases}$ 其中, $\forall \eta \in R$,

$$P\{|\xi| \leq 1\} \cdot P\{\eta \leq y | |\xi| \leq 1\} + P\{|\xi| > 1\} \cdot P\{\eta \leq y | |\xi| > 1\}$$

$$\begin{aligned} F_Y(y) &= P\{\eta \leq y\} = P\{|\xi| \leq 1\} \cdot P\{\xi \leq y | |\xi| \leq 1\} + P\{|\xi| > 1\} \cdot P\{\xi \geq -y | |\xi| > 1\} \\ &= P\{\xi \leq y, |\xi| \leq 1\} + P\{\xi \geq -y, |\xi| > 1\} \end{aligned}$$

$$= \begin{cases} P\{\emptyset\} + P\{\xi \geq -y\} = 1 - \Phi(-y) = \Phi(y), & y < -1; \\ P\{-1 \leq \xi \leq y\} + P\{\xi > 1\} = \Phi(y) - \Phi(-1) + 1 - \Phi(1) = \Phi(y), & -1 \leq y \leq 1; \\ P\{-1 \leq \xi \leq 1\} + P\{-y \leq \xi < -1\} + P\{\xi > 1\} \\ = \Phi(1) - \Phi(-1) + \Phi(-1) - \Phi(-y) + 1 - \Phi(1) = \Phi(y), & y > 1 \end{cases}$$



8. 设随机变量 $X \sim f(x) = \begin{cases} \frac{1}{9}x^2, & 0 < x < 3; \\ 0, & \text{其他}; \end{cases}$ 且 $Y = \begin{cases} 2, & X \leq 1; \\ X, & 1 < X < 2; \\ 1, & X \geq 2; \end{cases}$ 试求:

(1) Y 的分布函数; (2) $P(X \leq Y)$.

解: 设 $g(x) = \begin{cases} 2, & x \leq 1; \\ x, & 1 < x < 2; \\ 1, & x \geq 2. \end{cases}$ 则 $Y = g(X)$, $\forall g(x), x \in (0, 3)$, 则有: $R(Y) = [1, 2]$;

当 $y < 1$, $F_Y(y) = 0$; 当 $y \geq 2$, $F_Y(y) = 1$; $\forall y \in [1, 2]$, $F_Y(y) = P(Y \leq y) = P(X \leq 1) \cdot P(Y \leq y | X \leq 1) + P(1 < X < 2) \cdot P(Y \leq y | 1 < X < 2) + P(X \geq 2) \cdot P(Y \leq y | X \geq 2) = P(X \leq 1) \cdot P(2 \leq y | X \leq 1) + P(1 < X < 2) \cdot P(X \leq y | 1 < X < 2) + P(X \geq 2) \cdot P(1 \leq y | X \geq 2) = 0 + P(X \leq y, 1 < X < 2) + P(X \geq 2) = P(1 < X \leq y) + P(X \geq 2) = \int_1^y \frac{1}{9}x^2 dx + \int_2^3 \frac{1}{9}x^2 dx = \frac{y^3 + 18}{27}$, 即: $F_Y(y) = \begin{cases} 0, & y < 1; \\ \frac{y^3 + 18}{27}, & 1 \leq y < 2; \\ 1, & y \geq 2. \end{cases}$
 $F_Y(1) - F_Y(1-) = P(Y=1) = \frac{19}{27}$, $F_Y(2) - F_Y(2-) = P(Y=2) = \frac{1}{27}$, Y 是连续型随机变量.

(2) $P(X \leq Y) = P(X \leq 1) \cdot P(X \leq Y | X \leq 1) + P(1 < X < 2) \cdot P(X \leq Y | 1 < X < 2) + P(X \geq 2) \cdot P(X \leq Y | X \geq 2) = P(X \leq 1) \cdot P(X \leq 2 | X \leq 1) + P(1 < X < 2) \cdot P(X \leq X | 1 < X < 2) + P(X \geq 2) \cdot P(X \leq 1 | X \geq 2) = P(X \leq 1) \cdot 1 + P(1 < X < 2) \cdot 1 + 0 = P(X < 2) = \int_0^2 \frac{1}{9}x^2 dx = \frac{8}{27}$

9. 随机变量 X 的概率密度函数为 $f_X(x) = \begin{cases} 0, & x \leq 0; \\ \frac{1}{2}, & 0 < x < 1; \\ \frac{1}{2}x^2, & x \geq 1; \end{cases}$ 试求随机变量

$Y = \frac{1}{X}$ 的概率密度函数 $f_Y(y)$.

法1: 易见, $R(X) = (0, +\infty)$, $R(Y) = (0, +\infty)$, $\forall y \leq 0$, $F_Y(y) = P(Y \leq y) = P(\emptyset) = 0$; $\forall y > 0$, $F_Y(y) = P(Y \leq y) = P(\frac{1}{X} \leq y) = P(X \geq \frac{1}{y}) = \int_{\frac{1}{y}}^{+\infty} f_X(x) dx = \begin{cases} \int_{\frac{1}{y}}^{+\infty} \frac{1}{2x^2} dx = \frac{y}{2}, & 0 < y < 1; \\ \int_{\frac{1}{y}}^1 \frac{1}{2} dx + \int_1^{+\infty} \frac{1}{2x^2} dx = \frac{1}{2} - \frac{1}{2y}, & y \geq 1; \end{cases}$
 即: $F_Y(y) = \begin{cases} 0, & y \leq 0; \\ \frac{y}{2}, & 0 < y < 1; \\ 1 - \frac{1}{2y}, & y \geq 1; \end{cases}$ $f_Y(y) = \frac{d}{dy} F_Y(y) = \begin{cases} \frac{1}{2}, & 0 < y < 1; \\ \frac{1}{2y^2}, & y \geq 1; \\ 0, & \text{其他}; \end{cases}$

法2: 记: $y = g(x) = \frac{1}{x}$, $x \in (0, +\infty)$, 则有反函数 $x = h(y) = \frac{1}{y}$, $y \in (0, +\infty)$, 则有

$f_Y(y) = f_X[h(y)] \cdot |h'(y)|$, $y > 0$:

$= f_X[\frac{1}{y}] \cdot \frac{1}{y^2}$, $y > 0$

$= \begin{cases} \frac{1}{2} \cdot \frac{1}{y^2} = \frac{1}{2y^2}, & 0 < y < 1; \\ \frac{1}{2} \cdot \frac{1}{y^2} = \frac{1}{2y^2}, & y \geq 1; \end{cases}$

10. 设随机变量 X 的密度函数为 $f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 0; \\ \frac{x}{4}, & 0 < x < 2; \\ 0, & \text{其他}; \end{cases}$ 令 $Y = \min\{2X, 1\}$, 试

求 Y 的分布函数及 $P(Y=1)$.

解: $R(X) = (-1, 2)$, $R(Y) = (-2, 1]$. $\forall y \leq -2, F_Y(y) = P(Y \leq y) = P(\emptyset) = 0$.
 $\forall y \geq 1, F_Y(y) = P(Y \leq y) = 1$; $\forall y \in (-2, 1)$, $F_Y(y) = P(Y \leq y) = P(2X \wedge 1 \leq y) = 1 - P(2X \wedge 1 > y)$
 $= 1 - P(2X > y, \frac{1}{2} > y) = 1 - P(X > \frac{y}{2}) = P(X \leq \frac{y}{2}) = \int_{-\infty}^{\frac{y}{2}} f_X(x) dx = \begin{cases} \int_{-1}^{\frac{y}{2}} \frac{1}{2} dx, & -2 < y < 0; \\ \int_{-1}^0 \frac{1}{2} dx + \int_0^{\frac{y}{2}} \frac{x}{4} dx, & 0 \leq y < 1; \end{cases}$
 $= \begin{cases} \frac{y}{4} + \frac{1}{2}, & -2 < y < 0; \\ \frac{1}{2} + \frac{y^2}{32}, & 0 \leq y < 1; \\ 1, & y \geq 1; \end{cases}$ 即: $F_Y(y) = \begin{cases} 0, & y \leq -2; \\ \frac{y}{4} + \frac{1}{2}, & -2 < y < 0; \\ \frac{y^2}{32} + \frac{1}{2}, & 0 \leq y < 1; \\ 1, & y \geq 1; \end{cases}$ Y - 离散型随机变量

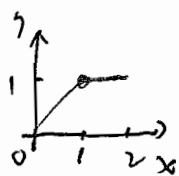
$P(Y=1) = F_Y(1) - F_Y(1-) = 1 - \lim_{y \rightarrow 1^-} (\frac{y^2}{32} + \frac{1}{2}) = \frac{15}{32}$.

11. (1) 设随机变量 $X \sim U[0, 2]$, $g(x) = \begin{cases} x, & 0 \leq x < 1; \\ 1, & 1 \leq x \leq 2; \end{cases}$ 试求随机变量 $Y = g(X)$

的分布函数, 并问: Y 是否为连续型随机变量?

解: $R(X) = [0, 2]$, $R(Y) = [0, 1]$. $\forall y < 0, F_Y(y) = P(Y \leq y) = P(\emptyset) = 0$;

$\forall y \geq 1, F_Y(y) = P(Y \leq y) = P(\Omega) = 1$; $\forall y \in [0, 1)$,



$F_Y(y) = P(Y \leq y) = P(0 \leq X < 1) \cdot P(Y \leq y | 0 \leq X < 1) + P(1 \leq X \leq 2) \cdot P(Y \leq y | 1 \leq X \leq 2)$
 $= P(0 \leq X < 1) \cdot P(X \leq y | 0 \leq X < 1) + P(1 \leq X \leq 2) \cdot P(1 \leq y | 1 \leq X)$
 $= P(0 \leq X < 1, X \leq y) + 0 = P(0 \leq X \leq y) = \frac{y}{2}$. 即:

$Y = g(X) = \begin{cases} X, & 0 \leq X < 1; \\ 1, & 1 \leq X \leq 2; \end{cases}$

$F_Y(y) = \begin{cases} 0, & y < 0; \\ \frac{y}{2}, & 0 \leq y < 1; \\ 1, & y \geq 1; \end{cases}$ 不是连续型.

$F_Y(1) - F_Y(1-) = P(Y=1) = 1 - \frac{1}{2} = \frac{1}{2}$

Y - 离散型随机变量

(2) 设随机变量 $X \sim E(\lambda)$, 证明: $Y = X \vee 2020 = \max\{X, 2020\}$ 的分布函数 $F_Y(y)$ 恰好有一个间断点;

解: $R(X) = (0, +\infty)$, $R(Y) = [2020, +\infty)$. $\forall y < 2020$, $\bar{F}_Y(y) = P\{Y \leq y\} = P\{\emptyset\} = 0$.
 $\forall y \geq 2020$, $\bar{F}_Y(y) = P\{Y \leq y\} = P\{X \vee 2020 \leq y\} = P\{X \leq y, \underbrace{2020 \leq y}_{\Omega}\} = P\{X \leq y\}$
 $= 1 - e^{-\lambda y}$, 即: $\bar{F}_Y(y) = \begin{cases} 0 & , y < 2020; \\ 1 - e^{-\lambda y} & , y \geq 2020. \end{cases}$ Y 不是连续型随机变量!

验证间断点: $y = 2020$. $\bar{F}_Y(2020) - \bar{F}_Y(2020-) = 1 - e^{-2020\lambda} > 0$.

(3) 假设一设备开机后无故障工作的时间 $X \sim E\left(\frac{1}{5}\right)$, 设备定时开机, 出现故障时自动关机; 且在无故障的情况下工作 2 小时便关机, 试求该设备每次开机无故障工作的时间 Y 的分布函数 $F_Y(y)$, 并指明 Y 是否为连续型随机变量?

由题意, $Y = X \wedge 2$, 解: $R(Y) = (0, 2]$, 即: $\forall y \leq 0$, $\bar{F}_Y(y) = P\{Y \leq y\} = P\{\emptyset\} = 0$;
 $\forall y > 2$, $\bar{F}_Y(y) = P\{Y \leq y\} = P(\Omega) = 1$; $\forall y \in (0, 2)$, $\bar{F}_Y(y) = P\{Y \leq y\} = 1 - P\{Y > y\}$
 $= 1 - P\{X \wedge 2 > y\} = 1 - P\{X > y, \underbrace{2 > y}_{\Omega}\} = 1 - P\{X > y\} = 1 - e^{-\frac{1}{5}y}$; 即:
 $\bar{F}_Y(y) = \begin{cases} 0 & , y \leq 0; \\ 1 - e^{-\frac{1}{5}y} & , 0 < y < 2; \\ 1 & , y \geq 2. \end{cases}$ $\bar{F}_Y(2) - \bar{F}_Y(2-) = 1 - (1 - e^{-\frac{2}{5}}) = e^{-\frac{2}{5}} > 0$
 $= P\{Y = 2\}$

Y 不是连续型随机变量, Y 是混合型随机变量!

12. 设随机变量 X 的分布函数 $F(x) = \begin{cases} 0, & x < 0; \\ x + \frac{1}{3}, & 0 \leq x < \frac{1}{2}; \\ 1, & x \geq \frac{1}{2}; \end{cases}$, 试求:

(1) $P\left(X = \frac{1}{4}\right), P\left(X = \frac{1}{2}\right)$; (2) $P\left(0 < X \leq \frac{1}{3}\right), P\left(0 \leq X \leq \frac{1}{3}\right)$; 问: X 是离散

型随机变量还是连续型随机变量? (为什么?)

(1) $P(X = \frac{1}{4}) = F(\frac{1}{4}) - F(\frac{1}{4}-) = 0$, $P(X = \frac{1}{2}) = F(\frac{1}{2}) - F(\frac{1}{2}-) = 1 - \frac{2}{3} = \frac{1}{3}$.
 (2) $P(0 < X \leq \frac{1}{3}) = F(\frac{1}{3}) - F(0) = \frac{1}{3} + \frac{1}{3} - \frac{1}{3} = \frac{1}{3}$, $P(0 \leq X \leq \frac{1}{3}) = F(\frac{1}{3}) - F(0-) = \frac{2}{3} - 0 = \frac{2}{3}$

X 不是 d.r.v., 也不是 c.r.v., 是混合型随机变量!

$F(0) - F(0-) = P(X = 0) = \frac{1}{3}$. $F(\frac{1}{2}) - F(\frac{1}{2}-) = \frac{1}{6} = P(X = \frac{1}{2})$

13. 假设随机变量 ξ 的绝对值不大于1, $P(\{\xi=1\})=2P(\{\xi=-1\})=\frac{1}{4}$, 在事件 $\{-1 < \xi < 1\}$ 出现的条件下, ξ 在 $(-1, 1)$ 内任一子区间上取值的条件概率与该区间的长度成正比, 试求 ξ 的分布函数.

由题设, $R(\xi) = [-1, 1]$. 证: $\forall x < -1, F_{\xi}(x) = P\{\xi \leq x\} = P(\emptyset) = 0$; $\forall x \geq 1, F_{\xi}(x) = P\{\xi \leq x\} = P(\Omega) = 1$; 易见, $\forall x \in (-1, 1)$,

$$P(-1 < \xi \leq x | -1 < \xi < 1) = k(x+1). \quad \text{从而, } \forall x \in (-1, 1), F_{\xi}(x) = P\{\xi \leq x\}$$

$$= P\{\xi = -1\} + P\{-1 < \xi \leq x\} = \frac{1}{8} + P\{-1 < \xi \leq x | -1 < \xi < 1\}$$

$$= \frac{1}{8} + P\{-1 < \xi < 1\} \cdot P(-1 < \xi \leq x | -1 < \xi < 1) = \frac{1}{8} + (1 - \frac{1}{4} - \frac{1}{8}) \cdot k(x+1)$$

$$= \frac{1}{8} + \frac{5}{8}k(x+1). \quad \text{证:}$$

$$F_{\xi}(x) = P\{\xi \leq x\} = \begin{cases} 0 & , x < -1; \\ \frac{1}{8} + \frac{5}{8}k(x+1) & , -1 \leq x \leq 1; \\ 1 & , x \geq 1; \end{cases}$$

$$\text{由 } P\{\xi=1\} = F_{\xi}(1) - F_{\xi}(1-) = 1 - [\frac{1}{8} + \frac{5}{8}k \cdot (1+1)] = \frac{7}{8} - \frac{5}{4}k = \frac{1}{4}$$

$$k = \frac{1}{2}. \quad \text{从而, } F_{\xi}(x) = \begin{cases} 0 & , x < -1; \\ \frac{5x+7}{16} & , -1 \leq x \leq 1; \\ 1 & , x \geq 1; \end{cases} \quad \xi - 1 \text{ 是含重随机变量.}$$

另法: 由 (*) 式, 令 $x \rightarrow 1-$, 证得:

$$\lim_{x \rightarrow 1-} P(-1 < \xi \leq x | -1 < \xi < 1) = \lim_{x \rightarrow 1-} k(x+1) = 1, \quad k = \frac{1}{2}.$$

习题 3.1

1. 袋中有1红2黑3白共6个球, 现有放回从袋中取两次, 每次取一球, 以 X, Y, Z 分别表示两次取到的红、黑、白球的个数,

(1) 求 $P(X=1|Z=0)$; (2) 求 (X, Y) 的概率分布.

$$(1) P(X=1|Z=0) = \frac{P(X=1, Z=0)}{P(Z=0)} = \frac{P(X=1, Y=1)}{P(Z=0)} = \frac{\frac{1}{6} \times \frac{2}{6} + \frac{2}{6} \times \frac{1}{6}}{\frac{3}{6} \times \frac{3}{6}} = \frac{4}{9};$$

(2) 解,

$X \backslash Y$	0	1	2
0	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{1}{9}$
1	$\frac{1}{6}$	$\frac{1}{6}$	0
2	$\frac{1}{36}$	0	0

$$\begin{aligned} \text{又, } P(X=0, Y=0) &= \frac{3}{6} \times \frac{3}{6} = \frac{1}{4}; \quad P(X=0, Y=1) = \frac{2}{6} \times \frac{3}{6} + \frac{3}{6} \times \frac{2}{6} \\ &= \frac{1}{3}; \quad P(X=0, Y=2) = \frac{2}{6} \times \frac{2}{6} = \frac{1}{9}; \quad P(X=1, Y=0) = \frac{1}{6} \times \frac{3}{6} + \frac{3}{6} \times \frac{1}{6} \\ &= \frac{1}{6}; \quad P(X=1, Y=1) = \frac{1}{6} \times \frac{2}{6} + \frac{2}{6} \times \frac{1}{6} = \frac{1}{9}; \end{aligned}$$

$$P(X=2, Y=0) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}.$$

4. (1) 假设 X, Y 同分布, 且 $X \sim \begin{pmatrix} -1 & 0 & 1 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{pmatrix}$, $P(XY=0)=1$, 试求 (X, Y) 的联合分布及 $P(|X|=|Y|)$;

由 $P(XY=0)=1$, 则有: $P(XY \neq 0) = P(\{X=-1, Y=-1\} \cup \{X=-1, Y=1\} \cup \{X=1, Y=-1\} \cup \{X=1, Y=1\}) = 0$; 则有:

$X \backslash Y$	-1	0	1
-1	0	$\frac{1}{4}$	0
0	$\frac{1}{4}$	0	$\frac{1}{4}$
1	0	$\frac{1}{4}$	0
	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

, 即为 (X, Y) 的联合分布.

$$\begin{aligned} P(|X|=|Y|) &= P(X=Y) + P(X=-Y) \\ &= P(X=-1, Y=-1) + P(X=1, Y=1) + P(X=-1, Y=1) \\ &\quad + P(X=1, Y=-1) = 0 + 0 + 0 + 0 = 0 \end{aligned}$$

(2) 设 X, Y 为离散型随机变量, 且 $X \sim \begin{pmatrix} -1 & 0 & 1 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{pmatrix}$, $Y \sim \begin{pmatrix} -1 & 0 & 1 \\ \frac{5}{12} & \frac{1}{4} & \frac{1}{3} \end{pmatrix}$, 已知

$P(X < Y) = 0$, $P(X > Y) = \frac{1}{4}$, 试求 (X, Y) 的联合分布.

由 $P(X < Y) = 0$, 则有: $P(X=-1, Y=0) + P(X=-1, Y=1) + P(X=0, Y=1) = 0$.

也即有:

$X \backslash Y$	-1	0	1
-1	$\frac{1}{4}$	0	0
0		0	$\frac{1}{4}$
1		$\frac{1}{3}$	$\frac{1}{2}$
	$\frac{5}{12}$	$\frac{1}{4}$	$\frac{1}{3}$

; 由 $P(X > Y) = \frac{1}{4}$, 则有: $P(X=Y) = 1 - P(X < Y) - P(X > Y) = \frac{3}{4}$.

$$\text{则有: } P(X=-1, Y=-1) + P(X=0, Y=0) + P(X=1, Y=1) = \frac{3}{4}$$

也即有:

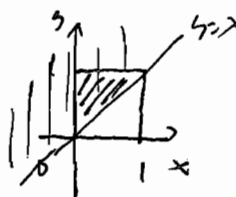
$X \backslash Y$	-1	0	1
-1	$\frac{1}{4}$	0	0
0	$\frac{1}{12}$	$\frac{1}{6}$	0
1	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{3}$
	$\frac{5}{12}$	$\frac{1}{4}$	$\frac{1}{3}$

, 即为 (X, Y) 的联合分布.

(4) 设二维随机变量 (X, Y) 具有密度 $f(x, y) = \begin{cases} 4xy, & 0 < x, y < 1; \\ 0, & \text{其他}; \end{cases}$, 试求:

$$P(X < Y), P(X + Y \geq 1), P\left(Y \geq X + \frac{1}{2}\right).$$

注: $D = [0, 1] \times [0, 1]$. 证: $f(x, y) = \begin{cases} 4xy, & (x, y) \in D; \\ 0, & (x, y) \notin D; \end{cases}$ 令 $D_1 = \{x < y\}$.



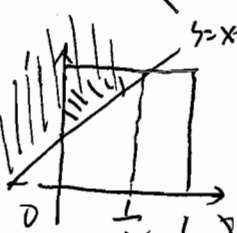
证: $P(X < Y) = P\{(X, Y) \in D_1\} = \iint_{D_1} f(x, y) dx dy = \left(\iint_{D_1 \cap D} + \iint_{D_1 \cap D^c} \right) f(x, y) dx dy$
 $= \iint_{D_1 \cap D} f(x, y) dx dy = \int_0^1 dx \int_x^1 4xy dy = \int_0^1 2x \cdot (1 - x^2) dx = 1 - \frac{1}{2} = \frac{1}{2}$. 可以验证:

$$P(X + Y \geq 1) = \iint_{x+y \geq 1} f(x, y) dx dy = \int_0^1 dx \int_{1-x}^1 4xy dy = \int_0^1 2x \cdot [1 - (1-x)^2] dx$$



$$= 1 - 2 \int_0^1 x \cdot (1-x)^2 dx = 1 - 2 \int_0^1 x^2 (1-x) dx = 1 - 2 \cdot \frac{1}{12} = \frac{5}{6};$$

$$P\left(Y \geq X + \frac{1}{2}\right) = \iint_{y \geq x + \frac{1}{2}} f(x, y) dx dy = \int_{\frac{1}{2}}^1 dx \int_{x+\frac{1}{2}}^1 4xy dy$$



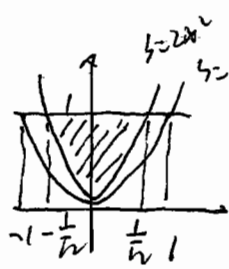
$$= \int_{\frac{1}{2}}^1 2x [1 - (x + \frac{1}{2})^2] dx = \frac{1}{4} - 2 \cdot \int_{\frac{1}{2}}^1 x \cdot (x + \frac{1}{2})^2 dx \quad \text{令 } x + \frac{1}{2} = t$$

$$\frac{1}{4} - 2 \int_{\frac{1}{2}}^1 (t - \frac{1}{2}) \cdot t^2 dt = \frac{1}{4} - \left[\frac{1}{3} t^3 - \frac{1}{2} t^2 \right] \Big|_{\frac{1}{2}}^1 = \frac{1}{4} - \left[\frac{15}{32} - \frac{7}{48} \right]$$

$$= \frac{1}{4} - \frac{17}{96} = \frac{7}{96}.$$

5. (1) 设 (X, Y) 的联合概率密度为 $f(x, y) = \begin{cases} cx^2y, & x^2 \leq y \leq 1; \\ 0, & \text{其他}; \end{cases}$

(i) 确定常数 c ; (ii) 求 $P((X, Y) \in D)$, $D: 2x^2 \leq y \leq 1$;



(i) 证: $D_1 = \{x^2 \leq y \leq 1\}$; 证: $\iint_{D_1} f(x, y) dx dy = 1$. 证: $\iint_{D_1} cx^2y dx dy = \int_{-1}^1 dx \int_{x^2}^1 cx^2y dy$
 $= \int_{-1}^1 \frac{c}{2} x^2 \cdot (1 - x^4) dx = \frac{c}{2} \int_{-1}^1 x^2 (1 - x^4) dx = c \cdot \int_0^1 x^2 (1 - x^4) dx = \frac{4}{21} c = 1.$

$$c = \frac{21}{4}; \quad (ii) \quad P((X, Y) \in D) = \iint_D f(x, y) dx dy = \frac{21}{4} \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} dx \int_{2x^2}^1 x^2y dy$$

$$= \frac{21}{8} \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} x^2 (1 - 4x^4) dx = \frac{21}{4} \int_0^{\frac{1}{\sqrt{2}}} x^2 (1 - 4x^4) dx = \frac{21}{4} \cdot \left[\frac{1}{3} \left(\frac{1}{\sqrt{2}}\right)^3 - \frac{4}{7} \left(\frac{1}{\sqrt{2}}\right)^7 \right]$$

$$= \frac{21}{4} \times \left[\frac{1}{6\sqrt{2}} - \frac{1}{14\sqrt{2}} \right] = \frac{21}{4} \cdot \frac{1}{\sqrt{2}} \cdot \frac{7-3}{42} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{4}.$$

(3) 设 $(X, Y) \sim f(x, y) = \begin{cases} ke^{-3x-4y}, & x, y > 0 \\ 0 & \text{其他} \end{cases}$, 试求:

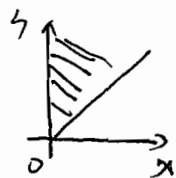
(i) (X, Y) 的联合分布函数; (ii) $P(X \leq Y), P(X+Y > 1)$;

$$\text{由 } \iint_{\mathbb{R}^2} f(x, y) dx dy = \iint_{x, y > 0} ke^{-3x-4y} dx dy = \int_0^{+\infty} dx \cdot \int_0^{+\infty} ke^{-3x} \cdot e^{-4y} dy = \frac{k}{12} = 1, \quad k=12.$$

$$(i) \forall (x, y) \in \mathbb{R}^2, F(x, y) = P(X \leq x, Y \leq y) = \begin{cases} P(\emptyset) = 0 & , x \leq 0 \text{ 或 } y \leq 0 \\ \int_{-\infty}^x ds \cdot \int_{-\infty}^y f(s, t) dt & , x, y > 0 \end{cases}$$

$$= \begin{cases} 0 & , x \leq 0 \text{ 或 } y \leq 0 \\ \int_0^x ds \int_0^y 12e^{-3s} \cdot e^{-4t} dt, & x, y > 0 \end{cases} = \begin{cases} 0 & , x \leq 0 \text{ 或 } y \leq 0 \\ (1-e^{-3x})(1-e^{-4y}), & x, y > 0 \end{cases}$$

$$(ii) P(X \leq Y) = \iint_{x \leq y} f(x, y) dx dy = \int_0^{+\infty} dy \int_0^y 12e^{-3x} \cdot e^{-4y} dx = \int_0^{+\infty} 3e^{-3x} \cdot e^{-4y} dx = \frac{3}{7};$$



$$\begin{aligned} P(X+Y > 1) &= \int_0^1 dx \int_{1-x}^{+\infty} 12e^{-3x} \cdot e^{-4y} dy + \int_1^{+\infty} dx \cdot \int_0^{+\infty} 12e^{-3x} \cdot e^{-4y} dy \\ &= \int_0^1 3e^{-3x} \cdot e^{-4+4x} dx + \int_1^{+\infty} 3e^{-3x} dx \\ &= 3e^{-4}(e-1) + e^{-3} = 4e^{-3} - 3e^{-4}. \end{aligned}$$

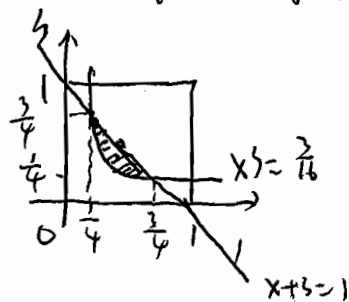


6. (1) 从 $(0, 1)$ 中随机地取两个数, 求“其积不小于 $\frac{3}{16}$ 且其和不大于一”的概率;

方法1: (几何法) $E: \text{向 } (0, 1) \times (0, 1) \text{ 内随机掷点, 记之为 } (x, y) \text{ (几何模型)}$

$$\Omega = (0, 1) \times (0, 1)$$

$$A = \{\text{两数之和(积)不大于一(不小于 } \frac{3}{16})\} = \{(x, y) | x+y \leq 1, xy \geq \frac{3}{16}, 0 < x, y < 1\}$$



$$\begin{aligned} P(A) &\stackrel{\text{几何法}}{=} \frac{S(A)}{S(\Omega)} = \int_{\frac{3}{16}}^{\frac{1}{4}} [1-x-\frac{3}{16x}] dx \\ &= \frac{1}{2} - \frac{1}{4} - \frac{3}{16} \ln 3 = \frac{1}{4} - \frac{3}{16} \ln 3. \end{aligned}$$

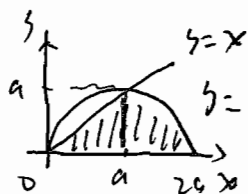
方法2: 设两数分别为 (X, Y) , 由题设 $(X, Y) \sim U(D)$, $D = (0, 1) \times (0, 1)$.

$$\text{令 } (X, Y) \sim f(x, y), \text{ 则 } f(x, y) = \begin{cases} \frac{1}{S(D)} = 1, & (x, y) \in D \\ 0, & (x, y) \notin D \end{cases}$$

$$P(A) = P(xy \geq \frac{3}{16}, x+y \leq 1) = \iint_{\substack{xy \geq \frac{3}{16} \\ x+y \leq 1}} f(x, y) dx dy = \dots = \frac{S(A)}{S(D)}$$

$$= \frac{1}{4} - \frac{3}{16} \ln 3.$$

(2) 向平面区域 $D = \{(x, y) | 0 < y < \sqrt{2ax - x^2}\}$ 内随机地投掷一点, 记其为 (X, Y) , 试求 $P(X \geq Y)$. ($a > 0$)



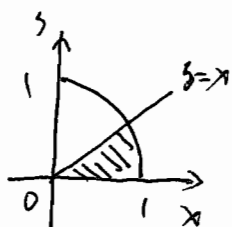
设 $(X, Y) \sim f(x, y)$, 则 $f(x, y) = \begin{cases} \frac{1}{S(D)} = \frac{2}{\pi a^2}, & (x, y) \in D; \\ 0, & (x, y) \notin D; \end{cases}$

也即 $P(X \geq Y) = \iint_{x \geq y} f(x, y) dx dy = \frac{\text{阴影面积}}{\text{总面积}} = \frac{\frac{1}{2}a^2 + \frac{3}{4}a^2}{\frac{\pi}{2}a^2} = \frac{2+\pi}{2\pi} = \frac{1}{2} + \frac{1}{\pi}$

(2) 设 $(X, Y) \sim f(x, y) = \begin{cases} \frac{2}{\pi} e^{-\sqrt{x^2+y^2}}, & x, y > 0; \\ 0, & \text{其他}; \end{cases}$ 令 $\begin{cases} U = \sqrt{X^2 + Y^2} \\ V = \arctan \frac{Y}{X} \end{cases}$, 记 $F(u, v)$ 为

(U, V) 的联合分布函数, 求 $F(1, \frac{\pi}{4})$. (U, V) - (X, Y) 的对应关系.

可见, $F(1, \frac{\pi}{4}) = P(U \leq 1, V \leq \frac{\pi}{4}) = P(\sqrt{X^2 + Y^2} \leq 1, \arctan \frac{Y}{X} \leq \frac{\pi}{4})$
 $= P(X^2 + Y^2 \leq 1, \arctan \frac{Y}{X} \leq \frac{\pi}{4}) = \iint_{\substack{x^2+y^2 \leq 1 \\ \arctan \frac{y}{x} \leq \frac{\pi}{4}}} f(x, y) dx dy = \iint_{\substack{x^2+y^2 \leq 1 \\ \arctan \frac{y}{x} \leq \frac{\pi}{4} \\ x, y > 0}} \frac{2}{\pi} e^{-\sqrt{x^2+y^2}} dx dy$



$= \int_0^{\frac{\pi}{4}} d\theta \int_0^1 \left(\frac{2}{\pi}\right) e^{-r} \cdot r dr = \frac{1}{2} \int_0^1 r \cdot e^{-r} dr = \frac{1}{2} (-r-1)e^{-r} \Big|_0^1$
 $= \frac{1}{2} (1 - 2e^{-1}) = \frac{1}{2} - \frac{1}{e}$

8. 设非负函数 $g(x)$ 满足 $\int_0^{+\infty} g(x) dx = 1$, 若

$$f(x, y) = \begin{cases} \frac{2g(\sqrt{x^2+y^2})}{\pi\sqrt{x^2+y^2}}, & 0 < x, y < +\infty; \\ 0, & \text{其他} \end{cases}$$

试问: $f(x, y)$ 是否为某二维连续型随机向量的联合概率密度?

可见, $\iint_{\mathbb{R}^2} f(x, y) dx dy = \iint_{x, y > 0} f(x, y) dx dy = \iint_{x, y > 0} \frac{2g(\sqrt{x^2+y^2})}{\pi\sqrt{x^2+y^2}} dx dy$

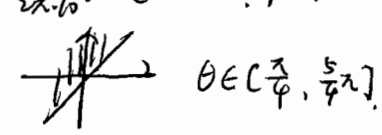
极坐标变换 $\int_0^{\frac{\pi}{2}} d\theta \int_0^{+\infty} \frac{2g(r)}{\pi r} \cdot r dr = \int_0^{\frac{\pi}{2}} \frac{2}{\pi} d\theta \cdot \int_0^{+\infty} g(r) dr = 1 \cdot \int_0^{+\infty} g(x) dx = 1$

故, $f(x, y)$ 可作为二维随机向量的联合密度!

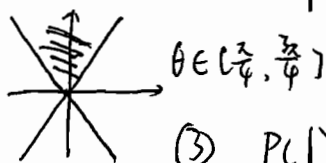
9. 设随机变量 (X, Y) 服从二维正态分布, 其联合密度函数为:

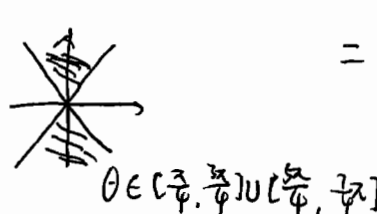
$$f(x, y) = \frac{1}{2\pi \times 10^2} e^{-\frac{x^2+y^2}{2 \times 10^2}}, \quad -\infty < x, y < +\infty;$$

试求: $P(Y \geq X)$, $P(Y \geq |X|)$, $P(|Y| \geq |X|)$.

$$\begin{aligned} \textcircled{1} P(Y \geq X) &= \iint_{y \geq x} f(x, y) dx dy \quad \text{极坐标变换} \quad \int_{\frac{3\pi}{4}}^{\frac{5\pi}{4}} d\theta \int_0^{+\infty} \frac{1}{2\pi \cdot 10^2} \cdot e^{-\frac{r^2}{2 \cdot 10^2}} \cdot r dr \\ &= \int_{\frac{3\pi}{4}}^{\frac{5\pi}{4}} \frac{1}{2\pi} d\theta = \frac{1}{2}; \end{aligned}$$


$$\textcircled{2} P(Y \geq |X|) = \iint_{y \geq |x|} f(x, y) dx dy = \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} d\theta \int_0^{+\infty} \frac{1}{2\pi \cdot 10^2} \cdot e^{-\frac{r^2}{2 \cdot 10^2}} \cdot r dr = \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{1}{2\pi} d\theta = \frac{1}{4};$$



$$\begin{aligned} \textcircled{3} P(|Y| \geq |X|) &= P(Y \geq |X|) + P(Y \leq -|X|) \\ &= \frac{1}{4} + \int_{\frac{5\pi}{4}}^{\frac{7\pi}{4}} d\theta \int_0^{+\infty} \frac{1}{2\pi \cdot 10^2} \cdot e^{-\frac{r^2}{2 \cdot 10^2}} \cdot r dr = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}. \end{aligned}$$


10. 设二维连续型随机变量 (X, Y) 的联合分布函数为:

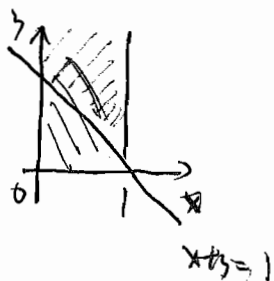
$$F(x, y) = \begin{cases} 0, & x < 0 \text{ 或 } y < 0; \\ x(1 - e^{-2y}), & 0 \leq x < 1, y \geq 0; \\ 1 - e^{-2y}, & x \geq 1, y \geq 0; \end{cases}$$

试求: (1) (X, Y) 的联合密度函数 $f(x, y)$; (2) $P\left(X \leq \frac{1}{2}, 1 < Y \leq 3\right)$, $P(X + Y \geq 1)$.

$$\begin{aligned} (1) \text{ 令 } (X, Y) \sim f(x, y), \text{ 则 } f(x, y) &= \frac{\partial^2 F}{\partial x \partial y}(x, y) \\ &= \begin{cases} 2e^{-2y}, & 0 \leq x < 1, y > 0; \\ 0, & \text{其他.} \end{cases} \end{aligned}$$

$$\begin{aligned} (2) P\left(X \leq \frac{1}{2}, 1 < Y \leq 3\right) &= P\left(X \leq \frac{1}{2}, Y \leq 3\right) - P\left(X \leq \frac{1}{2}, Y \leq 1\right) \\ &= F\left(\frac{1}{2}, 3\right) - F\left(\frac{1}{2}, 1\right) = \frac{1}{2}(1 - e^{-6}) - \frac{1}{2}(1 - e^{-2}) = \frac{1}{2}(e^{-2} - e^{-6}). \end{aligned}$$

$$\begin{aligned} P(X + Y \geq 1) &= \iint_{x+y \geq 1} f(x, y) dx dy = \int_0^1 dx \int_{1-x}^{+\infty} 2e^{-2y} dy = \int_0^1 e^{-2+2x} dx \\ &= e^{-2} \cdot \frac{1}{2} (e^2 - 1) = \frac{1}{2} (1 - e^{-2}), \end{aligned}$$



习题 3.2

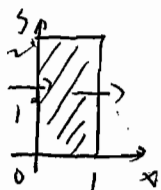
$a, 0 \leq a \leq 1$

1. 设 $(X, Y) \sim f(x, y) = \frac{1 + \sin x \cdot \sin y}{2\pi} e^{-\frac{1}{2}(x^2 + y^2)}$, $-\infty < x, y < +\infty$; 试求 (X, Y) 关于 X, Y 的边缘密度 $f_X(x), f_Y(y)$;

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{+\infty} f(x, y) dy = \int_{-\infty}^{+\infty} \frac{1 + \sin x \cdot \sin y}{2\pi} e^{-\frac{1}{2}(x^2 + y^2)} dy \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy + 0 = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, -\infty < x < +\infty, \quad \text{即: } X \sim N(0, 1); \\ \text{同理, } f_Y(y) &= \int_{-\infty}^{+\infty} f(x, y) dx = \dots = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}}, -\infty < y < +\infty, \quad \text{即: } Y \sim N(0, 1). \end{aligned}$$

2. (1) 设 $f(x, y) = \begin{cases} x^2 + \frac{xy}{3}, & (x, y) \in [0, 1] \times [0, 2]; \\ 0, & \text{其他;} \end{cases}$, 试求 X, Y 的边缘密度

$f_X(x), f_Y(y)$ 及 $P\left(Y < \frac{1}{2} \mid X < \frac{1}{2}\right)$;



$$\begin{aligned} f_X(x) &= \int_{-\infty}^{+\infty} f(x, y) dy = \begin{cases} \int_0^0 + \int_0^2 (x^2 + \frac{xy}{3}) dy & , 0 \leq x \leq 1; \\ \int_{-\infty}^{+\infty} 0 dy = 0 & , \text{其他;} \end{cases} \\ &= \begin{cases} \int_0^2 (x^2 + \frac{1}{3}xy) dy = 2x^2 + \frac{2}{3}x & , 0 \leq x \leq 1; \\ 0 & , \text{其他;} \end{cases} \end{aligned}$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \begin{cases} \int_0^1 (x^2 + \frac{1}{3}xy) dx = \frac{1}{3} + \frac{1}{6}y & , 0 \leq y \leq 2; \\ \int_{-\infty}^{+\infty} 0 dx = 0 & , \text{其他;} \end{cases}$$

$$\begin{aligned} P\left(Y < \frac{1}{2} \mid X < \frac{1}{2}\right) &= \frac{P\left(X < \frac{1}{2}, Y < \frac{1}{2}\right)}{P\left(X < \frac{1}{2}\right)} = \frac{\iint_{(-\infty, \frac{1}{2}) \times (-\infty, \frac{1}{2})} f(x, y) dx dy}{\int_{-\infty}^{\frac{1}{2}} f_X(x) dx} = \frac{\int_0^{\frac{1}{2}} dx \int_0^{\frac{1}{2}} (x^2 + \frac{1}{3}xy) dy}{\int_0^{\frac{1}{2}} (2x^2 + \frac{2}{3}x) dx} \\ &= \frac{\int_0^{\frac{1}{2}} [\frac{1}{2}x^2 + \frac{1}{6}x \cdot \frac{1}{4}] dx}{\frac{2}{3} \cdot \frac{1}{8} + \frac{1}{3} \cdot \frac{1}{4}} = \frac{\frac{1}{48} + \frac{1}{48} \cdot \frac{1}{4}}{\frac{1}{6}} = \frac{5}{32}. \end{aligned}$$

3. (1) 设 $(X, Y) \sim f(x, y) = \begin{cases} 2e^{-(x+y)}, & x, y > 0; \\ 0, & \text{其他}; \end{cases}$ 试求事件 $\{X < 3\}$ 的概率;

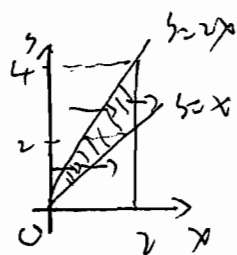
证1: $P(X < 3) = P(0 < X < 3) = P(0 < X < 3, 0 < Y < +\infty) = \iint_{(2,3) \times (0, +\infty)} f(x, y) dx dy$
 $= \int_0^3 dx \cdot \int_0^{+\infty} 2e^{-x} \cdot e^{-y} dy = \int_0^3 e^{-x} dx = 1 - e^{-3};$

证2: 设 $X \sim f_X(x)$, 则 $f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \begin{cases} \int_0^{+\infty} 2e^{-x} \cdot e^{-y} dy = e^{-x}, & x > 0; \\ \int_{-\infty}^{+\infty} 0 dy = 0, & x \leq 0; \end{cases}$ 证: $X \sim E(1)$

$P(X < 3) = 1 - e^{-3} = \int_0^3 e^{-x} dx$

(2) 设 $(X, Y) \sim f(x, y) = \begin{cases} \frac{3x}{8}, & 0 < x < 2, x < y < 2x; \\ 0, & \text{其他}; \end{cases}$ 试求: $P(Y > 3)$.

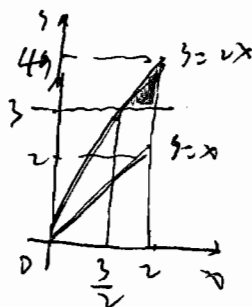
证1:



证 $Y \sim f_Y(y)$, 则 $f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \begin{cases} \int_{y/2}^{2y} \frac{3x}{8} dx, & y \in (2, 4); \\ \int_{y/2}^{2y} \frac{3x}{8} dx, & y \in (0, 2); \\ 0, & y \in (0, 4); \end{cases} = \begin{cases} \frac{9}{64} y^2, & y \in (0, 2); \\ \frac{3}{4} - \frac{3}{64} y^2, & y \in (2, 4); \\ 0, & y \in (0, 4); \end{cases}$

$P(Y > 3) = \int_3^{+\infty} f_Y(y) dy = \left(\int_3^4 + \int_4^{+\infty} \right) f_Y(y) dy = \int_3^4 \left(\frac{3}{4} - \frac{3}{64} y^2 \right) dy = \left[\frac{3}{4}y - \frac{3}{192} y^3 \right]_3^4 = \frac{11}{64}.$

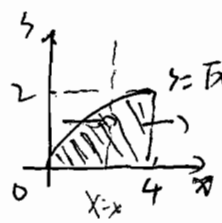
证2: $P(Y > 3) = P(0 < X < 2, Y > 3) = \iint_{(2,2) \times (3, +\infty)} f(x, y) dx dy$



$= \int_{3/2}^2 dx \int_3^{2x} \frac{3x}{8} dy = \int_{3/2}^2 \frac{3x}{8} (2x - 3) dx = \left[\frac{1}{4} x^3 - \frac{9}{16} x^2 \right]_{3/2}^2 = \frac{1}{4} \left(8 - \frac{27}{8} \right) - \frac{9}{16} \left(4 - \frac{9}{4} \right) = \frac{74}{64} - \frac{63}{64} = \frac{11}{64}.$

(3) 已知 (X, Y) 的联合密度函数 $f(x, y) = \begin{cases} Axy, & (x, y) \in D; \\ 0, & (x, y) \notin D; \end{cases}$ 其中

$D = \{(x, y) | 0 \leq x \leq 4, 0 \leq y \leq \sqrt{x}\}$, 试求 X 的边缘分布函数 $F_X(x)$ 及 Y 的边缘密度函数 $f_Y(y)$.



$$\iint_{R^2} f(x, y) dx dy = \iint_D Axy dx dy = \int_0^4 dx \cdot \int_0^{\sqrt{x}} Axy dy = \int_0^4 \frac{A}{2} x^2 dx = \frac{A}{6} \cdot 64 = 1$$

$$A = \frac{3}{32}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \begin{cases} \int_0^{y^2} \frac{3}{32} xy dx & , 0 \leq y \leq 2; \\ 0 & , \text{其他}; \end{cases}$$

$$= \begin{cases} \frac{3}{64} (16 - y^4), & 0 \leq y \leq 2; \\ 0 & , \text{其他}. \end{cases}$$

$\forall x \in \mathbb{R}, F_X(x) = P(X \leq x) = 0; \quad \forall x < 0, F_X(x) = P(X \leq x) = 0;$
 $\forall x \geq 4, F_X(x) = P(X \leq x) = 1; \quad \forall x \in (0, 4), F_X(x) = P(X \leq x) = P(X \leq x, 0 \leq Y \leq \sqrt{x})$

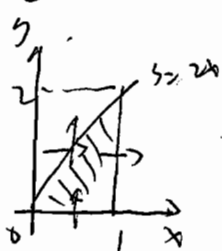
$$= \iint_{[-x, x] \times [0, \sqrt{x}]} f(s, t) ds dt = \int_0^x ds \int_0^{\sqrt{s}} \frac{3}{32} st dt = \int_0^x \frac{3}{64} s^2 ds = \frac{1}{64} x^3.$$

$$F_X(x) = \begin{cases} 0 & , x \leq 0; \\ \frac{1}{64} x^3 & , 0 \leq x < 4; \\ 1 & , x \geq 4; \end{cases}$$

5. 设 $(X, Y) \sim f(x, y) = \begin{cases} 1, & 0 < x < 1, 0 < y < 2x; \\ 0, & \text{其他}; \end{cases}$ 试求:

(1) $f_X(x), f_Y(y)$; (2) $Z = 2X - Y$ 的概率密度 $f_Z(z)$; (3) $P\left(Y \leq \frac{1}{2} \mid X \leq \frac{1}{2}\right)$.

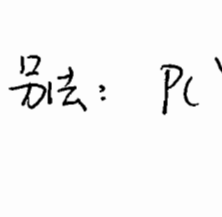
记 $D = \{0 < x < 1, 0 < y < 2x\}$, 则有: $(X, Y) \sim U(D)$.



$$(1) f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \begin{cases} \int_0^{2x} 1 dy = 2x & , 0 < x < 1; \\ 0 & , \text{其他}; \end{cases}$$

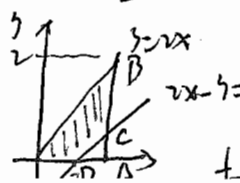
$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \begin{cases} \int_{y/2}^1 1 dx = 1 - \frac{y}{2} & , 0 < y < 2; \\ 0 & , \text{其他}; \end{cases}$$

$$(3) P\left(Y \leq \frac{1}{2} \mid X \leq \frac{1}{2}\right) = \frac{P\left(X \leq \frac{1}{2}, Y \leq \frac{1}{2}\right)}{P\left(X \leq \frac{1}{2}\right)} = \frac{\iint_{[-x, x] \times [0, \frac{1}{2}]} f(x, y) dx dy}{\int_{-\infty}^{\frac{1}{2}} f_X(x) dx} = \dots$$



$$\text{几何法: } P\left(Y \leq \frac{1}{2} \mid X \leq \frac{1}{2}\right) = \frac{S_{\triangle OAC}}{S_{\triangle OAB}} = \frac{3}{4}.$$

(2) 若 $Z = 2X - Y$, $(x, y) \in D$, 则有: $R(Z) = (0, 2)$; 又 $\forall z \leq 0, F_Z(z) = P(Z \leq z) = 0$; $\forall z \geq 2, F_Z(z) = P(Z \leq z) = 1$; $\forall z \in (0, 2), F_Z(z) = P(Z \leq z) = \frac{S_{\triangle ODC}}{S_{\triangle OAB}}$



$$= 1 - \left(1 - \frac{z}{2}\right)^2 = z - \frac{1}{4} z^2.$$

$$f_Z(z) = \begin{cases} 0, & z \leq 0; \\ z - \frac{1}{4} z^2, & 0 < z < 2; \\ 1, & z \geq 2; \end{cases}$$

$$f_Z(z) = \frac{d}{dz} F_Z(z) = \begin{cases} 1 - \frac{z}{2}, & 0 < z < 2; \\ 0, & \text{其他}. \end{cases}$$

习题 3.3

1. (1) 将2只球随机地放入3只盒中, 以 X, Y 分别表示1号盒与2号盒中的球数, 试求在 $Y=0$ 的条件下 X 的条件分布;

易见,
$$\begin{array}{c|ccc} X \backslash Y & 0 & 1 & 2 \\ \hline 0 & \frac{1}{9} & \frac{2}{9} & \frac{1}{9} \\ 1 & \frac{2}{9} & \frac{2}{9} & 0 \\ 2 & \frac{1}{9} & 0 & 0 \end{array}$$

这里, $P(X=0, Y=0) = \frac{1}{9}$, $P(X=0, Y=1) = \frac{2}{9}$,
 $P(X=0, Y=2) = \frac{1}{9}$, $P(X=1, Y=0) = \frac{2}{9}$, $P(X=1, Y=1) = \frac{2}{9}$,
 $P(X=2, Y=0) = \frac{1}{9}$; 且 $P(Y=0) = \frac{4}{9}$; 即有:

$$X|Y=0 \sim \begin{pmatrix} 0 & 1 & 2 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{pmatrix}$$

(2) 从1, 2, 3, 4中任取一个数, 记为 X ; 再从1, ..., X 中任取一个数记为 Y , 试求 (X, Y) 的联合分布及 Y 的边缘分布与给定 $X=i$ 时 Y 的条件分布;

易见, $P(X=i, Y=j) = P(X=i) \cdot P(Y=j|X=i) = \frac{1}{4} \cdot \frac{1}{i}$, $1 \leq j \leq i \leq 4$, 这里,

$$Y|X=i \sim \begin{pmatrix} 1 & 2 & \dots & i \\ \frac{1}{i} & \frac{1}{i} & \dots & \frac{1}{i} \end{pmatrix}; \text{ 即有: } \begin{array}{c|cccc} X \backslash Y & 1 & 2 & 3 & 4 \\ \hline 1 & \frac{1}{4} & 0 & 0 & 0 \\ 2 & \frac{1}{8} & \frac{1}{8} & 0 & 0 \\ 3 & \frac{1}{12} & \frac{1}{12} & \frac{1}{12} & 0 \\ 4 & \frac{1}{16} & \frac{1}{16} & \frac{1}{16} & \frac{1}{16} \end{array}$$

也即有:
$$Y \sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ \frac{25}{48} & \frac{13}{48} & \frac{7}{48} & \frac{1}{16} \end{pmatrix};$$

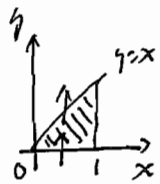
$$X|Y=1 \sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ \frac{12}{25} & \frac{6}{25} & \frac{4}{25} & \frac{3}{25} \end{pmatrix}$$

(3) 一射手进行射击, 已知其每次的命中率为 $p(0 < p < 1)$, 射击一直进行到击中目标两次为止; 令 X 表示其首次命中目标时所射击的次数, Y 表示总共射击的次数, 试求 (X, Y) 的联合分布、边缘分布和条件分布. 令 $A_i = \{\text{第 } i \text{ 次命中目标}\}$.

易见, $P(X=i, Y=j) = P(\bar{A}_1, \dots, \bar{A}_{i-1}, A_i, \bar{A}_{i+1}, \dots, \bar{A}_{j-1}, A_j)$
 $= P(\bar{A}_1) \cdots P(\bar{A}_{i-1}) P(A_i) P(\bar{A}_{i+1}) \cdots P(\bar{A}_{j-1}) P(A_j) = (1-p)^{j-2} \cdot p^2$, $i=1, 2, \dots; j=i+1, \dots$
 $P(X=i) = \sum_{j=i+1}^{\infty} P(X=i, Y=j) = \sum_{j=i+1}^{\infty} (1-p)^{j-2} \cdot p^2$
 $= \frac{(1-p)^{i-1} \cdot p^2}{1-(1-p)} = (1-p)^{i-1} \cdot p$, $i=1, 2, \dots; X \sim G(p)$.
 $P(Y=j) = \sum_{i=1}^{j-1} P(X=i, Y=j) = \sum_{i=1}^{j-1} (1-p)^{j-2} \cdot p^2 = (j-1) \cdot (1-p)^{j-2} \cdot p^2$, $j=2, 3, \dots$
 $P_{X|Y}(i|j) = P(X=i|Y=j) = \frac{1}{j-1}$, $i=1, 2, \dots, j-1$. 即: $X|Y=j \sim \begin{pmatrix} 1 & 2 & \dots & j-1 \\ \frac{1}{j-1} & \frac{1}{j-1} & \dots & \frac{1}{j-1} \end{pmatrix}$,
 $P_{Y|X}(j|i) = P(Y=j|X=i) = (1-p)^{j-i-1} \cdot p$, $j=i+1, i+2, \dots; j=2, 3, \dots$
 也即: $Y-i|X=i \sim G(p)$, $i=1, 2, \dots$.

2. (1) 设 $(X, Y) \sim f(x, y) = \begin{cases} 3x, & 0 < y < x < 1; \\ 0, & \text{其他}; \end{cases}$ 试求给定 $X=x$ ($0 < x < 1$) 时, Y

的条件密度函数 $f_{Y|X}(y|x)$ 及条件分布函数 $F_{Y|X}(y|x)$: 设 $X \sim f_X(x)$, 则有:



$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \begin{cases} \int_0^x 3x dy = 3x^2, & 0 < x < 1; \\ 0, & \text{其他}; \end{cases} \quad \forall x \in (0, 1).$$

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \begin{cases} \frac{1}{x}, & 0 < y < x; \\ 0, & \text{其他}; \end{cases} \quad \text{即: } Y|X=x \sim U(0, x), \quad 0 < x < 1;$$

$$\text{故: } \forall y \in \mathbb{R}, \quad F_{Y|X}(y|x) = P(Y \leq y | X=x) = \int_{-\infty}^y f_{Y|X}(y|x) dy = \begin{cases} 0, & y \leq 0; \\ \frac{y}{x}, & 0 \leq y < x; \\ 1, & y \geq x; \end{cases}$$

(2) 设 $(X, Y) \sim f(x, y) = \begin{cases} \frac{1}{y} e^{-y} \cdot e^{-\frac{x}{y}}, & x, y > 0; \\ 0, & \text{其他}; \end{cases}$, $\forall y > 0$, 试求给定 $Y=y$ 时, X 的

条件密度函数 $f_{X|Y}(x|y)$ 及 $P(X > 1 | Y=y)$: 设 $Y \sim f_Y(y)$, 则有: $f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx$

$$= \begin{cases} \int_0^{+\infty} \frac{1}{y} e^{-y} \cdot e^{-\frac{x}{y}} dx, & y > 0; \\ 0, & y \leq 0; \end{cases} = \begin{cases} e^{-y}, & y > 0; \\ 0, & y \leq 0; \end{cases} \quad \text{即: } Y \sim E(1); \quad \text{故: } \forall y > 0, \quad f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$$

$$= \begin{cases} \frac{1}{y} e^{-\frac{x}{y}}, & x > 0; \\ 0, & x \leq 0; \end{cases} \quad \text{即: } X|Y=y \sim E(\frac{1}{y}), \quad y > 0;$$

$$P(X > 1 | Y=y) = \int_1^{+\infty} f_{X|Y}(x|y) dx = \dots = e^{-\frac{1}{y}}.$$

(3) 设 $(X, Y) \sim f(x, y) = \begin{cases} \frac{21}{4} x^2 y, & x^2 \leq y \leq 1; \\ 0, & \text{其他}; \end{cases}$ 求条件概率 $P(Y \geq 0.75 | X=0.5)$.

注意到 $P(X=0.5) = 0$. 设 $X \sim f_X(x)$. 则有: $f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \begin{cases} \int_{x^2}^1 \frac{21}{4} x^2 y dy, & -1 < x < 1; \\ 0, & \text{其他}; \end{cases}$

$$= \begin{cases} \frac{21x^2(1-x^4)}{8}, & -1 < x < 1; \\ 0, & \text{其他}; \end{cases} \quad \text{故: } \forall x \in (-1, 1), \quad f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)}$$

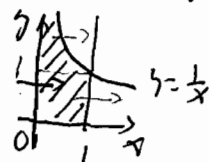
$$= \begin{cases} \frac{2y}{1-x^4}, & x^2 \leq y \leq 1 \text{ (或 } y \in [x^2, 1]) \\ 0, & \text{其他 (或 } y \notin [x^2, 1]) \end{cases} \quad \text{即: } f_{Y|X}(y|0.5) = \begin{cases} \frac{32}{15} y, & \frac{1}{4} \leq y \leq 1; \\ 0, & \text{其他}; \end{cases}$$

$$P(Y \geq 0.75 | X=0.5) = \int_{0.75}^{+\infty} f_{Y|X}(y|0.5) dy = \int_{\frac{3}{4}}^1 \frac{32}{15} y dy = \frac{16}{15} y \Big|_{\frac{3}{4}}^1 = \frac{7}{15}.$$

$$\text{或: } P(Y \geq 0.75 | X=0.5) = \lim_{\Delta x \rightarrow 0} P(Y \geq 0.75 | 0.5 - \Delta x < X \leq 0.5 + \Delta x) = \dots = \frac{7}{15}. \quad 52$$

3. (1) 设 $X \sim U(0,1)$, 已知 $X=x$ ($0 < x < 1$), $Y \sim U\left(0, \frac{1}{x}\right)$, 试求 Y 的概率密

度函数 $f_Y(y)$: 由题设, $Y|X=x \sim U\left(0, \frac{1}{x}\right)$, $0 < x < 1$; 记 $X \sim f_X(x)$, $Y|X=x \sim f_{Y|X}(y|x)$, $0 < x < 1$, 则有: $f_{Y|X}(y|x) = \begin{cases} x, & 0 < y < \frac{1}{x}; \\ 0, & \text{其他} \end{cases}$; 令 $(X, Y) \sim f_{X,Y}(x, y)$, 则有: $f_{X,Y}(x, y) = f_X(x) \cdot f_{Y|X}(y|x) = \begin{cases} x, & 0 < x < 1, 0 < y < \frac{1}{x}; \\ 0, & \text{其他} \end{cases}$



$$\begin{aligned} \text{从而, } f_Y(y) &= \int_{-\infty}^{+\infty} f_{X,Y}(x, y) dx \\ &= \begin{cases} \int_0^1 x dx = \frac{1}{2}, & 0 < y < 1; \\ \int_0^{1/y} x dx = \frac{1}{2y^2}, & 1 \leq y < +\infty; \\ 0, & y \leq 0; \end{cases} \end{aligned}$$

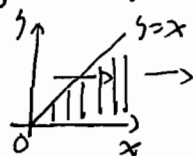
(2) 设 $\xi \sim f_\xi(x) = \begin{cases} \lambda^2 x e^{-\lambda x}, & x > 0; \\ 0, & x \leq 0; \end{cases}$, η 在 $(0, \xi)$ 上均匀分布, 试求 η 的密度函数.

由题设, $\eta \sim U(0, \xi)$, 也即: $\eta|\xi=x \sim U(0, x)$, $x > 0$; 令 $\eta|\xi=x \sim f_{\eta|\xi}(y|x)$, 则有:

$\forall x > 0$, $f_{\eta|\xi}(y|x) = \begin{cases} \frac{1}{x}, & 0 < y < x; \\ 0, & \text{其他} \end{cases}$; 记 $(\xi, \eta) \sim f_{\xi,\eta}(x, y)$, $\eta \sim f_\eta(y)$, 则有:

$f_{\xi,\eta}(x, y) = f_\xi(x) \cdot f_{\eta|\xi}(y|x) = \begin{cases} \lambda^2 x e^{-\lambda x}, & 0 < y < x; \\ 0, & \text{其他} \end{cases}$; 从而,

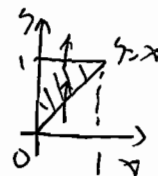
$$f_\eta(y) = \int_{-\infty}^{+\infty} f_{\xi,\eta}(x, y) dx = \begin{cases} \int_y^{+\infty} \lambda^2 x e^{-\lambda x} dx = \lambda e^{-\lambda y}, & y > 0; \\ 0, & y \leq 0; \end{cases} \quad \text{即: } \eta \sim E(\lambda).$$



4. (1) 设 $Y \sim f_Y(y) = \begin{cases} 5y^4, & 0 < y < 1; \\ 0, & \text{其他}; \end{cases}$, 给定 $Y=y$ ($0 < y < 1$) 时, X 的条件密度

为 $f_{X|Y}(x|y) = \begin{cases} \frac{3x^2}{y^3}, & 0 < x < y; \\ 0, & \text{其他}; \end{cases}$, 试求 $P(X > 0.5)$; 记 $(X, Y) \sim f_{X,Y}(x, y)$, $X \sim f_X(x)$, 则有:

$$f_{X,Y}(x, y) = f_Y(y) \cdot f_{X|Y}(x|y) = \begin{cases} 15x^2 y, & 0 < x < y < 1; \\ 0, & \text{其他} \end{cases}$$



$$f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x, y) dy = \begin{cases} \int_x^1 15x^2 y dy, & 0 < x < 1; \\ 0, & \text{其他}; \end{cases}$$

$$= \begin{cases} \frac{15}{2} x^2 (1-x^2), & 0 < x < 1; \\ 0, & \text{其他}; \end{cases} \quad \text{从而,}$$

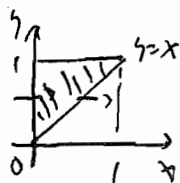
$$\begin{aligned} P(X > 0.5) &= \int_{\frac{1}{2}}^{+\infty} f_X(x) dx = \int_{\frac{1}{2}}^1 \frac{15}{2} x^2 (1-x^2) dx = \frac{15}{2} \cdot \left[\frac{1}{3} \cdot \frac{7}{8} - \frac{1}{5} \cdot \frac{31}{32} \right] \\ &= \frac{15}{2} \cdot \frac{140 - 93}{320} = \frac{47}{64}. \end{aligned}$$

(2) 已知随机变量 X 的密度 $f_X(x) = \begin{cases} 4x(1-x^2), & 0 < x < 1; \\ 0, & \text{其他}; \end{cases}$ 给定 $X=x$

($0 < x < 1$) 时, 随机变量 Y 的条件密度为 $f_{Y|X}(y|x) = \begin{cases} \frac{2y}{1-x^2}, & x < y < 1; \\ 0, & \text{其他}; \end{cases}$ 试求

$P(Y \geq 0.5)$. 注 $(X, Y) \sim f_{X,Y}$, $Y \sim f_Y(y)$. 即有: $f_{X,Y} = f_X(x) \cdot f_{Y|X}(y|x)$

$$= \begin{cases} 8xy, & 0 < x < y < 1; \\ 0, & \text{其他}; \end{cases} \quad f_Y(y) = \int_{-\infty}^{+\infty} f_{X,Y}(x,y) dx = \begin{cases} \int_0^y 8xy dx = 4y^3, & 0 < y < 1; \\ 0, & \text{其他}; \end{cases}$$



$$\text{从而, } P(Y \geq 0.5) = \int_{\frac{1}{2}}^{+\infty} f_Y(y) dy = \int_{\frac{1}{2}}^1 4y^3 dy = 1 - \frac{1}{16} = \frac{15}{16}.$$

5. 设二维随机变量 (X, Y) 的联合概率密度为 $f(x, y) = \begin{cases} xe^{-x(y+1)}, & x, y > 0; \\ 0, & \text{其他} \end{cases}$, 试求:

(1) 边缘密度函数 $f_X(x), f_Y(y)$:

$$f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x,y) dy = \begin{cases} \int_0^{+\infty} x \cdot e^{-x} \cdot e^{-xy} dy = e^{-x}, & x > 0; \\ 0, & x \leq 0; \end{cases} \quad \text{即: } X \sim E(1);$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f_{X,Y}(x,y) dx = \begin{cases} \int_0^{+\infty} x \cdot e^{-x(y+1)} dx, & y > 0; \\ 0, & y \leq 0; \end{cases} = \begin{cases} \frac{1}{(y+1)^2} \int_0^{+\infty} x(y+1) \cdot e^{-x(y+1)} d[x(y+1)], & y > 0; \\ 0, & y \leq 0; \end{cases}$$

(2) 条件密度函数 $f_{Y|X}(y|x), f_{X|Y}(x|y)$: $= \begin{cases} \frac{1}{(y+1)^2}, & y > 0; \\ 0, & y \leq 0; \end{cases}$

$$\forall x > 0, f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \begin{cases} xe^{-xy}, & y > 0; \\ 0, & y \leq 0; \end{cases} \quad \text{即: } Y|X=x \sim E(x), x > 0;$$

$$\forall y > 0, f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \begin{cases} x \cdot (y+1)^2 \cdot e^{-x(y+1)}, & x > 0; \\ 0, & x \leq 0. \end{cases}$$

(3) 条件概率 $P(Y \leq 1|X \leq 2), P(Y \leq 1|X=2)$.

注: $P(X \leq 2) > 0, P(X=2) = 0$.

$$\begin{aligned} P(Y \leq 1|X \leq 2) &= \frac{P(X \leq 2, Y \leq 1)}{P(X \leq 2)} = \frac{\int_0^2 dx \int_0^1 x \cdot e^{-x(y+1)} dy}{1 - e^{-2}} = \frac{\int_0^2 [e^{-x}(1 - e^{-x})] dx}{1 - e^{-2}} \\ &= \frac{1 - e^{-2} - \frac{1}{2}(1 - e^{-4})}{1 - e^{-2}} = 1 - \frac{1}{2}(1 + e^{-2}) = \frac{1}{2}(1 - e^{-2}). \end{aligned}$$

$$P(Y \leq 1|X=2) = \int_{-\infty}^1 f_{Y|X}(y|2) dy = \dots = 1 - e^{-2}.$$

注: $Y|X=2 \sim E(2)$

证1: 由 $\iint_{\mathbb{R}^2} f(x,y) dx dy = 1$. 证法: $A \cdot \iint_{\mathbb{R}^2} e^{-2x^2+2xy-y^2} dx dy = A \cdot \iint_{\mathbb{R}^2} e^{-(x-y)^2-x^2} dx dy$

6. 设二维随机变量 (X,Y) 的概率密度函数为

$$f(x,y) = Ae^{-2x^2+2xy-y^2}, -\infty < x, y < +\infty,$$

试求常数 A 及条件密度函数 $f_{Y|X}(y|x)$.

证2: 记 $X \sim f_X(x)$. 证法: $f_X(x) = \int_{-\infty}^{+\infty} f(x,y) dy = Ae^{-x^2} \int_{-\infty}^{+\infty} e^{-(y-x)^2} dy \xrightarrow{t=y-x} Ae^{-x^2} \int_{-\infty}^{+\infty} e^{-t^2} dt$
 (证: $\int_{-\infty}^{+\infty} e^{-t^2} dt \xrightarrow{t=\frac{x}{\sqrt{2}}} \frac{1}{\sqrt{2}} \int_{-\infty}^{+\infty} e^{-\frac{x^2}{2}} dx = \sqrt{2}$) $= A \cdot \sqrt{2} \cdot e^{-x^2}, -\infty < x < +\infty;$

由 $\int_{-\infty}^{+\infty} f_X(x) dx = A \cdot \sqrt{2} \cdot \int_{-\infty}^{+\infty} e^{-x^2} dx = A \cdot \sqrt{2} \cdot \sqrt{2} = 1, A = \frac{1}{2}.$

$\forall x \in (-\infty, +\infty), f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{1}{\sqrt{2}} e^{-(y-x)^2} = \frac{1}{\sqrt{2} \cdot \frac{1}{\sqrt{2}}} e^{-\frac{(y-x)^2}{2 \cdot (\frac{1}{\sqrt{2}})^2}}, -\infty < y < +\infty$

证: $Y|X=x \sim N(x, (\frac{1}{\sqrt{2}})^2), -\infty < x < +\infty.$

7. 设 $Y \sim U[2,4]$, 且给定 $Y=y$ ($2 \leq y \leq 4$) 时, $X \sim E(y)$, 试求:

(1) (X,Y) 的联合密度函数; (2) 试证: $XY \sim E(1)$. 证法: $X|Y=y \sim E(y), 2 \leq y \leq 4$

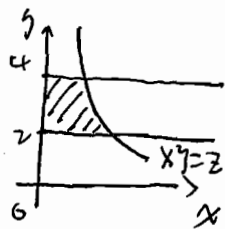
记 $Y \sim f_Y(y), X|Y=y \sim f_{X|Y}(x|y), 2 \leq y \leq 4. (X,Y) \sim f_{X,Y}(x,y)$. 证法:

$$f_{X|Y}(x|y) = \begin{cases} \frac{1}{y} \cdot e^{-\frac{x}{y}}, & x > 0; \\ 0, & x \leq 0. \end{cases} \quad 2 \leq y \leq 4.$$

$$f_{X,Y}(x,y) = f_Y(y) \cdot f_{X|Y}(x|y) = \begin{cases} \frac{1}{2} e^{-\frac{x}{y}}, & x > 0, 2 \leq y \leq 4; \\ 0, & \text{其他} \end{cases}$$

(2) 证: $Z = XY$. 证法: $R(Z) = (0, +\infty)$, 证法, $\forall z \leq 0, \bar{F}_Z(z) = P(Z \leq z) = P(\emptyset) = 0$

$$\forall z > 0, \bar{F}_Z(z) = P(Z \leq z) = P(XY \leq z) = \iint_{xy \leq z} f_{X,Y}(x,y) dx dy$$



$$= \int_2^4 dy \int_0^{\frac{z}{y}} \frac{1}{2} \cdot e^{-\frac{x}{y}} dx = \int_2^4 \frac{1}{2} \cdot (1 - e^{-z/y}) dy = 1 - e^{-z},$$

证: $\bar{F}_Z(z) = \begin{cases} 1 - e^{-z}, & z > 0; \\ 0, & z \leq 0. \end{cases}$ 也证:

$Z = XY \sim E(1).$

8. (1) 设 X, Y 为两个随机变量, $Y \sim \begin{pmatrix} 0 & 1 \\ 0.7 & 0.3 \end{pmatrix}$, 且给定 $Y=k$ 时, $X \sim N(k, 1^2)$,

$k=0, 1$; 试求 X 的分布; 由题设, $X|Y=0 \sim N(0, 1^2)$, $X|Y=1 \sim N(1, 1^2)$, 即有:
 $X-1|Y=1 \sim N(0, 1^2)$; $\forall x \in \mathbb{R}$,

$$\begin{aligned} F_X(x) &= P(X \leq x) = P(Y=0) \cdot P(X \leq x | Y=0) + P(Y=1) \cdot P(X \leq x | Y=1) \\ &= 0.7 \cdot \Phi(x) + 0.3 \cdot P(X-1 \leq x-1 | Y=1) \\ &= 0.7 \cdot \Phi(x) + 0.3 \Phi(x-1). \end{aligned}$$

(2) 设 $X \sim \begin{pmatrix} 1 & 2 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$, 且给定 $X=k$ 时, $Y \sim U(0, k)$, $k=1, 2$; 试求 Y 的分布.

由题设, $Y|X=1 \sim U(0, 1)$, $Y|X=2 \sim U(0, 2)$; 即有, $\forall y \in \mathbb{R}$,

$$F_Y(y) = P(Y \leq y) = P(X=1) \cdot P(Y \leq y | X=1) + P(X=2) \cdot P(Y \leq y | X=2)$$

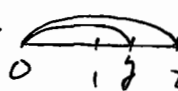
$$= \frac{1}{2} [P(Y \leq y | X=1) + P(Y \leq y | X=2)] = \begin{cases} \frac{1}{2} [0 + 0] = 0 & , y \leq 0 \\ \frac{1}{2} [y + \frac{y}{2}] = \frac{3}{4}y & , 0 < y < 1 \\ \frac{1}{2} [1 + \frac{y}{2}] = \frac{1}{2} + \frac{y}{4} & , 1 \leq y < 2 \\ \frac{1}{2} [1 + 1] = 1 & , y \geq 2 \end{cases}$$

, $y \leq 0$

, $0 < y < 1$

, $1 \leq y < 2$

, $y \geq 2$



令 $Y \sim f_Y(y)$, 即有:

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \begin{cases} \frac{3}{4} & , 0 < y < 1; \\ \frac{1}{4} & , 1 \leq y < 2; \\ 0 & , \text{其他} \end{cases}$$

9. (1) 设 $X \sim U(0, 2)$, 试求给定 $X > 1$ 时, X 的条件分布;

猜測: $\overline{\quad\quad\quad} \quad X|X>1 \sim U(1, 2)$. ? 验证:

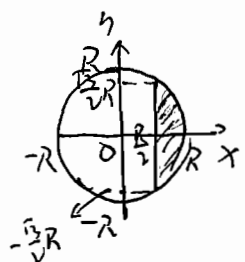
$\forall x \in \mathbb{R}$,

$$F_{X|X>1}(x) = P(X \leq x | X > 1) = \frac{P(\{X > 1, X \leq x\})}{P(X > 1)} = 2 \cdot P(X > 1, X \leq x)$$

$$= \begin{cases} 2 \cdot \phi = 0 & , x \leq 1; \\ 2 \cdot P(1 < X \leq x) = 2 \cdot \frac{x-1}{2} & , 1 < x < 2; \\ 2 \cdot P(X > 1) = 2 \cdot \frac{1}{2} = 1 & , x \geq 2; \end{cases} = \begin{cases} 0 & , x \leq 1; \\ x-1 & , 1 < x < 2; \\ 1 & , x \geq 2; \end{cases}$$

令 $X|X>1 \sim f_{X|X>1}(x)$, 即有: $f_{X|X>1}(x) = \frac{d}{dx} F_{X|X>1}(x) = \begin{cases} 1 & , 1 < x < 2; \\ 0 & , \text{其他}; \end{cases}$ ⁵⁶

即: $X|X>1 \sim U(1, 2)$



(2) 设 $(X, Y) \sim U(D)$, $D: x^2 + y^2 \leq R^2$, 求 $P(Y > 0 | X = \frac{R}{2})$, $P(Y > 0 | X > \frac{R}{2})$.

直观地, $Y | X = \frac{R}{2} \sim U[-\sqrt{R^2 - (\frac{R}{2})^2}, \sqrt{R^2 - (\frac{R}{2})^2}]$, $P(Y > 0 | X = \frac{R}{2}) = \frac{1}{2}$ ($\frac{\sqrt{R^2 - (\frac{R}{2})^2}}{\sqrt{R^2 - (\frac{R}{2})^2}}$).

$(X, Y) | X > \frac{R}{2} \sim U(D_1)$, $D_1: \frac{R}{2} < x \leq R, -\sqrt{R^2 - x^2} \leq y \leq \sqrt{R^2 - x^2}$

$P(Y > 0 | X > \frac{R}{2}) = \frac{1}{2}$ ($\frac{\text{面积}}{\text{面积}}$)

事实上, $P(Y > 0 | X = \frac{R}{2}) = \int_0^{\sqrt{R^2 - (\frac{R}{2})^2}} f_{Y|X}(\frac{R}{2}, y) dy = \dots = \frac{1}{2}$, $P(X = \frac{R}{2}) = 0$;

$P(Y > 0 | X > \frac{R}{2}) = \frac{P(X > \frac{R}{2}, Y > 0)}{P(X > \frac{R}{2})} = \frac{\int_{\frac{R}{2}}^R \int_0^{\sqrt{R^2 - x^2}} f_{X,Y}(x, y) dy dx}{\int_{\frac{R}{2}}^R \int_{-\sqrt{R^2 - x^2}}^{\sqrt{R^2 - x^2}} f_{X,Y}(x, y) dy dx} = \dots = \frac{1}{2}$, $P(X > \frac{R}{2}) > 0$.

10. 设一个人一年内患感冒的次数服从参数为 $\lambda_1 = 5$ 的 Poisson 分布; 现有某种预防感冒的药物对 75% 的人有效 (能将 Poisson 分布的参数减少为 $\lambda_2 = 3$); 对另外 25% 的人不起作用; 如果某人服用了此药, 一年内患了两次感冒, 那么该药对他有效的可能性是多少? 直观地挑这一人, 记 $A = \{\text{该药对此人有效}\}$.

记 X 为此人一年患感冒的次数; 由此设, $P(A) = \frac{3}{4}$, $P(\bar{A}) = \frac{1}{4}$; $X|A \sim P(3)$.

$X|\bar{A} \sim P(5)$. 则有:

$$P(A | X=2) = \frac{P(A \cap \{X=2\})}{P(X=2)} = \frac{P(A) \cdot P(X=2|A)}{P(A)P(X=2|A) + P(\bar{A}) \cdot P(X=2|\bar{A})}$$

$$= \frac{\frac{3}{4} \cdot \frac{3^2}{2!} \cdot e^{-3}}{\frac{3}{4} \cdot \frac{3^2}{2!} \cdot e^{-3} + \frac{1}{4} \cdot \frac{5^2}{2!} \cdot e^{-5}} = \frac{27}{27 + 25 \cdot e^{-2}}.$$

习题 3.4

1. 设随机变量 X, Y 独立同分布, 且 $X \sim \begin{pmatrix} 0 & 1 \\ 1-p & p \end{pmatrix}$, 令 $Z = \begin{cases} 1, & \text{若 } X+Y \text{ 为偶数;} \\ 0, & \text{若 } X+Y \text{ 为奇数;} \end{cases}$

问: p 取何值时, X, Z 相互独立?

直观, X, Y

$X \backslash Y$	0	1
0	$(1-p)^2$	$p(1-p)$
1	$p(1-p)$	p^2
	$1-p$	p

这里, $P(X=i, Y=j) = P(X=i) \cdot P(Y=j)$, $i, j = 0, 1$;

则有: $X+Y$

$X+Y$	0	1	2
P	$(1-p)^2$	$2p(1-p)$	p^2

其中, $P(X+Y=2)$

$= P(X=1, Y=1)$, $P(X+Y=0) = P(X=0, Y=0)$, 也则有: Z

Z	0	1
P	$2p(1-p)$	$p^2 + (1-p)^2$

若由 X, Z 独立, 则有, $\{X=0\}$ 与 $\{Z=0\}$ 独立, 即: $P(X=0, Z=0) = P(X=0) \cdot P(Z=0)$

则有: $P(X=0, Y=1) = P(X=0) \cdot P(Y=1)$ 也即: $p(1-p) = (1-p)p$ 即 $p = 1$ 或 $p = 0$

2. 设随机变量 X, Y 独立, 且 $X \sim P(\lambda_1), Y \sim P(\lambda_2)$, 试求给定 $X+Y=n$ 时, X 的条件分布. 证: $X \sim P(\lambda_1), Y \sim P(\lambda_2), X, Y$ 独立, $X+Y \sim P(\lambda_1+\lambda_2)$.

$$\forall k=0, 1, \dots, P(X+Y=k) = P\left(\bigcup_{i=0}^k \{X=i, Y=k-i\}\right) = \sum_{i=0}^k P(X=i, Y=k-i) = \sum_{i=0}^k P(X=i) \cdot P(Y=k-i) \\ = \sum_{i=0}^k \left(\frac{\lambda_1^i}{i!} e^{-\lambda_1}\right) \cdot \left(\frac{\lambda_2^{k-i}}{(k-i)!} e^{-\lambda_2}\right) = \frac{1}{k!} e^{-(\lambda_1+\lambda_2)} \sum_{i=0}^k C_k^i \lambda_1^i \lambda_2^{k-i} (= \text{二项式公式}) = \frac{(\lambda_1+\lambda_2)^k}{k!} e^{-(\lambda_1+\lambda_2)}$$

$$\text{证: } X+Y \sim P(\lambda_1+\lambda_2). \forall k \in \{0, 1, \dots, n\}, P_{X|X+Y}(k|n) = P(X=k|X+Y=n) = \frac{P(X=k, X+Y=n)}{P(X+Y=n)} \\ = \frac{P(X=k, Y=n-k)}{P(X+Y=n)} = \frac{P(X=k) \cdot P(Y=n-k)}{P(X+Y=n)} = \frac{\frac{\lambda_1^k}{k!} e^{-\lambda_1} \cdot \frac{\lambda_2^{n-k}}{(n-k)!} e^{-\lambda_2}}{\frac{(\lambda_1+\lambda_2)^n}{n!} e^{-(\lambda_1+\lambda_2)}} = C_n^k \left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^k \left(1 - \frac{\lambda_1}{\lambda_1+\lambda_2}\right)^{n-k}$$

3. 设随机向量 (X, Y) 具有如下的联合密度: 证: $X|X+Y=n \sim B(n, \frac{\lambda_1}{\lambda_1+\lambda_2})$.

(1) $f(x, y) = 4xy, 0 < x, y < 1$; (2) $f(x, y) = 8xy, 0 < x < y < 1$;

试讨论以上两种情形下, X, Y 是否独立? (直观上, $4xy$ 的 x, y 范围可分离为矩形, 且 $0 < x, y < 1$ 为矩形域. (1) 时 X, Y 独立), 证: 证 $X \sim f_X(x), Y \sim f_Y(y)$.

$$(1). f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \begin{cases} \int_0^1 4xy dy = 2x, & 0 < x < 1 \\ 0, & \text{其他} \end{cases} \quad \text{同理, } f_Y(y) = \begin{cases} 2y, & 0 < y < 1 \\ 0, & \text{其他} \end{cases}$$

证: $f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y), X, Y$ 独立!

$$(2). f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \begin{cases} \int_x^1 8xy dy = 4x(1-x^2), & 0 < x < 1 \\ 0, & \text{其他} \end{cases} \quad f_X(x) \cdot f_Y(y) \neq f_{X,Y}(x, y) \\ f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \begin{cases} \int_0^y 8xy dx = 4y^3, & 0 < y < 1 \\ 0, & \text{其他} \end{cases} \quad \text{证: } X, Y \text{ 不独立!}$$

4. (1) 设 $(X, Y) \sim U(D)$, 其中 $D: x^2 + y^2 \leq 1$, 试讨论 X, Y 的独立性;

(2) 设 $(X, Y) \sim U(G)$, 其中 $G = [0, 1] \times [0, 2]$, 试讨论 X, Y 的独立性.

(2) $X \sim U[0, 1], Y \sim U[0, 2]$. 且 X, Y 独立!

(1). $X \sim f_X(x), Y \sim f_Y(y)$,

$$f_X(x) = \begin{cases} \frac{2\sqrt{1-x^2}}{\pi}, & -1 \leq x \leq 1 \\ 0, & \text{其他} \end{cases} \quad f_Y(y) = \begin{cases} \frac{2\sqrt{1-y^2}}{\pi}, & -1 \leq y \leq 1 \\ 0, & \text{其他} \end{cases}$$

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{\pi}, & (x, y) \in D \\ 0, & \text{其他} \end{cases}$$

$f_{X,Y}(x, y) \neq f_X(x) \cdot f_Y(y), X, Y$ 不独立!

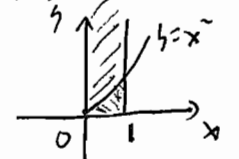
6. (1) 设随机变量 X, Y 独立, 且 $X \sim U[0, 1]$, $Y \sim E\left(\frac{1}{2}\right)$,

(i) 试写出 (X, Y) 的联合密度函数; 且 $X \sim f_X(x)$, $Y \sim f_Y(y)$, $(X, Y) \sim f_{X,Y}(x, y)$

证: $f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y) = \begin{cases} \frac{1}{2} e^{-\frac{1}{2}y}, & 0 \leq x \leq 1, y > 0; \\ 0, & \text{其他} \end{cases}$

(ii) 试求“方程 $t^2 + 2Xt + Y = 0$ 有实根”的概率;

$P\{\text{方程 } t^2 + 2Xt + Y = 0 \text{ 有实根}\} = P\{\Delta = 4X^2 - 4Y \geq 0\} = P\{Y \leq X^2\} = \iint_{y \leq x^2} f_{X,Y}(x, y) dx dy$

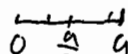


$$= \int_0^1 dx \int_0^{x^2} \frac{1}{2} e^{-\frac{1}{2}y} dy = \int_0^1 [1 - e^{-\frac{1}{2}x^2}] dx = 1 - \sqrt{2} \cdot \int_0^1 \frac{1}{\sqrt{2}} e^{-\frac{x^2}{2}} dx$$

$$= 1 - \sqrt{2} \cdot [\Phi(1) - \Phi(0)]$$

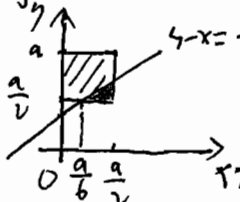
(2) 从长度为 a 的线段的中点两边随机各选取一点, 求“两点间距离小于 $\frac{a}{3}$ ”

的概率.



证: 几何概率 (几何方法计算) 证: 设两点分别为 X, Y , 由题意 $X \sim U[0, \frac{a}{2}]$, $Y \sim U[\frac{a}{2}, a]$ 且 X, Y 独立, 故令 $X \sim f_X(x)$, $Y \sim f_Y(y)$, $(X, Y) \sim f_{X,Y}(x, y)$, 证:

$f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y) = \begin{cases} \frac{4}{a^2}, & 0 \leq x \leq \frac{a}{2} \leq y \leq a; \\ 0, & \text{其他} \end{cases}$ $P\{\text{两点距离小于 } \frac{a}{3}\} = P\{Y - X < \frac{a}{3}\}$



$$= \iint_{y-x < \frac{a}{3}} f_{X,Y}(x, y) dx dy = \iint_{\substack{y-x < \frac{a}{3} \\ 0 \leq x \leq \frac{a}{2} \leq y \leq a}} \frac{4}{a^2} dx dy = \dots = \frac{\frac{a^2}{18}}{\frac{a^2}{4}} = \frac{2}{9} \quad (\frac{\text{面积}}{\text{面积}})$$

17. 试用概率方法证明: $\forall a > 0, \frac{1}{\sqrt{2\pi}} \int_{-a}^a e^{-\frac{x^2}{2}} dx \leq \sqrt{1 - e^{-a^2}}$.

证 X, Y 独立同 $N(0, 1)$ 分布, 证 $X \sim \varphi(x)$, $Y \sim \varphi(y)$

令 $(X, Y) \sim f_{X,Y}(x, y)$, 证: $f_{X,Y}(x, y) = \varphi(x) \cdot \varphi(y)$, $-\infty < x, y < \infty$.

$(\frac{1}{\sqrt{2\pi}} \int_{-a}^a e^{-\frac{x^2}{2}} dx)^2 = P(-a \leq X \leq a) \cdot P(-a \leq Y \leq a) = P((X, Y) \in [-a, a] \times [-a, a])$

$D: x^2 + y^2 \leq 2a^2$

$$P((X, Y) \in D) = \iint_D f_{X,Y}(x, y) dx dy = \iint_{x^2 + y^2 \leq 2a^2} \frac{1}{2\pi} e^{-\frac{x^2 + y^2}{2}} dx dy$$

$$= \int_0^{2\pi} d\theta \int_0^{\sqrt{2}a} \frac{1}{2\pi} e^{-\frac{r^2}{2}} r dr = \int_0^{2\pi} \frac{1}{2\pi} [1 - e^{-a^2}] d\theta = 1 - e^{-a^2}$$

证: $\frac{1}{\sqrt{2\pi}} \int_{-a}^a e^{-\frac{x^2}{2}} dx \leq \sqrt{1 - e^{-a^2}}$

8. 设随机向量 \$(X, Y)\$ 的联合密度为 $f(x, y) = \begin{cases} \frac{1+xy}{4}, & -1 < x, y < 1; \\ 0, & \text{其他}; \end{cases}$ 试证: X, Y 不独立, 但 X^2, Y^2 是独立地.

① 证明: X, Y 不独立! 证 $X \sim f_X(x), Y \sim f_Y(y)$, 即有

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \begin{cases} \int_{-1}^1 \frac{1+xy}{4} dy = \frac{1}{2}, & -1 < x < 1; \\ 0, & \text{其他}; \end{cases} \text{ 即: } X \sim U(-1, 1); \text{ 同理, } f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \begin{cases} \frac{1}{2}, & -1 < y < 1; \\ 0, & \text{其他}; \end{cases} \text{ 即: } Y \sim U(-1, 1).$$

即有: $f_X(x) \cdot f_Y(y) \neq f(x, y)$, 即: X, Y 不独立!

② 证: X^2, Y^2 独立! $\forall x, y \in \mathbb{R}, F_{X^2, Y^2}(x, y) = P(X^2 \leq x, Y^2 \leq y) = \begin{cases} P(\emptyset) = 0, & x < 0 \text{ 或 } y < 0; \\ P(-\sqrt{x} \leq X \leq \sqrt{x}) \cdot P(-\sqrt{y} \leq Y \leq \sqrt{y}), & 0 \leq x < 1, 0 \leq y < 1; \\ P(-\sqrt{x} \leq X \leq \sqrt{x}), & 0 \leq x < 1, 0 \leq y < 1; \\ P(X \leq \sqrt{x}), & 0 \leq x < 1, y \geq 1; \\ P(X \leq \sqrt{x}), & x \geq 1, y \geq 1; \end{cases}$

$$= \begin{cases} 0, & x < 0 \text{ 或 } y < 0; \\ \frac{\sqrt{x}}{\sqrt{y}}, & 0 \leq x < 1, 0 \leq y < 1; \\ \sqrt{x}, & 0 \leq x < 1, y \geq 1; \\ 1, & x \geq 1, y \geq 1; \end{cases}$$

$$* - \int_{-\sqrt{x}}^{\sqrt{x}} \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1+st}{4} dt ds = \int_{-\sqrt{x}}^{\sqrt{x}} \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{4} dt ds = \int_{-\sqrt{x}}^{\sqrt{x}} \frac{\sqrt{y}}{2} ds = \sqrt{x} \sqrt{y}$$

$$\lim_{x \rightarrow \infty} F_{X^2, Y^2}(x, y) = F_Y(y), \dots \text{ 或 } F_{X^2}(x) = P(X^2 \leq x) = \begin{cases} 0, & x < 0; \\ \sqrt{x}, & 0 \leq x < 1; \\ 1, & x \geq 1; \end{cases}$$

$$\text{同理, } F_Y(y) = P(Y^2 \leq y) = \begin{cases} 0, & y < 0; \\ \sqrt{y}, & 0 \leq y < 1; \\ 1, & y \geq 1; \end{cases} \text{ 且 } F_{X^2}(x) \cdot F_Y(y) = F_{X^2, Y^2}(x, y).$$

即有: X^2, Y^2 独立!

9. 设 X_1, X_2, X_3 独立, 且 $X_1, X_2 \sim N(0, 1^2)$, $X_3 \sim \begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 2 \end{pmatrix}$, $Y = X_3 X_1 + (1 - X_3) X_2$;

X_1, X_2 独立加权平均

试求 (X_1, Y) 的联合分布函数, 并证明: $Y \sim N(0, 1^2)$.

$$\forall (x, y) \in \mathbb{R}^2, F(x, y) = P(X_1 \leq x, Y \leq y) = P(X_1 \leq x, X_3 X_1 + (1 - X_3) X_2 \leq y) \quad (\text{令 } X_3 = 0, 1)$$

$$= P(X_3 = 0) P(X_1 \leq x, X_2 \leq y | X_3 = 0) + P(X_3 = 1) \cdot P(X_1 \leq x, X_1 \leq y | X_3 = 1)$$

$$= \frac{1}{2} [P(X_1 \leq x, X_2 \leq y | X_3 = 0) + P(X_1 \leq x, X_1 \leq y | X_3 = 1)]$$

$$= \frac{1}{2} [P(X_1 \leq x, X_2 \leq y) + P(X_1 \leq x, X_1 \leq y)] = \frac{1}{2} [P(X_1 \leq x) \cdot P(X_2 \leq y) + P(X_1 \leq x \wedge y)]$$

$$= \frac{1}{2} [\phi(x) \phi(y) + \phi(x \wedge y)] = \begin{cases} \frac{1}{2} [\phi(x) \phi(y) + \phi(x)], & x \leq y; \\ \frac{1}{2} [\phi(x) \phi(y) + \phi(y)], & x > y; \end{cases}$$

$$\forall y \in \mathbb{R}, F_Y(y) = \lim_{x \rightarrow +\infty} F(x, y) = \lim_{x \rightarrow +\infty} \frac{1}{2} [\phi(x) \phi(y) + \phi(x)] = \frac{1}{2} [\phi(+\infty) \phi(y) + \phi(y)] = \phi(y), \text{ 即: } Y \sim N(0, 1^2).$$

$$\text{证: } \forall y \in \mathbb{R}, F_Y(y) = P(Y \leq y) = P(X_3 X_1 + (1 - X_3) X_2 \leq y)$$

$$= P(X_3 = 0) \cdot P(X_2 \leq y | X_3 = 0) + P(X_3 = 1) \cdot P(X_1 \leq y | X_3 = 1)$$

$$= \frac{1}{2} [P(X_2 \leq y | X_3 = 0) + P(X_1 \leq y | X_3 = 1)]$$

$$= \frac{1}{2} [P(X_2 \leq y) + P(X_1 \leq y)] = \frac{1}{2} [\phi(y) + \phi(y)] = \phi(y), \text{ 即: } Y \sim N(0, 1^2)$$

10. (1) 设随机变量 X, Y 独立, 且 $X \sim E\left(\frac{1}{2}\right)$, $Y \sim \begin{pmatrix} -1 & 1 \\ 0.25 & 0.75 \end{pmatrix}$, 试求 $P(XY \leq 2)$;

$$\begin{aligned} P(XY \leq 2) &= P(Y = -1)P(XY \leq 2 | Y = -1) + P(Y = 1)P(XY \leq 2 | Y = 1) \\ &= \frac{1}{4} \cdot P(X \geq -2 | Y = -1) + \frac{3}{4} \cdot P(X \leq 2 | Y = 1) \\ &= \frac{1}{4} P(X \geq -2) + \frac{3}{4} P(X \leq 2) \\ &= \frac{1}{4} \cdot 1 + \frac{3}{4} \cdot [1 - e^{-1}] = 1 - \frac{3}{4}e^{-1} \end{aligned}$$

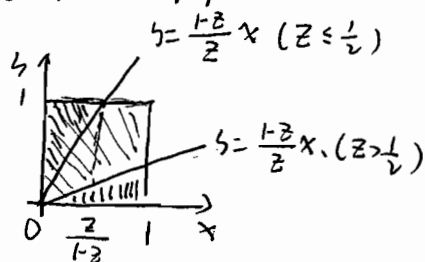
(2) 设随机变量 X, Y 独立同 $U(0,1)$ 分布, 试求 $Z = \frac{X}{X+Y}$ 的分布函数.

由 $R(X) = R(Y) = (0, 1)$, 证得: $R(Z) = (0, 1)$; 又, $\forall z \leq 0, F_Z(z) = P(Z \leq z) = 0$; $\forall z \geq 1, F_Z(z) = P(Z \leq z) = 1$; $\forall z \in (0, 1), F_Z(z) = P(Z \leq z) = P\left(\frac{X}{X+Y} \leq z\right) = P((1-z)X \leq zY)$

$$= P\left(Y \geq \frac{1-z}{z}X\right) = \begin{cases} \frac{z}{2(1-z)}, & z \leq \frac{1}{2}; \\ 1 - \frac{1-z}{2z}, & z > \frac{1}{2}; \end{cases}$$

证: $F_Z(z) = \begin{cases} 0 & z \leq 0; \\ \frac{z}{2(1-z)} & 0 < z \leq \frac{1}{2}; \\ 1 - \frac{1-z}{2z} & \frac{1}{2} < z < 1; \\ 1 & z \geq 1; \end{cases} \quad Z \sim C.Y.U.$

注: X, Y 独立同 $U(0,1)$ 分布, $(X, Y) \sim U(D), D = (0,1) \times (0,1)$



11. (1) 若随机向量 $(X, Y) \sim f(x, y) = ke^{-\frac{1}{2}(ax^2 + 2bxy + cy^2)}$, $-\infty < x, y < +\infty$, 在什么条件下, X 与 Y 相互独立? 证: $a, c > 0$.

设 $X \sim f_X(x), Y \sim f_Y(y)$, 证: $f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = k \cdot \int_{-\infty}^{+\infty} e^{-\frac{1}{2}(ax^2 + 2bxy + cy^2)} dy \cdot e^{-(a - \frac{b^2}{c})x^2}$

令 $\frac{1}{2}(ax^2 + 2bxy + cy^2) = u$

$$\frac{k}{\int_{-\infty}^{+\infty} e^{-u} du \cdot e^{-(a - \frac{b^2}{c})x^2}} = k \cdot \frac{\sqrt{\pi}}{\sqrt{c}} \cdot e^{-(a - \frac{b^2}{c})x^2}, \quad -\infty < x < +\infty; \text{ 同理,}$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = k \cdot \frac{\sqrt{\pi}}{\sqrt{a}} \cdot e^{-(c - \frac{b^2}{a})y^2}, \quad -\infty < y < +\infty. \text{ 由 } X, Y \text{ 独立, 证:}$$

$$f(x, y) = f_X(x) \cdot f_Y(y), \text{ 证得:}$$

$$k \cdot e^{-\frac{1}{2}(ax^2 + 2bxy + cy^2)} = k \cdot \frac{\sqrt{\pi}}{\sqrt{c}} \cdot e^{-(a - \frac{b^2}{c})x^2} \cdot k \cdot \frac{\sqrt{\pi}}{\sqrt{a}} \cdot e^{-(c - \frac{b^2}{a})y^2}, \quad -\infty < x, y < +\infty.$$

证得: $k = \frac{\sqrt{ac}}{\pi}, b = 0, a, c > 0$

(2) 试问在何条件下, 函数 $f(x, y) = k \exp\{-(ax^2 + 2bxy + cy^2)\}$ 为某二维随机变量的联合密度函数?

证1: 由 $\iint_{\mathbb{R}^2} f(x, y) dx dy = \iint_{\mathbb{R}^2} k \cdot \exp\{-(ax^2 + 2bxy + cy^2)\} dx dy = 1$. $\forall x, y \in \mathbb{R}$, $ax^2 + 2bxy + cy^2 > 0$

证: $a, c > 0$, $ac - b^2 > 0$, ... Jacobi 变换法!

证2: 设 $(X, Y) \sim f(x, y)$, 且 $X \sim f_X(x)$. 证: $c > 0$.

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = k \int_{-\infty}^{+\infty} e^{-(a(x+\frac{b}{c}y)^2)} dy \cdot e^{-(a-\frac{b^2}{c})x^2} = \frac{k}{\sqrt{c}} \cdot \sqrt{\pi} \cdot e^{-(a-\frac{b^2}{c})x^2} \quad -\infty < x < +\infty$$

由 $\int_{-\infty}^{+\infty} f_X(x) dx = 1$. 证: $c > 0$, $a - \frac{b^2}{c} > 0$. $\int_{-\infty}^{+\infty} f_X(x) dx = \frac{k}{\sqrt{c}} \cdot \sqrt{\pi} \cdot \int_{-\infty}^{+\infty} e^{-(a-\frac{b^2}{c})x^2} dx$

令 $t = \sqrt{a-\frac{b^2}{c}} x$ $\frac{k}{\sqrt{c}} \cdot \sqrt{\pi} \cdot \frac{1}{\sqrt{a-\frac{b^2}{c}}} \int_{-\infty}^{+\infty} e^{-t^2} dt = \frac{k \cdot \pi}{\sqrt{ac-b^2}} = 1$. 证: $a, c > 0$, $ac - b^2 > 0$, $k = \frac{\sqrt{ac-b^2}}{\pi}$.

习题 3.5

1. 设 X, Y 满足 $P(X \geq 0, Y \geq 0) = \frac{3}{7}$, 且 $P(X \geq 0) = P(Y \geq 0) = \frac{4}{7}$, 试求

$P(\max\{X, Y\} \geq 0)$. 证 $A = \{X \geq 0\}$, $B = \{Y \geq 0\}$. 证: $P(A \cap B) = \frac{3}{7}$, $P(A) = P(B) = \frac{4}{7}$.

$P(\max\{X, Y\} \geq 0) = P(\{X \geq 0\} \cup \{Y \geq 0\}) = P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{5}{7}$.

2. 设随机变量 X_1, X_2, X_3, X_4 独立同分布, 且 $P(X_i = 0) = 1 - P(X_i = 1) = 0.6$,

$i = 1, 2, 3, 4$, 试求行列式 $X = \begin{vmatrix} X_1 & X_2 \\ X_3 & X_4 \end{vmatrix}$ 的概率分布. 证: $X = X_1 X_4 - X_2 X_3$. 证:

$R(X) = \{-1, 0, 1\}$. $P(X = -1) = P(\{X_1 X_4 = 0\} \cap \{X_2 X_3 = 1\}) = P(X_1 X_4 = 0) \cdot P(X_2 X_3 = 1)$
 $= [P(X_1 = 0) + P(X_4 = 0) - P(X_1 = 0)P(X_4 = 0)] \cdot P(X_2 = 1) \cdot P(X_3 = 1)$ (独立性)
 $= 0.84 \times 0.16$.

$P(X = 1) = P(\{X_1 X_4 = 1\} \cap \{X_2 X_3 = 0\}) = \dots = 0.84 \times 0.16$

证:

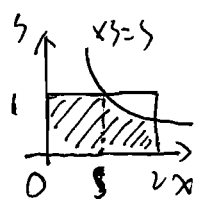
X	-1	0	1
P	0.84×0.16	$1 - 2 \times 0.84 \times 0.16$	0.84×0.16

4. 设某一设备装有三个同类的电器元件, 各元件工作相互独立, 且工作时间服从参数为 λ 的指数分布; 当三个元件都正常工作时, 设备才正常工作; 试求设备正常工作时间 T 的概率分布.

设三个元件的工作时间分别为 X_1, X_2, X_3 , 由题设 X_1, X_2, X_3 独立同 $E(\lambda)$ 分布, $T = \min\{X_1, X_2, X_3\} = \bigwedge_{i=1}^3 X_i$; 易见, $R(T) = (0, +\infty)$, 且 $\forall t \leq 0, F_T(t) = P(T \leq t) = 0$; $\forall t > 0, F_T(t) = P(T \leq t) = 1 - P(T > t) = 1 - P(\bigwedge_{i=1}^3 X_i > t)$
 $= 1 - P(\bigcap_{i=1}^3 \{X_i > t\}) = 1 - \prod_{i=1}^3 P(X_i > t) = 1 - \prod_{i=1}^3 e^{-\lambda t} = 1 - e^{-3\lambda t}$, 即:

$$F_T(t) = \begin{cases} 1 - e^{-3\lambda t} & , t > 0; \\ 0 & , t \leq 0; \end{cases} \quad \text{即: } T \sim E(3\lambda).$$

5. (1) 设 $(X, Y) \sim U(D)$, $D = \{(x, y) | 0 \leq x \leq 2, 0 \leq y \leq 1\}$, 试求边长为 X, Y 的矩形面积 S 的概率分布; 设 $(X, Y) \sim f_{X,Y}$, 则: $f_{X,Y} = \begin{cases} \frac{1}{2} & , (x, y) \in D; \\ 0 & , \text{其他} \end{cases}$.



$S = XY$, $R(S) = (0, 2]$. $\forall s \leq 0, F_S(s) = P(S \leq s) = 0$; $\forall s \geq 2, F_S(s) = 1$;
 $\forall s \in (0, 2)$, $F_S(s) = P(S \leq s) = P(XY \leq s) = \iint_{xy \leq s} f_{X,Y}(x,y) dx dy = \frac{S + \int_S^2 \frac{S}{x} dx}{2}$
 $= \frac{1}{2}[S + S(\ln 2 - \ln s)]$. 即: $F_S(s) = \begin{cases} 0 & , s \leq 0; \\ \frac{1}{2}[S + S(\ln 2 - \ln s)] & , 0 < s < 2; \\ 1 & , s \geq 2; \end{cases}$

或由公式 $S \sim f_{XY}(s) = \int_{-\infty}^{+\infty} f(x, \frac{s}{x}) \cdot \frac{1}{|x|} dx = \dots$ 求出 S 的概率密度!

(2) 设 X, Y 独立同 $N(0, 1^2)$ 分布, 则 $Z = \sqrt{X^2 + Y^2}$ 的分布称为瑞利 (Rayleigh)

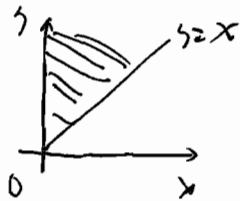
分布, 试求 $P(X^2 + Y^2 \leq 1)$. 设 $(X, Y) \sim f_{X,Y}$, 则: $f_{X,Y} = \phi(x)\phi(y) = \frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}}$, $-\infty < x < +\infty$

$$P(X^2 + Y^2 \leq 1) = \iint_{x^2+y^2 \leq 1} f_{X,Y}(x,y) dx dy \xrightarrow{\text{极坐标变换}} \int_0^{2\pi} d\theta \int_0^1 \frac{1}{2\pi} \cdot e^{-\frac{r^2}{2}} \cdot r dr$$

$$= \int_0^1 e^{-\frac{r^2}{2}} \cdot r dr = 1 - e^{-\frac{1}{2}}$$

6. 设 X, Y 独立, 且 $X \sim E(\lambda_1), Y \sim E(\lambda_2)$, 若 $P(\min\{X, Y\} > 1) = e^{-1}$, $P(X \leq Y) =$

$\frac{1}{3}$, 试求 λ_1, λ_2 . 由 $P(\min\{X, Y\} > 1) = P(\{X > 1\} \cap \{Y > 1\}) = P(X > 1) \cdot P(Y > 1)$
 $= e^{-\lambda_1} \cdot e^{-\lambda_2} = e^{-1}$. 证得: $\lambda_1 + \lambda_2 = 1$; 又 $(X, Y) \sim f_{X,Y}$, 由 X, Y 独立,
 $f_{X,Y} = \begin{cases} \lambda_1 e^{-\lambda_1 x} \cdot \lambda_2 e^{-\lambda_2 y}, & x > 0, y > 0 \\ 0, & \text{其他} \end{cases}$ 证得: $P(X \leq Y) = \iint_{x \leq y} f_{X,Y} dx dy$



$$= \int_0^{+\infty} dx \int_x^{+\infty} \lambda_1 e^{-\lambda_1 x} \cdot \lambda_2 e^{-\lambda_2 y} dy = \int_0^{+\infty} \lambda_1 e^{-\lambda_1 x} \cdot e^{-\lambda_2 x} dx$$

$$= \lambda_1 \int_0^{+\infty} e^{-(\lambda_1 + \lambda_2)x} dx = \frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{1}{3}; \lambda_1 = \frac{1}{3}, \lambda_2 = \frac{2}{3}.$$

7. (1) 设随机变量 X, Y 独立, 且 $P(X=i) = \frac{1}{3}, i = -1, 0, 1; Y \sim U[0, 1]$, 记:

$Z = X + Y$, 试求: $P(Z \leq \frac{1}{2} | X=0)$, Z 的概率密度 $f_Z(z)$; $P(Z \leq \frac{1}{2} | X=0) = P(X+Y \leq \frac{1}{2} | X=0)$
 $= P(Y \leq \frac{1}{2} | X=0) = P(Y \leq \frac{1}{2}) = \frac{1}{2}; \forall z \in \mathbb{R}, \bar{F}_Z(z) = P(Z \leq z) = P(X+Y \leq z) = P(X=-1) \cdot P(X+Y \leq z | X=-1)$
 $+ P(X=0) \cdot P(X+Y \leq z | X=0) + P(X=1) \cdot P(X+Y \leq z | X=1) = \frac{1}{3} [P(Y \leq z+1) + P(Y \leq z) + P(Y \leq z-1)]$ (若 $z < -1$)
 $= \begin{cases} \frac{1}{3}(0+0+0) = 0, & z < -1; \\ \frac{1}{3}(Z+1+0) = \frac{1}{3}(Z+1), & -1 \leq z < 0; \\ \frac{1}{3}(1+Z+0) = \frac{1}{3}(Z+1), & 0 \leq z < 1; \\ \frac{1}{3}(1+1+Z-1) = \frac{1}{3}(Z+1), & 1 \leq z < 2; \\ \frac{1}{3}(1+1+1) = 1, & z \geq 2 \end{cases}$
 $\bar{F}_Z(z) = \begin{cases} 0, & z < -1; \\ \frac{1}{3}(Z+1), & -1 \leq z < 2; \\ 1, & z \geq 2; \end{cases}$ 令 $Z \sim f_Z(z)$, 证得
 $f_Z(z) = \frac{d}{dz} \bar{F}_Z(z) = \begin{cases} \frac{1}{3}, & -1 \leq z < 2; \\ 0, & \text{其他} \end{cases}$ 证得: $Z \sim U[-1, 2]$

(2) 设随机变量 X, Y 独立, 且 $X \sim \begin{pmatrix} 1 & 2 \\ 0.3 & 0.7 \end{pmatrix}, Y \sim f_Y(y)$, 试求 $Z = X + Y$ 的概

率分布: $\forall z \in \mathbb{R}, \bar{F}_Z(z) = P(Z \leq z) = P(X+Y \leq z) = P(X=1) \cdot P(X+Y \leq z | X=1) +$
 $P(X=2) \cdot P(X+Y \leq z | X=2) = 0.3 \cdot P(Y \leq z-1 | X=1) + 0.7 \cdot P(Y \leq z-2 | X=2) = 0.3 \cdot P(Y \leq z-1) +$
 $0.7 \cdot P(Y \leq z-2) = 0.3 \int_{-\infty}^{z-1} f_Y(y) dy + 0.7 \int_{-\infty}^{z-2} f_Y(y) dy$ ① ②
 $\frac{\textcircled{1} \text{ 令 } y+1=u}{\textcircled{2} \text{ 令 } y+2=u} = 0.3 \cdot \int_{-\infty}^z f_Y(u-1) du + 0.7 \cdot \int_{-\infty}^z f_Y(u-2) du$
 $= \int_{-\infty}^z [0.3 \cdot f_Y(u-1) + 0.7 \cdot f_Y(u-2)] du$, 证得: 若令 $Z \sim f_{X+Y}(z)$,

证得: $f_{X+Y}(z) = 0.3 \cdot f_Y(z-1) + 0.7 \cdot f_Y(z-2)$.

(3) 设随机变量 X, Y (独立), 其分布函数分别为:

$$F_X(x) = \begin{cases} 0, & x < 0; \\ \frac{1}{4}, & 0 \leq x < 1; \\ 1, & x \geq 1; \end{cases} \quad F_Y(y) = \begin{cases} 0, & y < 0; \\ y, & 0 \leq y < 1; \\ 1, & y \geq 1; \end{cases}$$

试求 $Z = X + Y$ 的概率分布. 易见,

$$\begin{array}{c|cc} X & 0 & 1 \\ \hline P & \frac{1}{4} & \frac{3}{4} \end{array} \quad \forall z \in \mathbb{R}, F_Z(z) = P(Z \leq z) = P(X+Y \leq z) = P(X=0) \cdot P(X+Y \leq z | X=0) + P(X=1) \cdot P(X+Y \leq z | X=1)$$

$$= \frac{1}{4} \cdot P(Y \leq z | X=0) + \frac{3}{4} P(Y \leq z-1 | X=1) = \frac{1}{4} P(Y \leq z) + \frac{3}{4} P(Y \leq z-1) = \frac{1}{4} F_Y(z) + \frac{3}{4} F_Y(z-1)$$

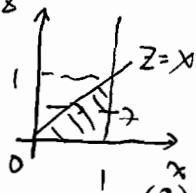
$$= \begin{cases} 0+0=0, & z < 0; \\ \frac{1}{4} F_Y(z) + 0 = \frac{1}{4} F_Y(z) = \frac{z}{4}, & 0 \leq z < 1; \\ \frac{1}{4} + \frac{3}{4} F_Y(z-1) = \frac{3}{4} z - \frac{1}{4}, & 1 \leq z < 2; \\ \frac{1}{4} + \frac{3}{4} = 1, & z \geq 2 \end{cases} \quad \text{即: } F_Z(z) = \begin{cases} 0, & z < 0; \\ \frac{z}{4}, & 0 \leq z < 1; \\ \frac{3}{4} z - \frac{1}{4}, & 1 \leq z < 2; \\ 1, & z \geq 2. \end{cases} \quad \text{若令 } Z \sim f_Z(z).$$

$$\text{则: } f_Z(z) = \frac{d}{dz} F_Z(z) = \begin{cases} \frac{1}{4}, & 0 < z < 1; \\ \frac{3}{4}, & 1 \leq z < 2; \\ 0, & \text{其他.} \end{cases}$$

8. (2) 设 $(X, Y) \sim f(x, y) = \begin{cases} 3x, & 0 < y < x < 1; \\ 0, & \text{其他;} \end{cases}$, 试求 $Z = X - Y$ 的密度;

$$\text{设 } Z = X - Y \sim f_{X-Y}(z), \text{ 则: } f_{X-Y}(z) = \int_{-\infty}^{+\infty} f(x, x-z) dx.$$

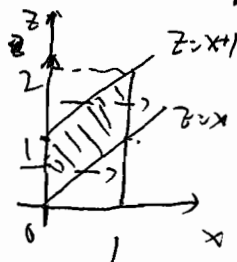
$$f(x, x-z) > 0 \Leftrightarrow 0 < x-z < x < 1 \quad = \begin{cases} \int_z^1 3x dx = \frac{3}{2}(1-z^2), & 0 < z < 1; \\ 0, & \text{其他;} \end{cases}$$



(3) 设 $(X, Y) \sim f(x, y) = \begin{cases} 2-x-y, & 0 < x, y < 1; \\ 0, & \text{其他;} \end{cases}$, 试求 $Z = X + Y$ 的密度;

$$\text{设 } Z = X + Y \sim f_{X+Y}(z), \text{ 则: } f_{X+Y}(z) = \int_{-\infty}^{+\infty} f(x, z-x) dx$$

$$f(x, z-x) > 0 \Leftrightarrow \begin{cases} 0 < x < 1 \\ 0 < z-x < 1 \end{cases}$$



$$= \begin{cases} \int_0^z (z-x) dx = z(2-z), & z \in (0, 1); \\ \int_{z-1}^1 (z-x) dx = (z-1)^2, & z \in [1, 2); \\ 0, & z \notin (0, 2); \end{cases}$$

称为 $U(0,1)$

9. 设 $X_1, X_2, \dots, X_n$ 独立, 若 $Y = \max(X_1, X_2, \dots, X_n)$, $Z = \min(X_1, X_2, \dots, X_n)$, 且 $X_i \sim E(\lambda)$, $i=1, 2, \dots, n$; 求 Y, Z 的分布及其联合分布.

解: $R(Y) = R(Z) = (0, 1)$, $\forall y, z \in (0, 1) \subset \mathbb{R}$

$$P(Y \leq y, Z \leq z) = \begin{cases} 0 & , y \leq 0 \text{ 或 } z \leq 0; \\ P(\bigcap_{i=1}^n \{X_i \leq y\}) = y^n & , 0 \leq y < 1 \text{ 且 } z \geq 1; \\ 1 - P(\bigcap_{i=1}^n \{X_i > z\}) = 1 - (1-z)^n & , y \geq 1 \text{ 且 } 0 < z < 1; \\ 1 & , 0 \leq y < 1 \text{ 且 } 0 < z < 1; \\ & , y \geq 1 \text{ 且 } z \geq 1; \end{cases} \quad \text{即, } 0 < y < 1, 0 < z < 1.$$

$$P(Y \leq y, Z \leq z) = P(Y \leq y) - P(Y \leq y, Z > z) = P(\bigcap_{i=1}^n \{X_i \leq y\}) - P(\bigcap_{i=1}^n \{z < X_i \leq y\})$$

$$= \begin{cases} y^n & , y \leq z; \\ y^n - (y-z)^n & , y > z; \end{cases} \quad \text{即: } F_{Y,Z}(y, z) = P(Y \leq y, Z \leq z) = \begin{cases} 0 & , y \leq 0 \text{ 或 } z \leq 0; \\ y^n & , 0 \leq y < 1 \text{ 且 } z \geq 1; \\ 1 - (1-z)^n & , y \geq 1 \text{ 且 } 0 < z < 1; \\ y^n - (y-z)^n & , 0 < y \leq z < 1; \\ 1 & , y \geq 1 \text{ 且 } z \geq 1; \end{cases}$$

$\forall y \in \mathbb{R}$.

$$F_Y(y) = \lim_{z \rightarrow \infty} F(y, z) = \begin{cases} \lim_{z \rightarrow \infty} 0 = 0 & , y \leq 0; \\ \lim_{z \rightarrow \infty} y^n = y^n & , 0 < y < 1; \\ \lim_{z \rightarrow \infty} 1 = 1 & , y \geq 1; \end{cases} \quad \text{即: } F_Y(y) = P(Y \leq y) = P(\bigvee_{i=1}^n X_i \leq y)$$

$$= P(\bigcap_{i=1}^n \{Y_i \leq y\}) = \prod_{i=1}^n P\{X_i \leq y\} = \begin{cases} 0 & , y \leq 0; \\ y^n & , 0 < y < 1; \\ 1 & , y \geq 1; \end{cases}$$

10. (1) 设 X, Y 独立同 $U[0,1]$ 分布, 若 $Z = \begin{cases} X+Y, & 0 \leq X+Y \leq 1; \\ (X+Y)-1, & 1 < X+Y \leq 2; \end{cases}$ 试问:

Z 服从什么分布?

$$\forall z \in \mathbb{R}, \quad F_Z(z) = \lim_{y \rightarrow \infty} F(y, z) = \begin{cases} \lim_{y \rightarrow \infty} 0 = 0 & , z \leq 0; \\ \lim_{y \rightarrow \infty} [1 - (1-z)^n] & , 0 < z < 1; \\ \lim_{y \rightarrow \infty} 1 = 1 & , z \geq 1. \end{cases} \quad \text{即:}$$

$$F_Z(z) = P(Z \leq z) = 1 - P(Z > z) = 1 - P(\bigwedge_{i=1}^n X_i > z) = 1 - P(\bigcap_{i=1}^n \{X_i > z\})$$

$$= 1 - \prod_{i=1}^n P\{X_i > z\} = 1 - (1-z)^n = \begin{cases} 1-1=0 & , z \leq 0; \\ 1 - (1-z)^n & , 0 < z < 1; \\ 1-0=1 & , z \geq 1; \end{cases}$$

(2) 设 X, Y 独立同 $N(0, 1^2)$ 分布, 若 $Z = \begin{cases} |Y|, & X \geq 0; \\ -|Y|, & X < 0; \end{cases}$ 则 $Z \sim N(0, 1^2)$.

$$\begin{aligned} \forall z \in \mathbb{R}, F_Z(z) &= P(Z \leq z) = P(X \geq 0) \cdot P(Z \leq z | X \geq 0) + P(X < 0) \cdot P(Z \leq z | X < 0) \\ &= \frac{1}{2} [P(|Y| \leq z | X \geq 0) + P(|Y| \geq -z | X < 0)] \\ &= \frac{1}{2} [P(|Y| \leq z) + P(|Y| \geq -z)] = \begin{cases} \frac{1}{2} [\Phi(z) + \Phi(-z)] + \frac{1}{2}, & z > 0; \\ \frac{1}{2} [1 - \Phi(-z)] + \frac{1}{2}, & z \leq 0; \end{cases} \\ &= \begin{cases} \Phi(z), & z > 0; \\ \Phi(z), & z \leq 0; \end{cases} \quad \text{即: } Z \sim N(0, 1^2) \end{aligned}$$

11. 设二维随机变量 (X, Y) 在区域 $D = \{(x, y) | 0 < x < 1, x^2 < y < \sqrt{x}\}$ 上服从均匀分布, 令 $U = \begin{cases} 1, & X \leq Y; \\ 0, & X > Y; \end{cases}$

布, 令 $U = \begin{cases} 1, & X \leq Y; \\ 0, & X > Y; \end{cases}$

(1) 写出 (X, Y) 的概率密度;

因 $(X, Y) \sim f(x, y)$, 即有:

$$f(x, y) = \begin{cases} \frac{1}{S(D)}, & (x, y) \in D; \\ 0, & \text{其他}; \end{cases} = \begin{cases} 3, & (x, y) \in D; \\ 0, & \text{其他}. \end{cases}$$

(2) 问 U 与 X 是否独立?

考虑: $\{U=1\}$ 与 $\{X \leq \frac{1}{4}\}$; 易见, $P(U=1) = \frac{1}{2}$, $P(X \leq \frac{1}{4}) = \frac{\int_0^{\frac{1}{4}} (\sqrt{x} - x^2) dx}{\frac{1}{3}}$

$$= 3 \cdot [\frac{2}{3} \cdot \frac{1}{8} - \frac{1}{3} \cdot \frac{1}{64}] = \frac{1}{4} - \frac{1}{64} = \frac{15}{64}, \quad P(U=1, X \leq \frac{1}{4}) = \frac{\int_0^{\frac{1}{4}} (\sqrt{x} - x^2) dx}{\frac{1}{3}} \quad (\text{面积之比})$$

$$= 3 \cdot [\frac{2}{3} \cdot \frac{1}{8} - \frac{1}{3} \cdot \frac{1}{16}] = \frac{1}{4} - \frac{3}{32} = \frac{5}{32} \neq \frac{1}{2} \cdot \frac{15}{64} = P(U=1) \cdot P(X \leq \frac{1}{4}), \quad \text{即: } \{U=1\} \text{ 与 } \{X \leq \frac{1}{4}\}$$

(3) 求 $Z = U + X$ 的分布函数 $F(z)$. 不独立, 即有: U 与 X 不独立!

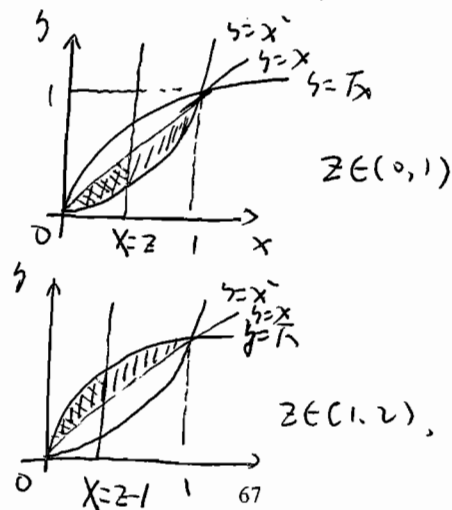
易见, $\forall z \leq 0, F_Z(z) = 0$; $\forall z \geq 2, F_Z(z) = 1$; $\forall z \in (0, 2)$.

$$F_Z(z) = P(Z \leq z) = P(U + X \leq z) = P(U=1) \cdot P(U+X \leq z | U=1) + P(U=0) \cdot P(U+X \leq z | U=0)$$

$$= \frac{1}{2} [P(X \leq z-1 | U=1) + P(X \leq z | U=0)]$$

$$= \begin{cases} \frac{1}{2} [0 + \frac{\int_0^z (x - x^2) dx}{\frac{1}{6}}], & z \in (0, 1); \\ \frac{1}{2} [0 + \frac{\int_0^1 (\sqrt{x} - x^2) dx}{\frac{1}{6}} + 1], & z \in (1, 2) \end{cases}$$

$$= 2 \cdot (z-1)^{\frac{3}{2}} - \frac{3}{2} (z-1)^{\frac{5}{2}} + \frac{1}{2}.$$



12. 设随机变量 X 与 Y 相互独立, X 服从参数为 1 的指数分布, Y 的概率分布为:

$$P(Y=-1)=p, P(Y=1)=1-p, 0 < p < 1; \text{ 令 } Z=XY,$$

(1) 求 Z 的概率密度: $\forall z \in \mathbb{R}, F_Z(z) = P(Z \leq z) = P(XY \leq z)$

$$\begin{aligned} &= P(Y=-1)P(XY \leq z | Y=-1) + P(Y=1)P(XY \leq z | Y=1) = p \cdot P(X > -z | Y=-1) + (1-p) \cdot P(X \leq z | Y=1) \\ &= p \cdot P(X > -z) + (1-p)P(X \leq z) \quad \text{即: } F_Z(z) = \begin{cases} p + (1-p)e^{-z}, & z > 0; \\ p \cdot e^{-z}, & z \leq 0; \end{cases} \quad \text{令 } Z \sim f_Z(z), \text{ 则:} \\ &= \begin{cases} p + (1-p)(1-e^{-z}) & , z > 0; \\ p \cdot e^{-z} & , z \leq 0; \end{cases} \quad f_Z(z) = \frac{d}{dz} F_Z(z) = \begin{cases} (1-p)e^{-z} & , z > 0; \\ p \cdot e^{-z} & , z \leq 0; \end{cases} \end{aligned}$$

(2) X 与 Z 是否独立?

考虑事件: $\{X > \ln 2\}$ 与 $\{Z > \ln 2\}$. $P(X > \ln 2) = \int_{\ln 2}^{+\infty} e^{-x} dx = \frac{1}{2}$,

$$P(Z > \ln 2) = \int_{\ln 2}^{+\infty} f_Z(z) dz = \int_{\ln 2}^{+\infty} (1-p) \cdot e^{-z} dz = (1-p) \cdot \frac{1}{2}.$$

$$\begin{aligned} P(X > \ln 2, Z > \ln 2) &= P(X > \ln 2, XY > \ln 2) = P(Y=-1)P(X > \ln 2, X > \ln 2 | Y=-1) + P(Y=1)P(X > \ln 2, X > \ln 2 | Y=1) \\ &= \frac{p}{2} \cdot P(X > \ln 2, X < -\ln 2 | Y=-1) + \frac{(1-p)}{2} \cdot P(X > \ln 2, X > \ln 2 | Y=1) \\ &= (1-p) \cdot P(X > \ln 2 | Y=1) = (1-p) \cdot P(X > \ln 2) = (1-p) \cdot \frac{1}{2} \neq P(X > \ln 2) \cdot P(Z > \ln 2), \quad 0 < p < 1; \end{aligned}$$

即: $\{X > \ln 2\}$ 与 $\{Z > \ln 2\}$ 不独立, 即 X, Z 不独立!

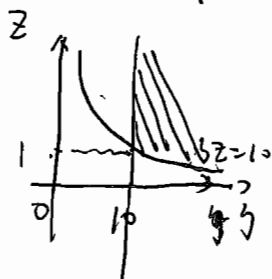
13. (1) 设随机向量 (X, Y) 的联合密度函数为 $f(x, y) = \begin{cases} \frac{100}{x^2 y^2}, & x, y > 10; \\ 0, & \text{其他}; \end{cases}$ 试求

$Z = \frac{X}{Y}$ 的密度函数 $f_Z(z)$:

则: $(X, Y) \sim f_{X,Y}$. $Z = \frac{X}{Y} \sim f_Z(z) = \int_{-\infty}^{+\infty} f(x, y) \cdot |y| dy$

$$f_Z(z) = \int_{-\infty}^{+\infty} f(x, y) \cdot |y| dy = \begin{cases} \int_{10}^{+\infty} \frac{100}{y^3 z} dy = \frac{1}{z} & , z \in (0, 1); \\ \int_{10}^{+\infty} \frac{100}{y^3 z} dy = \frac{1}{2z^2} & , z \in [1, +\infty); \\ 0 & , z \leq 0; \end{cases}$$

$$f(x, y) \cdot |y| > 0 \Leftrightarrow \begin{cases} y > 10 \\ x > 10 \end{cases}$$



例12: $(X, Y) \sim f(x, y)$. $XY \sim f_{XY}(z) = \int_{-\infty}^{+\infty} f(x, \frac{z}{x}) \cdot \frac{1}{|x|} dx \stackrel{\frac{z}{x} = t}{=} \int_{-\infty}^{+\infty} f(x) \cdot f(\frac{z}{x}) \cdot \frac{1}{|x|} dx$

$\hookrightarrow Z = XY \sim f_{XY}(z)$. 证明: $f_{XY}(z) = \int_{-\infty}^{+\infty} f(x) \cdot f(\frac{z}{x}) \cdot \frac{1}{|x|} dx$.

(2) 设随机变量 X, Y 相互独立, 且 $X \sim f_X(x) = \begin{cases} \frac{1}{\pi\sqrt{1-x^2}}, & -1 < x < 1; \\ 0, & \text{其他}; \end{cases}$

$\Leftrightarrow \begin{cases} -1 < x < 1; \\ \frac{z}{x} > 0 \end{cases}$

$Y \sim f_Y(y) = \begin{cases} ye^{-\frac{y^2}{2}}, & y > 0; \\ 0, & \text{其他}; \end{cases}$ 证明: $Z = XY \sim N(0, 1^2)$;

$z > 0$: $\int_0^1 \frac{1}{\sqrt{1-x^2}} \cdot \frac{z}{x} \cdot e^{-\frac{z^2}{2x^2}} \cdot \frac{1}{|x|} dx \stackrel{\frac{z}{x} = t}{=} \int_{\frac{z}{1}}^{\frac{z}{-1}} \frac{1}{\sqrt{1-\frac{z^2}{t^2}}} \cdot \frac{z}{t} \cdot e^{-\frac{z^2}{2t^2}} \cdot \frac{1}{|t|} dt$

$z \leq 0$: $\int_{-1}^0 \frac{1}{\sqrt{1-x^2}} \cdot \frac{z}{x} \cdot e^{-\frac{z^2}{2x^2}} \cdot \frac{1}{|x|} dx \stackrel{\frac{z}{x} = t}{=} \int_{\frac{z}{-1}}^{\frac{z}{1}} \frac{1}{\sqrt{1-\frac{z^2}{t^2}}} \cdot \frac{z}{t} \cdot e^{-\frac{z^2}{2t^2}} \cdot \frac{1}{|t|} dt$

(是正负)

(3) 设随机向量 (X, Y) 的联合密度函数为 $f(x, y) = \begin{cases} 2e^{-(x+2y)}, & x, y > 0; \\ 0, & \text{其他}; \end{cases}$ 试求 $\frac{X \sim U(1)}$

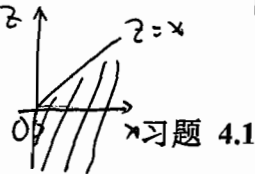
$Z = X - 2Y$ 的密度函数 $f_Z(z)$. 证: 公式法: $Z = aX + bY$ ($a, b \neq 0$)

$Z \sim f_{aX+bY}(z) = \int_{-\infty}^{+\infty} f(x, \frac{z-ax}{b}) \cdot \frac{1}{|b|} dx = \int_{-\infty}^{+\infty} f(\frac{z-bx}{a}) \cdot \frac{1}{|a|} dx$

例12: 设 X, Y 独立, 且 $X \sim U(1)$, $Y \sim f_Y(y)$. $ZY \sim \frac{1}{2} f_Y(\frac{z}{2})$. $X, 2Y$ 独立.

$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x) \cdot \frac{1}{2} f_Y(\frac{x-z}{2}) \cdot dx = \begin{cases} \int_0^1 e^{-x} \cdot \frac{1}{2} \cdot 2e^{-(x-z)} dx, & z > 0; \\ \int_0^1 e^{-x} \cdot \frac{1}{2} \cdot 2e^{-(x-z)} dx, & z \leq 0; \end{cases}$

$f_X(x) \cdot f_Y(\frac{x-z}{2}) > 0 \Leftrightarrow \begin{cases} x > 0 \\ \frac{x-z}{2} > 0 \end{cases}$



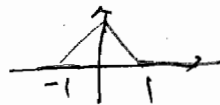
2. (1) 设随机变量 X 的概率密度 $f(x)$ 满足: $f(\mu+x) = f(\mu-x)$, $\forall x \in (-\infty, +\infty)$,

其中 μ 为常数, $\int_{-\infty}^{+\infty} |x| f(x) dx$ 收敛; 试证明: $EX = \mu$;

$EX = \int_{-\infty}^{+\infty} x \cdot f(x) dx = \int_{-\infty}^{+\infty} x \cdot f(2\mu-x) dx \stackrel{2\mu-x=t}{=} \int_{+\infty}^{-\infty} (2\mu-t) \cdot f(t) dt$

$= \int_{-\infty}^{+\infty} 2\mu f(t) dt - \int_{-\infty}^{+\infty} t f(t) dt = 2\mu \int_{-\infty}^{+\infty} f(x) dx - \int_{-\infty}^{+\infty} t f(t) dt = 2\mu - EX = \mu$.

(2) 设 $X \sim f(x) = \begin{cases} 1+x, & -1 \leq x < 0; \\ 1-x, & 0 \leq x \leq 1; \\ 0, & \text{其他}; \end{cases}$ 试求 EX ;



注意: ① EX 存在

② $f(x) = f(-x)$ 关于 y 轴对称, $EX = 0$

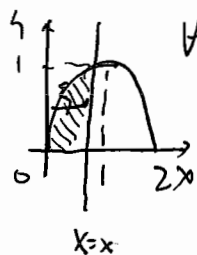
(3) 设 $X \sim f(x) = \frac{1}{2}e^{-|x-a|}$, $-\infty < x < +\infty$, 试求 EX ; $= \int_{-\infty}^{+\infty} x f(x) dx = \frac{1}{2} \int_{-\infty}^{+\infty} x \cdot e^{-|x-a|} dx$

$EX, EX^2, \dots, EX^n$ 存在

$\int_{-\infty}^{+\infty} (a+t) \cdot e^{-|t|} dt$

$f(a+x) = f(a-x)$ 关于 $x=a$ 对称, $EX = a$ $= \frac{1}{2}a \cdot \int_{-\infty}^{+\infty} e^{-|t|} dt = a \cdot \int_0^{+\infty} e^{-t} dt = a$

(4) 设 G 为曲线 $y = 2x - x^2$ 与 x 轴所围区域, 在区域 G 内任取一点, 该点到 y 轴的距离为 ξ , 求 $E\xi$. 知 $P(\xi) = (0, 2)$; $\forall x \leq 0, F_\xi(x) = P(\xi \leq x) = 0$; $\forall x \geq 2, F_\xi(x) = 1$;



$\forall x \in (0, 2), F_\xi(x) = P(\xi \leq x) = \frac{\int_0^x (2t-t^2) dt}{\int_0^2 (2t-t^2) dt} = \frac{x^2 - \frac{1}{3}x^3}{\frac{4}{3}} = \frac{3x^2 - x^3}{4}$

$E\xi = \int_{-\infty}^{+\infty} x \cdot f_\xi(x) dx = \int_0^2 x \cdot \frac{3x^2 - x^3}{4} dx = \dots = 1$ 由 $f_\xi(1+x) = f_\xi(1-x)$

3. (2) 设 $X \sim E(1)$, 试求 $E(X + e^{-2X})$;

$E(X + e^{-2X}) = EX + E(e^{-2X})$

$= \frac{1}{1} + \int_{-\infty}^{+\infty} e^{-2x} \cdot f_X(x) dx$

$= 1 + \int_0^{+\infty} e^{-2x} \cdot 1 \cdot e^{-x} dx = 1 + \frac{1}{3} = \frac{4}{3}$

$X \sim f_X(x)$
 $E[g(X)] = \int_{-\infty}^{+\infty} g(x) \cdot f_X(x) dx$

一般连续型随机变量的公式(定理)

(3) 设 $X \sim N(0, 1^2)$, 试求 $E(Xe^{2X})$;

$E(Xe^{2X}) = \int_{-\infty}^{+\infty} x \cdot e^{2x} \cdot \varphi(x) dx = \int_{-\infty}^{+\infty} x \cdot e^{2x} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$

$= \int_{-\infty}^{+\infty} x \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-2)^2}{2}} dx = 2 \cdot e^4$

注: 若 $X \sim N(\mu, \sigma^2)$, $X \sim f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < +\infty$.

则 $EX = \int_{-\infty}^{+\infty} x \cdot f_X(x) dx = \int_{-\infty}^{+\infty} x \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \mu$

类似地, $\int_{-\infty}^{+\infty} x \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = -\mu$

$$= \frac{2(a+4b)}{\sqrt{2}} \cdot \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = a+4b+16b=39. \quad b=2, a=-1.$$

(4) 设随机变量 X 有分布函数 $F(x) = a\Phi(x) + b\Phi\left(\frac{x-4}{2}\right)$, 其中 a, b 为常数,

$\Phi(x)$ 为标准正态分布函数, 且 $EX^2=39$, 试求 EX .

$$\begin{aligned} & \text{由 } F(+\infty)=1, \text{ 得 } a+b=1; \text{ 设 } X \sim f(x), \text{ 则 } f(x) = \frac{dF(x)}{dx} = a \cdot \phi(x) + b \cdot \phi\left(\frac{x-4}{2}\right) \cdot \frac{1}{2}. \\ EX^2 &= \int_{-\infty}^{+\infty} x^2 \cdot f(x) dx = a \cdot \int_{-\infty}^{+\infty} x^2 \cdot \phi(x) dx + \frac{b}{2} \cdot \int_{-\infty}^{+\infty} x^2 \cdot \phi\left(\frac{x-4}{2}\right) dx \quad \frac{x-4}{2} = t \\ &= \frac{b}{2} \int_{-\infty}^{+\infty} (2t+4)^2 \cdot \phi(t) \cdot 2 dt = a \cdot \int_{-\infty}^{+\infty} x^2 \cdot \phi(x) dx + b \cdot \int_{-\infty}^{+\infty} (2x+4)^2 \cdot \phi(x) dx \\ &= (a+4b) \cdot \int_{-\infty}^{+\infty} x^2 \cdot \phi(x) dx + 16b \cdot \int_{-\infty}^{+\infty} x \cdot \phi(x) dx + 16b \int_{-\infty}^{+\infty} \phi(x) dx = (a+4b) \cdot \int_{-\infty}^{+\infty} x^2 \cdot \phi(x) dx + 16b \quad +1 \\ &= 2(a+4b) \cdot \int_0^{+\infty} x^2 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx + 16b \cdot \frac{\sqrt{2\pi}}{2} = 2(a+4b) \cdot \int_0^{+\infty} t \cdot \frac{1}{\sqrt{2\pi}} e^{-t} \cdot \sqrt{2} dt + 16b = \frac{2(a+4b)}{\sqrt{2}} \int_0^{+\infty} t e^{-t} dt + 16b \\ &= \frac{2(a+4b)}{\sqrt{2}} \cdot 1 + 16b = 39 \end{aligned}$$

4. (1) 设 $X \sim f(x) = \frac{1}{\pi(1+x^2)}, -\infty < x < +\infty$, 试求: $E[\min\{|X|, 1\}] = \frac{2(a+4b)}{\sqrt{2}} \Gamma\left(\frac{1}{2}\right) + 16b(2+1)$

$$\begin{aligned} E[\min\{|X|, 1\}] &= \int_{-\infty}^{+\infty} \min\{|x|, 1\} \cdot f(x) dx = 2 \int_0^{+\infty} \min\{x, 1\} \cdot f(x) dx \\ &= 2 \int_0^1 \frac{x}{\pi(1+x^2)} dx + 2 \int_1^{+\infty} \frac{1}{\pi(1+x^2)} dx = \frac{1}{\pi} \ln(1+x^2) \Big|_0^1 + \frac{2}{\pi} \cdot \arctan x \Big|_1^{+\infty} = \frac{1}{\pi} \ln 2 + \frac{1}{\pi}. \end{aligned}$$

(2) 设 $X \sim f(x) = \frac{1}{2} e^{-|x|}, -\infty < x < +\infty$, 试求: $E[\max\{|X|, 1\}]$.

$$\begin{aligned} E[\max\{|X|, 1\}] &= \int_{-\infty}^{+\infty} \max\{|x|, 1\} \cdot f(x) dx = 2 \int_0^{+\infty} \max\{x, 1\} \cdot \frac{1}{2} e^{-x} dx \\ &= 2 \int_0^1 1 \cdot \frac{1}{2} e^{-x} dx + 2 \int_1^{+\infty} x \cdot \frac{1}{2} e^{-x} dx \\ &= (-e^{-x}) \Big|_0^1 + 2 \cdot (-x-1)e^{-x} \Big|_1^{+\infty} = 1 - e^{-1} + 2e^{-1} = 1 + e^{-1}. \end{aligned}$$

5. (2) 游客乘电梯从底层到电视塔顶层观光. 电梯于每个整点的第5分钟、第25分钟和第55分钟从底层起行, 假设一游客在早八点的第 X 分钟到达底层候梯处, 且 X 在 $[0, 60]$ 上服从均匀分布, 试求该游客的平均等候时间.

设游客的等候时间为 Y (min). 则

$$Y = \begin{cases} 5-X & 0 \leq X \leq 5; \\ 25-X & 5 < X \leq 25; \\ 55-X & 25 < X \leq 55; \\ 60-X+5 & 55 < X \leq 60 \end{cases} \quad X \sim f(x) = \begin{cases} \frac{1}{60}, & 0 \leq x \leq 60; \\ 0, & \text{其他} \end{cases}$$

$$\text{记: } g(x) = \begin{cases} 5-x & 0 \leq x \leq 5; \\ 25-x & 5 < x \leq 25; \\ 55-x & 25 < x \leq 55; \\ 65-x & 55 < x \leq 60 \end{cases} \quad \text{则有: } Y = g(X).$$

$$\begin{aligned} EY &= E[g(X)] = \int_{-\infty}^{+\infty} g(x) \cdot f(x) dx = \int_0^{60} g(x) \cdot \frac{1}{60} dx = \frac{1}{60} \left(\int_0^5 (5-x) dx \right. \\ &\quad \left. + \int_5^{25} (25-x) dx + \int_{25}^{55} (55-x) dx + \int_{55}^{60} (65-x) dx \right) \end{aligned}$$

6. (2) 设某种商品每周的需求量 X 是服从区间 $[10, 30]$ 上均匀分布的随机变量, 而经销商进货数量为区间 $[10, 30]$ 中的某一整数, 其每销售一单位该商品可获利 500 元; 若供大于求则削价处理, 每处理一单位亏损 100 元; 若供不应求, 则从外部调剂供应, 此时每一单位商品仅获利 300 元; 为使经销商所获利润期望值不 少于 9280 元, 试确定最少进货量.

设经销商为 k ($k \in [10, 30]$), 记 Y 为经销商所获利润. 由题设 $X \sim U[10, 30]$,

$$Y = \begin{cases} 500k + 300(X - k), & X \geq k; \\ 500X - 100(k - X), & X < k; \end{cases} = \begin{cases} 300X + 200k, & X \geq k; \\ 600X - 100k, & X < k; \end{cases} \quad \text{记 } g(x) = \begin{cases} 300x + 200k, & x \geq k; \\ 600x - 100k, & x < k. \end{cases} \quad \text{则有}$$

$$\begin{aligned} Y &= g(X). \quad E(Y) = E[g(X)] = \int_{-\infty}^{+\infty} g(x) \cdot f_x(x) dx = \int_{10}^{30} g(x) \cdot \frac{1}{20} dx = \frac{1}{20} \left(\int_{10}^k g(x) dx + \int_k^{30} g(x) dx \right) \\ &= \frac{1}{20} \left[\int_{10}^k (600x - 100k) dx + \int_k^{30} (300x + 200k) dx \right] = 15 \cdot (k^2 - 10^2) - 5k(k - 10) + \frac{15}{2} \cdot (900 - k^2) + 10k(30 - k) \\ &= -\frac{15}{2}k^2 + 350k - 1500 + 15 \cdot 450 \triangleq g(k) \quad \text{记为 } g(k) \geq 9280. \quad \text{则有} \\ &-\frac{15}{2}k^2 + 350k - 1500 + 15 \cdot 450 - 9280 \geq 0. \end{aligned}$$

7. (1) 一项保险规定最高理赔额为 10 万元, 假定投保人的损失 Y (单位: 万元)

具有密度函数: $f(y) = \frac{2}{y^3}, y > 1$; 试求其平均理赔额.

设保险公司理赔额为 X (万元); 由题设 $X = \begin{cases} Y & , Y \leq 10; \\ 10 & , Y > 10; \end{cases}$

记: $g(y) = \begin{cases} y & , y \leq 10; \\ 10 & , y > 10; \end{cases} \quad \text{则有 } X = g(Y)$

$$\begin{aligned} EX &= E[g(Y)] = \int_{-\infty}^{+\infty} g(y) \cdot f(y) dy = \int_1^{+\infty} g(y) \cdot \frac{2}{y^3} dy = \int_1^{10} y \cdot \frac{2}{y^3} dy + \int_{10}^{+\infty} 10 \cdot \frac{2}{y^3} dy \\ &= 2 \cdot \left[-\frac{1}{y^2} \right]_1^{10} + 10 \cdot \left[\frac{1}{y^2} \right]_{10}^{+\infty} = 2 - 0.2 + 0.1 = 1.9. \quad (\text{万元}) \end{aligned}$$

(2) 一台仪器连续地测量与记录遥控地区的地震波，仪器寿命 T 是均值为 3 年的指数随机变量；由于前两年仪器没得到监控，实际发现它失效的时间是

$U = T \vee 2 = \max(T, 2)$ ，试求 EU 。 已知 $T \sim f_T(t)$ ， $f_T(t) = \begin{cases} \frac{1}{3}e^{-\frac{1}{3}t}, & t > 0 \\ 0, & t \leq 0 \end{cases}$

$$\begin{aligned} \text{解: } EU &= E(T \vee 2) = \int_{-\infty}^{+\infty} (t \vee 2) \cdot f_T(t) dt = \int_0^{+\infty} (t \vee 2) \cdot \frac{1}{3}e^{-\frac{1}{3}t} dt \\ &= \int_0^2 2 \cdot \frac{1}{3}e^{-\frac{1}{3}t} dt + \int_2^{+\infty} t \cdot \frac{1}{3}e^{-\frac{1}{3}t} dt = 2 \cdot (1 - e^{-\frac{2}{3}}) + 3 \cdot (-\frac{t}{3} - 1)e^{-\frac{1}{3}t} \Big|_2^{+\infty} \\ &= 2(1 - e^{-\frac{2}{3}}) + 3 \cdot (\frac{2}{3} + 1)e^{-\frac{2}{3}} = 2 + 3e^{-\frac{2}{3}} \end{aligned}$$

8. (1) 设 $X \sim f(x) = \begin{cases} \frac{1}{2} \cos \frac{x}{2}, & 0 \leq x \leq \pi; \\ 0, & \text{其他}; \end{cases}$ ，对 X 进行独立重复观测四次，记 Y 为

观测值大于 $(\frac{\pi}{3})$ 的次数，试求 EY^2 ； $P(X > \frac{\pi}{3}) = \int_{\frac{\pi}{3}}^{\pi} f(x) dx = \int_{\frac{\pi}{3}}^{\pi} \frac{1}{2} \cos \frac{x}{2} dx = \sin \frac{x}{2} \Big|_{\frac{\pi}{3}}^{\pi}$

$$= \frac{1}{2}, \text{ 故 } Y \sim B(4, \frac{1}{2}). \quad EY = 4 \cdot \frac{1}{2} = 2, \quad DY = 4 \cdot \frac{1}{2} \cdot \frac{1}{2} = 1.$$

$$EY^2 = DY + (EY)^2 = 1 + 4 = 5.$$

(2) 设 $X \sim f(x) = \begin{cases} 2^{-x} \cdot \ln 2, & x > 0; \\ 0, & x \leq 0; \end{cases}$ ，对 X 进行独立重复地观测，直到第二个大

于 3 的观测值出现时停止，记 Y 为观测的次数，试求 Y 的分布及 EY

$$\text{解: } P(X > 3) = \int_3^{+\infty} f(x) dx = \int_3^{+\infty} 2^{-x} \ln 2 dx = (-2^{-x}) \Big|_3^{+\infty} = \frac{1}{8}. \text{ 故 } R(Y) = \{2, 3, \dots\}$$

$$\forall k \in R(Y), \quad P(Y = k) = \underbrace{C_{k-1}^1 \left(\frac{1}{8}\right)^1 \left(1 - \frac{1}{8}\right)^{k-1}}_{\text{前 } k-1 \text{ 次观测值均小于 3}} \cdot \left(\frac{1}{8}\right) = (k-1) \cdot \left(\frac{1}{8}\right)^2 \cdot \left(\frac{7}{8}\right)^{k-2}$$

$$EY = \sum_{k=2}^{+\infty} k \cdot P(Y = k) = \sum_{k=2}^{+\infty} k \cdot (k-1) \cdot \left(\frac{1}{8}\right)^2 \cdot \left(\frac{7}{8}\right)^{k-2}$$

$$\text{于是可得: } \sum_{k=2}^{+\infty} k(k-1)x^{k-2}, \text{ 其中 } x = \frac{7}{8}. \text{ 记 } S(x) = \sum_{k=2}^{+\infty} k(k-1)x^{k-2} = \sum_{k=2}^{+\infty} (x^k)''$$

$$= \left(\sum_{k=2}^{+\infty} x^k\right)'' = \left(\frac{x^2}{1-x}\right)'' = \left[\frac{1 - (1-x^2)}{1-x}\right]'' = \left[\frac{1}{1-x} - 1 - x\right]'' = \left(\frac{1}{1-x}\right)'' = \left(\frac{1}{(1-x)^2}\right)'$$

$$= \frac{2}{(1-x)^3}, \text{ 取 } x = \frac{7}{8}, \text{ 则 } \sum_{k=2}^{+\infty} k(k-1) \cdot \left(\frac{7}{8}\right)^{k-2} = 2 \cdot 8^3, \quad EY = 2 \cdot 8 = 16.$$

另法：利用几何分布的可加性。 $Y = Y_1 + Y_2$ ， Y_1 为第一次观测时“ $X > 3$ ”时观测的次数。

Y_1, Y_2 独立同 $G(\frac{1}{8})$ 。 $EY = E(Y_1 + Y_2) = EY_1 + EY_2$

Y_2 为第一次与第二次观测时“ $X > 3$ ”的

9. 设随机变量 X 有概率分布: $P(X=1)=P(X=2)=\frac{1}{2}$, 且在给定 $X=i$ 时, 随机变量 $Y \sim U(0, i)$, $i=1, 2$; 试求 EY . 由此, $Y|X=i \sim U(0, i)$, $i=1, 2$:

$$\forall y \in \mathbb{R}, F_Y(y) = P(Y \leq y) = P(X=1) \cdot P(Y \leq y|X=1) + P(X=2) \cdot P(Y \leq y|X=2) = \frac{1}{2} [P(Y \leq y|X=1) + P(Y \leq y|X=2)]$$

$$= \begin{cases} \frac{1}{2}(0+0)=0 & , y \leq 0; \\ \frac{1}{2}(y + \frac{y^2}{2}) = \frac{y}{4} + \frac{y^2}{4} & , 0 < y < 1; \\ \frac{1}{2}(1 + \frac{y^2}{2}) = \frac{1}{2} + \frac{y^2}{4} & , 1 \leq y < 2; \\ \frac{1}{2}(1+1)=1 & , y \geq 2; \end{cases}$$

令 $Y \sim f_Y(y)$, 则 $f_Y(y) = \frac{d}{dy} F_Y(y) = \begin{cases} \frac{3}{4} & , 0 < y < 1; \\ \frac{1}{2} & , 1 \leq y < 2; \\ 0 & , \text{其他}; \end{cases}$

$$\text{从而, } EY = \int_{-\infty}^{+\infty} y \cdot f_Y(y) dy = (\int_{-\infty}^0 + \int_0^1 + \int_1^2 + \int_2^{+\infty}) y \cdot f_Y(y) dy = \int_0^1 y \cdot \frac{3}{4} dy + \int_1^2 y \cdot \frac{1}{2} dy = \frac{3}{8} + \frac{3}{8} = \frac{3}{4}$$

10. (1) 将 n 只球独立地放入 M 只盒子中, 若每只球放入各只盒子是等可能的, 问: 平均有多少只盒子有球? 记 X_i 为有球的盒子数, 求 $E X_i$.

做法一: 通过分布求期望. 令 $X_i = \begin{cases} 1, & \text{若第 } i \text{ 个盒子有球}; \\ 0, & \text{否则}; \end{cases} \quad i=1, 2, \dots, M;$

即有: $X = X_1 + X_2 + \dots + X_M$, $E X = E(X_1 + X_2 + \dots + X_M) = E X_1 + E X_2 + \dots + E X_M$

求 $E X_i$: X_i 的分布律为: $P(X_i=1) = 1 - (1 - \frac{1}{M})^n$, $P(X_i=0) = (1 - \frac{1}{M})^n$

从而: $E X_i = 1 \cdot (1 - (1 - \frac{1}{M})^n) + 0 \cdot (1 - \frac{1}{M})^n = 1 - (1 - \frac{1}{M})^n$

从而: $E X = M \cdot [1 - (1 - \frac{1}{M})^n]$

(2) 一袋中装有 60 只黑球和 40 只红球, 现从中任取 20 只, 则平均取到多少只红球? 记 X_i 为取到的红球数, 求 $E X$.

记: $X_i = \begin{cases} 1, & \text{若第 } i \text{ 次取到红球}; \\ 0, & \text{否则}; \end{cases} \quad i=1, 2, \dots, 20;$ 即有: $X = X_1 + X_2 + \dots + X_{20}$, $E X = E(X_1 + X_2 + \dots + X_{20}) = E X_1 + E X_2 + \dots + E X_{20}$

这里, $P(X_i=1) = \frac{40}{100} = \frac{2}{5}$, $P(X_i=0) = \frac{3}{5}$

从而: $E X_i = 1 \times \frac{2}{5} + 0 \times \frac{3}{5} = \frac{2}{5}$, $E X = 20 \times \frac{2}{5} = 8$

(3) 设随机变量 $X \sim B(n, p)$, 试由“随机变量的分解法”求 $E X$:

即做了 n 次独立重复试验, 每次试验“成功”的概率为 p , 记成功的次数为 X . 即有:

$X \sim B(n, p)$, 令 $X_i = \begin{cases} 1, & \text{若第 } i \text{ 次试验“成功”}; \\ 0, & \text{否则}; \end{cases} \quad i=1, 2, \dots, n;$ 即有:

$$X = X_1 + X_2 + \dots + X_n, \quad E X = E(X_1 + X_2 + \dots + X_n) = E X_1 + E X_2 + \dots + E X_n;$$

从而: $E X_i = 1 \cdot p + 0 \cdot (1-p) = p$; 从而: $E X = np$

(4) 求连续独立地掷100颗骰子所得点数之和的数学期望。

记 X_i 为第 i 颗骰子掷出的点数, $i=1, 2, \dots, 100$; 易知, $X_1, X_2, \dots, X_{100}$ 独立同分布 $\left(\frac{1}{6}, \frac{2}{6}, \dots, \frac{6}{6}\right)$

分布, 则有: $E(X_1 + X_2 + \dots + X_{100}) = EX_1 + EX_2 + \dots + EX_{100} = 100 \times (1+2+\dots+6) \times \frac{1}{6} = 100 \times \frac{7}{2} = 350$.

11. (1) 设随机变量 $X_1, X_2, \dots, X_n$ 独立同 $U(0, \theta)$ 分布, 记: $Y = \max(X_1, X_2, \dots, X_n)$,

$Z = \min(X_1, X_2, \dots, X_n)$, 试求 EY, EZ ;

猜测一下: $\begin{array}{c} 0 \quad x_1 \quad x_2 \quad \dots \quad x_n \quad \theta \\ \hline \end{array}$ $X_1, X_2, \dots, X_n$ 在 $(0, \theta)$ 平均分割成 $n+1$ 份, $EZ = \frac{1}{n+1}\theta$, $EY = \frac{n}{n+1}\theta$.

证: $\forall y \in \mathbb{R}$, $F_Y(y) = P(Y \leq y) = P(\bigvee_{i=1}^n X_i \leq y) = P(\bigwedge_{i=1}^n \{X_i \leq y\}) = \prod_{i=1}^n P(X_i \leq y) = \begin{cases} 0 & y \leq 0; \\ (\frac{y}{\theta})^n & 0 < y < \theta; \\ 1 & y \geq \theta; \end{cases}$

记 $Y \sim f_Y(y)$, 则有: $f_Y(y) = \frac{d}{dy} F_Y(y) = \begin{cases} \frac{n}{\theta} (\frac{y}{\theta})^{n-1} & 0 < y < \theta; \\ 0 & \text{其他} \end{cases}$, $EY = \int_{-\infty}^{+\infty} y \cdot f_Y(y) dy = \int_0^\theta y \cdot \frac{n}{\theta} (\frac{y}{\theta})^{n-1} dy$

令 $\frac{y}{\theta} = t$ $n\theta \int_0^1 t^n dt = \frac{n}{n+1}\theta$.

$\forall z \in \mathbb{R}$, $F_Z(z) = P(Z \leq z) = 1 - P(Z > z) = 1 - P(\bigwedge_{i=1}^n X_i > z) = 1 - P(\bigwedge_{i=1}^n \{X_i > z\})$
 $= 1 - \prod_{i=1}^n P(X_i > z) = \begin{cases} 0 & z \leq 0; \\ 1 - (1 - \frac{z}{\theta})^n & 0 < z < \theta; \\ 1 & z \geq \theta; \end{cases}$ 令 $Z \sim f_Z(z)$, 则有 $f_Z(z) = \frac{d}{dz} F_Z(z)$

$= \begin{cases} \frac{n}{\theta} (1 - \frac{z}{\theta})^{n-1} & 0 < z < \theta; \\ 0 & \text{其他} \end{cases}$ $EZ = \int_{-\infty}^{+\infty} z \cdot f_Z(z) dz = \int_0^\theta z \cdot \frac{n}{\theta} (1 - \frac{z}{\theta})^{n-1} dz$ 令 $\frac{z}{\theta} = t$ $n\theta \int_0^1 t \cdot (1-t)^{n-1} dt$

令 $t = 1-x$ $n\theta \cdot \int_0^1 (1-x) x^{n-1} dx = n\theta \cdot [\frac{1}{n} - \frac{1}{n+1}] = \frac{1}{n+1}\theta$.

(2) 在区间 $(0, 1)$ 上随机地取 n 个点, 求相距最远的两点间距离的数学期望;

由 (1) 的结论: $EX_{(n)} = \frac{n}{n+1} \cdot 1$, $EX_{(1)} = \frac{1}{n+1} \cdot 1$, $E[X_{(n)} - X_{(1)}] = EX_{(n)} - EX_{(1)} = \frac{n-1}{n+1}$.

另法: (提示). 求 (Y, Z) 的联合密度, 再求 $Y-Z$ 的分布, 最后求 $E(Y-Z)$.

$Y = X_{(n)}, Z = X_{(1)}$,

见 (75) 附

(3) 系统有 n 个部件组成, 记 X_i 为第 i 个部件能持续工作的时间, 若 $X_1, X_2, \dots, X_n$ 独立同 $E(\lambda)$ 分布, 试在以下情况下求系统持续工作的平均时间;

(i) 如果一个部件停止工作, 系统就不工作; 记 T 为系统工作~~的~~时间.

易知, $T = X_{(1)} = \min\{X_1, X_2, \dots, X_n\}$.

是 P76 附

(ii) 如果至少有一个部件在工作, 系统就工作.

易知, $T = X_{(n)} = \max\{X_1, X_2, \dots, X_n\}$

12. (1) 设 X, Y 独立同 $N(0, 1^2)$ 分布, 试求 $E[\max(X, Y)]$;

证1: $(X, Y) \sim f_{X,Y}$, 则 $f_{X,Y} = \phi(x)\phi(y) = \frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}}$, $-\infty < x, y < +\infty$.

$$E(X \vee Y) = \iint_{\mathbb{R}^2} (x \vee y) \cdot f_{X,Y}(x,y) dx dy = 2 \iint_{x \geq y} (x \vee y) \cdot f_{X,Y}(x,y) dx dy = 2 \iint_{x \geq y} x \cdot f_{X,Y}(x,y) dx dy.$$

$$= 2 \cdot \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} d\theta \int_0^{+\infty} r \cos\theta \cdot \frac{1}{2\pi} e^{-\frac{r^2}{2}} \cdot r dr = \frac{\sqrt{2}}{\pi} \int_0^{+\infty} r^2 e^{-\frac{r^2}{2}} dr = \frac{\sqrt{2}}{\pi} \int_0^{+\infty} r \cdot e^{-\frac{r^2}{2}} d\left(\frac{r^2}{2}\right) \stackrel{\text{令 } \frac{r^2}{2} = x}{=} \frac{\sqrt{2}}{\pi} \int_0^{+\infty} x \cdot e^{-x} dx$$

$$= \frac{2}{\pi} \int_0^{+\infty} x^{\frac{1}{2}} \cdot e^{-x} dx = \frac{\sqrt{2}}{\pi} \Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{2}}{\pi} \cdot \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{2}}{\pi} \cdot \frac{1}{2} \cdot \sqrt{\pi} = \frac{1}{\sqrt{2}}$$

证2: $\max(X, Y) = \frac{X+Y+|X-Y|}{2}$. 则 $E[\max(X, Y)] = E\left[\frac{X+Y+|X-Y|}{2}\right]$

$= \frac{1}{2} E|X-Y|$. 易知, $X-Y \sim N(0, 2)$, $\frac{X-Y}{\sqrt{2}} \sim N(0, 1^2)$. 记 $Z = \frac{X-Y}{\sqrt{2}}$.

$$Z \sim N(0, 1^2), E|Z| = \int_{-\infty}^{+\infty} |z| \phi(z) dz = 2 \int_0^{+\infty} z \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = \frac{\sqrt{2}}{\pi} \int_0^{+\infty} z \cdot e^{-\frac{z^2}{2}} dz = \frac{\sqrt{2}}{\pi}.$$

$$E|X-Y| = \sqrt{2} \cdot \frac{\sqrt{2}}{\pi} = \frac{2}{\pi}, E[\max(X, Y)] = \frac{1}{\sqrt{2}}.$$

证3: 求 $X \vee Y$ 的分布函数, 求导得密度函数, 再积分求期望.

$N(0, 1^2)$

(2) 设 X, Y 独立同 $N(\mu, \sigma^2)$ 分布, 试求 $E[\max(X, Y)], E[\min(X, Y)]$.

易知, $\frac{X-\mu}{\sigma}, \frac{Y-\mu}{\sigma}$ 独立同 $N(0, 1^2)$ 分布,

记: $X_1 = \frac{X-\mu}{\sigma}, Y_1 = \frac{Y-\mu}{\sigma}, X_1, Y_1$ 独立同 $N(0, 1^2)$

$$E\left[\max\left(\frac{X-\mu}{\sigma}, \frac{Y-\mu}{\sigma}\right) \cdot \sigma + \mu\right] = E[\max(X, Y)] \quad (*)$$

由 (1) 的结论, $E(X_1 \vee Y_1) = \frac{1}{\sqrt{2}}$

且 $X_1 \vee Y_1 + X_1 \wedge Y_1 = X_1 + Y_1$,

$$E\left[\min\left(\frac{X-\mu}{\sigma}, \frac{Y-\mu}{\sigma}\right) \cdot \sigma + \mu\right] = E[\min(X, Y)]$$

则 $E[X_1 \vee Y_1] + E[X_1 \wedge Y_1] = E[X_1 + Y_1] = 0$,

$$E(X_1 \wedge Y_1) = -\frac{1}{\sqrt{2}}.$$

代入 (*) 式, 则 $E[\max(X, Y)] = \frac{\sigma}{\sqrt{2}} + \mu$

$$E[\min(X, Y)] = -\frac{\sigma}{\sqrt{2}} + \mu.$$

习题 4.2

2. (1) 试证: $\forall c \neq EX, DX = E(X - EX)^2 < E(X - c)^2$;

$$\begin{aligned} \forall c \in \mathbb{R}, E(X - c)^2 &= E[(X - EX) + (EX - c)]^2 = E(X - EX)^2 + E(EX - c)^2 + 2E[(EX - c)(X - EX)] \\ &= E(X - EX)^2 + (EX - c)^2 + 2(EX - c)E(X - EX) \\ &= E(X - EX)^2 + (EX - c)^2 > E(X - EX)^2 = DX, \quad c \neq EX \end{aligned}$$

(3) 设随机变量 X 仅在 $[a, b]$ 上取值, 试证: $a \leq EX \leq b, DX \leq \left(\frac{b-a}{2}\right)^2$.

不妨设 X 为 C.V., $X \sim f(x)$, 则有: $f(x) = 0, x \notin [a, b]$

$$EX = \int_{-\infty}^{+\infty} x f(x) dx = \int_a^b x f(x) dx \leq \int_a^b b f(x) dx = b \cdot \int_a^b f(x) dx = b \cdot \int_{-\infty}^{+\infty} f(x) dx = b$$

同理, 可知 $EX \geq a$. 由 (1) 的结论, $DX \leq E(X - c)^2$, 取 $c = \frac{a+b}{2}$.

$$\text{则有: } E(X - c)^2 = E\left(X - \frac{a+b}{2}\right)^2 \leq E\left(b - \frac{a+b}{2}\right)^2 = \left(\frac{b-a}{2}\right)^2$$

注: $DX \leq \left(\frac{b-a}{2}\right)^2$. 当且仅当 X 满足什么条件? $\begin{array}{c|cc} X & a & b \\ \hline P & \frac{1}{2} & \frac{1}{2} \end{array}$ $\begin{aligned} EX &= \frac{a+b}{2} \\ EX^2 &= \frac{a^2+b^2}{2} \\ DX &= EX^2 - (EX)^2 = \frac{a^2+b^2}{2} - \left(\frac{a+b}{2}\right)^2 \\ &= \frac{(b-a)^2}{4} \end{aligned}$

3. 设 $X \sim U(-1, 2)$, $Y = \begin{cases} 1, & X > 0; \\ 0, & X = 0; \\ -1, & X < 0; \end{cases}$ 试求 DY .

易见, $\begin{array}{c|ccc} Y & -1 & 0 & 1 \\ \hline P & \frac{1}{3} & 0 & \frac{2}{3} \end{array}$ $\begin{aligned} P(Y=0) &= P(X=0) = 0 \\ P(Y=-1) &= P(X < 0) = \frac{1}{3} \end{aligned}$

$$\begin{aligned} EY &= (-1) \times \frac{1}{3} + 1 \times \frac{2}{3} = \frac{1}{3}, \quad EY^2 = (-1)^2 \times \frac{1}{3} + 1^2 \times \frac{2}{3} = \frac{2}{3} \\ DY &= EY^2 - (EY)^2 = \frac{2}{3} - \frac{1}{9} = \frac{5}{9} \end{aligned}$$

4. (1) 设 $X \sim f(x) = \begin{cases} 1+x, & -1 \leq x < 0; \\ 1-x, & 0 \leq x < 1; \\ 0, & \text{其他}; \end{cases}$ 试求 $D(3X+2)$;

易见, $D(3X+2) = D(3X) = 9 \cdot DX$. 由 $f(x) = f(1-x)$, $EX = 0$.

$$EX^2 = \int_{-\infty}^{+\infty} x^2 f(x) dx = 2 \int_{-1}^1 x^2 f(x) dx = 2 \int_0^1 x^2 f(x) dx = 2 \int_0^1 x^2 (1-x) dx = 2 \cdot \left(\frac{1}{3} - \frac{1}{4}\right) = \frac{1}{6}$$

$$DX = EX^2 - (EX)^2 = \frac{1}{6}, \quad D(3X+2) = 9 \cdot DX = \frac{3}{2}$$

5. 设 X 有分布函数 $F(x) = \begin{cases} \frac{1}{2}e^x, & x < 0; \\ \frac{1}{2}, & 0 \leq x < 1; \\ 1 - \frac{1}{2}e^{-\frac{1}{2}(x-1)}, & x \geq 1; \end{cases}$, 试求 DX .

设 $X \sim f(x)$, 则 $f(x) = \frac{d}{dx} F(x)$

$$= \begin{cases} \frac{1}{2}e^x, & x < 0; \\ \frac{1}{4}e^{-\frac{1}{2}(x-1)}, & x \geq 1; \\ 0, & \text{其他}; \end{cases} \quad EX = \int_{-\infty}^{+\infty} x \cdot f(x) dx = \int_{-\infty}^0 x \cdot \frac{1}{2}e^x dx + \int_1^{+\infty} x \cdot \frac{1}{4}e^{-\frac{1}{2}(x-1)} dx$$

① 令 $x = -t$ ② 令 $\frac{x-1}{2} = t$

$$= -\frac{1}{2} \int_0^{+\infty} t \cdot e^{-t} dt + \int_0^{+\infty} (t+1) \cdot \frac{1}{4} e^{-t} dt = -\frac{1}{2} \Gamma(2) + \Gamma(2) + \frac{1}{4} \Gamma(1) = 1$$

$$EX^2 = \int_{-\infty}^{+\infty} x^2 \cdot f(x) dx = \int_{-\infty}^0 x^2 \cdot \frac{1}{2}e^x dx + \int_1^{+\infty} x^2 \cdot \frac{1}{4}e^{-\frac{1}{2}(x-1)} dx$$

③ 令 $x = -t$ ④ 令 $\frac{x-1}{2} = t$

$$= \frac{1}{2} \Gamma(3) + \frac{1}{4} [4 \Gamma(3) + 4 \Gamma(2) + \Gamma(1)] = 1 + \frac{1}{4} [8 + 4 + 1] = \frac{15}{4}$$

$$DX = EX^2 - (EX)^2 = \frac{13}{4}$$

6. (1) 设 $(X, Y) \sim U(D)$, 其中 D 是以点 $(0,1), (1,0), (1,1)$ 为顶点的三角形区域,

试求 $D(X+Y)$; 方法有多种, 选择: ① 求 $X+Y$ 分布; ② 求 $D(X+Y)$:



设 $(X, Y) \sim f(x, y)$, 则 $f(x, y) = \begin{cases} 2, & (x, y) \in D; \\ 0, & \text{其他}; \end{cases}$ 令 $X+Y \sim f_{X+Y}(z)$, 则

$$D: \begin{cases} 0 \leq x, y \leq 1, \\ x+y \geq 1 \end{cases} \quad f_{X+Y}(z) = \int_{-\infty}^{+\infty} f(x, z-x) dx = \begin{cases} \int_{z-1}^1 2 dx = 2(z-1), & z \in (1, 2); \\ 0, & z \notin (1, 2); \end{cases}$$

$$f(x, z-x) > 0 \Leftrightarrow \begin{cases} 0 \leq x \leq 1, \\ 0 \leq z-x \leq 1, \\ x+z-x \geq 1 \end{cases}$$

$$E(X+Y) = \int_{-\infty}^{+\infty} z \cdot f_{X+Y}(z) dz = \int_1^2 z \cdot 2(z-1) dz$$

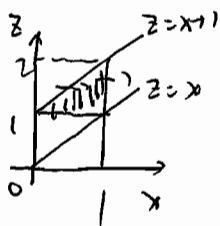
$$= \frac{2}{3} (2z^2 - \frac{2}{3}z^3) \Big|_1^2 = 2 \cdot 3 - \frac{2}{3} \cdot 7 = 6 - \frac{14}{3} = \frac{4}{3}$$

$$E(X+Y)^2 = \int_{-\infty}^{+\infty} z^2 \cdot f_{X+Y}(z) dz = \int_1^2 z^2 \cdot 2(z-1) dz$$

$$= (\frac{2}{3}z^3 - \frac{1}{2}z^4) \Big|_1^2 = \frac{2}{3} \cdot 8 - \frac{1}{2} \cdot 16 = \frac{16}{3} - 8 = \frac{11}{6}$$

$$D(X+Y) = E(X+Y)^2 - [E(X+Y)]^2$$

$$= \frac{11}{6} - \frac{16}{9} = \frac{33-32}{18} = \frac{1}{18}$$



(2) 设 $(X, Y) \sim f(x, y) = \begin{cases} 8xy, & 0 \leq y \leq x \leq 1; \\ 0, & \text{其他;} \end{cases}$, 试求 $DX, DY, E(X+Y)^2$.

$$EX = \iint_{R^2} x \cdot f(x, y) dx dy = \int_0^1 dx \int_0^x x \cdot 8xy dy = \int_0^1 4x^2 \cdot x^2 dx = \frac{4}{5}.$$

$$EY = \iint_{R^2} y \cdot f(x, y) dx dy = \int_0^1 dx \int_0^x y \cdot 8xy dy = \int_0^1 8x \cdot \frac{1}{3} x^3 dx = \frac{8}{15}.$$

$$EX^2 = \iint_{R^2} x^2 \cdot f(x, y) dx dy = \int_0^1 dx \int_0^x x^2 \cdot 8xy dy = \int_0^1 4x^3 \cdot x^2 dx = \frac{2}{3}.$$

$$EY^2 = \iint_{R^2} y^2 \cdot f(x, y) dx dy = \int_0^1 dx \int_0^x y^2 \cdot 8xy dy = \int_0^1 2x \cdot x^4 dx = \frac{1}{3}.$$

$$E(XY) = \iint_{R^2} xy \cdot f(x, y) dx dy = \int_0^1 dx \int_0^x xy \cdot 8xy dy = \int_0^1 8x^2 \cdot \frac{1}{3} x^3 dx = \frac{4}{9}.$$

$$DX = EX^2 - (EX)^2 = \frac{2}{3} - \frac{16}{25} = \frac{2}{75}, \quad DY = EY^2 - (EY)^2 = \frac{1}{3} - \left(\frac{8}{15}\right)^2 = \frac{1}{3} - \frac{64}{225} = \frac{11}{225}.$$

$$E(X+Y)^2 = E(X^2 + 2XY + Y^2) = EX^2 + 2E(XY) + EY^2 = \frac{2}{3} + \frac{8}{9} + \frac{1}{3} = \frac{17}{9}.$$

7. (1) 设 X, Y 独立同 $N\left(0, \frac{1}{2}\right)$ 分布, 试求 $|X-Y|$ 的方差:

解: $X-Y \sim N\left(0, \frac{1}{2} + \frac{1}{2}\right) = N(0, 1)$. 记: $Z = X-Y$. $Z \sim N(0, 1)$

$$E|X-Y| = E|Z| = \int_{-\infty}^{+\infty} |z| \cdot \varphi(z) dz = 2 \int_0^{+\infty} z \cdot \frac{1}{\sqrt{\pi}} e^{-\frac{z^2}{2}} dz = \sqrt{\frac{2}{\pi}}.$$

$$E|X-Y|^2 = E|Z|^2 = EZ^2 = DZ + (EZ)^2 = 1,$$

$$D|X-Y| = E|X-Y|^2 - [E|X-Y|]^2 = 1 - \frac{2}{\pi}.$$

(2) 设随机向量 $(X, Y) \sim f(x, y) = \begin{cases} 2e^{-(2x+y)}, & x, y > 0; \\ 0, & \text{其他;} \end{cases}$, 若 $Z = 2X+Y$, 试求 DZ .

解: X, Y 独立, 且 $X \sim E(2), Y \sim E(1)$. 从而

$$DZ = D(2X+Y) = D(2X) + D(Y) = 4DX + DY = 4 \cdot \frac{1}{4} + 1 = 2.$$

8. (1) 设 $X \sim E(1)$, 且 $Y = X + e^{-2X}$, 试求: DY ; 记 $X \sim f(x) = \begin{cases} e^{-x}, & x > 0; \\ 0, & x \leq 0. \end{cases}$

$$\begin{aligned} EY &= E(X + e^{-2X}) = EX + E(e^{-2X}) = 1 + \int_{-\infty}^{+\infty} e^{-2x} \cdot f(x) dx \\ &= 1 + \int_0^{+\infty} e^{-2x} \cdot e^{-x} dx = 1 + \frac{1}{3} = \frac{4}{3}. \end{aligned}$$

$$\begin{aligned} EY^2 &= E(X + e^{-2X})^2 = E(X^2 + e^{-4X} + 2X \cdot e^{-2X}) = EX^2 + E(e^{-4X}) + 2E(X \cdot e^{-2X}) \\ &= 2 + \int_{-\infty}^{+\infty} e^{-4x} \cdot f(x) dx + 2 \cdot \int_{-\infty}^{+\infty} x \cdot e^{-2x} \cdot f(x) dx \\ &= 2 + \int_0^{+\infty} e^{-4x} \cdot e^{-x} dx + 2 \int_0^{+\infty} x \cdot e^{-2x} \cdot e^{-x} dx \\ &= 2 + \frac{1}{5} + \frac{2}{3} \Gamma(2) = 2 + \frac{1}{5} + \frac{2}{9}. \end{aligned}$$

$$\begin{aligned} DY &= EY^2 - (EY)^2 \\ &= 2 + \frac{1}{5} + \frac{2}{9} - \frac{16}{9} = \frac{29}{45}. \end{aligned}$$

(2) 设 $X \sim E(1)$, 且 $Y_1 = \begin{cases} 1, & X > 1 \\ 0, & X \leq 1 \end{cases}, Y_2 = \begin{cases} 1, & X > 2 \\ 0, & X \leq 2 \end{cases}$, 试求: $D(Y_1 + Y_2)$.

解.

$Y_1 \backslash Y_2$	0	1
0	$1 - e^{-1}$	0
1	$e^{-1} - e^{-2}$	e^{-2}

即 $\{Y_1=0, Y_2=0\} = \{X \leq 1, X \leq 2\} = \{X \leq 1\}$

$\{Y_1=0, Y_2=1\} = \{X \leq 1, X > 2\} = \emptyset$

$\{Y_1=1, Y_2=0\} = \{X > 1, X \leq 2\} = \{1 < X \leq 2\}$

从而, $E(Y_1 + Y_2) = e^{-1} - e^{-2} + 2e^{-2} = e^{-1} + e^{-2}$

$E(Y_1 + Y_2)^2 = 1 \cdot (e^{-1} - e^{-2}) + 4e^{-2} = e^{-1} + 3e^{-2}$

$D(Y_1 + Y_2) = E(Y_1 + Y_2)^2 - [E(Y_1 + Y_2)]^2$
 $= e^{-1} + 3e^{-2} - (e^{-1} + e^{-2})^2$
 $= e^{-1} + 2e^{-2} - 2e^{-3} - e^{-4}$

证:

$Y_1 + Y_2$	0	1	2
P	$1 - e^{-1}$	$e^{-1} - e^{-2}$	e^{-2}

9. 设随机变量 $X \sim f(x) = ae^{4x-2x^2}$, $-\infty < x < +\infty$, 试确定 a , 并求 EX, DX 及

$P(X \leq k), k=1, 2.$

由 $f(x) = ae^{4x-2x^2} = a \cdot e^{-2(x-1)^2} \cdot e^2 = ae^2 \cdot e^{-\frac{(x-1)^2}{2(\frac{1}{2})^2}}, -\infty < x < +\infty.$

证: $X \sim N(1, \frac{1}{2})$. 从而, $a \cdot e^2 = \frac{1}{\sqrt{2\pi} \cdot \frac{1}{2}}$, $a = \sqrt{\frac{2}{\pi}} e^{-2}$. 从而: $EX = 1, DX = \frac{1}{4}.$

$P(X \leq 1) = P(\frac{X-1}{\frac{1}{2}} \leq 0) = \Phi(0) = \frac{1}{2}.$

$P(X \leq 2) = P(\frac{X-1}{\frac{1}{2}} \leq \frac{1}{\frac{1}{2}}) = \Phi(2).$

10. 设随机变量 X 服从拉普拉斯 (Laplace) 分布, 且具有概率密度: $f(x) = \frac{1}{2} e^{-|x|}$,

$-\infty < x < +\infty$; (1) 试求 EX, DX ; (2) 若 $Y = |X|$, 试求 EY, DY .

(1) $EX = \int_{-\infty}^{+\infty} x \cdot f(x) dx = 0, EX^2 = \int_{-\infty}^{+\infty} x^2 \cdot f(x) dx = \int_0^{+\infty} x^2 \cdot e^{-x} dx = \Gamma(3) = 2.$

$DX = EX^2 - (EX)^2 = 2;$

(2) $EY = E(|X|) = \int_{-\infty}^{+\infty} |x| \cdot f(x) dx = 2 \int_0^{+\infty} x \cdot \frac{1}{2} e^{-x} dx = \Gamma(2) = 1.$

$EY^2 = E(X^2) = 2.$

$DY = EY^2 - (EY)^2 = 1.$

习题 4.3

1. (1) 连续独立地掷两次骰子, 试求其点数之和与点数之差的协方差;

记 X_i 为第 i 次掷出的点数, $i=1, 2$; 由题设, X_1, X_2 独立同 $(\frac{1}{6}, \frac{2}{6}, \frac{3}{6}, \frac{4}{6}, \frac{5}{6}, \frac{6}{6})$ 分布. 证:

$$\begin{aligned} \text{Cov}(X_1+X_2, X_1-X_2) &= \text{Cov}(X_1, X_1) - \text{Cov}(X_1, X_2) + \text{Cov}(X_2, X_1) - \text{Cov}(X_2, X_2) \\ &= \text{Cov}(X_1, X_1) - \text{Cov}(X_2, X_2) = DX_1 - DX_2 = 0. \end{aligned}$$

(2) 将一枚均匀硬币重复掷 n 次, 记 X, Y 分别为正面朝上、反面朝上的次数, 试求的协方差与相关系数. 证: $X, Y \sim B(n, \frac{1}{2})$, 且 $X+Y=n$.

证: $EX=EY=\frac{n}{2}$, $DX=DY=\frac{n}{4}$; 证:

$$\text{Cov}(X, Y) = \text{Cov}(X, n-X) = \text{Cov}(X, n) - \text{Cov}(X, X) = -DX = -\frac{n}{4}; \text{证:}$$

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{DX \cdot DY}} = \frac{-\frac{n}{4}}{\frac{n}{4}} = -1.$$

另证: 由 $X+Y=n$, 证: $Y=-X+n$, 证: $P\{Y=-X+n\}=1$. 证: $\rho(X, Y)=-1$.

2. (1) 设随机变量 X, Y 独立同参数为 0.6 的 0-1 分布, 即: $B(1, 0.6)$: 随机变量

$U=X+Y, V=X-Y$ 不相关也不独立;

① U, V 不相关: 证: $EX=EY=0.6$, $DX=DY=0.24$, 证:

$$\begin{aligned} \text{Cov}(U, V) &= \text{Cov}(X+Y, X-Y) = \text{Cov}(X, X) - \text{Cov}(X, Y) + \text{Cov}(Y, X) - \text{Cov}(Y, Y) \\ &= \text{Cov}(X, X) - \text{Cov}(Y, Y) = DX - DY = 0. \end{aligned}$$

② U, V 不独立: 考虑事件: $\{U=2\}$ 与 $\{V=0\}$.

$$P(U=2) = P(X+Y=2) = P(X=1, Y=1) = P(X=1) \cdot P(Y=1) = 0.6 \cdot 0.6 = 0.36$$

$$\begin{aligned} P(V=0) &= P(X-Y=0) = P\{X=1, Y=1\} \cup \{X=0, Y=0\} = P(X=1, Y=1) + P(X=0, Y=0) \\ &= P(X=1)P(Y=1) + P(X=0)P(Y=0) = 0.6 \cdot 0.6 + 0.4 \cdot 0.4 = 0.52. \end{aligned}$$

$$P(U=2, V=0) = P(X+Y=2, X-Y=0) = P(X=1, Y=1) = 0.36 \neq P(U=2) \cdot P(V=0)$$

证: $\{U=2\}$ 与 $\{V=0\}$ 不独立, 证: U, V 不独立!

(2) 设随机变量 $X \sim f(x) = \frac{1}{2}e^{-|x|}$, $-\infty < x < +\infty$, 试证: X 与 $|X|$ 不相关也不独立.

立.

$$\text{① } X \text{ 与 } |X| \text{ 不相关: } EX = \int_{-\infty}^{+\infty} x \cdot f(x) dx = \int_{-\infty}^{+\infty} x \cdot \frac{1}{2} e^{-|x|} dx = 0, \text{ (对称性)}$$

$$E(X \cdot |X|) = \int_{-\infty}^{+\infty} x \cdot |x| \cdot f(x) dx = \int_{-\infty}^{+\infty} x \cdot |x| \cdot \frac{1}{2} e^{-|x|} dx = 0.$$

$$\text{Cov}(X, |X|) = E(X \cdot |X|) - EX \cdot E(|X|) = 0 - 0 = 0. \text{ 证: } X, |X| \text{ 不相关!}$$

② X 与 $|X|$ 不独立: 考虑 $\{X > \ln 2\}$ 与 $\{|X| > \ln 2\}$.

$$P(X > \ln 2) = \int_{\ln 2}^{+\infty} f(x) dx = \int_{\ln 2}^{+\infty} \frac{1}{2} e^{-x} dx = \frac{1}{4},$$

$$P(|X| > \ln 2) = P(X > \ln 2) + P(X < -\ln 2) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

$$\text{证: } P(X > \ln 2, |X| > \ln 2) = P(X > \ln 2) = \frac{1}{4} \neq P(X > \ln 2) \cdot P(|X| > \ln 2)$$

证: $\{X > \ln 2\}$ 与 $\{|X| > \ln 2\}$ 不独立; 证: $X, |X|$ 不独立!

3. (1) 设 $(X, Y) \sim N(\mu, \mu; \sigma^2, \sigma^2; 0)$, 试求: $P(X < Y)$;

由 $\rho=0$, X, Y 独立, 且同 $N(\mu, \sigma^2)$ 分布; 从而 $X-Y \sim N(0, 2\sigma^2)$; 进而,
 $E(X-Y) = EX - EY = 0$, $D(X-Y) = DX + DY = 2\sigma^2$; 从而, $P(X < Y) = P(X-Y < 0)$
 $= P(\frac{X-Y}{\sqrt{2}\sigma} < 0) = \Phi(0) = \frac{1}{2}$.

(2) 设 $(X, Y) \sim N(1, 0; 1, 1; 0)$, 试求: $P(XY - Y < 0)$.

由 $\rho=0$, 即 X, Y 独立, 且 $X \sim N(1, 1)$, $Y \sim N(0, 1)$. $X-1 \sim N(0, 1)$

$$\begin{aligned} P(XY - Y < 0) &= P((X-1)Y < 0) = P(X-1 < 0, Y > 0) + P(X-1 > 0, Y < 0) \\ &= P(X-1 < 0) \cdot P(Y > 0) + P(X-1 > 0) \cdot P(Y < 0) \\ &= \Phi(0) [1 - \Phi(0)] + [1 - \Phi(0)] \Phi(0) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \end{aligned}$$

4. (1) 设随机变量 X, Y 的 $DX = DY = 2$, 相关系数 $\rho = \rho(X, Y) = 0.25$, 试求随机变量 $U = 2X + Y$ 和 $V = 2X - Y$ 的相关系数;

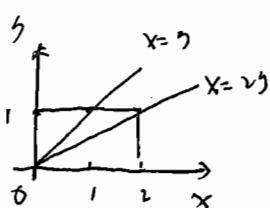
$$\begin{aligned} \text{Cov}(U, V) &= \text{Cov}(2X+Y, 2X-Y) = 4\text{Cov}(X, X) - \text{Cov}(Y, Y) = 4DX - DY = 6. \\ DU &= D(2X+Y) = 4DX + DY + 4\text{Cov}(X, Y) = 10 + 4 \cdot \rho(X, Y) \cdot \sqrt{DX} \cdot \sqrt{DY} = 12. \\ DV &= D(2X-Y) = 4DX + DY - 4\text{Cov}(X, Y) = 10 - 2 = 8. \end{aligned}$$

$$\rho(U, V) = \frac{\text{Cov}(U, V)}{\sqrt{DU} \cdot \sqrt{DV}} = \frac{6}{\sqrt{12} \cdot \sqrt{8}} = \frac{6}{4\sqrt{6}} = \frac{\sqrt{6}}{4}.$$

(2) 设二维随机向量 $(X, Y) \sim U(G)$, $G = \{(x, y) | 0 \leq x \leq 2, 0 \leq y \leq 1\}$, 记:

$$U = \begin{cases} 1, & X > Y; \\ 0, & X \leq Y; \end{cases} \quad V = \begin{cases} 1, & X > 2Y; \\ 0, & X \leq 2Y; \end{cases}$$

试求 U, V 的相关系数.



从而,

	U	0	1	
V	0	$\frac{1}{4}$	0	$\frac{1}{4}$
	1	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$
		$\frac{1}{2}$	$\frac{1}{2}$	

其中, $P(U=0, V=0)$
 $= P(X \leq Y, X \leq 2Y)$
 $= P(X \leq Y) = \frac{1}{4}$

从而,

U	0	1
P	$\frac{1}{4}$	$\frac{3}{4}$

$EU = \frac{3}{4}$, $DU = \frac{3}{16}$;

V	0	1
P	$\frac{1}{2}$	$\frac{1}{2}$

 $EV = \frac{1}{2}$, $DV = \frac{1}{4}$.

UV	0	1
P	$\frac{1}{2}$	$\frac{1}{2}$

, 进而, $\{UV=1\} = \{U=1\} \cap \{V=1\}$, $E(UV) = \frac{1}{2}$;

从而, $\rho(U, V) = \frac{\text{Cov}(U, V)}{\sqrt{DU} \cdot \sqrt{DV}} = \frac{E(UV) - EU \cdot EV}{\sqrt{\frac{3}{16}} \cdot \sqrt{\frac{1}{4}}} = \frac{\frac{1}{8}}{\sqrt{\frac{3}{16}} \cdot \sqrt{\frac{1}{4}}} = \frac{1}{\sqrt{3}}$ 82

5. 设 a 为 $(0,1)$ 内的一个定点, 随机变量 $X \sim U(0,1)$, 以 Y 表示 X 到点 a 的距离,

试问: a 为何值时, X 与 Y 不相关? 解: $Y = |X-a|$

$$EY = \int_{-\infty}^{+\infty} E|X-a| \cdot f_X(x) dx = \int_0^1 |x-a| dx = \int_0^a (a-x) dx + \int_a^1 (x-a) dx$$

$$= a^2 - \frac{1}{2}a^2 + \frac{1}{2} - \frac{1}{2}a^2 - a + a^2 = a^2 - a + \frac{1}{2};$$

$$E(XY) = E(X \cdot |X-a|) = \int_{-\infty}^{+\infty} x \cdot |x-a| \cdot f_X(x) dx = \int_0^1 x \cdot |x-a| \cdot dx = \int_0^a x(a-x) dx + \int_a^1 x(x-a) dx$$

$$= \frac{a^3}{2} - \frac{1}{3}a^3 + \frac{1}{3} - \frac{1}{3}a^3 + -\frac{1}{2}a + \frac{1}{2}a^3 = \frac{1}{3}a^3 - \frac{1}{2}a + \frac{1}{3}. \quad \text{由 } X, Y \text{ 不相关, 有:}$$

$$E(XY) = EX \cdot EY. \quad \text{则有: } \frac{1}{3}a^3 - \frac{1}{2}a + \frac{1}{3} = \frac{1}{2} \cdot (a^2 - a + \frac{1}{2}). \quad \text{即: } \frac{1}{3}a^3 - \frac{1}{2}a^2 + \frac{1}{6} = 0$$

$$\text{即: } 4a^3 - 6a^2 + 1 = 0. \quad 4a^3 - 2a^2 - (4a^2 - 1) = 2a^2(2a-1) - (2a+1)(2a-1) = (2a^2 - 2a - 1)(2a-1) = 0$$

6. (2) 设 (X,Y) 的联合密度为 $f(x,y) = \begin{cases} \frac{6}{7} \left(x^2 + \frac{1}{2}xy \right), & 0 < x < 1, 0 < y < 2; \\ 0, & \text{其他;} \end{cases}$ 试求 X, Y 的协方差与相关系数. $a = \frac{1}{2}$ (其他舍去)

求 X, Y 的协方差与相关系数.

$$EX = \iint_{\mathbb{R}^2} x \cdot f(x,y) dx dy = \frac{6}{7} \int_0^1 x^2 dx \int_0^2 (x + \frac{1}{2}y) dy = \frac{6}{7} \int_0^1 x^2 (2x+1) dx = \frac{6}{7} \times [\frac{1}{2} + \frac{1}{3}] = \frac{5}{7};$$

$$EY = \iint_{\mathbb{R}^2} y \cdot f(x,y) dx dy = \frac{6}{7} \int_0^1 x dx \int_0^2 y \cdot (x + \frac{1}{2}y) dy = \frac{6}{7} \int_0^1 x \cdot (2x + \frac{4}{3}) dx = \frac{6}{7} \times [\frac{2}{3} + \frac{2}{3}] = \frac{8}{7};$$

$$EX^2 = \iint_{\mathbb{R}^2} x^2 \cdot f(x,y) dx dy = \frac{6}{7} \int_0^1 x^3 dx \int_0^2 (x + \frac{1}{2}y) dy = \frac{6}{7} \int_0^1 x^3 (2x+1) dx = \frac{6}{7} \times [\frac{2}{5} + \frac{1}{4}] = \frac{39}{70};$$

$$EY^2 = \iint_{\mathbb{R}^2} y^2 \cdot f(x,y) dx dy = \frac{6}{7} \int_0^1 x dx \int_0^2 y^2 (x + \frac{1}{2}y) dy = \frac{6}{7} \int_0^1 x (\frac{8}{3}x + 2) dx = \frac{6}{7} \times [\frac{8}{9} + 1] = \frac{34}{21};$$

$$E(XY) = \iint_{\mathbb{R}^2} xy \cdot f(x,y) dx dy = \frac{6}{7} \int_0^1 x^2 dx \int_0^2 y (x + \frac{1}{2}y) dy = \frac{6}{7} \int_0^1 x^2 (2x + \frac{4}{3}) dx = \frac{6}{7} \times [\frac{1}{2} + \frac{4}{9}] = \frac{17}{21};$$

$$\text{Cov}(X,Y) = E(XY) - EX \cdot EY = \frac{17}{21} - \frac{5}{7} \times \frac{8}{7} = \frac{17 \times 7 - 120}{3 \times 7 \times 7} = -\frac{1}{147};$$

$$\rho(X,Y) = \frac{\text{Cov}(X,Y)}{\sqrt{DX} \cdot \sqrt{DY}} = \frac{-\frac{1}{147}}{\sqrt{\frac{23}{7 \times 7 \times 10}} \times \sqrt{\frac{46}{7 \times 7 \times 3}}} = -\frac{\sqrt{5}}{23}, \quad \begin{aligned} \text{即: } DX &= EX^2 - (EX)^2 = \frac{39}{70} - (\frac{5}{7})^2 = \frac{23}{7 \times 7 \times 10} \\ DY &= EY^2 - (EY)^2 = \frac{34}{21} - (\frac{8}{7})^2 = \frac{46}{7 \times 7 \times 3} \end{aligned}$$

7. (1) 设随机变量 X, Y 间有线性函数关系: $Y = aX + b$, 且 X 的方差存在, 试求 X, Y 的相关系数 ρ ;

$$\text{由 } Y = aX + b, \text{ 则有: } \text{Cov}(X,Y) = \text{Cov}(X, aX+b) = a \text{Cov}(X,X) + \text{Cov}(X,b) = aDX;$$

$$DY = D(aX+b) = D(aX) = a^2 DX. \quad \text{从而}$$

$$\rho(X,Y) = \frac{\text{Cov}(X,Y)}{\sqrt{DX} \cdot \sqrt{DY}} = \frac{aDX}{\sqrt{DX} \cdot \sqrt{a^2 DX}} = \frac{a}{|a|} = \begin{cases} 1, & a > 0; \\ -1, & a < 0; \end{cases}$$

另证: $Y = aX + b, a > 0 \Rightarrow \rho = 1. \quad \text{即: } \rho = 1 \Leftrightarrow Y = aX + b, a.s. \quad a > 0,$

$$Y = aX + b, a < 0 \Rightarrow \rho = -1.$$

$$\rho = -1 \Leftrightarrow Y = aX + b, a.s. \quad a < 0$$

(2) 设随机变量 $X \sim U\left(-\frac{1}{2}, \frac{1}{2}\right)$, $Y = \cos X$, 则 Y 与 X 有 (非线性) 函数关系,

试证: X, Y 不 (线性) 相关, 即无线性关系. $\because X \sim f_X(x)$, 且

$$EX = 0, E(XY) = E(X \cos X) = \int_{-\infty}^{\infty} x \cos x \cdot f_X(x) dx = \int_{-\frac{1}{2}}^{\frac{1}{2}} x \cos x \cdot \frac{1}{\frac{1}{2}} dx = 0.$$

则有: $\text{Cov}(X, Y) = E(XY) - EX \cdot EY = 0 - 0 = 0$. 证: X, Y 不 (线性) 相关!

11. 相关!

9. 设随机变量 $X \sim N(0, 1^2)$, $Y \sim \begin{pmatrix} -1 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$, 且 X, Y 相互独立, 令 $Z = XY$, 试证:

(1) $Z \sim N(0, 1^2)$; (2) X, Z 不相关也不独立.

(1) $\forall z \in \mathbb{R}, F_Z(z) = P(Z \leq z) = P(XY \leq z) = P(Y = -1) \cdot P(XY \leq z | Y = -1) + P(Y = 1) \cdot P(XY \leq z | Y = 1)$

$$= \frac{1}{2} [P(X \geq -z | Y = -1) + P(X \leq z | Y = 1)]$$

$$= \frac{1}{2} [P(X \geq -z) + P(X \leq z)]$$

$$= \frac{1}{2} [1 - \Phi(-z)] + \frac{1}{2} \Phi(z) = \Phi(z), \text{ 证: } Z \sim N(0, 1^2)$$

(2) ① X, Z 不相关: $EX = 0, EZ = 0, E(XZ) = E(X \cdot XY) = E(X^2 Y) = E(X^2) E(Y) = 1 \cdot 0 = 0$. 证: $\text{Cov}(X, Z) = E(XZ) - EX \cdot EZ = 0 - 0 = 0$

$$= EX^2 - EY = 1 - 0 = 0. \text{ 证: } \text{Cov}(X, Z) = E(XZ) - EX \cdot EZ = 0 - 0 = 0$$

证: X, Z 不相关!

② X, Z 不独立: 考虑: $\{X \leq 1\}$ 与 $\{Z \leq 1\}$

由 $X, Z \sim N(0, 1^2)$, $P(X \leq 1) = P(Z \leq 1) = 0.8413$

⑩ 设随机变量 $X_1, X_2, \dots, X_n$ 中任意两个的相关系数都是 ρ , 试证: $\rho \geq -\frac{1}{n-1}$. (是 P84 附页)

$$P(X \leq 1, Z \leq 1) = P(X \leq 1, XY \leq 1)$$

$$= P(Y = -1) \cdot P(X \leq 1, XY \leq 1 | Y = -1) + P(Y = 1) \cdot P(X \leq 1, XY \leq 1 | Y = 1)$$

$$= \frac{1}{2} [P(X \leq 1, X \geq -1 | Y = -1) + P(X \leq 1 | Y = 1)]$$

$$= \frac{1}{2} [P(-1 \leq X \leq 1) + P(X \leq 1)] = \frac{1}{2} [\Phi(1) - \Phi(-1) + \Phi(1)]$$

$$= \frac{1}{2} [3\Phi(1) - 1] = \frac{1}{2} [3 \cdot 0.8413 - 1] = 0.7813 \neq P(X \leq 1) \cdot P(Z \leq 1)$$

可以如下证明: 若记 $\Phi(1) = x$, 若 $P(X \leq 1, Z \leq 1) = P(X \leq 1) \cdot P(Z \leq 1)$, 证:

$$\frac{1}{2}(3x - 1) = x^2, \text{ 证: } 2x^2 - 3x + 1 = (2x - 1)(x - 1) = 0$$

$$x = \frac{1}{2} \text{ 或 } x = 1. \text{ 与 } \Phi(1) = 0.8413 \text{ 矛盾!}$$

证: $\{X \leq 1\}$ 与 $\{Z \leq 1\}$ 不独立 证: X, Z 不独立!

11. (1) 某班级共有 n 名新生, 班长从辅导员处领来全班所有的学生证, 随机地发给每一名学生, 试求恰好拿到自己学生证的人数 X 的数学期望与方差;

设 $X_i = \begin{cases} 1, & \text{若第 } i \text{ 名新生拿到自己学生证;} \\ 0, & \text{否则;} \end{cases} \quad i=1, 2, \dots, n;$ 即有: $X = X_1 + X_2 + \dots + X_n$, $EX = E(X_1 + X_2 + \dots + X_n)$

$= EX_1 + EX_2 + \dots + EX_n$; 这里, $\begin{array}{c|cc} X_i & 1 & 0 \\ \hline P & \frac{1}{n} & \frac{n-1}{n} \end{array}$, 其中, $P(X_i=1) = \frac{(n-1)!}{n!} = \frac{1}{n}$, $EX_i = \frac{1}{n}$, $EX = 1$.

$$DX = D(\sum_{i=1}^n X_i) = \sum_{i=1}^n DX_i + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j) = n \cdot \frac{n-1}{n^2} + 2 \sum_{1 \leq i < j \leq n} (E(X_i X_j) - EX_i \cdot EX_j)$$

这里, $\begin{array}{c|cc} X_i X_j & 1 & 0 \\ \hline P & \frac{1}{n(n-1)} & * \end{array}$, $\begin{array}{c|cc} X_i X_j & 1 & 0 \\ \hline P & \frac{1}{n(n-1)} & * \end{array}$, 其中, $P(X_i X_j = 1) = P(X_i = 1, X_j = 1) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}$

$$E(X_i X_j) = \frac{1}{n(n-1)}; \text{ 从而, } DX = \frac{n-1}{n} + 2 \sum_{1 \leq i < j \leq n} \left[\frac{1}{n(n-1)} - \frac{1}{n^2} \right]$$

$$= \frac{n-1}{n} + n(n-1) \left[\frac{1}{n(n-1)} - \frac{1}{n^2} \right] = 1 - \frac{1}{n} + 1 - \frac{n-1}{n} = 1$$

(2) 袋中有 n 张卡片, 分别标有号码 $1, 2, \dots, n$; 从中不放回地抽出 k 张卡片, 设 ξ 表示所抽出的号码之和, 试求 $E\xi$ 与 $D\xi$.

记: ξ_i 为第 i 次抽出的卡片号码, $i=1, 2, \dots, k$; 即有: $\xi = \xi_1 + \xi_2 + \dots + \xi_k$; 且由

抽签原理 (Polya 模型), $\xi_i \sim \left(\frac{1}{n}, \frac{2}{n}, \dots, \frac{n}{n} \right)$; 从而, $E\xi_i = (1+2+\dots+n) \cdot \frac{1}{n} = \frac{n+1}{2}$,

$$E\xi = \sum_{i=1}^k E\xi_i = \frac{k(n+1)}{2}; \text{ 由于 } P(\xi_i = s, \xi_j = t) = \begin{cases} \frac{1}{n(n-1)}, & s \neq t, \\ 0, & s = t; \end{cases}$$

即有: $E(\xi_i, \xi_j) = \sum_s \sum_{t \neq s} s \cdot t \cdot \frac{1}{n(n-1)} = \frac{1}{n(n-1)} \left[\sum_{s=1}^n \sum_{t=1}^n s t - \sum_{s=1}^n s^2 \right]$

$$= \frac{1}{n(n-1)} \left[\frac{n(n+1)^2}{4} - \frac{n(n+1)(2n+1)}{6} \right] = \frac{1}{n(n-1)} \cdot n \cdot (n+1) \cdot \frac{3n(n+1) - (2n+1)}{12} = \frac{(n+1)(3n+2)}{12}$$

$$\text{Cov}(\xi_i, \xi_j) = \frac{(n+1)(3n+2)}{12} - \left(\frac{n+1}{2} \right)^2 = -\frac{n+1}{12}; \text{ 从而,}$$

$$D\xi = D\left(\sum_{i=1}^k \xi_i\right) = \sum_{i=1}^k D\xi_i + 2 \sum_{1 \leq i < j \leq k} \text{Cov}(\xi_i, \xi_j)$$

$$= \sum_{i=1}^k \left[\frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} \right] + 2 \sum_{1 \leq i < j \leq k} \left(-\frac{n+1}{12} \right)$$

$$= \frac{(n+1)k(n-k)}{12}$$

2.3.21: $\max(a, b) = a \vee b = \frac{a+b+|a-b|}{2}$

12. 设二维随机向量 (X, Y) 满足: $EX=EY=0$, $DX=DY=1$, $\text{Cov}(X, Y)=\rho$, $EX^2=EY^2=1$

试证: $E[\max(X^2, Y^2)] \leq 1 + \sqrt{1-\rho^2}$.

$E[X^2 \vee Y^2] = E\left[\frac{X^2+Y^2+|X^2-Y^2|}{2}\right] = 1 + \frac{1}{2} E|X^2-Y^2| = 1 + \frac{1}{2} E(|X+Y| \cdot |X-Y|)$ (Cauchy 不等式)

$1 + \frac{1}{2} \cdot \sqrt{E(X+Y)^2 \cdot E(X-Y)^2} = 1 + \frac{1}{2} \sqrt{E(X+Y)^2 \cdot E(X-Y)^2} = 1 + \frac{1}{2} \sqrt{(EX^2+EY^2+2EXY)(EX^2+EY^2-2EXY)}$
 $= 1 + \frac{1}{2} \sqrt{2(1+\rho) \cdot 2(1-\rho)} = 1 + \sqrt{1-\rho^2}$

13. (1) 设 $DX, DY > 0$, $\rho = \rho(X, Y)$, 则

$\min_{a, b \in \mathbb{R}} E[Y - (aX + b)]^2 = DY \cdot (1 - \rho^2)$;

由此亦可见相关系数 ρ 刻画了随机变量 X, Y 的线性关系的强弱;

$E[Y - (aX + b)]^2 = E[(Y - EY) - (a(X - EX) + (EY - aEX - b))]^2 = E\{(X - EX)^2 + a^2(Y - EY)^2 + (EY - aEX - b)^2$
 $- 2a(X - EX)(Y - EY) + 2(EY - aEX - b)(X - EX) - 2a(EY - aEX - b)(Y - EY)\}$ (中间项为0)

$= E(X - EX)^2 + a^2 E(Y - EY)^2 + (EY - aEX - b)^2 - 2a \cdot \text{Cov}(X, Y)$

$= DX + a^2 DY + (EY - aEX - b)^2 - 2a \rho \cdot \sqrt{DX \cdot DY}$

$= DX \cdot (1 - \rho^2) + (a \cdot \sqrt{DX} - \rho \cdot \sqrt{DY})^2 + (EY - aEX - b)^2 \geq DX \cdot (1 - \rho^2)$

当且仅当 $\begin{cases} a \cdot \sqrt{DX} = \rho \cdot \sqrt{DY} \\ EY = aEX + b \end{cases}$

(4) 设 $(X, Y) \sim f(x, y) = cxe^{-y}$, $0 < x < y < +\infty$; 试求: $P(X < 1 | Y = 2)$ 及

$E[Y - (aX + b)]^2$ 的最小值.

由上述 (1), $\min_{a, b \in \mathbb{R}} E[Y - (aX + b)]^2 = DY \cdot (1 - \rho^2)$. 由 $\iint_{\mathbb{R}^2} f(x, y) dx dy = \iint_{0 < x < y} cxe^{-y} dx dy$

$= \int_0^{+\infty} dy \int_x^{+\infty} cxe^{-y} dy = \int_0^{+\infty} cxe^{-x} dx = 1$, $C = 1$, 故 $Y \sim f_Y(y)$, $f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx$

$= \begin{cases} \int_0^y xe^{-y} dx, & y > 0; \\ 0, & y \leq 0; \end{cases} = \begin{cases} \frac{y^2}{2} e^{-y}, & y > 0; \\ 0, & y \leq 0; \end{cases}$ $\forall y > 0$, 令 $X|Y=y \sim f_{X|Y}(x|y)$, 则

$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \begin{cases} \frac{2x}{y}, & 0 < x < y; \\ 0, & \text{其他}; \end{cases}$ 也即有: $f_{X|Y}(x|2) = \begin{cases} \frac{x}{2}, & 0 < x < 2; \\ 0, & \text{其他}. \end{cases}$ 故

$P(X < 1 | Y = 2) = \int_{-\infty}^1 f_{X|Y}(x|2) dx = \int_0^1 \frac{x}{2} dx = \frac{1}{4}$. $EY = \int_{-\infty}^{+\infty} y \cdot f_Y(y) dy = \int_0^{+\infty} y \cdot \frac{y^2}{2} e^{-y} dy = \frac{1}{2} \Gamma(4) = 3$.

$EY^2 = \int_{-\infty}^{+\infty} y^2 \cdot f_Y(y) dy = \int_0^{+\infty} y^2 \cdot \frac{y^2}{2} e^{-y} dy = \frac{1}{2} \Gamma(5) = 12$, $DY = EY^2 - (EY)^2 = 3$. 又 $EX = \iint_{\mathbb{R}^2} x \cdot f(x, y) dx dy = \iint_{0 < x < y} xe^{-y} dx dy$

$= \int_0^{+\infty} dy \int_x^{+\infty} xe^{-y} dy = \int_0^{+\infty} xe^{-x} dx = 2$, $EX^2 = \iint_{\mathbb{R}^2} x^2 \cdot f(x, y) dx dy = \iint_{0 < x < y} x^2 \cdot xe^{-y} dx dy = \int_0^{+\infty} dy \int_x^{+\infty} x^3 e^{-y} dy = \int_0^{+\infty} \frac{y^3}{6} e^{-y} dy = 6$

$DX = EX^2 - (EX)^2 = 2$. $E(XY) = \iint_{\mathbb{R}^2} xy \cdot f(x, y) dx dy = \int_0^{+\infty} dy \int_x^{+\infty} xy^2 e^{-y} dx dy = \int_0^{+\infty} y^2 dy \int_0^y x^2 e^{-y} dx = \frac{1}{3} \int_0^{+\infty} y^4 e^{-y} dy = 8$

14. (1) 掷一颗均匀的骰子直到所有六个点数全部出现为止, 试求所需投掷次数 Y 的数学期望与方差;

记: Y_1 为获得第一个点所需掷的次数, Y_i 为掷出第 $i-1$ 个不同点所需第 i 个不同点出现所需次数, $i=2, 3, 4, 5, 6$; 易见, $Y_1=1$, $Y_2 \sim G(\frac{5}{6})$, $Y_3 \sim G(\frac{4}{6})$, $Y_4 \sim G(\frac{3}{6})$, $Y_5 \sim G(\frac{2}{6})$, $Y_6 \sim G(\frac{1}{6})$ 且 $Y_1, Y_2, Y_3, Y_4, Y_5, Y_6$ 独立, $Y = Y_1 + Y_2 + Y_3 + Y_4 + Y_5 + Y_6$. 证有:

$$EY = EY_1 + EY_2 + EY_3 + EY_4 + EY_5 + EY_6 = 1 + \frac{6}{5} + \frac{6}{4} + \frac{6}{3} + \frac{6}{2} + 6 = 1 + 1.2 + 1.5 + 2 + 3 + 6 = 14.7,$$

$$DY = DY_1 + DY_2 + DY_3 + DY_4 + DY_5 + DY_6 = 0 + \frac{6}{25} + \frac{3}{4} + 2 + 6 + 30 = 38.99.$$

(2) 易见, $X \sim B(n, \frac{1}{6})$, $Y \sim B(n, \frac{1}{6})$, $EX = EY = \frac{n}{6}$, $DX = DY = \frac{5n}{36}$, $\text{Cov}(X, Y) = E(XY) - EX \cdot EY$.

令 $X_i = \begin{cases} 1, & \text{若第 } i \text{ 次掷出 "3" 点} \\ 0, & \text{否则} \end{cases}$; $Y_j = \begin{cases} 1, & \text{若第 } j \text{ 次掷出 "5" 点} \\ 0, & \text{否则} \end{cases}$; $i, j = 1, 2, \dots, n$; 证有:

$$X = \sum_{i=1}^n X_i, Y = \sum_{j=1}^n Y_j, \text{ 从而, } E(XY) = \sum_{i=1}^n \sum_{j=1}^n E(X_i Y_j). \text{ 若 } i=j, X_i Y_i = 0.$$

$$E(X_i Y_i) = 0, \text{ 故 } E(XY) = 2 \cdot \sum_{1 \leq i < j \leq n} E(X_i Y_j). \text{ 证: } \begin{array}{|c|c|c|} \hline X_i & Y_j & \\ \hline 1 & 1 & 0 \\ \hline 1 & 0 & 0 \\ \hline 0 & 1 & 0 \\ \hline 0 & 0 & 0 \\ \hline \end{array}$$

$$= 2 \cdot \sum_{1 \leq i < j \leq n} EX_i \cdot EY_j = 2 \cdot \sum_{1 \leq i < j \leq n} \frac{1}{6} \cdot \frac{1}{6}$$

$$= \frac{n(n-1)}{36}. \text{ 从而, } \text{Cov}(X, Y) = \frac{n(n-1)}{36} - \frac{n}{6} \cdot \frac{n}{6} = -\frac{n}{36}.$$

(2) 连续独立地掷一颗均匀的骰子 n 次, 试求 "3" 点和 "5" 点出现次数 X, Y 的协方差及相关系数.

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{DX} \cdot \sqrt{DY}} = \frac{-\frac{n}{36}}{\frac{5n}{36}} = -\frac{1}{5}.$$

习题 4.4

3. (1) 设随机向量 (X, Y) 的协方差矩阵为 $\begin{pmatrix} 1 & 1 \\ 1 & 4 \end{pmatrix}$, 试求 $U = X - 2Y$ 和 $V = 2X - Y$

的相关系数; 易见, $DX = 1, DY = 4, \text{Cov}(X, Y) = 1$, 故

$$\begin{aligned} \text{Cov}(U, V) &= \text{Cov}(X - 2Y, 2X - Y) = 2\text{Cov}(X, X) - \text{Cov}(X, Y) - 4\text{Cov}(Y, X) + 2\text{Cov}(Y, Y) \\ &= 2DX - 5\text{Cov}(X, Y) + 2DY = 5 \end{aligned}$$

$$DU = D(X - 2Y) = DX + 4DY - 4\text{Cov}(X, Y) = 13.$$

$$DV = D(2X - Y) = 4DX + DY - 4\text{Cov}(X, Y) = 4.$$

$$\rho(U, V) = \frac{\text{Cov}(U, V)}{\sqrt{DU} \cdot \sqrt{DV}} = \frac{5}{2\sqrt{13}}.$$

(2) 设随机向量 $(X, Y) \sim f(x, y) = \begin{cases} 6xy^2, & 0 < x, y < 1; \\ 0, & \text{其他}; \end{cases}$ 试求 (X, Y) 的协方差矩

阵.

$$EX = \iint_{R^2} x \cdot f(x, y) dx dy = \int_0^1 dx \int_0^1 6x^2 y^2 dy = \int_0^1 2x^2 dx = \frac{2}{3},$$

$$EX^2 = \iint_{R^2} x^2 \cdot f(x, y) dx dy = \int_0^1 dx \int_0^1 6x^3 y^2 dy = \int_0^1 2x^3 dx = \frac{1}{2}, \quad DX = EX^2 - (EX)^2 = \frac{1}{18};$$

$$EY = \iint_{R^2} y \cdot f(x, y) dx dy = \int_0^1 dx \int_0^1 6x y^3 dy = \int_0^1 \frac{3}{2} x dx = \frac{3}{4},$$

$$EY^2 = \iint_{R^2} y^2 \cdot f(x, y) dx dy = \int_0^1 dx \int_0^1 6x y^4 dy = \int_0^1 \frac{6}{5} x dx = \frac{3}{5}, \quad DY = EY^2 - (EY)^2 = \frac{3}{80}$$

$$E(XY) = \iint_{R^2} xy \cdot f(x, y) dx dy = \int_0^1 dx \int_0^1 6x^2 y^3 dy = \frac{1}{2} = EX \cdot EY \quad (\text{可以证明: } X, Y \text{ 独立}).$$

$$\text{Cov}(X, Y) = E(XY) - EX \cdot EY = 0. \quad \text{故协方差矩阵 } \Sigma = \begin{pmatrix} \frac{1}{18} & 0 \\ 0 & \frac{3}{80} \end{pmatrix}.$$

4. 两支股票 A 和 B , 在一个给定时期内的收益率 R_A, R_B 均为随机变量, 且 R_A, R_B

的协方差阵为: $V = \begin{pmatrix} 16 & 6 \\ 6 & 9 \end{pmatrix}$, 现将一笔资金按比例 $x, 1-x$ 分别投资于股票 A, B ,

从而形成一个投资组合 Π , 记其收益率为 R_Π ;

(1) 求 R_A, R_B 的相关系数; (2) 求 $D(R_\Pi)$.

(1) 由 $D(R_A) = 16, D(R_B) = 9, \text{Cov}(R_A, R_B) = 6$, 即有:

$$\rho(R_A, R_B) = \frac{\text{Cov}(R_A, R_B)}{\sqrt{DR_A} \cdot \sqrt{DR_B}} = \frac{6}{4 \cdot 3} = \frac{1}{2}.$$

(2) 由题设, $R_\Pi = x \cdot R_A + (1-x) \cdot R_B$, 即有:

$$\begin{aligned} D(R_\Pi) &= D[x \cdot R_A + (1-x) \cdot R_B] \\ &= x^2 D(R_A) + (1-x)^2 D(R_B) + 2x(1-x) \cdot \text{Cov}(R_A, R_B) \\ &= 16x^2 + 9(1-x)^2 + 2x(1-x) \cdot 6 \\ &= 13x^2 + 6x + 9. \end{aligned}$$

5. 设随机向量 $(X, Y) \sim f(x, y) = \frac{2}{\pi(1+x^2+y^2)^2}$, $-\infty < x, y < +\infty$, 试求 EX, EY 及

协方差阵. $EX = \iint_{R^2} x \cdot f(x, y) dx dy$ (由对称性) $= 0$. 同理, $EY = \iint_{R^2} y \cdot f(x, y) dx dy = 0$.

$EX^2 = \iint_{R^2} x^2 \cdot f(x, y) dx dy$, $EY^2 = \iint_{R^2} y^2 \cdot f(x, y) dx dy$. 由轮换对称性,

$$EX^2 = EY^2 = E\left(\frac{x^2+y^2}{2}\right) = \iint_{R^2} \frac{x^2+y^2}{2} \cdot f(x, y) dx dy = \frac{1}{\pi} \iint_{R^2} \frac{x^2+y^2}{(1+x^2+y^2)^2} dx dy$$

$$= \frac{1}{\pi} \int_0^{2\pi} d\theta \int_0^{+\infty} \frac{r^3}{(1+r^2)^2} dr = 2 \cdot \int_0^{+\infty} \frac{r^3}{(1+r^2)^2} dr \stackrel{\text{令 } x=r^2}{=} \int_0^{+\infty} \frac{x}{(1+x)^2} dx = \int_0^{+\infty} \left(\frac{1}{(1+x)} - \frac{1}{(1+x)^2}\right) dx$$

$$= 1 - \frac{1}{2} = \frac{1}{2}. \quad DX = DY = EY^2 - (EY)^2 = \frac{1}{2}.$$

$$E(XY) = \iint_{R^2} xy \cdot f(x, y) dx dy \text{ (由对称性) } = 0. \quad \text{Cov}(X, Y) = E(XY) - EX \cdot EY = 0.$$

习题 5.1 协方差矩阵 $\Sigma = \begin{pmatrix} DX & \text{Cov}(X, Y) \\ \text{Cov}(X, Y) & DY \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}.$

1. (3) 设 $X \sim N(1, 1^2)$, $Y \sim N(0, 1^2)$, 且 $E(XY) = -0.1$, 试由切比雪夫不等式估

计 $P(-4 < X+2Y < 6)$; $E(X+2Y) = EX+2EY = 1$; $D(X+2Y) = DX+4DY+4\text{Cov}(X, Y)$

$$= DX+4DY+4E(XY) = 4.6. \text{ 由 Chebyshev 不等式,}$$

$$P(-4 < X+2Y < 6) = P(|(X+2Y) - E(X+2Y)| < 5) = 1 - P(|(X+2Y) - E(X+2Y)| \geq 5) \\ \geq 1 - \frac{D(X+2Y)}{5^2} = 1 - \frac{4.6}{25} = 1 - 0.184 = 0.816.$$

(4) 设随机变量 $X_1, X_2, \dots, X_n$ 独立, 且 $EX_i = \mu$, $DX_i = \sigma^2$, $i=1, 2, \dots, n$; 对于

$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, 试由切比雪夫不等式估计 $P(\mu-2 < \bar{X} < \mu+2)$.

$$\text{易见, } E(\bar{X}) = \frac{1}{n} \sum_{i=1}^n EX_i = \mu, \quad D(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^n DX_i = \frac{\sigma^2}{n}; \quad P(\mu-2 < \bar{X} < \mu+2) = P(|\bar{X} - \mu| < 2) \\ = 1 - P(|\bar{X} - \mu| \geq 2) = 1 - P(|\bar{X} - E(\bar{X})| \geq 2) \geq 1 - \frac{D\bar{X}}{2^2} = 1 - \frac{\sigma^2}{4n}.$$

2. 设随机变量 $X \sim f(x) = \begin{cases} \frac{x^m}{m!} e^{-x}, & x > 0; \\ 0, & x \leq 0; \end{cases} m \in N$, 试证:

$$P(0 < X < 2(m+1)) \geq \frac{m}{m+1}.$$

$$EX = \int_{-\infty}^{+\infty} x \cdot f(x) dx = \int_0^{+\infty} x \cdot \frac{x^m}{m!} e^{-x} dx = \frac{1}{m!} \int_0^{+\infty} x^{m+1} e^{-x} dx = \frac{1}{m!} \Gamma(m+2) = \frac{(m+1)!}{m!} = m+1.$$

$$EX^2 = \int_{-\infty}^{+\infty} x^2 \cdot f(x) dx = \int_0^{+\infty} x^2 \cdot \frac{x^m}{m!} e^{-x} dx = \frac{1}{m!} \int_0^{+\infty} x^{m+2} e^{-x} dx = \frac{1}{m!} \Gamma(m+3) = \frac{(m+2)!}{m!} = (m+2)(m+1), \\ DX = EX^2 - (EX)^2 = m+1; \text{ 由 Chebyshev 不等式,}$$

$$P(0 < X < 2(m+1)) = P(|X - (m+1)| < m+1) = 1 - P(|X - EX| \geq m+1) \geq 1 - \frac{DX}{(m+1)^2} \\ = 1 - \frac{m+1}{(m+1)^2} = 1 - \frac{1}{m+1} = \frac{m}{m+1}.$$

3. 设在每次试验中, 事件 A 发生的概率均为 0.75, 试求 n 需多大时才能使得 n 次独立重复试验中事件 A 发生的频率在 0.74~0.76 之间的概率至少为 0.90?

记 X 为 n 次试验中 A 发生的次数; 易见, $X \sim B(n, \frac{3}{4})$. $EX = 0.75n$, $DX = 0.75n \times 0.25$

$$\text{从而, 欲 } P(0.74 < \frac{X}{n} < 0.76) = P(|X - 0.75n| < 0.01n) = 1 - P(|X - EX| \geq 0.01n)$$

$$\geq 1 - \frac{DX}{(0.01n)^2} = 1 - \frac{75 \times 25}{n} \geq 0.90$$

$$\text{即有: } n \geq 75 \times 250 = 18750$$

4. 设随机序列 $\{X_n, n \geq 1\}$ 独立同 $U(0, a)$ 分布, $a > 0$ 为常数, 则有:

$$Y_n = \max\{X_1, X_2, \dots, X_n\} \xrightarrow{P} a.$$

$$\begin{aligned} \forall \varepsilon > 0 (< a), \quad P(|Y_n - a| \geq \varepsilon) &= P(\{Y_n \leq a - \varepsilon\} \cup \{Y_n \geq a + \varepsilon\}) = P(Y_n \leq a - \varepsilon) = 1 - P(Y_n \geq a + \varepsilon) \\ &= 1 - P(\bigcap_{i=1}^n X_i \geq a + \varepsilon) = P(\{Y_n \geq a + \varepsilon\} \cup \{Y_n \leq a - \varepsilon\}) = P(Y_n \leq a - \varepsilon) = P(\bigcap_{i=1}^n X_i \leq a - \varepsilon) \\ &= P(\bigcap_{i=1}^n \{X_i \leq a - \varepsilon\}) = \prod_{i=1}^n P(X_i \leq a - \varepsilon) = \prod_{i=1}^n \left(\frac{a - \varepsilon}{a}\right) = \left(1 - \frac{\varepsilon}{a}\right)^n \rightarrow 0, \quad n \rightarrow \infty \end{aligned}$$

$$\text{即有: } \lim_{n \rightarrow \infty} P(|Y_n - a| \geq \varepsilon) = 0, \text{ 也即: } Y_n \xrightarrow{P} a.$$

5. (1) 如果 $X_n \xrightarrow{P} a$, 则 $\forall c \in R, cX_n \xrightarrow{P} ca$; 不妨设 $c > 0$ ($c < 0$ 时类似).

$$\forall \varepsilon > 0, \text{ 由 } P(|cX_n - ca| \geq \varepsilon) = P(c|X_n - a| \geq \varepsilon) = P(|X_n - a| \geq \frac{\varepsilon}{c}) \rightarrow 0, \quad n \rightarrow \infty$$

$$\text{即: } cX_n \xrightarrow{P} ca.$$

(2) 如果 $X_n \xrightarrow{P} X$ 且 $X_n \xrightarrow{P} Y$, 则 $P(X=Y)=1$, 即: $X=Y, a.s.$;

$$\begin{aligned} \forall \varepsilon > 0, \quad P(|X - Y| \geq \varepsilon) &= P(|X - X_n + X_n - Y| \geq \varepsilon) \leq P(\{|X - X_n| \geq \frac{\varepsilon}{2}\} \cup \{|X_n - Y| \geq \frac{\varepsilon}{2}\}) \\ &\leq P(|X_n - X| \geq \frac{\varepsilon}{2}) + P(|X_n - Y| \geq \frac{\varepsilon}{2}) \rightarrow 0 + 0 = 0, \quad n \rightarrow \infty \end{aligned}$$

由 ε 的任意性, $P(|X - Y| \neq 0) = P(X \neq Y) = 0$, 即: $P(X=Y)=1$, 也即: $X=Y, a.s.$

注: $\forall c > 0$, 若 $|a+b| > c$, 则必有 $|a| > \frac{c}{2}$ 或 $|b| > \frac{c}{2}$.

(3) 如果 $X_n \xrightarrow{P} X, Y_n \xrightarrow{P} Y$, 则 $X_n + Y_n \xrightarrow{P} X + Y$;

$$\begin{aligned} \forall \varepsilon > 0, \text{ 由 } P(|X_n + Y_n - (X + Y)| \geq \varepsilon) &= P(|(X_n - X) + (Y_n - Y)| \geq \varepsilon) \\ &\leq P(|X_n - X| \geq \frac{\varepsilon}{2}) + P(|Y_n - Y| \geq \frac{\varepsilon}{2}) \rightarrow 0 + 0 = 0, \quad n \rightarrow \infty \end{aligned}$$

$$\text{即有: } X_n + Y_n \xrightarrow{P} X + Y.$$

(4) 如果 $X_n \xrightarrow{P} X$, $g(x)$ 是直线上的连续函数, 则 $g(X_n) \xrightarrow{P} g(X)$.

这里运用到“一致连续”的性质! 将问题减弱为: $X_n \xrightarrow{P} c$, $g(x)$ 在 $x=c$ 点连续, 则 $g(X_n) \xrightarrow{P} g(c)$.
由题设, $\forall \varepsilon > 0, \exists \delta > 0$, 当 $0 < |x - c| < \delta$ 时, 必有: $|g(x) - g(c)| < \varepsilon$; 从而, 对于上述的 ε, δ , 必有: $\{|g(X_n) - g(c)| \geq \varepsilon\} \subset \{|X_n - c| \geq \delta\}$; 也即有:

$$P(|g(X_n) - g(c)| \geq \varepsilon) \leq P(|X_n - c| \geq \delta) \rightarrow 0, \quad n \rightarrow \infty, \text{ 即有:}$$

$$\lim_{n \rightarrow \infty} P(|g(X_n) - g(c)| \geq \varepsilon) = 0, \text{ 即: } g(X_n) \xrightarrow{P} g(c).$$

6. (1) 设 $\{X_n, n \geq 1\}$ 为独立随机序列, 且 $P(X_n = \pm 2^n) = \frac{1}{2^{2n+1}}, P(X_n = 0) = 1 - \frac{1}{2^{2n}},$

$n = 1, 2, \dots$; 则 $\{X_n, n \geq 1\}$ 服从大数定律;

$$\text{由 } \begin{array}{c|ccc} X_n & -2^n & 0 & 2^n \\ \hline P & \frac{1}{2^{2n+1}} & 1 - \frac{1}{2^{2n}} & \frac{1}{2^{2n+1}} \end{array}, \text{ 即有: } EX_n = 0, EX_n^2 = 1, DX_n = 1, n \geq 1; \text{ 从而,}$$

$$D(\frac{1}{n} \sum_{i=1}^n X_i) = \frac{1}{n^2} D(\sum_{i=1}^n X_i) = \frac{1}{n^2} \sum_{i=1}^n DX_i^2 = \frac{1}{n} \rightarrow 0, n \rightarrow \infty, \text{ 由 Chebyshev 不等式,}$$

$$\frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{n} \sum_{i=1}^n EX_i = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} 0, \text{ 即: } \{X_n, n \geq 1\} \text{ 服从大数定律!}$$

(2) 设随机序列 $\{X_n, n \geq 1\}$ 独立同分布, 且有共同的分布函数

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan \frac{x}{a}, \quad -\infty < x < +\infty;$$

试问: 辛钦大数定律对此随机序列是否适用?

$$\text{设 } X_n \sim f(x), \text{ 即有: } f(x) = \frac{dF(x)}{dx} = \frac{1}{\pi} \frac{a}{a^2 + x^2}, \quad -\infty < x < +\infty; \text{ 由 } \int_{-\infty}^{+\infty} |x| \cdot f(x) dx$$

$$= \int_{-\infty}^{+\infty} \frac{a \cdot |x|}{\pi(a^2 + x^2)} dx = \infty, \text{ 即有: } EX_n \text{ 不存在, 故 Khinchine 大数定律对此随机序列}$$

不适用!

7. (1) 设随机序列 $\{X_n, n \geq 1\}$ 独立同 $E(2)$ 分布, 则当 $n \rightarrow \infty$ 时,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \frac{1}{n} \sum_{i=1}^n X_i^2, \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \text{ 分别依概率收敛于什么?}$$

① 由 $\{X_n, n \geq 1\}$ i.i.d., 且 $EX_n = \frac{1}{2}, n \geq 1$, 依 Khinchine 大数定律,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \frac{1}{2};$$

② 由 $\{X_n, n \geq 1\}$ i.i.d., 且 $EX_n^2 = DX_n + (EX_n)^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$, 依 Khinchine 大数定律,

$$\frac{1}{n} \sum_{i=1}^n X_i^2 \xrightarrow{P} \frac{1}{2};$$

③ 由 $\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - (\bar{X})^2$, 及依概率收敛的性质, 基于 ①、②

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \xrightarrow{P} \frac{1}{2} - \left(\frac{1}{2}\right)^2 = \frac{1}{4},$$

(2) 设随机序列 $\{X_n, n \geq 1\}$ 独立同 $U(-1, 1)$ 分布, 试求 $\lim_{n \rightarrow \infty} P\left(\frac{\sum_{i=1}^n X_i}{\sqrt{n}} \leq 1\right)$;
 易见, $EX_n = 0, DX_n = \frac{1}{3}$; 由 Lindeberg-Levy 中心极限定理,

$$\forall x \in \mathbb{R}, \lim_{n \rightarrow \infty} P\left(\frac{\sum_{i=1}^n X_i - E(\sum_{i=1}^n X_i)}{\sqrt{D(\sum_{i=1}^n X_i)}} \leq x\right) = \lim_{n \rightarrow \infty} P\left(\frac{\sum_{i=1}^n X_i}{\sqrt{n}} \leq \frac{x}{\sqrt{1/3}}\right) = \Phi(x), \text{ 取 } x = \sqrt{3}, \text{ 即有.}$$

$$\lim_{n \rightarrow \infty} P\left(\frac{\sum_{i=1}^n X_i}{\sqrt{n}} \leq 1\right) = \Phi(\sqrt{3}).$$

(3) 设随机序列 $\{X_n, n \geq 1\}$ 独立同参数为 $\frac{1}{2}$ 的 0-1 分布, 若

$$\lim_{n \rightarrow \infty} P\left(\frac{C \sum_{i=1}^n (X_{2i} - X_{2i-1})}{\sqrt{n}} \leq x\right) = \Phi(x), \text{ 试求常数 } C.$$

易见, $X_{2i} - X_{2i-1}, i = 1, 2, \dots, n$ 独立同分布, 且 $E(X_{2i} - X_{2i-1}) = EX_{2i} - EX_{2i-1} = \frac{1}{2} - \frac{1}{2} = 0, D(X_{2i} - X_{2i-1}) = DX_{2i} + DX_{2i-1} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.

由 Lindeberg-Levy 中心极限定理, $\forall x \in \mathbb{R}$, 记: $Y_i = X_{2i} - X_{2i-1}$.

$$\lim_{n \rightarrow \infty} P\left(\frac{\sum_{i=1}^n Y_i - E(\sum_{i=1}^n Y_i)}{\sqrt{D(\sum_{i=1}^n Y_i)}} \leq x\right) = \lim_{n \rightarrow \infty} P\left(\frac{\sum_{i=1}^n Y_i}{\sqrt{n}} \leq x\right) = \Phi(x), \text{ 故常数 } C = \sqrt{2}.$$

习题 5.2

1. (3) 某餐厅每天接待 400 名顾客, 假设每位顾客的消费额 (元) 服从 $(20, 100)$

上的均匀分布, 且顾客的消费额是相互独立的, 试求:

(i) 该餐厅每天的平均营业额; 设 X_i 为第 i 位顾客的消费额, 即: $X_i \sim U(20, 100), i = 1, \dots, 400$

$$\text{即有: } E\left(\sum_{i=1}^{400} X_i\right) = \sum_{i=1}^{400} EX_i = \sum_{i=1}^{400} 60 = 24000 \text{ (元)}$$

$$\text{又 } D\left(\sum_{i=1}^{400} X_i\right) = \sum_{i=1}^{400} DX_i = 400 \times \frac{6400}{12} = 160000$$

(ii) “该餐厅每天的营业额在平均营业额 ± 760 内” 的概率.

$$P\left(\left|\sum_{i=1}^{400} X_i - E\left(\sum_{i=1}^{400} X_i\right)\right| < 760\right) = P\left(\left|\frac{\sum_{i=1}^{400} X_i - E\left(\sum_{i=1}^{400} X_i\right)}{\sqrt{D\left(\sum_{i=1}^{400} X_i\right)}}\right| \leq \frac{760}{400}\right) \approx 2\Phi(1.9) - 1 = 0.90.$$

2. (2) 独立重复地对某物体的长度 l 进行 n 次测量, 假设每次测量的结果 X_i 服

从正态分布 $N(l, 0.2^2)$; 记 $\bar{X}$ 为 n 次测量结果的算术平均值, 为保证有 95% 的把

握使平均值与实际值 l 的差异小于 0.1, 试问至少需要测量多少次?

$$\text{易见, } \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \sim N\left(l, \frac{0.04}{n}\right), \text{ 这里, } E\bar{X} = \frac{1}{n} \sum_{i=1}^n EX_i = l, D(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^n DX_i = \frac{0.04}{n};$$

$$\text{欲使 } P(|\bar{X} - l| < 0.1) = P\left(\left|\frac{\bar{X} - l}{\frac{0.2}{\sqrt{n}}}\right| < \frac{\sqrt{n}}{2}\right) = \Phi\left(\frac{\sqrt{n}}{2}\right) - \Phi\left(-\frac{\sqrt{n}}{2}\right)$$

$$= 2\Phi\left(\frac{\sqrt{n}}{2}\right) - 1 > 0.95, \text{ 即: } \Phi\left(\frac{\sqrt{n}}{2}\right) > 0.975 = \Phi(1.96), \text{ 又知 } \frac{\sqrt{n}}{2} > 1.96,$$

$$n > 15.366, \text{ 取 } n = 16, \text{ 即可!}$$

3. (1) 设有 2500 个同一年龄段和同一社会阶层的人参加了某保险公司的人寿保险, 假设在一年中每个人死亡的概率为 0.002, 每个人在年初向保险公司缴纳保费 1200 元, 而在死亡时保险受益人可以从保险公司领到保险金 200000 元, 问:

(i) “保险公司亏本”的概率是多少? 记 X 为死亡投保人数, 由题设, $X \sim B(2500, 0.002)$,

即有: $EX = 5$, $DX = 4.99$, 从而, $P(\text{保险公司亏本}) = P(2500 \times 1200 - 2 \times 10^5 X < 0) = P(X > 15)$
 $= 1 - P(X \leq 15) = 1 - P\left(\frac{X-5}{\sqrt{4.99}} \leq \frac{10}{\sqrt{4.99}}\right) \approx 1 - \Phi\left(\frac{10}{\sqrt{4.99}}\right) \approx 0.000069 \approx 0$.

(ii) “保险公司获利不少于 1000000 元”的概率是多少?

$P(\text{保险公司获利不少于 } 10^6 \text{ 元}) = P(2500 \times 1200 - 2 \times 10^5 X \geq 10^6) = P(X \leq 10) = P\left(\frac{X-5}{\sqrt{4.99}} \leq \frac{5}{\sqrt{4.99}}\right)$
 $\approx \Phi\left(\frac{5}{\sqrt{4.99}}\right) = 0.9863$.

(2) 银行为支付某日即将到期的债券须准备一笔现金, 已知这批债券共发行了 500 张, 每张须支付本息 1000 元, 假设“持券人 (一人一券) 到期日到银行领取本息”的概率为 0.4, 试问: 银行于该日应准备多少现金才能以 99.9% 的把握满足客户的兑换?

设 $X_i = \begin{cases} 1, & \text{若第 } i \text{ 个持券人到期去银行兑换;} \\ 0, & \text{否则;} \end{cases} \quad i = 1, 2, \dots, 500$; 即有: $EX_i = 0.4$, $DX_i = 0.4 \cdot 0.6 = 0.24$;

由 Lindeberg-Levy 中心极限定理, 使 $P\left(\sum_{i=1}^{500} X_i \leq x\right) = P\left(\frac{\sum_{i=1}^{500} X_i - E\left(\sum_{i=1}^{500} X_i\right)}{\sqrt{D\left(\sum_{i=1}^{500} X_i\right)}} \leq \frac{x-200}{\sqrt{120}}\right)$

$\approx \Phi\left(\frac{x-200}{\sqrt{120}}\right) \geq 0.999$, 且 $\frac{x-200}{\sqrt{120}} \geq 3.1$, 即: $x \geq 233.96$; 故银行至少应准备 234000 元.

注: 本题也可依 De Moivre-Laplace 中心极限定理计算!

4. (1) 一复杂系统由 100 个相互独立工作的部件组成, 每个部件正常工作的概率为 0.9; 已知整个系统中至少有 85 个部件正常工作, 系统才能正常工作, 试求“系统正常工作”的概率;

设 X 为正常工作的部件数, 由题设, $X \sim B(100, 0.9)$, $EX = 90$, $DX = 9$.

即有: $P(\text{系统正常工作}) = P(X \geq 85) = 1 - P(X < 85) = 1 - P\left(\frac{X-90}{\sqrt{9}} < \frac{85-90}{3}\right)$
 $\approx 1 - \Phi\left(-\frac{5}{3}\right) = \Phi\left(\frac{5}{3}\right) = 0.9525$.

注: 教材后所附习题的参考答案为 0.9664, 系用正态分布作为二项分布的近似计算中, 为提高精度所作的修正!

(2) 某车间有同型号的机床 200 台，在一小时内每台机床约有 70% 的时间是工作的；假定各机床工作是相互独立的，工作时每台机床要消耗电能 15kW，问：

至少需要多少电能，才可以有 95% 的可能性保证此车间正常生产？记 X 为同时工作的机床数，

由题设， $X \sim B(200, 0.7)$ ， $EX = 140$ ， $DX = 42$ ，由 $P(15X \leq x) = P(X \leq \frac{x}{15}) = P(\frac{X-140}{\sqrt{42}} \leq \frac{\frac{x}{15}-140}{\sqrt{42}}) \approx \Phi(\frac{\frac{x}{15}-140}{\sqrt{42}}) > 0.95 = \Phi(1.645)$ ，即： $\frac{\frac{x}{15}-140}{\sqrt{42}} > 1.645$ ， $x > 2259.5 \text{ kW}$ 。

注：教材后所附习题参考答案，仍为提高正态分布近似计算精度所作的修正！

(3) 电视台做关于某节目收视率的调查，在每天该节目播出时随机地向当地居民做电话问询；问其是否在看电视，若在看是否在看此节目；设回答在看电视的居民数为 n ，问为保证以 95% 的概率使调查误差在 10% 之内， n 应取多大？

设 X 为回答看电视的居民中收看节目的人数；由题设， $X \sim B(n, p)$ ， p 为收视率， $EX = np$ ， $DX = np(1-p)$ ，欲使 $P(|\frac{X}{n} - p| < 0.1) \geq 0.95$ ，即： $P(|\frac{X-np}{\sqrt{np(1-p)}}| < \frac{0.1n}{\sqrt{np(1-p)}}) = \frac{\sqrt{n}}{10\sqrt{p(1-p)}} \geq 1.96$ ，

$\approx 2\Phi(\frac{\sqrt{n}}{10\sqrt{p(1-p)}}) - 1 \geq 0.95$ ，只须

$\frac{\sqrt{n}}{10\sqrt{p(1-p)}} \geq 1.96$ ，即 $n \geq (19.6)^2 \cdot p(1-p)$ ；为 2、 $\forall p \in (0, 1)$ ， $p(1-p) \leq \frac{1}{4}$ ，

只须 $n \geq (19.6)^2 \cdot \frac{1}{4} = 96.04$ ，取 $n = 97$ 即可。

5. 设随机变量 $X_1, X_2, \dots, X_{100}$ 独立同 $U(0, 1)$ 分布， $Y = \prod_{i=1}^{100} X_i$ ，试由中心极限定理估计概率 $P(Y < 10^{-40})$ 。易见， $-\ln X_i \sim E(1)$ ， $-\ln Y = \sum_{i=1}^{100} (-\ln X_i)$ 。

$$\begin{aligned} P(Y < 10^{-40}) &= P(-\ln Y > 40 \ln 10) = P(\sum_{i=1}^{100} (-\ln X_i) > 40 \ln 10) \\ &= P(\frac{\sum_{i=1}^{100} (-\ln X_i) - E(\sum_{i=1}^{100} (-\ln X_i))}{\sqrt{D(\sum_{i=1}^{100} (-\ln X_i))}} > \frac{40 \ln 10 - 100}{10}) \\ &\approx 1 - \Phi(\frac{40 \ln 10 - 100}{10}) = \Phi(10 - 4 \ln 10) = \Phi(0.79) = 0.7852. \end{aligned}$$

6. 一中学有师生 1600 人，到学校食堂就餐人数约占师生总人数的 $\frac{3}{4}$ ，试由中心极限定理确定：记 X 为到食堂就餐的人数，由题设 $X \sim B(1600, \frac{3}{4})$ 。

(1) 学校食堂应最多安排多少座位，使“空座位超过 100 个”的概率不超过 0.01；
 $EX = 1600 \times \frac{3}{4} = 1200$ ， $DX = 1600 \times \frac{3}{4} \times \frac{1}{4} = 300$ ，若设安排 n 个座位，由

$$\begin{aligned} P(0 < X < n - 100) &= P(\frac{-1200}{\sqrt{300}} < \frac{X-1200}{\sqrt{300}} < \frac{n-1300}{\sqrt{300}}) \approx \Phi(\frac{n-1300}{\sqrt{300}}) - \Phi(-\frac{1200}{\sqrt{300}}) \\ &\approx \Phi(\frac{n-1300}{\sqrt{300}}) \leq 0.01, \text{ 即: } \Phi(\frac{1300-n}{\sqrt{300}}) \geq 0.99 = \Phi(2.33), n \leq 1300 - 2.33 \cdot \sqrt{300} \end{aligned}$$

(2) 在此安排下，求“有师生就餐无座位”的概率。 ≈ 1259

$$\text{由(1), } P(X > 1259) = 1 - P(X \leq 1259) = 1 - P(\frac{X-1200}{\sqrt{300}} \leq \frac{59}{\sqrt{300}})$$

$$\approx 1 - \Phi(\frac{59}{\sqrt{300}}) = 1 - \Phi(3.4) = 1 - 0.9997 = 0.0003.$$

7. 用概率论的方法证明: $\lim_{n \rightarrow \infty} \left(1 + n + \frac{n^2}{2!} + \cdots + \frac{n^n}{n!} \right) e^{-n} = \frac{1}{2}$;

设 $\{X_n, n \geq 1\}$ 独立同 $P(1)$ 分布, $EX_i = DX_i = 1$, $X_1 + X_2 + \cdots + X_n \sim P(n)$ (Poisson 分布的可加性),

由 Lindeberg-Levy 中心极限定理, $\forall x \in R$,

$$\lim_{n \rightarrow \infty} P\left(\frac{\sum_{i=1}^n X_i - E(\sum_{i=1}^n X_i)}{\sqrt{D(\sum_{i=1}^n X_i)}} \leq x\right) = \Phi(x); \text{ 取 } x=0, \text{ 即有.}$$

$$\lim_{n \rightarrow \infty} P\left(\frac{\sum_{i=1}^n X_i \leq n\right) = \Phi(0) = \frac{1}{2}; \text{ 这里, 记: } Y = \sum_{i=1}^n X_i, \text{ 由 } Y \sim P(n),$$

$$P\left(\frac{\sum_{i=1}^n X_i \leq n\right) = P(Y \leq n) = \sum_{k=0}^n P(Y=k) = \sum_{k=0}^n \frac{n^k}{k!} e^{-n}. \text{ 即有:}$$

习题 6.1

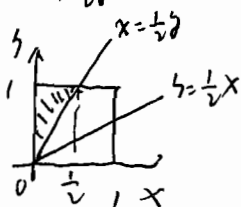
$$\lim_{n \rightarrow \infty} \left(\sum_{k=0}^n \frac{n^k}{k!} \right) e^{-n} = \frac{1}{2}.$$

2. (1) 设总体 X 的密度函数为 $f(x) = \begin{cases} 2x, & 0 < x < 1; \\ 0, & \text{其他;} \end{cases}$, (X_1, X_2) 为取自 X 的样

本, 求 $P\left(\frac{X_1}{X_2} \leq \frac{1}{2}\right)$; 易见, X_1, X_2 独立, 且与 X 同分布, 故令 $(X_1, X_2) \sim f(x, y)$, 则

$$f(x, y) = f_{X_1}(x) \cdot f_{X_2}(y) = \begin{cases} 2x \cdot 2y, & 0 < x, y < 1; \\ 0, & \text{其他;} \end{cases} \quad \text{从而, } P\left(\frac{X_1}{X_2} \leq \frac{1}{2}\right) = P(X_1 \leq \frac{1}{2} X_2)$$

$$= \iint_{x \leq \frac{1}{2}y} f(x, y) dx dy = \int_0^1 dx \int_{2x}^1 2x \cdot 2y dy = \int_0^{\frac{1}{2}} 2x \cdot (1 - 4x^2) dx = \frac{1}{4} - 2\left(\frac{1}{2}\right)^3 = \frac{1}{4} - \frac{1}{8} = \frac{1}{8}.$$



(2) 设 (X_1, X_2) 为取自 $X \sim E(\lambda)$ 的样本, $Y = \sqrt{X_1 X_2}$, 则 $E\left(\frac{4Y}{\pi}\right) = \frac{1}{\lambda}$.

易见, X_1, X_2 独立同 $E(\lambda)$ 分布, 令 $X \sim f(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0; \\ 0, & x \leq 0. \end{cases}$

$$\text{从而: } E\left(\frac{4Y}{\pi}\right) = \frac{4}{\pi} E(\sqrt{X_1 X_2}) = \frac{4}{\pi} E(\sqrt{X_1} \cdot \sqrt{X_2}) = \frac{4}{\pi} E(\sqrt{X_1}) E(\sqrt{X_2}) \text{ (独立性)}$$

$$= \frac{4}{\pi} E(\sqrt{X}) \cdot E(\sqrt{X}) = \frac{4}{\pi} [E(\sqrt{X})]^2 = \frac{4}{\pi} \left[\int_0^{\infty} \sqrt{x} \cdot f(x) dx \right]^2 = \frac{4}{\pi} \left[\int_0^{\infty} \sqrt{x} \cdot \lambda e^{-\lambda x} dx \right]^2$$

$$\stackrel{\text{(代表性)}}{=} \frac{4}{\pi} \left[\frac{1}{\lambda} \int_0^{\infty} \sqrt{\lambda x} \cdot e^{-\lambda x} d(\lambda x) \right]^2 \stackrel{\text{令 } \lambda x = t}{=} \frac{4}{\pi} \left[\frac{1}{\lambda} \cdot \int_0^{\infty} t^{\frac{1}{2}} \cdot e^{-t} dt \right]^2$$

$$= \frac{4}{\pi} \left[\frac{1}{\lambda} \Gamma\left(\frac{3}{2}\right) \right]^2 = \frac{4}{\pi} \left[\frac{1}{\lambda} \cdot \frac{1}{2} \Gamma\left(\frac{1}{2}\right) \right]^2 = \frac{4}{\pi} \left[\frac{1}{\lambda} \cdot \frac{1}{2} \cdot \sqrt{\pi} \right]^2 = \frac{1}{\lambda}.$$

习题 6.2

记总体为 X , 则: $X \sim N(8, 2^2)$
 $\frac{X-8}{2} \sim N(0, 1)$

1. (2) 设 $(X_1, X_2, \dots, X_{16})$ 是取自总体 $N(8, 2^2)$ 的样本, 试求以下概率:

$$\begin{aligned} P(X_{(16)} > 10), P(X_{(1)} > 5). \\ P(X_{(16)} > 10) &= 1 - P(X_{(16)} \leq 10) = 1 - P(\bigcap_{i=1}^{16} \{X_i \leq 10\}) = 1 - \prod_{i=1}^{16} P(X_i \leq 10) = 1 - \prod_{i=1}^{16} P(X \leq 10) \\ &= 1 - \prod_{i=1}^{16} P\left(\frac{X-8}{2} \leq \frac{2}{2}\right) = 1 - [\Phi(1)]^{16}; \quad P(X_{(1)} > 5) = P(\bigcap_{i=1}^{16} \{X_i > 5\}) = \prod_{i=1}^{16} P(X_i > 5) = \prod_{i=1}^{16} P(X > 5) \\ &= \prod_{i=1}^{16} [1 - P(X \leq 5)] = \prod_{i=1}^{16} [1 - P\left(\frac{X-8}{2} \leq \frac{-3}{2}\right)] = \prod_{i=1}^{16} [1 - \Phi(-\frac{3}{2})] = [\Phi(\frac{3}{2})]^{16}. \end{aligned}$$

2. 设 $(X_1, X_2, \dots, X_n)$ 是取自总体 $X \sim U(0, 1)$ 的样本, $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ 是样本 $(X_1, X_2, \dots, X_n)$ 的顺序统计量, 试求 $E[X_{(1)}], E[X_{(n)}]$ 以及样本极差 $X_{(n)} - X_{(1)}$ 的分布. 法1: 由 $X_{(1)}, X_{(n)}$ 的分布求 $E(X_{(1)}), E(X_{(n)})$;

法2 (微元密度法): $\forall 0 < y < x < 1$, 取 $\Delta x, \Delta y > 0$ 充分小, 且 $0 \leq y - \Delta y < y \leq x - \Delta x < x < 1$,

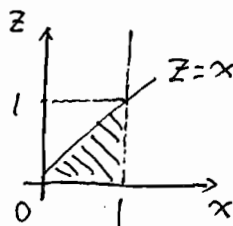
$$\begin{aligned} \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{P(x - \Delta x < X_{(n)} \leq x, y - \Delta y < X_{(1)} \leq y)}{\Delta x \cdot \Delta y} &= \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{[n \cdot (y - (y - \Delta y))] \cdot [n \cdot (x - \Delta x - y)]^{n-2} \cdot [x - (x - \Delta x)] + \Delta x \cdot \Delta y \cdot \alpha}{\Delta x \cdot \Delta y} \\ &= \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{n \cdot (n-1) \cdot \Delta x \cdot \Delta y \cdot (x - \Delta x - y)^{n-2} + \Delta x \cdot \Delta y \cdot 0}{\Delta x \cdot \Delta y} = n \cdot (n-1) \cdot (x - y)^{n-2}; \end{aligned}$$

令 $(X_{(n)}, X_{(1)}) \sim f(x, y)$, 且有: $f(x, y) = \begin{cases} n \cdot (n-1) \cdot (x-y)^{n-2}, & 0 < y < x < 1; \\ 0, & \text{其他}; \end{cases}$ $X_{(n)} \sim f_{X_{(n)}}(x), X_{(1)} \sim f_{X_{(1)}}(y)$, 即有:

$$\begin{aligned} f_{X_{(1)}}(y) &= \int_{-\infty}^{+\infty} f(x, y) dx = \begin{cases} \int_y^1 n \cdot (n-1) \cdot (x-y)^{n-2} dx, & 0 < y < 1; \\ 0, & \text{其他}; \end{cases} = \begin{cases} n \cdot (1-y)^{n-1}, & 0 < y < 1; \\ 0, & \text{其他}; \end{cases} \\ E X_{(1)} &= \int_{-\infty}^{+\infty} y \cdot f_{X_{(1)}}(y) dy = \int_0^1 y \cdot n \cdot (1-y)^{n-1} dy = \int_0^1 n \cdot (1-x) \cdot x^{n-1} dx = 1 - \frac{n}{n+1} = \frac{1}{n+1}; \\ f_{X_{(n)}}(x) &= \int_{-\infty}^{+\infty} f(x, y) dy = \int_0^x n \cdot (n-1) \cdot (x-y)^{n-2} dy = \begin{cases} n \cdot x^{n-1}, & 0 < x < 1; \\ 0, & \text{其他}; \end{cases} \\ E X_{(n)} &= \int_{-\infty}^{+\infty} x \cdot f_{X_{(n)}}(x) dx = \int_0^1 x \cdot n \cdot x^{n-1} dx = \frac{n}{n+1} \end{aligned}$$

令 $X_{(n)} - X_{(1)} \sim f(z)$, 由差密度公式, $f_{X_{(n)}-X_{(1)}}(z) = \int_{-\infty}^{+\infty} f(x, x-z) dx$

$$= \begin{cases} \int_z^1 n \cdot (n-1) \cdot z^{n-2} dx = n \cdot (n-1) \cdot z^{n-2} \cdot (1-z), & 0 < z < 1; \\ 0, & \text{其他}; \end{cases}$$



3. 设 $X_1, X_2, \dots, X_5$ 独立同分布, 且 $X_i \sim f(x) = \begin{cases} 2x, & 0 < x < 1; \\ 0, & \text{其他}; \end{cases}$; 试求:

$P(X_{(2)} < \frac{1}{2})$. 方法1: 可由联合分布求 $X_{(2)}$ 的分布, 再求 $P(X_{(2)} < \frac{1}{2})$ (见P97附页)

方法2: $P(X_i < \frac{1}{2}) = \int_{-\infty}^{\frac{1}{2}} f(x) dx = \int_0^{\frac{1}{2}} 2x dx = \frac{1}{4}$, 记 Y 为 $X_1, X_2, \dots, X_5$

~~$P(X_{(2)} < \frac{1}{2})$~~ = 中位数大于 $\frac{1}{2}$ 的概率, 易见, $Y \sim B(5, \frac{1}{4})$.

$P(X_{(2)} < \frac{1}{2}) = P(X_1, X_2, \dots, X_5 \text{ 中至少有2个大于 } \frac{1}{2}) = P(Y \geq 2)$

$$= 1 - P(Y < 2) = 1 - C_5^0 (\frac{1}{4})^0 \cdot (1 - \frac{1}{4})^5 - C_5^1 (\frac{1}{4})^1 \cdot (1 - \frac{1}{4})^4 = 1 - \frac{3^5}{4^5} - \frac{5 \cdot 3^4}{4^5}$$

$$= 1 - \frac{3^5 \cdot 8}{4^5} = 1 - \frac{3^4}{4^4} \cdot 2 = 1 - \frac{81}{128} = \frac{47}{128}.$$

习题 6.3

2. (1) 设总体 $X \sim N(\mu, 10^2)$, 现抽取一容量为 n 的样本, 样本均值记为 $\bar{X}$, 欲

使 $P(\mu - 5 < \bar{X} < \mu + 5) = 0.954$, 试问 n 取何值? 易见, $\bar{X} \sim N(\mu, \frac{10^2}{n})$. 从而,

$$P(\mu - 5 < \bar{X} < \mu + 5) = P(|\bar{X} - \mu| < 5) = P\left(\left|\frac{\bar{X} - \mu}{\frac{10}{\sqrt{n}}}\right| < \frac{\sqrt{n}}{2}\right) = \Phi\left(\frac{\sqrt{n}}{2}\right) - \Phi\left(-\frac{\sqrt{n}}{2}\right)$$

$$= 2\Phi\left(\frac{\sqrt{n}}{2}\right) - 1 = 0.954. \text{ 从而: } \Phi\left(\frac{\sqrt{n}}{2}\right) = 0.9770 = \Phi(2.0)$$

从而: $n = 16.$

(2) 从正态总体 $X \sim N(\mu, \sigma^2)$ 中抽取一个样本 $(X_1, X_2, \dots, X_n)$, 求 k 使得

$$P(\bar{X} > \mu + k S_n) = 1 - \alpha, \text{ 其中 } \alpha \text{ 很小; 这里, } S_n = \sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}. \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$\text{易见, } \bar{X} \sim N(\mu, \frac{\sigma^2}{n}), \quad \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1), \quad \frac{n S_n^2}{\sigma^2} = \frac{(n-1) S^2}{\sigma^2} \sim \chi^2(n-1). \quad \text{且}$$

$$\bar{X} \text{ 与 } S_n^2 \text{ 独立, 从而: } \frac{\bar{X} - \mu}{\frac{S_n}{\sqrt{n}}} = \frac{\bar{X} - \mu}{S_n} \cdot \sqrt{n-1} \sim t(n-1), \quad \text{从而}$$

$$P(\bar{X} > \mu + k S_n) = P\left(\frac{\bar{X} - \mu}{S_n} > k\right) = P\left(\frac{\bar{X} - \mu}{S_n} \cdot \sqrt{n-1} > k \cdot \sqrt{n-1}\right) = 1 - \alpha, \quad \text{从而}$$

$$k \cdot \sqrt{n-1} = t_{1-\alpha}(n-1), \quad \text{从而: } k = \frac{1}{\sqrt{n-1}} \cdot t_{1-\alpha}(n-1).$$

3. 设在正态总体 $X \sim N(\mu, \sigma^2)$ 中抽取一容量为 n 的样本 $(X_1, X_2, \dots, X_n)$, μ, σ^2 未知; (1) 求 $E(S^2), D(S^2)$; 由 $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$. 证:

$$E\left[\frac{(n-1)S^2}{\sigma^2}\right] = n-1, \quad D\left[\frac{(n-1)S^2}{\sigma^2}\right] = 2(n-1). \text{ 也证得:}$$

$$E S^2 = \sigma^2, \quad D S^2 = \frac{2\sigma^4}{n-1} \quad (2) \text{ 当 } n=16 \text{ 时, 求 } P\left(\frac{S^2}{\sigma^2} \leq 2.04\right); \text{ 这里, } S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

$$\text{当 } n=16, \quad \frac{(n-1)S^2}{\sigma^2} = \frac{15S^2}{\sigma^2} \sim \chi^2(15). \text{ 由 } P\left(\frac{15S^2}{\sigma^2} > 15 \cdot 2.04 = 30.6\right) = 0.01$$

$$\text{证得: } P\left(\frac{S^2}{\sigma^2} \leq 2.04\right) = P\left(\frac{15S^2}{\sigma^2} \leq 30.6\right) = 1 - 0.01 = 0.99.$$

4. (1) 设 $(X_1, X_2, \dots, X_{10})$ 是取自正态总体 $X \sim N(1, \sigma^2)$ 的简单随机样本, $\bar{X}$ 为样

本均值, S 为样本标准差; 若 $P(\bar{X} \leq 1, S^2 \leq \sigma^2) = \frac{1}{3}$, 试求 $P(S^2 \leq \sigma^2)$;

易知, $\bar{X}$ 与 S^2 独立, 且 $\bar{X} \sim N(1, \frac{\sigma^2}{10})$. 证得: $P(\bar{X} \leq 1) = \frac{1}{2}$, 且

$$P(\bar{X} \leq 1, S^2 \leq \sigma^2) = P(\bar{X} \leq 1) \cdot P(S^2 \leq \sigma^2) = \frac{1}{3}. \text{ 证得: } P(S^2 \leq \sigma^2) = \frac{2}{3}.$$

(2) 设 $(X_1, X_2, \dots, X_n)$ 是取自正态总体 $X \sim N(\mu, \sigma^2)$ 的样本, $\bar{X}$ 为样本均值,

若 $P(|X - \mu| < a) = P(|\bar{X} - \mu| < b)$, 试求 $\frac{a}{b}$. $\frac{X - \mu}{\sigma} \sim N(0, 1)$.

$$P(|X - \mu| < a) = P\left|\frac{X - \mu}{\sigma}\right| < \frac{a}{\sigma} \quad \text{由 } \bar{X} \sim N(\mu, \frac{\sigma^2}{n})$$

$$P(|\bar{X} - \mu| < b) = P\left|\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}\right| < \frac{b}{\frac{\sigma}{\sqrt{n}}} \quad \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

$$\text{证得: } \frac{a}{\sigma} = \frac{b}{\frac{\sigma}{\sqrt{n}}}, \quad \frac{a}{b} = \sqrt{n}.$$

5. 设 $(X_1, X_2, \dots, X_n)$ 是取自正态总体 $X \sim N(\mu, \sigma^2)$ 的样本, $\bar{X}$ 为样本均值, S^2 为

样本方差, 计算: $E[\bar{X} \cdot S^2], D\left[(\bar{X} - \mu)^2 + \left(1 - \frac{1}{n}\right)S^2\right]$.

由 $\bar{X}$ 与 S^2 独立, 证得: $\bar{X}^2$ 与 $(S^2)^2$ 独立, 从而 $E[\bar{X} \cdot S^2] = E(\bar{X} \cdot S^2) = E\bar{X} \cdot E S^2$,
由 $E\bar{X} = \mu, D\bar{X} = \frac{\sigma^2}{n}$, 证得: $E\bar{X}^2 = D\bar{X} + (E\bar{X})^2 = \frac{\sigma^2}{n} + \mu^2$,

$$\text{由 } \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1), \text{ 且 } D\left[\frac{(n-1)S^2}{\sigma^2}\right] = 2(n-1), \text{ 证得: } D S^2 = \frac{2\sigma^4}{n-1}.$$

$$E S^4 = E(S^2)^2 = (E S^2)^2 + D S^2 = (\sigma^2)^2 + \frac{2\sigma^4}{n-1} = \frac{n+1}{n-1} \sigma^4.$$

$$D[(\bar{X} - \mu)^2 + (1 - \frac{1}{n})S^2] = D(\bar{X} - \mu)^2 + D[(1 - \frac{1}{n})S^2] = D(\bar{X} - \mu)^2 + (1 - \frac{1}{n})^2 \cdot D S^2 \quad 98$$

$$\text{由 } \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1), \quad \frac{(\bar{X} - \mu)^2}{\frac{\sigma^2}{n}} \sim \chi^2(1). \quad D\left[\frac{(\bar{X} - \mu)^2}{\frac{\sigma^2}{n}}\right] = 2, \quad D(\bar{X} - \mu)^2 = \frac{2}{n^2} \sigma^4.$$

6. 设 $(X_1, X_2, \dots, X_{10})$ 是取自正态总体 $X \sim N(\mu, 0.5^2)$ 的简单随机样本, $\bar{X}$ 为样本

均值; (1) 若 $\mu=0$, 求 $P\left(\sum_{i=1}^{10} X_i^2 \geq 4\right)$; (2) 若 μ 未知, 求 $P\left(\sum_{i=1}^{10} (X_i - \bar{X})^2 \geq 2.85\right)$.

(1) $\mu=0$, 由 $X_i \sim N(0, 0.5)$, $i=1, 2, \dots, 10$ 独立
同 $N(0, 1)$ 分布, 即 $\sum_{i=1}^{10} (2X_i)^2 \sim \chi^2(10)$, 从而,

$$P\left(\sum_{i=1}^{10} X_i^2 \geq 4\right) = P\left(\sum_{i=1}^{10} (2X_i)^2 \geq 16\right)$$

$$= 0.1 \text{ (查表)}$$

$$\text{(查表)} \quad \chi_{0.1}^2(10) = 15.987$$

$$(2) \text{ 由 } \frac{(10-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^{10} (X_i - \bar{X})^2}{0.5^2} \sim \chi^2(9).$$

$$\text{即有: } P\left(\sum_{i=1}^{10} (X_i - \bar{X})^2 \geq 2.85\right)$$

$$= P\left(\frac{\sum_{i=1}^{10} (X_i - \bar{X})^2}{0.5^2} \geq 11.4\right) = 0.25$$

$$\text{(查表, } \chi_{0.25}^2(9) = 11.389)$$

8. 设总体 $X \sim N(\mu, \sigma^2)$, $(X_1, X_2, \dots, X_{2n})$ 为取自 X 的样本, $\bar{X} = \frac{1}{2n} \sum_{i=1}^{2n} X_i$, 试求

统计量 $T = \sum_{i=1}^n (X_i + X_{i+n} - 2\bar{X})^2$ 的数学期望. $E T = E\left\{\sum_{i=1}^n [(X_i - \bar{X}) + (X_{i+n} - \bar{X})]^2\right\}$

$$= \sum_{i=1}^n E(X_i - \bar{X})^2 + \sum_{i=1}^n E(X_{i+n} - \bar{X})^2 + 2 \sum_{i=1}^n [E(X_i X_{i+n} - X_i \bar{X} - X_{i+n} \bar{X} + \bar{X}^2)]$$

$$= \sum_{i=1}^{2n} E(X_i - \bar{X})^2 + 2 \sum_{i=1}^n E(X_i X_{i+n}) - 2 E\left(\sum_{i=1}^n X_i \bar{X}\right) - 2 E\left(\sum_{i=1}^n X_{i+n} \bar{X}\right) + 2n E(\bar{X}^2)$$

$$= (2n-1) \cdot E\left[\frac{1}{2n-1} \sum_{i=1}^{2n} (X_i - \bar{X})^2\right] + 2 \sum_{i=1}^n E X_i \cdot E X_{i+n} - 2 E\left(\sum_{i=1}^n X_i \bar{X}\right) + 2n E(\bar{X}^2)$$

$$= (2n-1) \sigma^2 + 2n \mu^2 - 4n \cdot E(\bar{X}^2) + 2n \cdot E(\bar{X}^2)$$

$$= (2n-1) \sigma^2 + 2n \mu^2 - 2n \cdot E(\bar{X}^2) = (2n-1) \sigma^2 + 2n \mu^2 - 2n \left(\mu^2 + \frac{\sigma^2}{2n}\right)$$

$$= 2(n-1) \sigma^2,$$

9. 设 (X_1, X_2) 是取自正态总体 $N(0, \sigma^2)$ 的样本,

(1) 求 $\frac{(X_1 - X_2)^2}{(X_1 + X_2)^2}$ 的分布; 易见, $(X_1 - X_2)$ 与 $(X_1 + X_2)$ 独立同 $N(0, 2\sigma^2)$ 分布,

$$\text{从而, } \frac{(X_1 - X_2)^2}{(X_1 + X_2)^2} = \frac{\left(\frac{X_1 - X_2}{\sqrt{2}\sigma}\right)^2}{\left(\frac{X_1 + X_2}{\sqrt{2}\sigma}\right)^2} \sim F(1, 1),$$

$$\text{记: } F = \frac{(X_1 - X_2)^2}{(X_1 + X_2)^2}, \text{ 由 } P\left(\frac{(X_1 + X_2)^2}{(X_1 + X_2)^2 + (X_1 - X_2)^2} > k\right)$$

$$= P\left(\frac{1}{1+F} > k\right) = P\left(1+F < \frac{1}{k}\right) = P\left(F < \frac{1}{k} - 1\right) = P\left(\frac{1}{F} > \frac{1}{\frac{1}{k} - 1}\right) = 0.1,$$

(2) 求常数 k , 使得 $P\left(\frac{(X_1 + X_2)^2}{(X_1 + X_2)^2 + (X_1 - X_2)^2} > k\right) = 0.1$. 且 $F_{0.1}(1, 1) = 39.86$, 记 $\frac{1}{F} \sim F(1, 1)$.

$$P\left(\frac{1}{F} > 39.86\right) = 0.1, \text{ 即有: } \frac{1}{\frac{1}{k} - 1} = 39.86, \quad k = 0.976.$$

习题 7.1

1. 设总体 X 的概率密度为 $f(x; \theta) = \begin{cases} \frac{3x^2}{\theta^3}, & 0 < x < \theta; \\ 0, & \text{其他;} \end{cases}$, 其中 $\theta \in (0, +\infty)$ 为未知参

数, (X_1, X_2, X_3) 为取自总体 X 的简单随机样本, 令 $T = \max\{X_1, X_2, X_3\} = X_{(3)}$,

(1) 求 T 的概率密度; (2) 确定 a , 使得 $E(aT) = \theta$. (1) 易见, $R(T) = (0, \theta)$; 故 $\forall t \leq 0$,

$$\begin{aligned} \bar{F}_T(t) &= P(T \leq t) = 0; \quad \forall t > \theta, \bar{F}_T(t) = P(T \leq t) = 1; \quad \forall t \in (0, \theta), \bar{F}_T(t) = P(T \leq t) = P(\bigvee_{i=1}^3 X_i \leq t) = P(\bigcap_{i=1}^3 \{X_i \leq t\}) \\ &= \prod_{i=1}^3 P(X_i \leq t) = \left(\frac{3}{\theta^3} \int_0^t x^2 dx\right)^3 = \left(\frac{3}{\theta^3} \cdot \frac{t^3}{3}\right)^3 = \left(\frac{t}{\theta}\right)^9. \quad \text{即: } \bar{F}_T(t) = \begin{cases} 0, & t \leq 0; \\ \left(\frac{t}{\theta}\right)^9, & 0 < t < \theta; \\ 1, & t \geq \theta; \end{cases} \end{aligned}$$

$$= \begin{cases} 0, & t \leq 0; \\ \frac{9}{\theta} \left(\frac{t}{\theta}\right)^8, & 0 < t < \theta; \\ 0, & \text{其他;} \end{cases} \quad (2) E(aT) = aET = a \cdot \int_{-\infty}^{+\infty} t f_T(t) dt = a \cdot \int_0^\theta t \cdot \frac{9}{\theta} \left(\frac{t}{\theta}\right)^8 dt$$

$$= a \cdot \theta \int_0^1 x^9 dx = a \cdot \theta \cdot \frac{1}{10} = \frac{a\theta}{10} = \theta. \quad \text{从而: } a = \frac{10}{9}.$$

2. 设总体 X 的概率密度为 $f(x; \theta) = \begin{cases} \frac{2x}{3\theta^2}, & \theta < x < 2\theta; \\ 0, & \text{其他;} \end{cases}$, 其中 θ 是未知参数, 设

$(X_1, X_2, \dots, X_n)$ 为取自总体 X 的简单随机样本, 若 $E\left(c \sum_{i=1}^n X_i^2\right) = \theta^2$, 试求 c .

$$\begin{aligned} \text{由 } E\left(c \cdot \sum_{i=1}^n X_i^2\right) &= c \cdot \sum_{i=1}^n E(X_i^2) = c \cdot \sum_{i=1}^n E(X^2) = nc \cdot E(X^2) = nc \cdot \int_{-\infty}^{+\infty} x^2 \cdot f(x; \theta) dx \\ &= nc \cdot \int_\theta^{2\theta} x^2 \cdot \frac{2x}{3\theta^2} dx = nc \cdot \frac{2}{3\theta^2} \cdot \frac{1}{4} [(2\theta)^4 - \theta^4] = \frac{5}{2} nc \cdot \theta^2 = \theta^2. \end{aligned}$$

$$\text{从而: } c = \frac{2}{5n}.$$

5. 设 $(X_1, X_2, \dots, X_n) (n > 2)$ 是取自总体 $X \sim N(0, \sigma^2)$ 的一个样本, $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$,

$Y_i = X_i - \bar{X}, i=1, 2, \dots, n$; 试求: $D(Y_i), i=1, 2, \dots, n$ 及 $Cov(Y_i, Y_n)$; 若 $C(Y_1 + Y_n)^2$ 是 σ^2 的无偏估计量, 求常数 C .

$$\textcircled{1} DY_i = D(X_i - \bar{X}) = D[-\frac{1}{n}X_1 - \dots - \frac{1}{n}X_{i-1} + (1-\frac{1}{n})X_i - \frac{1}{n}X_{i+1} - \dots - \frac{1}{n}X_n] = \frac{1}{n^2}DX_1 + \dots + \frac{1}{n^2}DX_{i-1} + (1-\frac{1}{n})^2DX_i + \frac{1}{n^2}DX_{i+1} + \dots + \frac{1}{n^2}DX_n = \sigma^2 \cdot [\frac{1}{n^2} \times (n-1) + (1-\frac{1}{n})^2] = (1-\frac{1}{n})\sigma^2;$$

$$\textcircled{2} Cov(Y_i, Y_n) = Cov(X_i - \bar{X}, X_n - \bar{X}) = Cov(X_i, X_n) - Cov(X_i, \bar{X}) - Cov(X_n, \bar{X}) + Cov(\bar{X}, \bar{X}) = 0 - 2Cov(X_i, \bar{X}) + D\bar{X} = -2 \sum_{j=1}^n \frac{1}{n} Cov(X_i, X_j) + \frac{1}{n}\sigma^2 = -\frac{1}{n}\sigma^2;$$

$$\textcircled{3} \text{由 } E[C(Y_1 + Y_n)^2] = C \cdot E[(X_1 + X_n - 2\bar{X})^2] = C \cdot E[Y_1^2 + Y_n^2 + 2Y_1Y_n] = C[DY_1 + DY_n + 2Cov(Y_1, Y_n)] = C \cdot [(1-\frac{1}{n})\sigma^2 + (1-\frac{1}{n})\sigma^2 + (-\frac{2}{n}\sigma^2)] = 2C \cdot (1-\frac{2}{n})\sigma^2 = \sigma^2, \text{ 即有:}$$

$$C = \frac{n}{2(n-2)}.$$

6. (1) 设总体 $X \sim N(\mu, \sigma^2)$, $(X_1, X_2, \dots, X_n)$ 为 X 的一个样本, 试确定常数 C ,

$$\text{使得 } C \sum_{i=1}^{n-1} (X_{i+1} - X_i)^2 \text{ 为 } \sigma^2 \text{ 的无偏估计; 由 } E[C \sum_{i=1}^{n-1} (X_{i+1} - X_i)^2] = C \cdot \sum_{i=1}^{n-1} [EX_{i+1}^2 + EX_i^2 - 2EX_i \cdot EX_{i+1}] = C \cdot [\sum_{i=1}^{n-1} EX^2 + \sum_{i=1}^{n-1} EX^2 - 2 \sum_{i=1}^{n-1} EX \cdot EX] = C \cdot [2(n-1) \cdot (\mu^2 + \sigma^2) - 2(n-1)\mu^2] = C \cdot 2(n-1) \cdot \sigma^2 = \sigma^2, \text{ 即有: } C = \frac{1}{2(n-1)}.$$

(2) 设 $(X_1, X_2, \dots, X_n)$ 是正态总体 $N(\mu, \sigma^2)$ 的一个简单随机样本, 试求常数 k ,

使得 $\sigma = k \sum_{i=1}^n \sum_{j=1}^n |X_i - X_j|$ 为 σ 的无偏估计.

$$\text{由 } E\sigma = k \cdot E[2 \sum_{1 \leq i < j \leq n} \frac{|X_i - X_j|}{\sqrt{2}}] \cdot \sqrt{2}\sigma, \text{ 且 } X_i - X_j \sim N(0, 2\sigma^2), i < j:$$

$$\text{从而, } E\sigma = 2k \cdot [\sum_{1 \leq i < j \leq n} E \frac{|X_i - X_j|}{\sqrt{2}}] \cdot \sqrt{2}\sigma = 2k \cdot \sum_{1 \leq i < j \leq n} \frac{\sqrt{2}}{2} \cdot \sqrt{2}\sigma$$

$$= 2k \cdot \frac{n(n-1)}{2} \cdot \frac{\sqrt{2}}{2} \cdot \sqrt{2}\sigma = \frac{2k \cdot n \cdot (n-1)}{2} \sigma = \sigma, \text{ 即有 } k = \frac{\sqrt{2}}{2n(n-1)}; \text{ 其中,}$$

$$\text{若 } Y = \frac{X_i - X_j}{\sqrt{2}} \sim N(0, 1), \text{ 则 } E|Y| = \int_{-\infty}^{+\infty} |y| \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy = 2 \cdot \int_0^{+\infty} y \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy$$

$$= \sqrt{\frac{2}{\pi}} \cdot (-e^{-\frac{y^2}{2}}) \Big|_0^{+\infty} = \sqrt{\frac{2}{\pi}}.$$

$$(1) \forall x \in \mathbb{R}, F_{X_{(3)}}(x) = P(X_{(3)} \leq x) = \sum_{i=1}^3 P(X_i \leq x) = \begin{cases} 0, & x \leq 0; \\ \frac{x}{\theta}, & 0 < x < \theta; \\ 1, & x \geq \theta; \end{cases} \quad \text{令 } X_{(3)} \sim f_{X_{(3)}}(x), \text{ 则有:}$$

$$f_{X_{(3)}}(x) = \frac{d}{dx} F_{X_{(3)}}(x) = \begin{cases} \frac{3}{\theta} \left(\frac{x}{\theta}\right)^2, & 0 < x < \theta; \\ 0, & \text{其他}; \end{cases} \quad \text{从而, } E\left[\frac{4}{3}X_{(3)}\right] = \frac{4}{3}E[X_{(3)}] = \frac{4}{3} \cdot \int_{-\infty}^{+\infty} x f_{X_{(3)}}(x) dx = \frac{4}{3} \int_0^{\theta} x \cdot \frac{3}{\theta} \left(\frac{x}{\theta}\right)^2 dx = \theta;$$

7. (1) 设 X_1, X_2, X_3 独立同均匀分布 $U(0, \theta)$, 试证: $\frac{4}{3}X_{(3)}$ 与 $4X_{(1)}$ 均是 θ 的无偏估计量, 哪个更有效?

$$\forall x \in \mathbb{R}, F_{X_{(1)}}(x) = P(X_{(1)} \leq x) = 1 - P(X_{(1)} > x) = 1 - \prod_{i=1}^3 P(X_i > x) = \begin{cases} 0, & x \leq 0; \\ 1 - \left(1 - \frac{x}{\theta}\right)^3, & 0 < x < \theta; \\ 1, & x \geq \theta; \end{cases} \quad \text{令 } X_{(1)} \sim f_{X_{(1)}}(x)$$

$$\text{则有: } f_{X_{(1)}}(x) = \frac{d}{dx} F_{X_{(1)}}(x) = \begin{cases} \frac{3}{\theta} \left(1 - \frac{x}{\theta}\right)^2, & 0 < x < \theta; \\ 0, & \text{其他}; \end{cases} \quad \text{从而, } E[4X_{(1)}] = 4E[X_{(1)}] = 4 \cdot \int_{-\infty}^{+\infty} x f_{X_{(1)}}(x) dx$$

$$= 4 \cdot \int_0^{\theta} x \cdot \frac{3}{\theta} \left(1 - \frac{x}{\theta}\right)^2 dx = \theta. \quad \text{即: } \frac{4}{3}X_{(3)}, 4X_{(1)} \text{ 均为 } \theta \text{ 的无偏估计量! 由}$$

$$E\left[\frac{4}{3}X_{(3)}\right]^2 = \frac{16}{9}E[X_{(3)}^2] = \frac{16}{9} \int_0^{\theta} x^2 \cdot \frac{3}{\theta} \left(\frac{x}{\theta}\right)^2 dx = \frac{16}{3}\theta^2 \int_0^1 t^4 dt = \frac{16}{15}\theta^2.$$

$$E[4X_{(1)}]^2 = 16 \cdot E[X_{(1)}^2] = 16 \cdot \int_0^{\theta} x^2 \cdot \frac{3}{\theta} \left(1 - \frac{x}{\theta}\right)^2 dx = 48\theta^2 \int_0^1 t^2(1-t)^2 dt = \frac{8}{5}\theta^2 > \frac{16}{15}\theta^2$$

$$\text{即: } D[4X_{(1)}] > D\left[\frac{4}{3}X_{(3)}\right], \text{ 故 } \frac{4}{3}X_{(3)} \text{ 比 } 4X_{(1)} \text{ 更有效!}$$

习题 7.2

1. 设总体 X 服从对数正态分布 $LN(\mu, \sigma^2)$, 即 X 的概率密度函数为:

$$f(x; \mu, \sigma^2) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma x} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}, & x > 0; \\ 0, & x \leq 0; \end{cases}$$

其中 $\mu \in (-\infty, +\infty)$, $\sigma^2 \in (0, +\infty)$ 均未知;

(1) 求 μ, σ^2 的矩估计量: 设 $(X_1, X_2, \dots, X_n)$ 为取自 X 的一个样本, $(x_1, x_2, \dots, x_n)$ 为样本值,

$$EX = \int_{-\infty}^{+\infty} x f(x; \mu, \sigma^2) dx = \frac{1}{\sqrt{2\pi}\sigma} \int_0^{+\infty} x \cdot e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}} dx \quad \frac{\ln x - \mu}{\sigma} = t \quad \frac{1}{\sqrt{2\pi}} e^{\mu} \cdot \int_{-\infty}^{+\infty} e^{-\frac{t^2}{2}} \cdot e^{\sigma t} dt =$$

$$\frac{1}{\sqrt{2\pi}} e^{\mu + \frac{\sigma^2}{2}} \cdot \int_{-\infty}^{+\infty} e^{-\frac{t^2 - 2\sigma t + \sigma^2}{2}} dt \quad \text{令 } u = t - \sigma \quad \frac{1}{\sqrt{2\pi}} e^{\mu + \frac{\sigma^2}{2}} \cdot \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du = e^{\mu + \frac{\sigma^2}{2}};$$

$$EX^2 = \int_{-\infty}^{+\infty} x^2 f(x; \mu, \sigma^2) dx = \frac{1}{\sqrt{2\pi}\sigma} \int_0^{+\infty} x^2 \cdot e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}} dx \quad \frac{\ln x - \mu}{\sigma} = t \quad \frac{1}{\sqrt{2\pi}} e^{2\mu} \cdot \int_{-\infty}^{+\infty} e^{-\frac{t^2}{2}} \cdot e^{2\sigma t} dt =$$

$$= e^{2\mu + 2\sigma^2} \cdot \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(t-2\sigma)^2}{2}} dt \quad \frac{t-2\sigma}{\sigma} = u \quad e^{2\mu + 2\sigma^2} \cdot \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du = e^{2\mu + 2\sigma^2}.$$

$$\text{由上述, } \mu + \frac{\sigma^2}{2} = \ln(EX), \quad 2\mu + 2\sigma^2 = \ln(EX^2). \quad \text{即有: } \mu = 2 \cdot \ln(EX) - \frac{1}{2} \ln(EX^2),$$

$$\sigma^2 = \ln(EX^2) - 2 \ln(EX), \text{ 用 } \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \text{ 替换 } EX, \text{ 用 } \frac{1}{n} \sum_{i=1}^n X_i^2 \text{ 替换 } EX^2, \text{ 即有:}$$

$$\hat{\mu}_{ME} = 2 \cdot \ln(\bar{X}) - \frac{1}{2} \ln\left(\frac{1}{n} \sum_{i=1}^n X_i^2\right), \quad \hat{\sigma}_{ME}^2 = \ln\left(\frac{1}{n} \sum_{i=1}^n X_i^2\right) - 2 \cdot \ln(\bar{X}).$$

(2) 求 μ, σ^2 的最大似然估计量: 考虑 μ, σ^2 的似然函数 $L(\mu, \sigma^2) = \prod_{i=1}^n f(x_i; \mu, \sigma^2)$

$$= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(\ln x_i - \mu)^2}{2\sigma^2}}, x_i > 0; \text{ 即有: } \ln L(\mu, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (\ln x_i - \mu)^2$$

令 $\begin{cases} \frac{\partial \ln L(\mu, \sigma^2)}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (\ln x_i - \mu) = 0, \\ \frac{\partial \ln L(\mu, \sigma^2)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\sum_{i=1}^n (\ln x_i - \mu)^2}{2\sigma^4} = 0, \end{cases}$ 即有: $\begin{cases} \hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n \ln x_i, \\ \hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (\ln x_i - \hat{\mu}_{MLE})^2 = \frac{1}{n} \sum_{i=1}^n \ln^2 x_i - \hat{\mu}_{MLE}^2 \end{cases}$

从而, μ, σ^2 的最大似然估计量为: $\hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n \ln x_i$,

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n \ln^2 x_i - \hat{\mu}_{MLE}^2.$$

(3) 求 EX, DX 的最大似然估计量.

由 $EX = e^{\mu + \frac{\sigma^2}{2}}$, $DX = EX^2 - (EX)^2 = e^{2\mu + 2\sigma^2} - (e^{\mu + \frac{\sigma^2}{2}})^2 = e^{2\mu + \sigma^2} (e^{\sigma^2} - 1)$,

依最大似然估计的不变性, $\hat{EX} = e^{\hat{\mu}_{MLE} + \frac{1}{2} \hat{\sigma}_{MLE}^2}$,

$$\hat{DX} = e^{2\hat{\mu}_{MLE} + \hat{\sigma}_{MLE}^2} (e^{\hat{\sigma}_{MLE}^2} - 1).$$

2. 设总体 X 的概率密度为: $f(x; \theta, \lambda) = \begin{cases} \theta e^{-\theta(x-\lambda)}, & x > \lambda; \\ 0, & x \leq \lambda; \end{cases}$ 其中 $\lambda \in R, \theta > 0$ 均

未知, 试求 θ, λ 的最大似然估计量.

设 $(X_1, X_2, \dots, X_n)$ 为取自 X 的样本, $(x_1, x_2, \dots, x_n)$ 为样本值; 考虑 θ, λ 的似然函数

$$L(\theta, \lambda) = \prod_{i=1}^n f(x_i; \theta, \lambda) = \prod_{i=1}^n \theta \cdot e^{-\theta(x_i - \lambda)} = \theta^n \cdot e^{-\theta(\sum_{i=1}^n x_i - n\lambda)}, x_i > \lambda, i=1, 2, \dots, n;$$

$$= \theta^n \cdot e^{-\theta(\sum_{i=1}^n x_i)} \cdot e^{n\theta\lambda}, x_{(1)} > \lambda, x_{(1)} = \min\{x_1, x_2, \dots, x_n\},$$

$$\ln L(\theta, \lambda) = n \ln \theta - \theta(\sum_{i=1}^n x_i) + n\theta\lambda, x_{(1)} > \lambda$$

由 $\frac{\partial \ln L(\theta, \lambda)}{\partial \lambda} = n\theta > 0$, 即有 λ 的最大似然估计值 $\hat{\lambda}_{MLE} = x_{(1)}$; 令

$$\frac{d \ln L(\theta, \hat{\lambda}_{MLE})}{d\theta} = \frac{n}{\theta} - \sum_{i=1}^n x_i + n\hat{\lambda}_{MLE} = 0, \text{ 即有: } \hat{\theta}_{MLE} = \frac{1}{\bar{x} - \hat{\lambda}_{MLE}} = \frac{1}{\bar{x} - x_{(1)}}, \text{ 这里,}$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \text{ 且 } \left. \frac{d^2 \ln L(\theta, \hat{\lambda}_{MLE})}{d\theta^2} \right|_{\theta = \hat{\theta}_{MLE}} = -\frac{n}{\hat{\theta}_{MLE}^2} < 0; \text{ 故 } \theta, \lambda \text{ 的最大似然估计量分}$$

$$\text{别为: } \hat{\theta}_{MLE} = \frac{1}{\bar{x} - x_{(1)}}, \hat{\lambda}_{MLE} = x_{(1)}, x_{(1)} = \min\{x_1, x_2, \dots, x_n\},$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

3. (1) 设 $(X_1, X_2, \dots, X_n)$ 是取自总体 X 的样本, $(x_1, x_2, \dots, x_n)$ 为样本值,

① 求 θ 的矩估计: $EX = 0$ 与 θ 无关, $DX = \frac{\theta^2}{3}$, $\theta = \sqrt{3DX}$, 用 $S_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ 替换 DX ,

3. (1) 设总体 $X \sim U[-\theta, \theta]$, $\theta > 0$ 未知; 试求 θ 的矩估计与最大似然估计;

即得 $\hat{\theta}_{ME} = \sqrt{3} S_n$. 或由 $EX^2 = \frac{\theta^2}{3}$, $\theta = \sqrt{3EX^2}$, 用 $\frac{1}{n} \sum_{i=1}^n X_i^2$ 替换 EX^2 ,

也可得 $\hat{\theta}_{ME} = \sqrt{\frac{3}{n} \sum_{i=1}^n X_i^2}$.

② 求 θ 的最大似然估计: 设 $X \sim f(x; \theta) = \begin{cases} \frac{1}{2\theta}, & -\theta \leq x \leq \theta; \\ 0, & \text{其他.} \end{cases}$ 考虑 θ 的似然函数

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta) = \left(\frac{1}{2\theta}\right)^n, \quad -\theta \leq x_i \leq \theta$$

$$= \left(\frac{1}{2\theta}\right)^n, \quad -\theta \leq x_{(1)} \leq x_{(n)} \leq \theta, |x_i| \leq \theta$$

$$= \left(\frac{1}{2\theta}\right)^n, \quad \max\{|x_1|, |x_2|, \dots, |x_n|\} \leq \theta, \quad \text{由 } \frac{dL(\theta)}{d\theta} < 0, \text{ 即有:}$$

$$\hat{\theta}_{MLE} = \max\{|x_1|, |x_2|, \dots, |x_n|\}; \quad \text{即最大似然估计量 } \hat{\theta}_{MLE} = \max\{|x_1|, |x_2|, \dots, |x_n|\}$$

(2) 设总体 $X \sim U[\theta_1, \theta_1 + \theta_2]$, $\theta_1, \theta_2 > 0$ 为未知参数, 试求参数 θ_1, θ_2 的矩估计与

最大似然估计. 设 $(X_1, X_2, \dots, X_n)$ 为取自 X 的样本, $(x_1, x_2, \dots, x_n)$ 为样本值,

① 求矩估计: 由 $EX = \theta_1 + \frac{\theta_2}{2}$, $DX = \frac{\theta_2^2}{12}$, 即有: $\theta_1 = EX - \sqrt{3DX}$, $\theta_2 = 2\sqrt{3DX}$; 用 $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ 替换 EX , 用 $S_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ 替换 DX , 即有: $\hat{\theta}_{1ME} = \bar{X} - \sqrt{3} S_n$, $\hat{\theta}_{2ME} = 2\sqrt{3} S_n$;

② 求最大似然估计: 设 $X \sim f(x; \theta_1, \theta_2) = \begin{cases} \frac{1}{\theta_2}, & \theta_1 \leq x \leq \theta_1 + \theta_2; \\ 0, & \text{其他.} \end{cases}$ 考虑 θ_1, θ_2 的似然函数

$$L(\theta_1, \theta_2) = \prod_{i=1}^n f(x_i; \theta_1, \theta_2) = \frac{1}{\theta_2^n}, \quad \theta_1 \leq x_i \leq \theta_1 + \theta_2$$

$$= \frac{1}{\theta_2^n}, \quad \theta_1 \leq x_{(1)}, \theta_2 \geq x_{(n)} - \theta_1$$

4 设离散型总体 X 有如下分布: $X \sim \begin{pmatrix} 0 & 1 & 2 & 3 \\ \theta^2 & 2\theta(1-\theta) & \theta^2 & 1-2\theta \end{pmatrix}$, 其中

$\theta \in \Theta = \left(0, \frac{1}{2}\right)$ 是未知参数; 由总体得如下样本值: 3, 1, 3, 0, 3, 1, 2, 3,

求未知参数 θ 的矩估计与最大似然估计. 设 $(X_1, X_2, X_3, X_4, X_5, X_6, X_7, X_8)$ 为取自 X 的样本,

$(x_1, x_2, \dots, x_8) = (3, 1, \dots, 3)$ 为样本值;

(1) 求矩估计 $\hat{\theta}_{ME}$: 由 $EX = 1 \cdot 2\theta(1-\theta) + 2 \cdot \theta^2 + 3 \cdot (1-2\theta) = 3 - 4\theta$, 即: $\theta = \frac{3-EX}{4}$, 用 $\bar{X} =$

$\frac{1}{8} \sum_{i=1}^8 X_i$ 替换 EX , 即有矩估计量 $\hat{\theta}_{ME} = \frac{3-\bar{X}}{4}$; 即, 矩估计量 $\hat{\theta}_{ME} = \frac{3-\bar{X}}{4} = \frac{3-2}{4} = \frac{1}{4}$

(2) 求最大似然估计 $\hat{\theta}_{MLE}$: 考虑 θ 的似然函数 $L(\theta) = P(X_1=3, X_2=1, X_3=3, X_4=0, X_5=3, X_6=1,$

$$X_7=2, X_8=3) = P(X_1=3) \cdot P(X_2=1) \cdot P(X_3=3) \cdot P(X_4=0) \cdot P(X_5=3) \cdot P(X_6=1) \cdot P(X_7=2) \cdot P(X_8=3);$$

$$= (1-2\theta)^4 \cdot [2\theta(1-\theta)]^2 \cdot \theta^2 \cdot \theta^2 = 4 \cdot \theta^6 \cdot (1-\theta)^2 \cdot (1-2\theta)^4.$$

$$\text{由 } \ln L(\theta) = \ln 4 + 6 \ln \theta + 2 \ln(1-\theta) + 4 \ln(1-2\theta), \quad \frac{d \ln L(\theta)}{d\theta} = \frac{6}{\theta} - \frac{2}{1-\theta} - \frac{8}{1-2\theta} = 0, \text{ 即:}$$

$$8\left(\frac{1}{\theta} - \frac{1}{1-2\theta}\right) = 2\left(\frac{1}{\theta} + \frac{1}{1-\theta}\right), \text{ 也即: } 4 \cdot \frac{1-3\theta}{\theta(1-2\theta)} = \frac{1}{\theta(1-\theta)}, \text{ 即得 } \hat{\theta}_{MLE} = \frac{7-\sqrt{13}}{12}$$

5. (1) 设总体为 X , 且 $X \sim \left(\frac{1}{R+1} \right)$ $X = \begin{cases} 1 & \text{取出白球} \\ 0 & \text{取出黑球} \end{cases}$

$X \sim \left(\frac{1}{R+1} \right)$, 设 $(X_1, X_2, \dots, X_n)$ 为取自 X 的一个样本, $(x_1, x_2, \dots, x_n)$ 为样本值.

5. (1) 一个袋有白球与黑球, 有放回地抽取一个容量为 n 的样本, 其中有 k 个白球, 试求袋中黑球数与白球数之比 R 的最大似然估计.

考虑 R 的似然函数 $L(R) = P(X_1, X_2, \dots, X_n = (x_1, x_2, \dots, x_n)) = \prod_{i=1}^n P(X_i = x_i) = \prod_{i=1}^n P(X = x_i)$

$$= \prod_{i=1}^n \left(\frac{1}{R+1} \right)^{x_i} \left(\frac{R}{R+1} \right)^{1-x_i}, \quad x_i = 0 \text{ 或 } 1.$$

$$= \frac{R^{n-\sum_{i=1}^n x_i}}{(R+1)^n}, \quad \ln L(R) = (n - \sum_{i=1}^n x_i) \ln R - n \ln(R+1), \quad \text{令 } \frac{d \ln L(R)}{dR} = \frac{n - \sum_{i=1}^n x_i}{R} - \frac{n}{R+1} = 0$$

$$\text{即 } R \text{ 的最大似然估计值 } \hat{R}_{MLE} = \frac{1}{\bar{x}} - 1, \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i; \quad \text{从而,}$$

$$R \text{ 的最大似然估计量 } \hat{R}_{MLE} = \frac{1}{\bar{x}} - 1, \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

(2) 甲、乙两人独立地校对一书稿的清样, 他们分别发现了 k_1, k_2 个错误, 其中

有 $k_{12} > 0$ 个错误是共同的, 求总的错误个数 n 的估计. 设清样中有 n 个错误, 校对者读到有

错误处可能发现错误, 也可能发现不了; 令甲、乙发现错误的概率为 p_1, p_2 , 且甲、乙发现共同错误的概率为 p_{12} ; 记 X, Y 为甲、乙发现清样的错误数; 易见, $X \sim B(n, p_1), Y \sim B(n, p_2)$. 令 Z 为甲、乙发现相同错误数, 即有: $Z \sim B(n, p_{12})$; 由于甲、乙两人是独立校对的, 故 $p_{12} = p_1 \cdot p_2$.

于是由 $EZ = np_{12}$, 即有: $n = \frac{EX \cdot EY}{EZ}$; 由矩估计法, 即有 n 的矩估计:

$$\hat{n} = \frac{\bar{X} \cdot \bar{Y}}{\bar{Z}}; \quad \text{视 } k_1, k_2, k_{12} \text{ 为取自总体 } X, Y, Z \text{ 的容量为 } 1 \text{ 的样本值, 即:}$$

$$\bar{x} = k_1, \bar{y} = k_2, \bar{z} = k_{12}, \quad \text{即有 } n \text{ 的矩估计值: } \hat{n} = \frac{k_1 k_2}{k_{12}}.$$

6. 设总体 X 的概率密度为 $f(x; \theta) = \begin{cases} \frac{\theta^2}{x^3} e^{-\frac{\theta}{x}}, & x > 0; \\ 0, & x \leq 0; \end{cases}$ 其中 $\theta > 0$ 未知,

$(X_1, X_2, \dots, X_n)$ 为取自总体 X 的简单随机样本,

(1) 求 θ 的矩估计量; (2) 求 θ 的最大似然估计量.

(1) 易见, $EX = \int_0^{\infty} x f(x; \theta) dx = \int_0^{\infty} \frac{\theta^2}{x^2} e^{-\frac{\theta}{x}} dx = \theta e^{-\frac{\theta}{x}} \Big|_{0^+}^{\infty} = \theta$, 即: $\theta = EX$, 用

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n x_i \text{ 替换 } EX, \text{ 即有 } \theta \text{ 的矩估计量 } \hat{\theta} \text{ (或 } \hat{\theta}_{ME}) = \bar{X};$$

(2) 设 $(x_1, x_2, \dots, x_n)$ 为样本值, 考虑 θ 的似然函数

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta) = \left(\frac{\theta^2}{\prod_{i=1}^n x_i^3} \right) e^{-\theta \sum_{i=1}^n \frac{1}{x_i}}, \quad x_i > 0,$$

$$\ln L(\theta) = 2n \ln \theta - 3 \sum_{i=1}^n \ln x_i - \theta \cdot \sum_{i=1}^n \frac{1}{x_i}, \quad \text{令 } \frac{d \ln L(\theta)}{d\theta} = \frac{2n}{\theta} - \sum_{i=1}^n \frac{1}{x_i} = 0$$

$$\text{即有 } \theta \text{ 的最大似然估计值 } \hat{\theta} \text{ (或 } \hat{\theta}_{MLE}) = \frac{2n}{\sum_{i=1}^n \frac{1}{x_i}}, \quad \text{从而,}$$

$$\theta \text{ 的最大似然估计量 } \hat{\theta} \text{ (或 } \hat{\theta}_{MLE}) = \frac{2n}{\sum_{i=1}^n \frac{1}{X_i}}$$

7. 设总体 X 的概率密度为 $f(x; \theta) = \begin{cases} \frac{1}{1-\theta}, & \theta \leq x \leq 1; \\ 0, & \text{其他;} \end{cases}$ 其中 θ 为未知参数,

$(X_1, X_2, \dots, X_n)$ 为取自总体 X 的简单随机样本,

(1) 求 θ 的矩估计量; (2) 求 θ 的最大似然估计量.

(1) 由 $EX = \int_{-\infty}^{+\infty} x \cdot f(x; \theta) dx = \int_{\theta}^1 x \cdot \frac{1}{1-\theta} dx = \frac{1+\theta}{2}$, 即: $\theta = 2EX - 1$; 用 $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ 替换 EX , 即有 θ 的矩估计量 $\hat{\theta}$ (或 $\hat{\theta}_{ME}$) $= 2\bar{X} - 1$;

(2) 设 $(x_1, x_2, \dots, x_n)$ 为样本值, 考虑 θ 的似然函数, $L(\theta) = \prod_{i=1}^n f(x_i; \theta) = \frac{1}{(1-\theta)^n}$, $\theta \leq x_i \leq 1$
 $= \frac{1}{(1-\theta)^n}$, $\theta \leq x_{(n)}$; $L(\theta)$ 越大, 则 θ 越大, 故 $\hat{\theta}_{MLE} = x_{(n)}$; 从而最大似然估计量
 $\hat{\theta}_{MLE} = X_{(n)}$; 这里, $X_{(n)} = \max\{X_1, X_2, \dots, X_n\}$, $X_{(n)} = \min\{X_1, X_2, \dots, X_n\}$.

8. 设总体 X 的概率密度为 $f(x; \sigma^2) = \begin{cases} \frac{A}{\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, & x \geq \mu; \\ 0, & x < \mu; \end{cases}$ 其中 μ 是已知参数,

$\sigma > 0$ 是未知参数, A 是常数, $(X_1, X_2, \dots, X_n)$ 为取自总体 X 的简单随机样本,

(1) 求 A ; (2) 求 σ^2 的最大似然估计量.

(1) 由 $\int_{-\infty}^{+\infty} f(x; \sigma^2) dx = \int_{\mu}^{+\infty} \frac{A}{\sigma} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} dx = A \cdot \int_{\mu}^{+\infty} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} d(\frac{x-\mu}{\sigma}) \xrightarrow{\text{令 } \frac{x-\mu}{\sigma} = t} A \cdot \int_0^{+\infty} e^{-\frac{1}{2}t^2} dt$
 $= \frac{A}{2} \cdot \sqrt{2\pi} \cdot \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{t^2}{2}} dt = A \cdot \frac{\sqrt{2}}{2} = 1$, 即有: $A = \frac{\sqrt{2}}{2}$.

(2) 设 $(x_1, x_2, \dots, x_n)$ 为样本值, 考虑 σ^2 的似然函数,

$$L(\sigma^2) = \prod_{i=1}^n f(x_i; \sigma^2) = A^n (\sigma^2)^{-\frac{n}{2}} \cdot e^{-\frac{\sum_{i=1}^n (x_i - \mu)^2}{2\sigma^2}},$$

$$\ln L(\sigma^2) = n \cdot \ln A - \frac{n}{2} \cdot \ln(\sigma^2) - \frac{\sum_{i=1}^n (x_i - \mu)^2}{2\sigma^2}; \quad \text{令 } \frac{d \ln L(\sigma^2)}{d \sigma^2}$$

$$= -\frac{n}{2} \cdot \frac{1}{\sigma^2} + \frac{1}{2} \cdot \frac{\sum_{i=1}^n (x_i - \mu)^2}{\sigma^4} = 0, \quad \text{即有: } \sigma^2 \text{ 的最大似然估计量}$$

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2; \quad \text{从而, } \sigma^2 \text{ 的最大似然估计量 } \hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$$

9. (1) 设 $(x_1, x_2, \dots, x_n)$ 为样本值, 考虑似然函数 $L(\theta) = \prod_{i=1}^n f(x_i; \theta) = \frac{1}{2^n \theta^n} \cdot e^{-\frac{1}{\theta} \sum_{i=1}^n |x_i|}$,
 $\ln L(\theta) = -n \ln 2 - n \ln \theta - \frac{1}{\theta} \sum_{i=1}^n |x_i|$; 令 $\frac{d \ln L(\theta)}{d\theta} = -\frac{n}{\theta} + \frac{1}{\theta^2} \sum_{i=1}^n |x_i| = 0$, 即有: θ 的最大似然估计
 值 $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n |x_i|$; 从而, θ 的最大似然估计量 $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n |X_i|$;

(2) $E(\hat{\theta}) = \frac{1}{n} \sum_{i=1}^n E|X_i|$. 设总体 X 的概率密度为: $f(x; \sigma) = \frac{1}{2\sigma} e^{-\frac{|x|}{\sigma}}, -\infty < x < +\infty$, 其中 $\sigma \in (0, +\infty)$

为未知参数, $(X_1, X_2, \dots, X_n)$ 为取自总体 X 的简单随机样本; 记 σ 的最大似然估计为 $\hat{\sigma}$, (1) 求 $\hat{\sigma}$; (2) 求 $E(\hat{\sigma}), D(\hat{\sigma})$; (3) 证明 $\hat{\sigma}$ 是 σ 的相合

估计量.
 $= \int_{-\infty}^{+\infty} |x| \cdot \frac{1}{2\sigma} e^{-\frac{|x|}{\sigma}} dx$
 $= \sigma \int_0^{+\infty} \frac{x}{\sigma} \cdot e^{-\frac{x}{\sigma}} d(\frac{x}{\sigma}) \xrightarrow{\text{令 } \frac{x}{\sigma} = t} \sigma \cdot \int_0^{+\infty} t \cdot e^{-t} dt = \sigma \cdot \Gamma(2) = \sigma$, $D(\hat{\theta}) = \frac{1}{n} \sum_{i=1}^n D|X_i| = \frac{1}{n} \sum_{i=1}^n D|X_i| = \frac{1}{n} D(|X|)$,
 这里, $E(|X|^2) = E(X^2) = \int_{-\infty}^{+\infty} x^2 \cdot f(x; \sigma) dx = \int_{-\infty}^{+\infty} x^2 \cdot \frac{1}{2\sigma} e^{-\frac{|x|}{\sigma}} dx = \sigma^2 \int_0^{+\infty} (\frac{x}{\sigma})^2 \cdot e^{-\frac{x}{\sigma}} d(\frac{x}{\sigma}) \xrightarrow{\text{令 } \frac{x}{\sigma} = t} \sigma^2 \int_0^{+\infty} t^2 \cdot e^{-t} dt$
 $= \sigma^2 \cdot \Gamma(3) = 2\sigma^2$, 从而, $D(\hat{\theta}) = \frac{1}{n} D(|X|) = \frac{1}{n} [E(|X|^2) - (E|X|)^2] = \frac{1}{n} \sigma^2$;

(3) 由(2), $\forall \varepsilon > 0$, $P(|\hat{\theta} - \theta| \geq \varepsilon) \leq \frac{D(\hat{\theta})}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0, n \rightarrow \infty$, 即:

$\lim_{n \rightarrow \infty} P(|\hat{\theta} - E(\hat{\theta})| \geq \varepsilon) = 0$, 也即: $\hat{\theta} \xrightarrow{P} E(\hat{\theta}) = \theta$, 即:

$\hat{\theta}$ 是 θ 的相合(一致)估计量!

10. 某工程师为了解一台天平的精度, 用该天平对一物体的质量做 n 次测量, 该物体的质量 μ 是已知的, 设 n 次测量结果 $X_1, X_2, \dots, X_n$ 相互独立且均服从正态分布 $N(\mu, \sigma^2)$. 该工程师记录的是 n 次测量的绝对误差 $Z_i = |X_i - \mu| (i=1, 2, \dots, n)$, 利用 $Z_1, Z_2, \dots, Z_n$ 估计 σ ;

(1) 求 Z_1 的概率密度; (2) 利用一阶矩求 σ 的矩估计量;

(3) 求 σ 的最大似然估计量. (1) $\forall z \leq 0, F_{Z_1}(z) = P(Z_1 \leq z) = 0$; $\forall z > 0, F_{Z_1}(z) = P(Z_1 \leq z) = P(|X_1 - \mu| \leq z) = P(-\frac{z}{\sigma} \leq \frac{X_1 - \mu}{\sigma} \leq \frac{z}{\sigma}) = 2\phi(\frac{z}{\sigma}) - 1$; 即: $F_Z(z) = \begin{cases} 0, & z \leq 0; \\ 2\phi(\frac{z}{\sigma}) - 1, & z > 0; \end{cases}$ 令 $Z_1 \sim f_{Z_1}(z)$, 即:
 $f_{Z_1}(z) = \frac{d}{dz} F_{Z_1}(z) = \begin{cases} 0, & z \leq 0; \\ \frac{2}{\sqrt{2\pi}\sigma} e^{-\frac{z^2}{2\sigma^2}}, & z > 0. \end{cases}$ (2) 视 $(Z_1, Z_2, \dots, Z_n)$ 为取自总体 $Z \sim f_Z(z)$ 的样本,

由 $EZ = \int_{-\infty}^{+\infty} z f_Z(z) dz = \int_0^{+\infty} z \cdot \frac{2}{\sqrt{2\pi}\sigma} e^{-\frac{z^2}{2\sigma^2}} dz \xrightarrow{\frac{z^2}{2\sigma^2} = x} \frac{\sqrt{2}}{\sigma} \cdot \int_0^{+\infty} x e^{-x} dx \xrightarrow{\text{令 } x = \frac{t^2}{2}} \frac{\sqrt{2}}{\sigma} \cdot \int_0^{+\infty} \frac{t^2}{2} e^{-\frac{t^2}{2}} \cdot t dt$
 $= \frac{\sqrt{2}}{\sigma} \cdot (-\frac{1}{2} e^{-\frac{t^2}{2}}) \Big|_0^{+\infty} = \frac{\sqrt{2}}{\sigma}$, 即: $\sigma = \sqrt{2} EZ$, 用 $\bar{Z} = \frac{1}{n} \sum_{i=1}^n Z_i$ 替换 EZ , 即有 σ 的矩估计量

$\hat{\sigma}_{ME} = \sqrt{2} \bar{Z}$; (3) 设 $(Z_1, Z_2, \dots, Z_n)$ 为取自 Z 的样本值, 考虑 σ 的似然函数 $L(\sigma) = \prod_{i=1}^n f(Z_i; \sigma)$

这里, $f(Z_i; \sigma) = f_Z(Z_i) = \begin{cases} 0, & Z_i \leq 0; \\ \frac{2}{\sqrt{2\pi}\sigma} e^{-\frac{Z_i^2}{2\sigma^2}}, & Z_i > 0. \end{cases}$ 即有: $L(\sigma) = \prod_{i=1}^n \frac{2}{\sqrt{2\pi}\sigma} e^{-\frac{Z_i^2}{2\sigma^2}}, Z_i > 0$
 $= (\frac{2}{\sqrt{2\pi}})^n \cdot \frac{1}{\sigma^n} \cdot e^{-\frac{\sum_{i=1}^n Z_i^2}{2\sigma^2}}$, $\ln L(\sigma) = \frac{n}{2} \ln(\frac{2}{\sqrt{2\pi}}) - n \ln \sigma - \frac{\sum_{i=1}^n Z_i^2}{2\sigma^2}$, 令 $\frac{d \ln L(\sigma)}{d\sigma} = -\frac{n}{\sigma} + \frac{\sum_{i=1}^n Z_i^2}{\sigma^3} = 0$.

即有: σ 的最大似然估计值: $\hat{\sigma}_{MLE} = \sqrt{\frac{1}{n} \sum_{i=1}^n Z_i^2}$; 从而, σ 的最大似然估计量为

$\hat{\sigma}_{MLE} = \sqrt{\frac{1}{n} \sum_{i=1}^n Z_i^2}$.

11. 设 $X \sim f(x; \theta)$, 即有: $f(x; \theta) = \frac{d}{dx} F(x; \theta) = \begin{cases} \frac{2x}{\theta} e^{-\frac{x^2}{\theta}}, & x > 0; \\ 0, & x \leq 0. \end{cases}$

$$(1) EX = \int_{-\infty}^{\infty} x \cdot f(x; \theta) dx = \int_0^{\infty} x \cdot \frac{2x}{\theta} e^{-\frac{x^2}{\theta}} dx \xrightarrow{\text{令 } x = \sqrt{\theta} t} \int_0^{\infty} 2t \cdot e^{-t^2} \cdot \sqrt{\theta} \cdot \frac{1}{\sqrt{\theta}} dt = \sqrt{\theta} \cdot \int_0^{\infty} t^2 \cdot e^{-t^2} dt$$

$$= \sqrt{\theta} \cdot \Gamma(\frac{3}{2}) = \sqrt{\theta} \cdot \frac{1}{2} \cdot \sqrt{\pi} = \frac{1}{2} \sqrt{\pi \theta};$$

11. 设总体 X 的分布函数为 $F(x; \theta) = \begin{cases} 1 - e^{-\frac{x^2}{\theta}}, & x > 0; \\ 0, & x \leq 0; \end{cases}$ 其中 $\theta > 0$ 未知,

$(X_1, X_2, \dots, X_n)$ 为取自总体 X 的简单随机样本,

(1) 求 $EX, E(X^2)$; (2) 求 θ 的最大似然估计量 $\hat{\theta}_n$;

$$EX^2 = \int_{-\infty}^{\infty} x^2 \cdot f(x; \theta) dx = \int_0^{\infty} x^2 \cdot \frac{2x}{\theta} e^{-\frac{x^2}{\theta}} dx \xrightarrow{\text{令 } x = \sqrt{\theta} t} \int_0^{\infty} \sqrt{\theta} t \cdot 2t \cdot e^{-t^2} \cdot \sqrt{\theta} \cdot \frac{1}{\sqrt{\theta}} dt = \theta \cdot \int_0^{\infty} t^2 e^{-t^2} dt = \theta \cdot \Gamma(2)$$

$$= \theta;$$

(2) 设 $(x_1, x_2, \dots, x_n)$ 为样本值, 考虑 θ 的似然函数 $L(\theta) = \prod_{i=1}^n f(x_i; \theta) = \frac{2^n x_1 x_2 \dots x_n}{\theta^n} e^{-\frac{1}{\theta} \sum_{i=1}^n x_i^2}$,

$$\ln L(\theta) = n \ln 2 + \ln(x_1 x_2 \dots x_n) - n \ln \theta - \frac{1}{\theta} \sum_{i=1}^n x_i^2, \text{ 令 } \frac{d \ln L(\theta)}{d \theta} = -\frac{n}{\theta} + \frac{1}{\theta^2} \sum_{i=1}^n x_i^2 = 0, \text{ 即有 } \theta \text{ 的}$$

最大似然估计值 $\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n x_i^2$; 从而, θ 的最大似然估计量 $\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n X_i^2$;

(2) 是否存在实数 a , 使得 $\forall \varepsilon > 0$, 都有 $\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - a| \geq \varepsilon) = 0$.

$$(3) \text{ 由 } E(\hat{\theta}_n) = E(\frac{1}{n} \sum_{i=1}^n X_i^2) = \frac{1}{n} \sum_{i=1}^n E(X_i^2) = \frac{1}{n} \sum_{i=1}^n EX^2 = EX^2 = \theta, D(\hat{\theta}_n) = D(\frac{1}{n} \sum_{i=1}^n X_i^2) = \frac{1}{n^2} \sum_{i=1}^n D(X_i^2)$$

$$= \frac{1}{n^2} \sum_{i=1}^n D(X^2) = \frac{1}{n} D(X^2), \text{ 这里, } EX^4 = \int_{-\infty}^{\infty} x^4 \cdot f(x; \theta) dx = \int_0^{\infty} x^4 \cdot \frac{2x}{\theta} e^{-\frac{x^2}{\theta}} dx \xrightarrow{\text{令 } x = \sqrt{\theta} t}$$

$$\int_0^{\infty} \theta^2 t^2 \cdot e^{-t^2} dt = \theta^2 \Gamma(3) = 2\theta^2, D(X^2) = EX^4 - (EX^2)^2 = \theta^2, \text{ 由 Chebyshev 不等式, } \forall \varepsilon > 0,$$

$$P(|\hat{\theta}_n - \theta| \geq \varepsilon) = P(|\hat{\theta}_n - E(\hat{\theta}_n)| \geq \varepsilon) \leq \frac{D(\hat{\theta}_n)}{\varepsilon^2} = \frac{\theta^2}{n \varepsilon^2} \rightarrow 0, n \rightarrow \infty, \text{ 即: } \hat{\theta}_n \xrightarrow{P} a = \theta$$

或由 Khinchine 大数定律, $X_1^2, X_2^2, \dots, X_n^2$ 独立同分布, 且 $EX_i^2 = EX^2 = \theta$, 即有:

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n X_i^2 \xrightarrow{P} EX^2 = \theta,$$

1. 从正态总体 $N(3.4, 6^2)$ 中抽取一容量为 n 的样本, 如果要求其样本均值位于

$(1.4, 5.4)$ 内的概率不小于 0.95, 问: 样本容量 n 至少应为多大?

易见, 样本均值 $\bar{X} \sim N(3.4, \frac{6^2}{n})$, 由 $P(\bar{X} \in (1.4, 5.4)) = P(-2 < \bar{X} - 3.4 < 2)$

$$= P(-\frac{2}{\frac{6}{\sqrt{n}}} < \frac{\bar{X} - 3.4}{\frac{6}{\sqrt{n}}} < \frac{2}{\frac{6}{\sqrt{n}}}) = \Phi(\frac{\sqrt{n}}{3}) - \Phi(-\frac{\sqrt{n}}{3}) = 2\Phi(\frac{\sqrt{n}}{3}) - 1 \geq 0.95, \text{ 即: } \Phi(\frac{\sqrt{n}}{3}) \geq 0.975 = \Phi(1.96)$$

$$\text{即有: } \frac{\sqrt{n}}{3} \geq 1.96, n \geq (3 \times 1.96)^2, \text{ 也即有: } n \geq 35.$$

2. 已知一批零件的长度 (单位: cm) 服从正态分布 $N(\mu, 1^2)$, 从中随机抽取 16

个, 得到长度的平均值为 40cm, 试求 μ 的置信度为 0.95 的置信区间.

易见, 样本均值 $\bar{X} \sim N(\mu, \frac{1}{16})$, 即有: $\frac{\bar{X} - \mu}{\frac{1}{4}} \sim N(0, 1^2)$; 对于给定的置信度 $1 - \alpha = 0.95$

$$\text{由 } P(|\frac{\bar{X} - \mu}{\frac{1}{4}}| < u_{0.975}) = P(\bar{X} - \frac{1}{4} u_{0.975} < \mu < \bar{X} + \frac{1}{4} u_{0.975}) = 0.95, \text{ 即: 随机区间 } (\bar{X} - \frac{1}{4} u_{0.975}, \bar{X} + \frac{1}{4} u_{0.975})$$

以 0.95 的概率包含未知参数 μ 的真值, 即为 μ 的置信度为 0.95 的置信区间.

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$$\text{对于 } \bar{x} = 40, u_{0.975} = 1.96, \text{ 即得置信区间 } (40 - \frac{1}{4} \cdot 1.96, 40 + \frac{1}{4} \cdot 1.96) = (40 - 0.49, 40 + 0.49).$$

4. 设总体 $X \sim N(\mu, 2.8^2)$, 现抽取一容量为10的样本, 且有 $\bar{x} = \frac{1}{10} \sum_{i=1}^{10} x_i = 1500$;

(1) 试求 μ 的置信水平为0.95的置信区间; 由 $\bar{X} \sim N(\mu, \frac{2.8^2}{n})$, 且

$$P(u_{0.05} < \frac{\bar{X} - \mu}{\frac{2.8}{\sqrt{n}}} < u_{0.05}) = 0.95, \text{ 即有: } \mu \text{ 的 } 0.95 \text{ 的置信区间 } (\bar{X} - \frac{2.8}{\sqrt{n}} u_{0.05}, \bar{X} + \frac{2.8}{\sqrt{n}} u_{0.05}),$$

$$u_{0.05} = 1.96, \bar{x} = 1500, \text{ 即得置信水平 } (1500 - \frac{2.8}{\sqrt{10}} \cdot 1.96, 1500 + \frac{2.8}{\sqrt{10}} \cdot 1.96)$$

(2) 欲使置信水平为0.95的置信区间长度小于1, 样本容量 n 至少为多少?

由 $\bar{X} \sim N(\mu, \frac{2.8^2}{n})$, 且 $P(u_{0.05} < \frac{\bar{X} - \mu}{\frac{2.8}{\sqrt{n}}} < u_{0.05}) = 0.95$, 欲使 μ 的0.95的置信区间 $(\bar{X} - \frac{2.8}{\sqrt{n}} u_{0.05}, \bar{X} + \frac{2.8}{\sqrt{n}} u_{0.05})$ 的长度 $2 \cdot \frac{2.8}{\sqrt{n}} u_{0.05} = 2 \cdot \frac{2.8}{\sqrt{n}} \cdot 1.96 < 1$, 则须 $n \geq 121$.

(3) 若 $n=100$, 随机区间 $(\bar{X} - \frac{1}{2}, \bar{X} + \frac{1}{2})$ 作为 μ 的置信区间, 其置信水平为多

少? $n=100, \bar{X} \sim N(\mu, \frac{2.8^2}{100})$, 即有:

$$P(\bar{X} - \frac{1}{2} < \mu < \bar{X} + \frac{1}{2}) = P(-\frac{1}{2} < \bar{X} - \mu < \frac{1}{2}) = P(\frac{-\frac{1}{2}}{\frac{2.8}{10}} < \frac{\bar{X} - \mu}{\frac{2.8}{10}} < \frac{\frac{1}{2}}{\frac{2.8}{10}}) = \Phi(\frac{5}{2.8}) - \Phi(-\frac{5}{2.8}) \\ = 2\Phi(\frac{5}{2.8}) - 1 = 0.925.$$

5. 设总体 X 有概率密度 $f(x; \theta) = \begin{cases} e^{-(x-\theta)}, & x > \theta; \\ 0, & x \leq \theta; \end{cases} -\infty < \theta < +\infty; (X_1, X_2, \dots, X_n)$

为取自该总体的样本;

(1) 证明: $X_{(1)} - \theta$ 的分布与 θ 无关, 并求此分布;

$$\forall x \leq 0, F(x) = P(X_{(1)} - \theta \leq x) = 0; \forall x > 0, F(x) = P(X_{(1)} - \theta \leq x) = P(X_{(1)} \leq \theta + x) \\ = 1 - P(X_{(1)} > \theta + x) = 1 - P(\bigwedge_{i=1}^n X_i > \theta + x) = 1 - P(\bigcap_{i=1}^n \{X_i > \theta + x\}) = 1 - \prod_{i=1}^n P(X_i > \theta + x) \\ = 1 - \prod_{i=1}^n \int_{\theta+x}^{+\infty} e^{-(x-\theta)} dx = 1 - e^{-nx}; \text{ 即: } F(x) = \begin{cases} 1 - e^{-nx} & , x > 0. \\ 0 & , x \leq 0. \end{cases} \text{ 也即:}$$

$$X_{(1)} - \theta \sim E(n).$$

(2) 求 θ 的置信水平为 $1-\alpha$ 的置信区间.

选择 $X_{(1)} - \theta$ 作为区间估计的枢轴量, 由 $P(0 < X_{(1)} - \theta < \frac{1}{n} \ln \alpha) = F(\frac{1}{n} \ln \alpha) \\ = 1 - e^{-n \cdot (\frac{1}{n} \ln \alpha)} = 1 - \alpha$, 即: $P(X_{(1)} + \frac{1}{n} \ln \alpha < \theta < X_{(1)}) = 1 - \alpha$, 即有 θ 的置信水平为 $1-\alpha$ 的置信区间: $(X_{(1)} + \frac{1}{n} \ln \alpha, X_{(1)})$.

